

*Life Histories of North American
Cuckoos, Goatsuckers, Hummingbirds,
and Their Allies*

by
Arthur Cleveland Bent

in two parts

PART I

DOVER PUBLICATIONS, INC.

NEW YORK

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This Dover edition, first published in 1964, is an unabridged and unaltered republication of the work first published in 1940 by the United States Government Printing Office, as Smithsonian Institution United States National Museum *Bulletin 176*.

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ADVERTISEMENT

The scientific publications of the National Museum include two series, known, respectively, as *Proceedings* and *Bulletin*.

The *Proceedings* series, begun in 1878, is intended primarily as a medium for the publication of original papers, based on the collections of the National Museum, that set forth newly acquired facts in biology, anthropology, and geology, with descriptions of new forms and revisions of limited groups. Copies of each paper, in pamphlet form, are distributed as published to libraries and scientific organizations and to specialists and others interested in the different subjects. The dates at which these separate papers are published are recorded in the table of contents of each of the volumes.

The series of *Bulletins*, the first of which was issued in 1875, contains separate publications comprising monographs of large zoological groups and other general systematic treatises (occasionally in several volumes), faunal works, reports of expeditions, catalogs of type specimens, special collections, and other material of similar nature. The majority of the volumes are octavo in size, but a quarto size has been adopted in a few instances in which large plates were regarded as indispensable. In the *Bulletin* series appear volumes under the heading *Contributions from the United States National Herbarium*, in octavo form, published by the National Museum since 1902, which contain papers relating to the botanical collections of the Museum.

The present work forms No. 176 of the *Bulletin* series.

ALEXANDER WETMORE,
Assistant Secretary, Smithsonian Institution.

WASHINGTON, D. C., May 3, 1940.

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INTRODUCTION

This is the thirteenth in a series of bulletins of the United States National Museum on the life histories of North American birds. Previous numbers have been issued as follows:

- 107. Life Histories of North American Diving Birds, August 1, 1919.
- 113. Life Histories of North American Gulls and Terns, August 27, 1921.
- 121. Life Histories of North American Petrels and Pelicans and their Allies, October 19, 1922.
- 126. Life Histories of North American Wild Fowl (part), May 25, 1923.
- 130. Life Histories of North American Wild Fowl (part), June 27, 1925.
- 135. Life Histories of North American Marsh Birds, March 11, 1927.
- 142. Life Histories of North American Shore Birds (pt. 1), December 31, 1927.
- 146. Life Histories of North American Shore Birds (pt. 2), March 24, 1929.
- 162. Life Histories of North American Gallinaceous Birds, May 25, 1932.
- 167. Life Histories of North American Birds of Prey (pt. 1), May 3, 1937.
- 170. Life Histories of North American Birds of Prey (pt. 2), August 8, 1938.
- 174. Life Histories of North American Woodpeckers, May 23, 1939.

The same general plan has been followed, as explained in previous bulletins, and the same sources of information have been utilized. The

nomenclature of the 1931 check-list of the American Ornithologists' Union has been followed.

An attempt has been made to give as full a life history as possible of the best-known subspecies of each species and to avoid duplication by writing briefly of the others, giving only the characters of the subspecies, its range, and any habits peculiar to it. In many cases certain habits, probably common to the species as a whole have been recorded for only one subspecies; such habits are mentioned under the subspecies on which the observations were made. The distribution gives the range of the species as a whole, with only rough outlines of the ranges of the subspecies, which in many cases cannot be accurately defined.

The egg dates are the condensed results from a mass of records taken from the data in a large number of the best collections in the country, as well as from contributed field notes and from a few published sources. They indicate the dates on which eggs have been found in various parts of the country, showing the earliest and latest dates and the limits between which half the dates fall, indicating the height of the season.

The plumages are described in only enough detail to enable the reader to trace the sequence of molts and plumages from birth to maturity and to recognize the birds in the different stages and at the different seasons.

No attempt has been made to describe fully the adult plumages; this has been well done already in the many manuals. The names of colors, when in quotation marks, are taken from Ridgway's Color Standards and Nomenclature (1912). In the measurements of eggs the four extremes are printed in boldface type.

Many who have contributed material for previous volumes have continued to cooperate. Receipt of material from over 450 contributors has been acknowledged previously. In addition to these, our thanks are due to the following new contributors: E. C. Aldrich, Mrs. H. P. Bracelin, Maurice Brooks, Mildred Campbell, R. T. Congdon, Murl Deusing, S. S. Dickey, H. E. Edgerton, W. G. F. Harris, L. B. Howsley, Charles Macnamara, R. F. Mason, Jr., R. T. Moore, A. R. Phillips, A. J. Pinckney, O. P. Silliman, A. F. Skutch, Emily Smith, W. P. Steinbeck, R. R. Talmadge, H. O. Todd, Jr., B. P. Tyler, L. O. Williams, G. R. Wilson, and H. B. Wood. As the demand for these bulletins is greater than the supply, the names of those who have not

contributed to the work during the previous ten years may be dropped from the author's mailing list.

Dr. Winsor M. Tyler rendered valuable assistance in reading and indexing, for these groups, a large part of the literature on North American birds, and contributed three complete life histories. E. C. Stuart Baker and Alexander F. Skutch each contributed two complete life histories; and Dr. Alfred O. Gross, the Rev. F. C. R. Jourdain, Alexander Sprunt, Jr., Dr. George M. Sutton, and Robert S. Woods contributed one each. Our thanks are also due F. Seymour Hersey for figuring egg measurements. Egg measurements were furnished, especially for this volume, by Dean Amadon, A. M. Bailey, American Museum of Natural History, Griffing Bancroft, R. M. Barnes, I. McT. Cowan, H. G. Deignan, C. E. Doe, J. H. Gillin, W. C. Hanna, R. C. Harlow, R. T. Moore, R. T. Orr, P. B. Philipp, M. S. Ray, J. H. Riley, G. H. Stuart, 3d, and Miss M. W. Wythe.

Through the courtesy of the Bureau of Biological Survey, the services of Frederick C. Lincoln were again obtained to compile the distribution paragraphs. With the matchless files of the Survey at his disposal, his many hours of careful work have produced results far more satisfactory than could have been attained by the author, who claims no credit and assumes no responsibility for this part of the work.

The manuscript for this bulletin was completed in March 1939. Contributions received since then will be acknowledged later. Only information of great importance could be added. The reader is reminded again that this is a cooperative work; if he fails to find in these volumes anything that he knows about the birds, he can blame himself for not having sent the information to—

THE AUTHOR.

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LIFE HISTORIES OF NORTH AMERICAN CUCKOOS, GOAT-SUCKERS, HUMMINGBIRDS, AND THEIR ALLIES.

ORDERS PSITTACIFORMES, CUCULIFORMES, TROGONIFORMES, CORACIIFORMES,
CAPRIMULGIFORMES, AND MICROPODIIFORMES

By ARTHUR CLEVELAND BENT

Taunton, Mass.

Order PSITTACIFORMES

Family PSITTACIDAE: Parrots and Parakeets

CONUOPSIS CAROLINENSIS CAROLINENSIS (Linnaeus)

CAROLINA PARAKEET

HABITS

MANY of the glories of North American bird life have gone, never to return. The spread of civilization, the selfish greed of human interests, and the lust to kill have wiped out some of the most spectacular and beautiful features in our formerly abundant bird life. The countless millions of passenger pigeons that formerly darkened the sky in their seasonal migrations are gone forever. And the great flocks of gorgeous parakeets that formerly roamed over nearly all the eastern part of our country will be seen no more. This was the only representative of the parrot family that lived and bred within the United States; it gave a touch of tropical character to our avifauna and a vivid tinge of color to the landscape; its loss is much to be regretted. Never again may be seen the glorious sights witnessed by Wilson, Audubon, and other early writers, as great flocks of these gorgeous birds

wheeled through the air, in close formation, their long tails streaming out in straight flight or spreading as they turned, and their brilliant colors, red, yellow, bright green, and soft blue, gleaming in the sunlight. As Wilson (1832) says: "They came screaming through the woods in the morning, about an hour after sunrise, to drink the salt water, of which they, as well as the pigeons, are remarkably fond. When they alighted on the ground, it appeared at a distance as if covered with a carpet of richest green, orange, and yellow: they afterwards settled, in one body, on a neighbouring tree, which stood detached from any other, covering almost every twig of it, and the sun, shining strongly on their gay and glossy plumage, produced a very beautiful and splendid appearance."

Conuropsis carolinensis, as a species, covered a wide range in eastern North America, from the vicinity of the Great Lakes southward to Florida and the Gulf States, and from Colorado (rarely) to the Atlantic coast. For a full account, given in detail, of the former range of the species and its gradual disappearance, the reader is referred to a comprehensive article on the subject by Edwin M. Hasbrouck (1891). As the species has been divided into two subspecies since this was written, we shall consider here only the former distribution of the eastern race, *Conuropsis c. carolinensis*. The oldest and northernmost records, of what was probably this race, appeared in Bartram's Fragments (1799) in the following words:

The two first of these birds were seen in the neighbourhood of Philadelphia, between thirty and forty years ago. The Psittacus, most probably the Psittacus pertinax, Illinois Parrot, or the Psittacus carolinensis, Carolina Parrot, has been occasionally observed in Shareman's Valley, on Shareman's Creek, a branch of the river Susquehanna, within twenty miles of the town of Carlisle. This last fact seems to contradict the observation of Mr. William Bartram, who says, "The parakeets (Psittacus carolinensis) never reach so far north as Pennsylvania, which to me is unaccountable, considering they are a bird of such singular rapid flight, they could easily perform the journey in ten or twelve hours from North-Carolina, where they are very numerous, and we abound with all the fruits which they delight in." * * * I may add, that a very large flight of parakeets, which came from the westward, was seen a few years ago, about twenty-five miles to the north-west of Albany, in the State of New-York.

The arrival of these birds in the depth of winter (January, 1780) was, indeed, a very remarkable circumstance. The more ignorant Dutch settlers were exceedingly alarmed. They imagined, in dreadful consternation, that it portended nothing less calamitous than the destruction of the world.

DeKay (1844) places this New York record as occurring in 1795. The only record we have for New Jersey is one recently published by Warren F. Eaton (1936); Albert E. Hedden (1841–1915) told his son and nephew "of

the occurrence of this species in East Orange, Essex County, New Jersey, when he was a boy. They placed the time between 1850 and 1860, and both recall exactly the same story. The Paroquets occurred probably twice at least in hot weather (I suspect September) and were considered very destructive to the small household apple orchards, maintained by the family at that time. The birds occurred in flocks and tore the apple fruit apart, extracting the seeds." There seems to be no record for Delaware, but in Audubon's time they were found as far north as the boundary line between Virginia and Maryland, where a flock was seen and specimens were shot as recently as September 1865 (Smith and Palmer, 1888). They apparently were common in the Carolinas up to 1850, or perhaps 1860, but must have disappeared from there soon after that. For Georgia, there seem to be no records since 1849. In Florida, the species made its last stand; parakeets were evidently common throughout the State up to the 1860's, but during the next 20 years all observers reported them as becoming rarer and more restricted in range. In the early 1890's it was still common in certain remote localities in Florida. Arthur H. Howell (1932) has this to say about the last records of this vanishing bird:

E. J. Brown reported the birds plentiful in March, 1896, near Campbell, Osceola County. Dr. E. A. Mearns took 6 specimens on Padget Creek, Brevard County, April 18, 1901. Apparently the last stronghold was in the vicinity of Taylor Creek, on the northeastern side of Okeechobee Lake. Here on February 29, 1896, Robert Ridgway collected 13 specimens, and in April, 1904, Frank M. Chapman saw two flocks aggregating 13 birds (1912, p. 318). W. W. Worthington hunted along both sides of Taylor Creek on March 26, 1907, without seeing any Paroquets. * * * Capt. F. W. Sams, an old resident of Florida, told Dr. Amos W. Butler that he saw a flock of 8 or 10 Paroquets in 1909 at Cabbage Slough, on the west side of Turnbull Hammock, about 12 or 15 miles southwest of New Smyrna. E. Stewart Hyer, of Orlando, reports seeing one bird at Istokpoga Lake on February 16, 1910. A late and apparently authentic record is published by Chapman (Bird Lore, 1915, p. 453) on the authority of W. J. F. McCormick, who claims to have seen about a dozen birds in March and April, 1915. Henry Redding, who knows the birds well, reported a flock of about 30 seen on Fort Drum Creek in February, 1920.

The causes that led to the extermination of the parakeet are not hard to find. It was a bad actor, regarded by fruit growers and agriculturists as a destructive pest, doing extensive damage to their crops. Consequently it was slaughtered in enormous numbers on every opportunity. It was more or less hunted as a game bird, for it was abundant and its flesh was said to be very palatable. It was shot in enormous numbers for mere sport, or for practice. Hundreds were captured by professional bird catchers and sent north, as cage birds or pets, and many were killed for their plumage.

Furthermore, it has always retreated before the spread of civilization and seemed incapable of surviving in settled regions, probably for the reasons mentioned above.

W. E. D. Scott (1889) says that “they were wantonly mischievous and cut hundreds of young *green oranges*, peaches, and the like, from the trees almost as soon as the fruit was formed.” Many were shot by farmers in their cornfields, where the birds had formed the bad habit of feeding on the tender corn on the ears, thus destroying, or injuring, a large part of the crop.

According to Audubon (1842) they ate or destroyed almost every kind of fruit indiscriminately and on this account were always unwelcome visitors “to the planter, the farmer, or the gardener.” He says:

The stacks of grain put up in the field are resorted to by flocks of these birds, which frequently cover them so entirely, that they present to the eye the same effect as if a brilliantly coloured carpet had been thrown over them. They cling around the whole stack, pull out the straws, and destroy twice as much of the grain as would suffice to satisfy their hunger. They assail the pear and apple-trees, when the fruit is yet very small and far from being ripe, and this merely for the sake of the seeds. As on the stalks of corn, they alight on the apple-trees of our orchards, or the pear-trees in the gardens, in great numbers; and, as if through mere mischief, pluck off the fruits, open them up to the core, and, disappointed at the sight of the seeds, which are yet soft and of a milky consistence, drop the apple or pear, and pluck another, passing from branch to branch, until the trees which were before so promising, are left completely stripped. * * * They visit the mulberries, pecan-nuts, grapes, and even the seeds of the dog-wood, before they are ripe, and on all commit similar depredations. * * *

Do not imagine, reader, that all these outrages are borne without severe retaliation on the part of the planters. So far from this, the Parakeets are destroyed in great numbers, for whilst busily engaged in plucking off the fruits or tearing the grain from the stacks, the husbandman approaches them with perfect ease, and commits great slaughter among them. All the survivors rise, shriek, fly around about for a few minutes, and again alight on the very place of most imminent danger. The gun is kept at work; eight or ten, or even twenty, are killed at every discharge. The living birds, as if conscious of the death of their companions, sweep over their bodies, screaming as loud as ever, but still return to the stack to be shot at, until so few remain alive, that the farmer does not consider it worth his while to spend more of his ammunition. I have seen several hundreds destroyed in this manner in the course of a few hours.

This fatal habit of hovering over their fallen companions has helped, more than any one thing, to bring about their extermination. Their social disposition has been their undoing. C. J. Maynard (1896) says of this trait: “This is not a mere liking for company, as they are actually fond of one another, for, if one out of a flock be wounded, the survivors attracted by its screams, will return to hover over it and, even if constantly shot at, will not

leave as long as their distressed friend calls for assistance; in fact, I have seen every individual in a flock killed one after the other, and the last bird betrayed as much anxiety for the fate of its prostrate friends which were strewn upon the ground, as it did when the first fell.”

Nesting.—Nothing very definite seems to be known about the nesting habits of the Carolina parakeet. No competent ornithologist has ever seen a nest. Even Wilson and Audubon, who lived in the days when these birds were so abundant, never saw a nest; and all they wrote about it was based on hearsay. Most observers seemed to agree that the parakeets nested in hollow trees, but some of the accounts were rather fantastic. For example, Wilson (1832) wrote: “One man assured me that he cut down a large beech tree, which was hollow, and in which he found the broken fragments of upwards of twenty parakeets’ eggs, which were of a greenish yellow colour. The nests, though destroyed in their texture by the falling of the tree, appeared, he said, to be formed of small twigs glued to each other, and to the side of the tree, in the manner of the Chimney Swallow.” Audubon (1842) says: “Their nest, or the place in which they deposit their eggs, is simply the bottom of such cavities in trees as those to which they usually retire at night. Many females deposit their eggs together.”

Maynard (1896) was told by some cedar hunters that a large number of parakeets nested in a hollow in a huge cypress tree in the depths of a great cypress swamp. He offered them a good sum to procure the eggs, which they attempted to do; but, on opening the tree, about which they saw a large number of the parakeets, they were disappointed to find only young birds. H. B. Bailey (1883) had in his collection a set of two eggs, which he felt sure were eggs of the Carolina parakeet. “The eggs, which were taken April 26, 1855, were deposited in a hollow tree, on the chips at bottom. One of them was sent to Mr. Ridgway who has kindly compared it with identified eggs, and who confirms the identification.” There was an apparently authentic set in the John Lewis Childs (1906b) collection, taken in the wild, of which he writes:

The set consists of three eggs which were taken on April 2, 1896, by Dr. H. H. Pendry. They were found in a cavity of a sycamore tree forty feet up on the outskirts of the Great Swamp near the head of the Caloosahatchee River and west of Lake Okechobee, De Soto County, Florida. Dr. Pendry was not sure of the identity of these eggs, as he saw no Paroquets at the nest, but they were in the swamp and he had frequently seen and taken young birds in the same locality. * * * The eggs were sent to us for identification, and there

seems to be not the slightest doubt but that they are genuine. They measure as follows: 1.35×1.06 – 1.26×1.06 , 1.25×1.05 [34.3 by 27.1, 32.1 by 27.1, and 31.8 by 26.8 millimeters].

William Brewster (1889) published the following account, which seemed to him “to rest on evidence sufficiently good to warrant its publication.” He questioned everybody he met about the nesting of the parakeet and was told by two professional hunters of alligators and plume birds that they had “seen Parrakeets’ nests, which they described as flimsy structures built of twigs and placed on the branches of cypress trees.” He goes on to say:

This account was so widely at variance with what has been previously recorded regarding the manner of nesting of this species that I considered it, at the time, as a mere fabrication, but afterwards it was unexpectedly and most strongly corroborated by Judge E. L. Long of Tallahassee. The latter gentleman, who, by the way, has a very good general knowledge of the birds of our Northern States, assured me that he had examined many nests of the Parrakeet built precisely as above described. Formerly, when the birds were abundant in the surrounding region, he used to find them breeding in large colonies in the cypress swamps. Several of these colonies contained at least a thousand birds each. They nested invariably in small cypress trees, the favorite position being on a fork near the end of a slender horizontal branch. Every fork would be occupied, and he has seen as many as forty or fifty nests in one small tree. Their nests closely resembled those of the Carolina Dove, being similarly composed of cypress twigs put together so loosely that the eggs were often visible from the ground beneath. The twigs of the cypress seemed to be preferred to those of any other kind of tree. The height at which the nests were placed varied from five or six feet to twenty or thirty feet. Mr. Long described the eggs as being of a greenish white color, unspotted. He did not remember the maximum number which he had found in one set, but thought it was at least four or five. He had often taken young birds from the nests to rear or to give to his friends.

Several times parakeets have been known to breed in captivity or attempt to do so. Robert Ridgway brought several birds from Florida that laid at least 13 eggs in captivity; most of the eggs now in American collections are the product of these birds. Dr. William C. Herman writes to me that parakeets bred successfully in the Cincinnati Zoological Garden, where some were kept for 20 or more years; the last one died in September 1914. “Some of these birds bred in captivity. Dozens of young birds were raised, especially when others recently captured were added.” Dr. Nowotny (1898) purchased a pair of Carolina parakeets in Vienna and tried to breed them in captivity. The female laid in all ten eggs; the first five were put in a breeding box, but never hatched, as they were “picked and sucked,” presumably by the birds. Two more eggs were placed under a hen but were destroyed through carelessness. The other three were placed in the breeding

box and incubated by the parakeets; three young were hatched, but they did not live to maturity.

Eggs.—It is not definitely known how many eggs were laid by the Carolina parakeet in a normal set, but indications point to two and three as being the commonest numbers. Bendire (1895) says, of the eggs laid by Mr. Ridgway's birds in captivity:

None of these eggs can be called round; they vary from ovate to short ovate, and are rather pointed. They are white, with the faintest yellowish tint, ivory-like and quite glossy; the shell is rather thick, close grained, and deeply pitted, not unlike the eggs of the African Ostrich (*Struthio camelus*), but of course not as noticeable. Holding the eggs in a strong light, the inside appears to be pale yellow. * * *

The deep pitting is noticeable in every specimen, and there can be no possible doubt about the identity of these eggs. * * * There is no difficulty whatever in distinguishing these eggs from those of the Burrowing Owl or the Kingfisher, both of which are occasionally substituted for them.

Mr. Childs (1905) figures the three eggs sent to him by Mr. Ridgway and describes them as "color pure white with ivory gloss surpassing that of the Ivory-billed Woodpecker." One of Mr. Ridgway's eggs, in the John E. Thayer collection, I should describe as ovate in shape and dull white in color, with a very slight gloss. The measurements of 24 eggs average 34.23 by 27.80 millimeters; the eggs showing the four extremes measure **37** by 38, 33 by **30.2**, **32.1** by 27.1, and 34.4 by **25.8** millimeters.

Young.—We do not know much about the development and care of the young in the wild state, but in captivity the birds seem to be very careless or indifferent in the care and feeding of the young. Dr. Nowotny's eggs were hatched, after continuous incubation, in about 19 or 20 days, but the young all eventually died from neglect.

Mr. Ridgway wrote to Mr. Childs (1905), under date of November 13, 1902: "My female Parakeet laid only six eggs the past summer and I shall never get any more, as the bird is now dead. The first she laid is the one I sent you. The remaining five hatched, but I have only two young ones left, a rat having carried off one, another was starved by the parents when half grown, and the third I gave to a friend who had time and disposition to take care of it in order to save it from starving."

Plumages.—I have never seen the downy young or nestling plumages and doubt if there are any such in collections. Audubon (1840) says that

“the young are at first covered with soft down, such as is seen on young Owls. During the first season, the whole plumage is green; but towards autumn a frontlet of carmine appears. Two years, however, are passed before the male or female are in full plumage.” His plate shows a bird with a wholly green head.

Dr. Nowotny's (1898) young birds, when between five and six weeks old, “had already attained green wings and tail; the older one also had red feathers above the bill and on the under parts. * * * The oldest young one had already attained many dense strong red feathers above the bill at the age of eight weeks”. This does not agree with Audubon's account, or with C. J. Maynard's (1896) who says, of the young of the year: “Head and neck, wholly green, and the tail is short.” He says, of the nestling: “One of my collectors, who found the young in the nest, informs me that they are covered with a grayish down.” I have never seen a young bird with a wholly green head; those that I have seen, some nine in number, were in first winter plumage, and were collected between October 7 and April 3, indicating that this plumage is worn all through the first winter. These are much like adults, but with no yellow on the head, thighs, or anal region, and only greenish yellow on the edge of the wing; the forehead and front of the crown up to the front edges of the eyes are “flame scarlet” or “cadmium orange,” shading off to dull brownish orange on the lores and to dull brown on the cheeks. Specimens taken in February show yellow feathers coming in on the head, and progressive changes toward maturity continue all through the spring months until, by summer, the yellow head is fully acquired. Audubon (1840) says that the young bird requires two years to attain its full plumage, and Maynard (1896) says that it is acquired during the third year, but the material examined does not indicate this.

Adults have a complete molt in fall, from September to November. Dr. Frank M. Chapman's (1890) captive bird began to molt in September, “and by November had acquired an entirely new plumage.” Arthur T. Wayne (1910) says that they “commence to molt about October 5, and require at least six weeks to acquire their perfect plumage.”

Food.—Much has been said about the food of the Carolina parakeet in the earlier part of this account, as its feeding habits were so destructive in fruit orchards and grain fields that the birds were unmercifully slaughtered by the planters, thus hastening the extermination of the species. But before

the lands were so extensively cultivated, the parakeets lived on their natural, wild food. C. J. Maynard (1896) writes:

I observed a large flock of Parakeets moving along the ground. * * * At first, I could not make out what they were doing but soon found that they were busily engaged in eating cockspurs, the seeds of a species of grass which grows very abundantly in old fields. They walked quite well for birds having such short legs and, in pressing forward, moved side by side in a long rank, looking exactly like miniature soldiers. After a few moments, something startled them and they arose, wheeled about, darting rapidly up and down, precisely like pigeons, at the same time, uttering loud cries; then settled quietly down again and resumed their meal, as composedly as if nothing had occurred to interrupt.

This is the only time that I ever chanced to see the Parakeets feed on the ground but I have been informed by the inhabitants of Florida, that they are very fond of cockspurs and will frequently alight in the fields in order to eat them. Early in winter, they visit the swamps, where they feed upon the cypress balls. Then it is very difficult to find them as they often remain for weeks in the impenetrable fastnesses of the vast wooded tracts which, at this season, are submerged in water. Later, about the first of February, the Parakeets emerge from the swamps in small flocks and enter the open woods to search for the seeds of the pine cones which are then ripe. At this time, they may be met with quite frequently but the best opportunity to procure specimens occurs about the middle of February, when they may be found in large companies, feeding upon the green seeds of the maples and elms which grow along the rivers.

Dr. Chapman (1890) found these parakeets feeding on the prairies near the Sebastian River in Florida, of which he says:

About these “prairies” and at the borders of small streams or low ground grew in abundance a species of thistle (*Cirsium Lecontei*, T. & G.) the seeds of which, so far as I could learn, constituted at this season [February] the entire food of *Conurus*. Not a patch of thistles did we find which had not been visited by them, the headless stalks showing clearly where the thistles had been neatly severed by the sharp chisel-like bill, while the ground beneath favorite trees would be strewn with the scattered down. * * *

Two days passed before I again met *Conurus*, and this time to better advantage. It was a wet and drizzling morning when we found a flock of six birds feeding on thistles at the edge of a “prairie.” Perched on the leafless branches of the tree before us, their brilliant green plumage showed to the best advantage, as we approached through the pines without difficulty. Several were skillfully dissecting the thistles they held in their feet, biting out the milky seed while the released fluffy down floated away beneath them. There was a sound of suppressed conversation; half articulate calls. * * *

There was an evident regularity in the habits of the birds we afterwards observed—in all about fifty, in flocks of from six to twenty. At an early hour they left their roost in the hummock bordering the river and passed out into the pines to feed, always, so far as I observed, selecting thistle patches, and eating the seeds only when in the milky stage. At about ten o’clock they returned to the hummock and apparently to some favorite tree, here to pass the rest of the morning and early afternoon, when they again started out to feed, returning to the roost just before sunset. A flock of these birds feeding among the thistles is a most beautiful and animated sight; one is almost persuaded not to disturb them. There is constant movement as they fly from plant to plant, or when securing thistles they fly with them in their bills to a neighboring tree, there to dissect them at their leisure. The loud

rolling call was apparently uttered only when on the wing, but when at rest, or feeding, there was a loud conversational murmur of half articulate, querulous notes and calls.

Cottam and Knappen (1939) examined the stomach and crop of one bird, of which they say: "Except for two rabbit hairs, two bits of the bird's own feathers, and two fragments of an indeterminable ant, which formed only traces, the entire content consisted of the remains of no fewer than thirty-two seeds of loblolly pine (*Pinus taeda*)."

Behavior.—Audubon (1842) writes:

The flight of the Parakeet is rapid, straight, and continued through the forests, or over fields and rivers, and is accompanied by inclinations of the body which enable the observer to see alternately their upper and under parts. They deviate from a direct course only when impediments occur, such as the trunks of trees or houses, in which case they glance aside in a very graceful manner, merely as much as may be necessary. A general cry is kept up by the party, and it is seldom that one of these birds is on wing for ever so short a space without uttering its cry. On reaching a spot which affords a supply of food, instead of alighting at once, as many other birds do, the Parakeets take a good survey of the neighborhood, passing over it in circles of great extent, first above the trees, and then gradually lowering until they almost touch the ground, when suddenly re-ascending they all settle on the tree that bears the fruit of which they are in quest, or on one close to the field in which they expect to regale themselves.

They are quite at ease on trees or any kind of plant, moving sidewise, climbing or hanging in every imaginable posture, assisting themselves very dexterously in all their motions with their bills. They usually alight extremely close together. I have seen branches of trees as completely covered by them as they could possibly be. If approached before they begin their plundering, they appear shy and distrustful, and often at a single cry from one of them, the whole take wing, and probably may not return to the same place that day.

Maynard (1896) says:

I have remarked that the Parakeets scream very loudly when flying; so loudly, in fact, that their shrill cries can be heard for miles. They come dashing along, moving in a most eccentric manner; now near the ground, then high over the tree tops, seeming about to alight a dozen times but still without settling, each in the company endeavoring to excel the other in producing the most discordant yells, when they will all pitch, at once, into some tree and a sudden silence ensues. So great had been the din but a second before that the comparative stillness is quite bewildering, then too, the large flock of highly colored birds, lately so conspicuous, have disappeared completely. I well remember my first experience of this nature; I stood, gun in hand, watching the evolutions of a large company as it wheeled about, awaiting an opportunity to shoot, when, of a sudden, they all alighted in a large live-oak which stood a few rods away. I cautiously approached the tree, ready to slaughter half the flock at a single discharge, if possible, when, what was my surprise upon going within a suitable distance, not to perceive a bird. Neither could I see so much as a feather of the desired game although I walked around the tree several times and even went beneath its branches to peer up between them. After spending some time in these fruitless efforts, my patience became quite exhausted and I threw a large oyster shell up into the tree. This certainly produced an effect, not just what I intended, however, for, in an instant,

out darted the entire body of screaming birds but on the opposite side of the thick tree; thus I could only stand and watch them as they disappeared in the neighboring swamp.

Audubon (1842) says:

Their roosting-place is in hollow trees, and the holes excavated by the larger species of Woodpeckers, as far as these can be filled by them. At dusk, a flock of Parakeets may be seen alighting against the trunk of a large sycamore or any other tree, when a considerable excavation exists within it. Immediately below the entrance the birds all cling to the bark, and crawl into the hole to pass the night. When such a hole does not prove sufficient to hold the whole flock, those around the entrance hook themselves on by their claws, and the tip of the upper mandible, and look as if hanging by the bill. I have frequently seen them in such positions by means of a glass, and am satisfied that the bill is not the only support used in such cases.

Dr. William C. Herman writes to me, of the parakeets in the Cincinnati Zoological Garden: "The parakeets were well adapted to being caged, some were in the zoo for 20 or more years. A hollow tree was provided for the birds for roosting. Here they hung for the night; that is, they used their beaks for holding to the interior of the tree trunk and so rested for the night."

DISTRIBUTION

Range.—Southeastern United States; probably extinct.

The range of the Carolina parakeet extended **north** to casually northeastern Colorado (Little Thompson River); eastern Nebraska (Omaha and Brownville); casually Iowa (Spirit Lake and Decatur County); casually southern Wisconsin (Lake Koskonong and Waukesha County); Ohio (Columbus and Summit County); and Pennsylvania (Juniata River and Shermans Valley). **East** to Pennsylvania (Shermans Valley); casually the District of Columbia (Washington); casually West Virginia (White Sulphur Springs); South Carolina (Pine Barrens and Edding Island); Georgia; and Florida (Oklawaha River, Wekiva River, and Micco); **south** to Florida (Micco, Lake Okeechobee, Tampa, Tarpon Springs, Old Town, and Tallahassee); southern Louisiana (Bayou Sara and St. Mary); and central Texas (Brownwood). **West** to central Texas (Brownwood); eastern Oklahoma (Caddo and Fort Gibson); and casually eastern Colorado (Fort Lyon, Denver, and the Little Thompson River).

Casual records.—Several of the records that figure in the range as above outlined can be considered as little more than casual occurrences, but this

status must be accorded a flock reported 25 miles northwest of Albany, N. Y., in January 1780, and to flocks observed “many years ago” at Buffalo and West Seneca, N. Y. There also is an indefinite record of this species in East Orange, Essex County, N. J., sometime between 1850 and 1860.

Systematists now consider the species as separable into two geographic races, the true Carolina parakeet, *C. c. carolinensis*, being the eastern form that ranged west to Alabama, while the Louisiana parakeet, *C. c. ludovicianus*, ranged westward from Mississippi, Tennessee, Kentucky, and Ohio.

Egg dates.—Florida: 2 records, April 2 and 26.

CONUOPSIS CAROLINENSIS LUDOVICIANUS Gmelin

LOUISIANA PARAKEET

HABITS

In describing this pale race of *Conuopsis*, Outram Bangs (1913) says: “For many years it has been common knowledge among the older set of American ornithologists that the Carolina paroquet divided into two very distinct geographical races.” He says of its characters: “A much paler bird than *Conuopsis c. carolinensis* (Linn.); yellow portions of head and neck pale lemon yellow or picric yellow, instead of lemon yellow or lemon chrome; green of upper parts much paler and more bluish, verdigris green to variscite green on wing coverts and sides of neck; under parts dull green-yellow glossed with variscite green; bend of wing and feathers of tibia paler, purer yellow, less orange.” Mr. Ridgway (1916) adds: “Greater wing-coverts, proximal secondaries, and basal portion of outer webs of primaries more pronouncedly and more extensively yellowish, contrasting more strongly with the general green color; size averaging decidedly greater.” The latter author says of its former range: “Formerly inhabiting the entire Mississippi Valley (except open prairies and plains), from West Virginia to eastern Colorado, north to the southern shores of Lake Erie and Lake Michigan, south to the Gulf coast.”

This parakeet is now, doubtless, quite extinct throughout all this wide range. Though formerly abundant over most of this region, it had begun to disappear even in Audubon’s time, for he (1842) says: “Our Parakeets are

very rapidly diminishing in number; and in some districts, where twenty-five years ago they were plentiful, scarcely any are now to be seen. At that period, they could be procured as far up the tributary waters of the Ohio as the Great Kenhawa, the Scioto, the heads of the Miami, the mouth of the Manimee at its junction with Lake Erie, on the Illinois River, and sometimes as far north-east as Lake Ontario. * * * At the present day, very few are to be found higher than Cincinnati, nor is it until you reach the mouth of the Ohio that Parakeets are met with in considerable numbers. I should think that along the Mississippi there is not now half the number that existed fifteen years ago.”

Myron H. Swenk (1934), in a comprehensive paper on this parakeet of the interior, sums up its disappearance in the following words: “By 1840 they were practically gone in West Virginia and Ohio. They disappeared from Indiana about 1858 and from Illinois about 1861. The Colorado birds were gone by about 1862. In Kansas they were gone by about 1867, and during the years 1875–1880 they disappeared from Kentucky, Tennessee, Louisiana, Mississippi and Alabama. Their last stand was made in Missouri and along the Arkansas River and its tributaries in Arkansas and central Oklahoma, but by 1890 they were practically gone in these localities also. * * * The very last records of living Interior Carolina Paroquets are of lone individuals shot at Atchison, Kansas, in 1904, and seen at Notch, Stone County, Missouri, in 1905 (*vide* Widmann, * * * 1907).”

Since that time there have been at least two sight records reported. Harry Harris (1919) says that “in some unaccountable manner a lone bird strayed into the Courtney bottoms in 1912 and was observed by Bush for several weeks before it finally disappeared.” Dr. Daniel S. Gage has sent me a letter from Prof. Elliot R. Downing, of the University of Chicago, reporting that he saw a Carolina parakeet in the sand-dune region on the shore of Lake Michigan, not far from Chicago, on June 11, 1912. His letter states that he saw the parakeet “on a Juneberry tree, a small one, on the margin of an interdunal pond. I remember the observation very clearly. I was within 20 feet of the bird and had a chance to observe it with my bird glasses for 10 or 15 minutes. I am therefore quite confident that there was no error in the observation.”

Both of these records might well be based on escaped cage birds, as there were a number in captivity at that time, and the wild birds had long

since disappeared.

Wilson (1832) suggests certain reasons why the inland parakeet enjoyed a wider and more northern distribution than its relative on the Atlantic coast. He writes:

The preference, however, which this bird gives to the western countries, lying in the same parallel of latitude with those eastward of the Alleghany mountains, which it rarely or never visits, is worthy of remark; and has been adduced, by different writers, as a proof of the superior mildness of climate in the former to that of the latter. But there are other reasons for this partiality equally powerful, though hitherto overlooked; namely, certain peculiar features of country to which these birds are particularly and strongly attached; these are, low, rich, alluvial bottoms, along the borders of creeks, covered with a gigantic growth of sycamore trees, or button-wood; deep, and almost impenetrable swamps, where the vast and tower-cypress lifts its still more majestic head; and those singular salines, or, as they are usually called, *licks*, so generally interspersed over that country, and which are regularly and eagerly visited by the Paroquets. A still greater inducement is the superior abundance of their favorite fruits. That food which the paroquet prefers to all others is the seeds of the cockle bur, a plant rarely found in the lower parts of Pennsylvania or New York; but which unfortunately grows in too great abundance along the shores of the Ohio and Mississippi.

Nesting.—We have no more positive information on the nesting habits of this parakeet than we have of the eastern race, beyond the following statement by Col. N. S. Goss (1891): “Their nests are usually placed in holes or hollow cavities of trees. In the spring of 1858, a small flock reared their young in a large hollow limb of a giant sycamore tree, on the banks of the Neosho River, near Neosho Falls, Kansas. I have never been able to procure their eggs.”

Eggs.—What few eggs of this race are in existence are indistinguishable from those of the Carolina parakeet. The measurements of the only four eggs that I have been able to locate are **36** by 27, 35 by **27.5**, 35 by **26.5**, and 36 by 26.5 millimeters.

Food.—Prof. Myron H. Swenk (1934) says of the food of this parakeet:

The food of the Interior Carolina Paroquet, though all vegetable, was highly varied, and they seemed to delight in the fruits of spiny or thorny plants. One of the most relished foods was the seeds of the cocklebur (*Xanthium canadense*), and they fed also on the seeds of the sand-bur grass (*Cenchrus tribuloides*) and of the various species of thistles (*Cirsium*). In the fall they ate the seeds of the honey locust (*Gleditsia triacanthos*) and the tender buds and fruit of the osage orange (*Maclura pomifera*). In the spring they ate the buds of the red maple (*Acer rubrum*) and birch (*Betula spp.*). During the summer they ate much fruit, especially mulberries, wild grapes, hackberries and pawpaws, and, after the planting of cultivated apple orchards, were likely to visit them and peck out the apple seeds in the fall, sometimes doing injury in this way. Corn in the milk was also sometimes

injured, but not extensively. Other favorite items of food were the seed balls of the sycamore and beech and pecan nuts. In the South cypress seeds were much eaten.

Dr. Amos W. Butler (1892), quoting from W. B. Seward, thus describes the parakeet's method of eating the cocklebur seeds: "In eating, the bird picked up a burr with its beak, this was then delivered to one foot raised to receive it. Then one end of the burr was cut off with the sharp-ended under beak, the burr being held with the foot and the under side of the upper beak while two small kernels were extracted with the assistance of the tongue and the husk was thrown away."

Dr. Butler, elsewhere (1898), adds to the items of food mentioned above cherries, persimmons, black-gum berries, haws, and acorns.

Voice.—Mr. Swenk (1934) says: "The common call notes consisted of a loud, shrill series of rapidly uttered, discordant cries, given incessantly when the birds were in flight, resembling *qui-qui, qui, qui, qui, qui-i-i-i*, with a rising inflection on each i and the last cry drawn out. Another call resembled the shrill cry of a goose and was frequently uttered for minutes at a time. When at rest they had a low, conversational chatter."

Winter.—The Carolina parakeet was evidently a very hardy bird, a remarkable quality, quite unique among parrots. Wilson (1832) saw them "in the month of February, along the banks of the Ohio, in a snow storm, flying about like pigeons, and in full cry." Dr. Butler (1898) was furnished the following note by Prof. John Collett:

In 1842, Return Richmond, of Lodi (Parke County), Ind., cut down, in the cold weather of winter, a sycamore tree some four feet in diameter. In its hollow trunk he found hundreds of Parakeets in a quiescent or semi-torpid condition. The weather was too cold for the birds to fly or even to make any exertion to escape. Mr. Richmond cut off with his saw a section of the hollow trunk some five feet long, cut out a doorway one foot by two in size, nailed it over a wire screen of his fanning mill, rolled this cumbersome cage into the house and placed in it a dozen of the birds. They soon began to enjoy the feed of fruit, huckleberries and nuts he gave them, and he had the pleasure of settling absolutely the disputed question of how they slept. At night they never rested on a perch, but suspended themselves by their beaks, and with their feet on the side of their cage. This was repeated night after night during their captivity.

RHYNCHOPSITTA PACHYRHYNCHA (Swainson)

THICK-BILLED PARROT

HABITS

This fine, large parrot is a Mexican species that claims a place on our list as an irregular visitor across our southwestern border in southern Arizona and New Mexico. Charles T. Vorhies (1934) has published an account of all the Arizona records that he was able to find, which probably tells the whole story. R. D. Lusk (1900) reported that a flock of nine or ten of these parrots came into the Chiricahua Mountains, just north of the Mexican border, about the middle of June 1900. He collected two specimens, and some polecutters killed several others. "They appeared to come up the large cañon, at the head of which I was encamped, to about midway of the mountain's height, where the oaks begin to give way to pine, and there they tarried."

Austin Paul Smith (1907) reported "an immense flock of this species, * * * observed by miners at Bonita Park, near Cochise head in the Chiricahua Mountains, during the month of August, 1904. * * * This flock was estimated at from 700 to 1,000, by those who observed the birds. Among these were a number of young birds, easily distinguished by plumage and small size."

The next occurrence of these parrots, and probably the last definite record, was in the winter of 1917–18, of which Percy Hands wrote to Professor Vorhies as follows: "The Mexican Thick-billed Parrots were first noticed on my ranch in lower Pinery Canyon in mid-September. About 250—counted over 200—in one flock. They have been on my ranch at irregular intervals since then up to March 27. They roosted here seven times during late October and early November. The longest period they were away was about three weeks before they appeared March 27. There were about 70, rough count, at that appearance."

The thick-billed parrot may have occurred elsewhere in Arizona, but Professor Vorhies was unable to find any definite records for any locality outside of the Chiricahuas. The New Mexico records are both hearsay, sight records. Mrs. Bailey (1928) reports: "In 1919, Dr. Alexander Wetmore was told by R. Winkler that in recent years his son had at times seen parrots on Animas Mountain, above Deer Creek. * * * A confirmatory record was given Aldo Leopold by Forest Ranger Don S. Sullivan, who said that in 1917 some large Parrots were seen near the Elvey Ranch. * * * on the Animas Division of the Coronado Forest near the Mexican boundary."

The main range of the thick-billed parrot is in the mountains bordering the Mexican tableland. Dr. W. H. Bergtold (1906) writes:

This bird is increasingly common from Chuichupa southward, and was especially an everyday sight during the trip, in 1904, to the mountains west of Parral. * * * In the higher mountains west of Parral, a region varying in altitude from 4,000 to 10,000 feet, the Thick-billed Parrot is far more common than northward in the country west of Casas Grandes; in fact it is the characteristic bird of these high places, as much so as is the Magpie part of the local color of our Western Plains. * * * In whatever section we saw them, these parrots were most abundant in the pines. They frequented the tops of dead pines, and were, a good part of the time, going in and out of abandoned woodpecker nests, nests which we took to be those of the Imperial Woodpecker (*Campephilus imperialis*), for this splendid woodpecker is relatively common in the same neighborhood, and is the only woodpecker which excavates such a large hole.

Nesting.—Practically all we know about the nesting habits of the thick-billed parrot is contained in the report of Col. John E. Thayer (1906) on some ten nests of this species examined by Wilmot W. Brown, Jr., in 1905, in the mountains near Colonia Pachaco and Colonia Garcia, Chihuahua, Mexico. The breeding grounds were at altitudes varying from 6,500 to 7,500 feet, and all the nests were in the tall pines, characteristic of the region. The parrots occupied the old nests of the imperial woodpecker, which had been excavated at heights ranging from 50 to 80 feet above ground; one nest was in a living longleaf pine, but all the others were in dead, dry, or rotten pines. The entrance to the hole was usually circular and 6 or 7 inches in diameter; the holes varied in depth from 18 to 24 inches, and the inner cavity measured from 8 to 10 inches in diameter. No nesting material had been brought in, the eggs having been laid on the bare wood dust left by woodpeckers. One nest, found on August 20, contained two big young ones and one fresh egg; another found on August 28, held two young ones. All the other nests, found between August 11 and 25, contained one or two eggs. Mr. Brown wrote to Colonel Thayer (1906) that the first tree he climbed was “located on a flat-topped mountain at an altitude of 7,500 feet above the sea. * * * The tree was about one hundred feet high and was thirty-six inches in diameter at the base and was so dry that the bark had all peeled off. It was very difficult to climb, not to mention the danger.” The nest was 80 feet from the ground.

Eggs.—I have examined the eggs referred to above, which, with the entire Thayer collection, are now in the Museum of Comparative Zoology, in Cambridge. They vary in shape from ovate to rounded-ovate, are pure white in color, and are decidedly glossy; the shell is hard and thick. The

measurements of 20 eggs average 39.49 by 30.63 millimeters; the eggs showing the four extremes measure 42 by 31. 39.2 by 32, 37.6 by 30.6, and 38.2 by 29 millimeters.

Plumages.—Aside from the alcoholic specimens in the Thayer collection, which are unsuitable for description, I have seen no nestlings of this species. The immature bird is similar to the adult, but the red of the forehead is more restricted, and the red is lacking on the bend of the wing and on the lower thighs. Nothing seems to be known about the length of time that the immature plumage is worn, or about subsequent molts and plumages.

Food.—The birds observed by R. D. Lusk (1900) “were very busily engaged with the pine cones, and investigation of their stomachs showed nothing but a plentiful quantity of very immature pinones wrested from their cavities in the hearts of the hard, green cones by their powerful beaks. The beaks, at their bases, as well as nearly the entire under parts of the birds, were more or less begummed with the resin of the cones.” The birds reported by Austin Paul Smith (1907) were first noticed when “feeding on pinyon nuts. Some of the birds were on the ground, searching for the fallen nuts.” During their sojourn in Arizona, these parrots have been reported as feeding on pinyon nuts and acorns, after they had exhausted the supply of pine cones.

The stomachs of two birds examined by Cottam and Knappen (1939) contained 136 and 284 or more seeds of the Chihuahua pine (*Pinus chihuahuana*). “The pine seeds were in all stages from entire seeds to a pulverized mast-like débris.”

Behavior.—All observers seem to agree that the thick-billed parrot is very tame and unsuspicious, probably owing to its unfamiliarity with man.

William Beebe (1905) says: “It is either a very stupid bird or controlled by its curiosity, for the flocks followed us everywhere.”

Dr. Bergtold (1906) writes:

It was a great surprise to see how different is a wild parrot from a tame one; one must need get an idea from the latter that a parrot is a slow, lumbering climber, able to use its wings perhaps, yet little given to prolonged and vigorous flight. On the contrary, this Thick-billed Parrot flew across deep barrancas, from mountain to mountain, as swift and strong on wing as a duck, going often in large flocks, which were noticeably divided in pairs, each couple flying one above another as closely as beating wings allowed. Its loud

squawk resounded overhead, across the barrancas, and in the pines all day long, from dawn till dusk; and many and many a time a flock could be heard long before it was in sight. The birds were not at all shy, as one could walk up under a tree and watch a pair climbing in it without disturbing them in the least. Here they seemed natural, at least to one whose previous knowledge of parrots came via the cage bird, for they climbed about precisely as does the domesticated species, using bill and feet in the familiar way; on the wing the birds seemed anything but parrots.

Dr. Alexander Wetmore (1935) obtained considerable information from reliable observers about the occurrence of this parrot in Arizona and its habits; regarding its movements and behavior, he writes:

According to all accounts Thick-billed Parrots gathered at night to roost in flocks and then spread out in small bands to feed during the day. In Pinery Canyon they roosted somewhere on the upper mountain slopes during summer and fall. Morning and evening they were seen in two large flocks. As the weather became colder the roosting place was changed to one at a lower altitude. In Rucker Canyon the birds came at night to the mountain side above the site of old Camp Rucker.

As there were a thousand or more here, their morning and evening flights were quite impressive. In the Dragoon Range the parrots roosted somewhere near the head of Cochise Stronghold and made a morning flight that often carried them directly out over the plains to the east.

In feeding, the large bands usually broke up into smaller parties. In winter such flocks at times came down to perch on broken sandstone ledges where they clambered about or basked in the sun. In Backer Canyon toward evening flocks often flew down to the river to drink before passing on to their roost. In the Chiricahua Mountains during late fall and winter the birds came down into the foothills to an altitude of between 5000 and 5500 feet though earlier they were confined to the higher basins.

The birds were noisy and their coming was heralded by their loud calls that were said to be readily audible at a distance of more than a mile. Like parrots elsewhere they were said to show much fear of hawks (though it seems strange that a bird with so powerful a bill and so muscular a body should show such fear); when a red-tail or a hawk of some other species appeared they rose in flocks and circled in the air, doubling the volume of their ordinary screeching calls.

He was told by F. Hands that "as cold weather came during the fall some of the birds disappeared. Others remained during the entire winter, although at one time the ground was covered by six inches of snow for over two weeks and the birds were forced to seek their food on the ground where this covering had blown partly away."

DISTRIBUTION

Range.—Southeastern Arizona and southwestern New Mexico (casually) and the mountainous area adjacent to the tableland of Mexico.

The range of the thick-billed parrot extends **north** to (casually) southeastern Arizona (probably near Mowry, Pinery Canyon, Cochise Head, and Paradise); (casually) southwestern New Mexico (Animas Mountain); eastern Chihuahua (Ciudad Chihuahua); and Veracruz (Perote). **East** to Veracruz (Perote and Jalapa). **South** to central Veracruz (Jalapa); Mexico (Popocatepetl); and southern Durango (Canyon Rio San Juan). **West** to Durango (Canyon Rio San Juan, Ciudad Durango, and Arroyo del Buey); western Chihuahua (Guadalupe y Calvo, Jesus Maria, and Colonia Garcia); and southeastern Arizona (Nogales and probably Mowry).

While the range as outlined includes the entire region over which the species is known to occur, it has actually been found breeding only in the Mexican States of Chihuahua and Durango. At irregular intervals flocks that sometimes are of large size travel northward and invade the mountain ranges of southeastern Arizona. Such invasions took place in 1900, 1904, and 1917.

Egg dates.—Mexico: 10 records, May 10 to August 25; 7 records, August 11 to 25.

Order CUCULIFORMES

Family CUCULIDAE: Cuckoos, Roadrunners, and Anis

CROTOPHAGA ANI Linnaeus

SMOOTH-BILLED ANI

HABITS

This curious member of the cuckoo family is a tropical species of rare, or perhaps only casual, occurrence within the limits of the United States; a few may occur more or less regularly in southern Florida or Louisiana, but to find it in abundance one must visit the West Indies or South America. It has a variety of local names, such as black ani, black witch, blackbird, savanna blackbird, and tickbird, the last from its habit of eating the ticks that infest cattle. Charles B. Taylor sent some interesting notes on this bird to W. E. D. Scott (1892); regarding its haunts in Jamaica, he says:

The Ani appears to be abundant in all parts of the island. It is one of the commonest birds near Kingston; and in most open and sparsely wooded lands, or in the vicinity of cultivated clearings, little groups or companies may nearly always be seen. Blackbirds are invariably present wherever cattle are pastured. I cannot recollect an instance in which I have noted a herd of cows at pasture without a flock of these birds appearing in company with them or in their immediate vicinity. This association is doubtless chiefly for the purpose of feeding on the ticks and other parasites on the animals, a good work largely shared by the Grackles (*Quiscalus crassirostris*). It is most interesting to watch a company of Blackbirds when thus engaged. Many are perched on the backs of the cattle (two or three sometimes on one cow), others are on the ground hopping about fearlessly among the grazing herd, searching for insects at the roots of the herbage or capturing those disturbed by the feet of the cattle.

Nesting.—He says on this subject:

Their nesting habits are exceedingly curious and interesting. Many individuals (possibly members of one flock) work together in the construction of a large nest in which all the females of the company lay their eggs. The number of eggs deposited in different nests varies greatly but is of course dependent on the number of birds in a company. Six and eight eggs are commonly found. I once took eleven, and in August last year I saw a clutch of twenty-one that had been taken from a single nest! It is probable that normally not more than two eggs are deposited by each bird, but nothing definite can be said on this point.

The nest, which is usually placed high up in a tall tree, very frequently in a clump of mistletoe on a “bastard cedar”, is a large, loosely constructed mass of twigs, entirely lined with dried leaves. But the most remarkable circumstance in connection with the nesting of these birds is the deposition of the eggs in *regular layers* with leaves between. This custom I had long heard of before an opportunity offered for personal observation. In the first nest I examined, the eggs were in two distinct layers, separated by a deep bed of dry leaves; the bottom layer consisted of four eggs and these, strange to say, were all infertile. I believe this singular habit is practised in all cases where a large number of birds resort to the same nest.

Dr. Alexander Wetmore (1927) writes:

Although this species often builds a communal nest, this is not always the case. Near Cayey [Porto Rico], January 22, two anis were seen constructing a nest in a tree about thirty feet above a small stream, the male sitting on a limb above while the female was in the nest, as yet only a loose mass of sticks and weeds. She moved and turned to shape it to her body, with her long tail sticking nearly straight up in the air. Near Bayamón, July 25, a single bird slipped quietly from a bulky nest in a clump of bamboos and only its mate appeared to join it. The nests were all large and bulky and were located from six to thirty feet above the ground. Bowdish reports a communal nest found near Aguadilla, August 13, built eight feet from the ground, in a thicket of bushes and trees. This nest contained twenty eggs, placed in layers of four or five, each layer being covered with dead leaves to separate it from the next lot of eggs above. Eight of the eggs were partly incubated and twelve were fresh.

John G. Wells (1902) says: “A flat nest is first built and about 6 or 7 eggs laid in it; then these are covered over and more eggs laid, and so on until four or five layers of nests have been constructed one over the other.

“I have seen four of these birds sitting on the nest together. When the top layer of eggs is hatched, and the young fledged, it is scraped off and incubation goes on with each succeeding layer, until all the eggs are hatched.”

Eggs.—On account of its communal nesting habits, the number of eggs laid by each individual ani does not seem to be known; nor is anything known about the period of incubation. The number of eggs found in a nest varies greatly, from 4 or 5 up to 20 or more, depending on the number of females laying in the nest. I have in my collection a set of 18, there is a set of 19 in the Thayer collection, and sets of 20 and 21 are mentioned above.

The eggs vary in shape from oval to elliptical-oval. Bendire (1895) says that the eggs “are glaucous-blue in color, and this is overlaid and hidden by a thin, chalky, white deposit; as incubation advances the eggs become more or less scratched and the blue underneath is then plainly visible in places, giving them a very peculiar appearance.” The underlying color of some of

the eggs that I have seen is “pale Nile blue”; others have described them as green.

The eggs are often very much nest stained. Mr. Taylor (Scott, 1892) makes the following interesting observation regarding the scratches: “What seems very singular is that comparatively little of this chalky covering gets rubbed off the sides, where from the turning over of the eggs in the nest we should expect to see the greatest extent of denudation, whereas *one* or *both ends* are nearly always *wholly* denuded. * * * So cleanly and evenly is it done, and to such an extent, that I feel confident it is the work of the birds themselves, their beaks alone being able to accomplish it. At the same time it is easy to see that the marks and scratches at the sides are the result of friction with the twigs and leaves of the nest.”

The measurements of 63 eggs average 35.03 by 26.27 millimeters; the eggs showing the four extremes measure **40.4** by 28, 36.5 by **30**, and **29.21** by **23.37** millimeters.

Young.—Alexander F. Skutch has sent me the following notes on the development of young smooth-billed anis, as observed by him on Barro Colorado Island, Panama: “The newly hatched ani is covered with black skin quite devoid of any trace of feathers. Its eyes are tightly closed. The development of this naked, helpless, little creature is amazingly rapid. By the second day the sheaths of the flight plumes have already begun to sprout. By the third day the eyes are open, and the sheaths of the contour feathers have begun to push out from the skin. The pinfeathers grow rapidly and become very long before they begin to release the true feathers which they enclose. When the young ani is 5 days old, the body feathers peep from the ends of their sheaths. At this age, the nestling could hang by one foot from my finger, and pull itself up by the use of its feet and bill, which was hooked over the support. When placed on the ground, it attempted to crawl away through the grass, and might have succeeded in escaping if I had not watched it carefully. Returned to the nest, it would not remain in the bowl, but climbed out to perch on the rim, where it uttered a little whine.

“When I approached the nest on the following day, the 6-day-old youngster hopped out and began to climb through the thorny branches of the orange tree, but soon it lost its hold and fell to the ground. When I picked it up, it uttered a weak imitation of the parents’ usual call. The flight

feathers, as well as the body feathers, were now pushing forth from their sheaths, and the latter were longer than on the preceding day. Much bare skin was, however, still visible between them. On the following morning the nest was empty; and I could not discover what had befallen the occupant."

Plumages.—The youngest bird I have seen is about half grown and fully feathered. The contour plumage is soft and short, "bone brown" on the under parts and somewhat darker above; the wings and tail are glossy, purplish black, much like those of the adult; the bill is smaller and less specialized than that of the adult. On account of the extended breeding season, it is difficult to give any definite dates for the molts. I have seen a young male, taken March 7, that was just completing the molt of the wings and tail, but the body plumage was still juvenal; from this I infer that the juvenal plumage may be worn for the whole of the first year. Maynard (1896) took some birds that had just completed molting on March 8; also some that had just begun to molt on April 24. I have seen young birds molting into adult plumage in December.

Food.—P. H. Gosse (1847) says of the food of the ani in Jamaica:

The food of our Blackbird, though consisting mainly of insects, is not confined to them. We usually find the stomach distended with caterpillars, moths, grasshoppers, beetles, and other insects, to such a degree that we wonder how the mass could have been forced in. But I have found these contents mixed up with, and stained by the berries of the snake-withe; and in July I have found the stomach crammed with the berries of the fiddle-wood, (*Cytharaxylon*) which had stained the whole inner surface a bright crimson. Flocks of these birds were at that time feeding on the glowing clusters profusely ripe upon the trees. Stationary insects are the staple food; to obtain which, they hop about grassy places, and are often seen to jump, or to run eagerly at their prey; on which occasions the long tail, continuing the given motion after the body has stopped, is thrown forward in an odd manner, sometimes nearly turning the bird head over heels. * * *

One day I noticed a cow lying down, around which were four or five Blackbirds, hopping on or off her neck, and eagerly picking the insects from her body; which service seemed in no wise unpleasing to her. I have also seen them leaping up on cows when grazing; and, on another occasion, jumping to and from a horse's back; and my lad Sam has repeatedly observed them clinging to a cow's tail, and picking insects from it, as far down as the terminal tuft. * * *

But stationary insects are not the only prey of the *Crotophaga*; in December, I have seen little groups of them engaged in the evenings, leaping up from the pasture about a yard into the air, doubtless after flying insects, which they seemed to catch. * * * I have seen one with a dragon-fly in its beak, which it had just caught, but it may have been while resting. At another time I saw that a Blackbird had actually made prey of one of our little nimble lizards (*Anolis*).

Maynard (1896) says that “anis live largely on locusts, especially a large species, which is quite common on the Bahamas, and which has a peculiar, rather disagreeable odor, which is imparted to the birds.” W. E. Clyde Todd (1916) says that, on the Isle of Pines, “it is fond of following in the wake of brush-fires, picking up the roasted lizards, snails, and insects.” Dr. Wetmore (1916) reports that the stomachs of 41 birds from Puerto Rico contained 91.3 percent animal matter, mostly harmful insects and arachnids, including mole crickets, other crickets, locusts, sugar-cane root-borers, leaf beetles, other beetles, squashbugs, other bugs, caterpillars, and spiders; the other 8.7 percent was vegetable matter, mostly seeds and fruits of 7 species of wild plants. He also says elsewhere (1927): “On May 20 near Yauco three anis were seen in a tree in which several mazambiques had nests. The anis were very near these nests, and the blackbirds, together with a pair of gray kingbirds, were much excited, but appeared to be unable to drive out the intruders. It was certain that the anis were bent on robbing the nests, and one was shot in the act of gulping down something which was later found to be an egg. No other instances of this evil habit were observed.”

Behavior.—Gosse (1847) writes:

Though its usual mode of progression on the ground is by hopping, or rather bounding, the feet being lifted together, the Blackbird is seen occasionally to *run* in a headlong manner for a short distance, moving the feet alternately. He is fond of sitting in the morning sun on a low tree with the wings expanded; remaining there perfectly still for a considerable time. In the heat of the day, in July and August, many may be seen in the lowland plains, sitting on the fences and logwood hedges with the beaks wide open, as if gasping for air; they then forget their usual loquacity and wariness. Often two or three will sit in the centre of a thick bush, overhung with a matted drapery of convolvulus, whence they utter their singular cry in a calling tone, as if they were playing at hide-and-seek, and requiring their fellows to come and find them.

Several observers have noted that anis roost at night huddled close together on a branch, like domestic fowls, and that they often bunch together in this way during a rain. Dr. Wetmore (1927) writes:

These strange birds are found in flocks that contain from half a dozen to twenty or more individuals, ranging mainly in pasture-lands, but going also into the cane-fields and orange groves to feed. In pastures they remain near the cattle, keeping ahead of them with long hops, in order to get the insects that the cattle scare up. Any intruder is greeted with a querulous call, and the whole flock flies in a straggling line across the fields to perch in a bush or low tree, where they crowd together and peer out curiously, their long tails and arched beaks giving them an odd appearance. In the early morning, when the grass is wet, they frequently sit in the sun with the wings extended in order to dry them or to absorb heat. The wings are small for the size of the bird, and the flight, accomplished by a series of steady wing beats alternating with short sails, is not strong. In a heavy wind the birds are

almost helpless, and they seldom rise high from the ground at any time. When on the wing, the back appears concave from the fact that the head and tail are held on a higher level.

C. J. Maynard (1896) says:

In flight they most nearly resemble a Canada or Florida Jay, alternately flopping and sailing, moving in a straight forward flight from tree to tree with great rapidity, uttering their mournful notes as they quickly disappear in the distance. * * *

A careful study of the Anis convinced me of the fact that a number of females are led by two or three males, and these males take great care of their charges. They utter cries of alarm when they perceive an intruder, and drive the females before them into a place of safety. I have even seen males fly against females or young birds which did not attempt to escape soon enough, and knock them off the limb on which they sat and then accompany them to a distant thicket.

I am inclined to think that the Ani is polygamous and this habit of the males taking care of a number of females would appear to confirm this idea.

Mr. Taylor (Scott, 1892) gives a somewhat different impression, thus:

The Blackbirds at their best have a very lean and shabby appearance, and are slow and awkward in their movements. I have watched an individual make several ineffectual attempts to alight on the frond of a cocoanut palm; but even among the branches of other trees their actions appear awkward. Their flight is slow and gliding, somewhat labored, and of little duration, the birds often appearing to fall short of the point originally aimed at. Yet they will chase the large yellow butterflies, and I was shown a large green locust that one of these birds was seen to capture in flight and afterwards drop. In the progress of a flock from place to place they do not usually fly all together, but move away in straggling groups or couples. One or more individuals first start off with their wailing call, followed soon after by two or three; after a little delay then two more go; another pause, then one, then three, and so on. If a tree has very dense foliage they alight (with much awkward scrambling) on the tops or extremities of the highest branches, where they may gain a clear and uninterrupted view, and this is usually the case when they are traversing very open country.

Voice.—The note of the ani has been called a wailing or a whining whistle; it has been said to resemble the notes of the wood duck. Dr. Wetmore (1927) says: “The ordinary call-notes are a low *kur-r-rk* and a querulous *quee ick, quee ick*, varied by low chuckling notes. When the birds are at all wild, they serve to alarm the entire country, as they begin to call on the slightest provocation.” It has also been expressed as *que-yuch, que-yuch, que-yuch* by Gosse (1847) and similarly by others.

Field marks.—The ani is such a peculiar and unique bird that it could hardly be mistaken for anything else. It is the only long, slender, black bird, with a long tail, short wings, and a huge bill, that is to be found within its range, so far as I know. Its shape and its manner of flight are quite different from those of the grackles; its concave back in flight, referred to above, is

distinctive. Gosse (1847) says: "The appearance of the bird in its sliding flights is unusual; the body is slender, the head large, and the beak enormous; and as in flying it assumes a perfectly straight form, with the long tail in the same line, without flapping the wings, it takes the aspect, on a side view, rather of a fish than of a bird."

Where our two small species of anis come together in Panama, they are likely to be confused, but the voices of the two are quite distinct; the call of the groove-billed is softer and higher in pitch, while that of the smooth-billed is more raucous and whining. The grooves in the bill of the former are not easily seen, except under favorable circumstances, but the culmen of the bill in the smooth-billed is much higher and sharper than in the groove-billed.

DISTRIBUTION

Range.—South America and the West Indies; casual in winter in eastern Central America and casual or accidental in Florida, Louisiana, North Carolina, and Pennsylvania.

The normal range of the smooth-billed ani extends **north** to western Cuba (Los Indios and Nueva Gerona); the Bahama Islands (Little Abaco and Nassau); Haiti (Jacmel); the Dominican Republic (Sesua and Ciudad Trujillo); Puerto Rico (Aguidillo and Rio Piedras); and the Virgin Islands (Cuelebra, Vieques, and St. Croix). **East** to the Virgin Islands (St. Croix); the Lesser Antilles (Grenada and Trinidad); British Guiana (Georgetown); Surinam (Paramaribo); and eastern Brazil (Ilha Mexiana, Maranhao, Bahia, Sapetiba, Canatgallo, Rio de Janeiro, and Iguape). **South** to southeastern Brazil (Iguape); and northern Argentina (Posadas, the Chaco District, and San Jose). **West** to northwestern Argentina (San Jose, Concepcion, and Salta); Peru (Huanuco and Iquitos); northwestern Colombia (the Cauca River Valley); Panama (Gatun, Perme, and Obaldia); Jamaica (Port Henderson, Spanish Town, Grand Cayman, and Little Cayman); and western Cuba (Los Indios).

This species occurs in other parts of Central America only in winter, when it has been recorded from Costa Rica (Rio Coto); Nicaragua (Great and Little Corn Islands); Honduras (Ruatan Island); and Quintana Roo, Mexico (Cozumel Island).

Casual records.—The following are among the several records the species has for Florida: Flamingo, a specimen taken in June 1916; Pass-a-Grille, a specimen collected on February 25, 1929; Dry Tortugas, three seen June 18 and 19, 1935; and Miami Beach, a small flock seen on February 24, 1937. Undated specimens also are available for Brevard County, Tortugas, Pepper Hammock near the head of the Banana River, and Charlotte Harbor.

A specimen was collected on July 18, 1893, at Diamond, and several have been reported from the vicinity of Buras, St. Bernard Parish, La. A specimen was collected previous to 1866 at Edenton, N. C., while individuals were reported from Piney Creek in the western part of that State on July 17, 1932, June 1, 1933, and June 25, 1934. There also is an old record of the occurrence of this species on Pettys Island in the Delaware River, opposite Kensington, Pa.

Egg dates.—Bahamas: 2 records, June 14.

Brazil: 3 records, October 1 to November 18.

British Guiana: 3 records, January 14 to July 2.

Cuba: 4 records, February 22 to August 20.

Jamaica: 7 records, May 29 to July 30.

Puerto Rico: 2 records, July 25 and August 13.

CROTOPHAGA SULCIROSTRIS SULCIROSTRIS Swainson

GROOVE-BILLED ANI

[Plates 1, 2](#)

HABITS

This Central and South American species was added to our fauna by George B. Sennett (1879), who secured a fine male on May 19, 1878, near Lomita, Tex., while it was “flying about the low bushes in open chaparral. It was very shy, flying in and about the bushes, and was shot on the wing.” The only one I have ever seen did not seem at all shy. I was sitting down, quietly watching some Texas sparrows that were hopping around on the ground near me, in some thick brush bordering a resaca near Brownsville, Tex., when one of these curious birds appeared. It seemed more curious

than shy, as it moved about slowly in the bushes, looking me over; it remained in my vicinity for some time and I could have shot it easily. It is said to show a preference for thick underbrush in the vicinity of water, or for lightly wooded swamps.

In his proposed work on the birds of the Caribbean lowlands, Alexander F. Skutch devotes two long and very interesting chapters to the home life of the groove-billed ani. He has kindly placed at my disposal his unpublished manuscript and allowed me to quote freely from it. As to its haunts, he writes: "The variety of the habitat of the anis is enormous and their only restriction seems to be that they do not tolerate the forest and are never seen there. They are birds of open country but seem nearly indifferent to its type. In the inhabited districts of the humid coastal regions they are one of the most conspicuous species. Their favorite haunts are bushy pastures, orchards, the lighter second growth, and even lawns and clearings about the native huts. Marshland is as acceptable to them as a well-drained hillside, and they are numerous in such extensive stands of sawgrass as that surrounding the Toloa Lagoon in Honduras, although it is probable that they do not venture far from some outstanding hummock or ridge which supports a few low bushes in which they can roost and nest. In the semidesert regions of the interior, where their associates of the coast lands, if present at all, are as a rule rare and restricted to the moist thickets along the rivers, they are among the most numerous of birds, and live among scattered cacti and acacias as successfully as amid the rankest vegetation of the districts watered by 12 feet of rainfall in the year. In altitude they range upward to 5,000 feet, but are not nearly so numerous in the elevated districts as in the lowlands."

Nesting.—Dr. Charles W. Richmond, drawing on his experience with it near Bluefields, Nicaragua, sent some elaborate notes on the groove-billed ani to Major Bendire (1895), from which I quote as follows:

It appears to breed at various times during the year, as I have found fresh eggs July 6, 1892, and young birds, recently from the nest, November 29, the breeding season spreading over seven months of the year at least, as it begins nesting earlier than the date of taking my first eggs. Nests are frequently built in the heart of a thick, thorny orange or lemon tree, and this appears to be a favorite situation. In this case the nest is from 4 to 7 feet from the ground, and, besides being difficult to get at, is somewhat protected from invasion by the wasps which almost invariably take up their abode in the same tree. In going through a small lemon grove I found an old nest of this species. In the cavity there were no eggs, but on poking the nest to pieces six badly decayed eggs rolled out.

One nest containing three eggs in the proper place and two others at the bottom, under the lining of green leaves, was located in a bamboo about 12 feet from the ground. The eggs were fresh, and more would probably have been deposited; the leaves forming the lining were still green. The parent birds were away at the time. Another nest was situated in some vines which had over-run an old tree stub, and was about 15 feet from the ground.

It may be that where numerous eggs are deposited in one nest only those eggs that are deposited in the proper place and directly influenced by the incubating bird are hatched, while those placed among the sticks forming the bulky exterior are left unhatched. It would be interesting to watch the progress of a large nestful of eggs and note results. The nests found by me were all composed of dead black twigs, rather loosely put together, very bulky and conspicuous structures, lined with green leaves, or, if old nests, with leaves that had the appearance of having been picked green.

One of Dr. Richmond's nests is described by Major Bendire (1895) as "a rather loose structure, about 10 inches in diameter and 4 inches in height. The inner cup measures 4 inches in diameter by 2½ inches in depth."

According to George K. Cherrie (1892) Señor Don Anastasio Alfaro says of the nests he collected in Costa Rica:

The structure is voluminous, composed chiefly of coarse dead twigs, but presents one peculiarity not observed in any other bird, namely the nest being lined with fresh green leaves. My three specimens were all placed in low trees, and neither was found at a greater height than three meters. One had been built above an old nest of one of the larger Tyrannidae.

It will not be without interest, I think, to insert my observations relative to one of these nests. On the 20th of May I noticed a Zopilotillo with a dry stick in its bill, which was immediately carried to a point in the hedge-row where it was deposited with three others. After assuring myself that the bird was building its nest there, I retired, with the intention of returning at a more opportune moment. And when one week later I returned to the same spot, what was my surprise to see not only the nest completed and containing six eggs, but more than this: in the thorns and leaves about it were scattered seven more eggs! As a consequence, if that collection was not the work of the Zopilotillos collectively, the poor owner would have had to deposit three eggs daily! In the finding of some of the eggs scattered in the leaves was revealed one of the architect's peculiarities. A hole had been left in the centre of the nest and only recently filled with leaves whose fresh green color testified that they had been cut and placed there later than the others forming the carpeting to the bottom of this common incubator.

Alden H. Miller (1932) writes of his experience with the nesting habits of the groove-billed ani in El Salvador:

Several nests were watched from the time there was one egg until there were nine, or in one case eleven, eggs. The eggs were deposited regularly at one day intervals and there was no certain indication in any of the sets, of two females contributing to the same nest as has been claimed by other observers. My findings, however, do not prove conclusively that community nests may not exist, at least occasionally. Incubation is uniform within a set and, correspondingly, birds were found to be incubating regularly only after the sets were

completed. In the set of thirteen the eggs were resting in three layers in the necessarily ample cup of the nest.

Nest sites were from two feet to twenty-five feet above ground in almost any kind of bush or tree. Usually they were located between six and twelve feet above ground in thorny tangles or close twiggery. One nest was found, however, in an open crotch of a fan of a royal palm. Adult birds are not much in evidence around nests which are being built or around sets of incomplete eggs. When flushed from sets of complete eggs, they may approach within five feet of the intruder and utter their feeble, squeaking notes of protest.

A. J. van Rossem (1938) found several nests, on July 29, 1925, "in the mimosa scrub in the marsh along the north shore of Lake Olomega," in El Salvador. "These nests were all in similar situations, that is, they were rather conspicuously placed in mimosa bushes and more than six feet above the mud or water."

Mr. Skutch devoted considerable time to studying the communal nesting habits of the groove-billed ani. He found them to be among the latest birds to breed, remaining together in small flocks from February to May while the other birds of the region were raising their broods. During June he watched the construction of a communal nest, on which three pairs worked in perfect harmony, operating in pairs and not as a unit. One of the pair remained on the nest while its mate brought in the material. There was not the least jealousy between the pairs, and "two or more pairs often perched quietly in the same bush. Each pair preferred to work alone at the nest, and if a second pair flew into the nesting tree, the first often quietly withdrew. This was not always their conduct, and sometimes one of the second pair (probably the female) took a place on the nest beside one of the first pair, while their two mates perched near by, or else brought them sticks.

"The normal set of eggs for each female is three, or more commonly four. Nests belonging to a single pair generally contain this number, and the nests belonging to two pairs, which I encountered, contained a maximum of eight. Once I found a nest with 12 eggs, which covered the bottom two layers deep, and six birds were interested in them. * * * Of ten nests which I found in Honduras and Guatemala and was able to watch for an adequate period, four were the property of single pairs, five of two pairs together, and one belonged to three pairs in common."

All the birds of both sexes took turns in incubating the eggs, but their shifts on the nest had no regular order and no fixed duration. "Just as the parents cooperate in incubating the eggs, they all join in the care of the

nestlings. I have watched three nests, each belonging to two pairs, during the time they contained young. Two of them, I made quite sure, were attended by four adults, but at the other I could not convince myself that there were more than three attendants. Possibly some calamity had befallen the fourth bird, or possibly also I failed to recognize it, since the anis at this nest were unmarked and indistinguishable."

Eggs.—The groove-billed ani lays 4 to 13 eggs, the smaller numbers being apparently commoner. The eggs vary in shape from oval to elliptical-oval, or rarely to elliptical-ovate or rounded-ovate. The ground color, when visible, varies from "glaucous-blue" to "Nile blue" or "pale Nile blue"; when first laid, the ground color is completely covered with a thin layer of dull white, chalky deposit, which eventually becomes somewhat discolored; as incubation progresses this chalky covering becomes more or less scratched, by contact with the twigs in the nest or by the action of the bird's feet in turning the eggs, or in relining the nest, so that ultimately much or nearly all of the blue ground color is visible; even then the shell is not glossy. The measurements of 51 eggs average 30.93 by 24.06 millimeters; the eggs showing the four extremes measure **35** by 25, 32 by **26**, and **27.68** by **21.84** millimeters.

Young.—Mr. Skutch determined, by close observation of marked birds and by noting which individual of a marked pair laid the eggs, that both sexes share in the duties of incubation. They were very impatient sitters, constantly changing about; 30 minutes was the longest time that he saw one incubate, during the early stages, and they sometimes left the nest unguarded for 10 or 20 minutes, while they enjoyed each other's company; during the last two days of incubation they lengthened their periods on the eggs to from 30 minutes to an hour. He describes the hatching process in detail:

"Fourteen days after the last egg had been laid, I held one in my hand while the birdling worked its way out of it. When I first took it up there was a gap in the larger end, which extended about a third of the way around the circumference. The little bird's short, thick bill was in this gap, and so pressed out of position that the lower mandible overshot the upper, only a temporary condition. At intervals the struggling prisoner drew its bill farther into the egg, then suddenly pushed it outward, bringing the keeled upper edge, armed with a rather insignificant egg-tooth or 'pip,' against the edge

of the shell at one end of the hole, and breaking off a small fragment at the outward thrust. In its squirmings the bird, impelling itself in some manner I could not determine, rotated imperceptibly slowly in the shell, in such a way that the head, turned under one wing, moved backward and the upper edge of the bill was constantly brought to bear upon a fresh portion of the shell, which was chipped off at the next outward thrust. Occasionally the struggling birdling emitted a weak cry. Thus bit by bit the ragged-margined aperture was lengthened until it extended about two-thirds of the way around the egg, when the struggles of the bird succeeded in cracking the remainder, and the large end of the shell fell off as a cap. Then the naked creature wormed its way out into my palm, where it lay exhausted by its continued effort.”

In another nest the last egg to be laid hatched in 13 days. The young, when first hatched, “were blind, black-skinned, and without any trace of feathers.” When the nestlings were 6 days old they were “both bristling with long pinfeathers. The plumes of the one that was the older by a few hours were already beginning to peep through the tips of these pinfeathers.” Later in the day, the feathers had burst their sheaths with amazing rapidity, and the young bird “was already well covered. The back and belly, save for a naked line down the middle of the latter, bore a soft, downy black plumage. Broad ends of the flight feathers of both wings and tail now showed.”

From this time on, the young birds became more lively and left the nest when approached, climbing about among the branches or down onto the ground to hide in the grass. They were brooded by one of the parents until they were a week old, returning to the nest at night. But “the following two nights they remained in the nesting tree but did not return to the nest to sleep. They could not yet fly and had entered a half scansorial, half terrestrial stage of existence. When they were 10 days old I tried to catch them for a photograph, but they hopped from limb to limb with such agility that, protected as they were by the sharp thorns, I was unable to secure them. When they were 10 days old they could make short flights from branch to branch of the same bush. Their bills were smooth, without any grooves, and their cheeks were bare of feathers.”

All through his account Mr. Skutch emphasizes the affection that is shown by all the members of the ani family for one another. While he was

watching the second nest of the pair on which the above observations were made, something very unusual, if not entirely unique, happened. One of the young from the previous brood, now nearly fully grown, “was the constant companion of his parents during the period in which they were busy with their second brood.” He frequently perched on the rim of the nest while one of his parents was incubating, and twice he was seen to offer food to the parent on the nest. After the young had hatched, “the youngster fed the nestlings regularly, but not so often as his parents.” During four hours and a quarter, he saw the male, always the more attentive parent, bring food to the nestlings 29 times, the female 14, and the young assistant 8 times. “The young bird not only fed the nestlings but was ardent in protecting them, flying up close to me and uttering an angry *grrr-rr-rr* whenever I came near them. In the absence of the parents he attempted to defend them alone.”

Plumages.—The young nestling and the development of its first plumage are described above. In this juvenal plumage the contour feathers are short and soft, “bone brown” below and darker above; the wings and tail are much like those of the adult; the bill is less specialized and not grooved.

Van Rossem (1938) says:

The postjuvenal plumage, attained by a complete body and tail and a partial wing molt, is not different from that of the adults except that the rectrices are noticeably narrower. An irregular molt of the primaries takes place at this time, although some of the juvenal quills (though their number and location varies) are held over till the following spring. In March and April of the next year there is a partial body, tail, and wing molt in which an irregular number of rectrices are renewed and such juvenile remiges as have been held over from the previous fall are replaced by new ones. The adult plumage with wide rectrices follows in the second fall, that is, at the first annual molt. The time of the annual molt extends from the middle of July to the first of October, the younger (one-year-old) birds molting earlier than the older ones. The spring molt of the adults includes some of the rectrices and secondaries.

Food.—Mr. Skutch writes (MS.): “The food of the anis consists largely of insects, which they secure both from the ground and among the foliage of bushes, and to a smaller extent of fruit and berries. Often they hunt grasshoppers and other creatures among the long grass or tall weeds, where they are completely hidden from view except when occasionally they leap a foot or so above the herbage to snatch up an insect which has tried to escape by flight. Perhaps their favorite method of foraging is beside a grazing cow or mule. Several together remain close to the head of the beast, moving along by awkward hops as it moves and just managing to escape its jaws

and forefeet, ever on the alert to snatch up the insects frightened from their retreat in the grass by the passage of the herbivore. It is frequently stated in books, and affirmed by the residents of the countries where the anis live, that they alight upon cattle and pluck ticks and other vermin from their skin—hence the name *garrapatero* (tick-eater) given them in Costa Rica.

“While this is doubtless true in certain parts of the ani’s range, I have watched them in the neighborhood of cattle from Panama to Guatemala and only once in three years have I seen an ani alight on a cow. Since the ani associates so much with cattle without alighting upon them, and the giant cowbird, another black bird of approximately the same size, does frequently perch upon them and relieve them of their vermin, it seems likely that the ani may be often credited with the acts of the cowbird, especially since the latter is shier and less known. I have occasionally questioned one who informed me that the ani plucks ticks from the grazing animals, only to find that he was unaware of the existence of the giant cowbird. At a little distance such a person might easily suppose that the birds upon the animal’s back were the same as those about its feet, and since his closer approach would leave only the latter, the illusion would probably persist.”

He has also seen a group of anis excitedly following a battalion of army ants, probably not to feed on such fiery morsels as the ants, but to pick up “the cockroaches, spiders, and other small creatures driven from their retreat among the dead leaves by the relentless hordes.” Again he says: “After the first heavy rain of the season has sent the winged brood of the termites forth from their nests in countless millions, one can watch the anis everywhere feeding like flycatchers, making ungraceful darts, not exceeding a few feet, from low twigs and fences; but the insects are so numerous on these occasions that they can catch many without quitting their perches.”

Major Bendire (1895) quotes Dr. Richmond as saying: “The food of those examined by me on banana plantations consisted almost entirely of small grasshoppers, the stomachs being much distended with these insects. From the fresh earth found on the bill and feet of these birds, I should judge they also feed on the ground.”

Prof. A. L. Herrera, of the City of Mexico, wrote to Bendire that “it is a social bird, being usually found in small companies of from six to fifteen

individuals, absolutely monogamous, sedentary, and of semidomesticated habits, frequenting the haciendas and the fields and pastures in their vicinity, and as it is considered very useful because of its habit of destroying large numbers of parasites infesting the cattle, it is not molested by the inhabitants, and becomes very tame. It extracts the *Ixodes* and other *Acaridans* with remarkable skill, without causing ulcerations which might result from the proboscis or sucker remaining in the fibres of the skin, and it must be regarded as one of the most useful birds of Mexico, especially in warm regions, so abounding in parasites of all kinds.”

Behavior.—Dr. Richmond (Bendire, 1895) says:

At Mr. Haymond’s plantation, on the Escondido River, above Bluefields, this species was unusually plentiful, owing, no doubt, to a large number of cattle kept there. The birds follow these animals as they meander over the pastures, hopping along on each side of an animal, catching grasshoppers and other insects which the cow disturbs as it moves along. Frequently the cow moves too rapidly and the birds lag behind, when they make short flights to the front again, passing over one another after the manner of the Grackles when feeding in a field. Only half a dozen birds or so follow a cow usually, and not many congregate in a flock, except when roosting. On this plantation, where the species is more abundant than usual, the birds appear to roost in numbers. An orange tree near the house was a favorite place where thirty or forty birds came to pass the night, flying in from the surrounding pasture about dusk, and after a few short flights from one tree to another, passed into the roost one or two at a time, hopping about as if seeking a favorable perch, uttering their peculiar note meanwhile. Out of this roost I shot seventeen birds one evening, and the males greatly predominated; there were only five females in the lot. The note of this species reminds one somewhat of the Flicker, *Colaptes auratus*, but may be better represented by the combination “plee-co,” rapidly repeated, with the accent usually on the first syllable, but sometimes on the last. I have frequently found one of the small flocks resting on a bush or bamboo along the water’s edge, perfectly silent, until my near approach started them off, one or two at a time, scolding as they went. Their flight is even, slow, as short as possible, and consists of a few flaps of the wings, followed by a short sail, then a few more flaps, etc.

Mr. Skutch writes (MS.): “Their flight is as perfectly characteristic of the birds as any other of their peculiar habits. A long journey, say anything much in excess of a hundred yards, is seldom made by a continuous flight, but the bird advances with frequent pauses in conveniently situated trees and bushes. As he alights on one of the lower branches, the momentum of the long tail carries it forward above the head with an abrupt jerk. Recovering his balance, he remains here for some moments, looking around with caution and calling in his high-pitched voice. Then, satisfied that the path ahead is clear, with a *tuc tuc tuc*, *pihuy pihuy pihuy*, he launches himself upon the next stage of his journey. A few rapid beats of his short

wings serve to impart the requisite momentum, and he sets them for a long glide. In this manner he can cover surprisingly long distances, on a slightly descending course, without further muscular effort. If his ultimate destination is a certain perch in a tree or bush, he will often arrest his flight on another considerably below it. By a few queer, rapid, sideways hops along the branch, and some bounds, or better bounces, from limb to limb, he gains the desired position where, as likely as not, he spreads his wings to the morning sun.”

Mr. van Rossem (1938) says: “They feed side by side with never a sign of friction or argument over the choicer insects, and at night roost in low trees or bushes, pressed shoulder to shoulder to the limit of available space. We not infrequently found them thus when hunting at night. During the time just preceding nesting it was noticeable that they were inclined to roost two and two instead of in a long line. In some cases the pair was sitting in close contact, even though there might be plenty of room to perch comfortably.”

Voice.—Dr. Richmond’s impression of the note of the groove-billed ani is mentioned above. Señor Alfaro says, in his notes given Mr. Cherrie (1892), that, in Costa Rica, it is “known as ‘Tijo, tijo’ (tée-ho) in imitation of its peculiar notes which seem to repeat the word *tijo* over and over again.” Dr. Chapman (1896) says: “Its note is a prolonged *chee-wyyah*, easily distinguishable from the single whining whistle of *C. ani*.” Mr. van Rossem (1938) says: “The ordinary or ‘conversational’ notes are a series of very liquid and what can best be described as ‘contented’ bubblings and cluckings. The louder, often repeated ‘chee-múy-o-chee-múy-o’ is the alarm note.”

DISTRIBUTION

Range.—Lower Rio Grande Valley of Texas and Baja California, south to northern South America.

The range of the groove-billed ani extends **north** to southern Baja California (San Pedro and Santiago); southern Sinaloa (Mazatlan); southern Texas (Lomita and Corpus Christi); and Yucatan (Progreso and Chichen-Itza). **East** to Yucatan (Chichen-Itza); British Honduras (Cayo); Honduras (La Ceiba); Colombia (Santa Marta); and Venezuela (Altigracia, Curaçao,

and Caicara). **South** to Venezuela (Caicara); and Peru (Chachapoyas and Lima). **West** to Peru (Lima and Cutervo); Ecuador (Tumbez); Costa Rica (San Jose and La Palma); El Salvador (Sonsonate); western Guatemala (Lake Amatitlan); southwestern Chiapas (Tapachula); Nayarit (San Blas); and southern Baja California (San Jose del Cabo and San Pedro).

The birds found in the cape district of Baja California have been separated subspecifically and are known as the San Lucas ani, *C. s. pallidula*.

Casual records.—In Louisiana a specimen was collected near New Orleans about 1890; another was taken at Ostrica during the winter of 1919; a third was obtained without date of collection near Houma; a fourth was taken at Cottonport on December 11, 1932; while a fifth was collected on Grand Isle on April 23, 1935. A specimen was taken at Huachucas, Ariz., 10 miles from the Mexican border, in May 1888, and it was noted at a point 20 miles north of Tucson on August 21, 1932. About November 1, 1904, a specimen was taken near Emporia, Kans., and on October 12, 1913, one was killed on an island about 9 miles north of Red Wing, Minn. A specimen was obtained at Jupiter Inlet, Fla., during the first week in January 1891, and there is also a somewhat indefinite record of its occurrence in the early part of January in the vicinity of Kingston, Jamaica.

Egg dates.—Central America: 4 records, May 27 to July 29.

Baja California: 6 records, April 1 to September 3.

Mexico: 40 records, March 20 to August 14; 20 records, May 16 to June 30, indicating the height of the season.

Texas: 7 records, March 17 to July 15.

CROTOPHAGA SULCIROSTRIS PALLIDULA Bangs and Penard

SAN LUCAS ANI

HABITS

The groove-billed ani of the Cape region of Lower California, Mexico, was described and given the above name by Bangs and Penard (1921), based on a series of 18 specimens from San José del Cabo. It is said to be—

similar to *Crotophaga sulcirostris sulcirostris* Swainson of Mexico, and of about the same size, but much paler and with less purplish iridescence; the U-shaped iridescent markings of the back and breast paler and duller greenish, not so brilliant; the dull purplish bronze of the head and neck of true *sulcirostris* replaced by paler, more grayish bronze; the lustreless parts of the body-feathers grayish brownish black instead of dull black. * * *

Eighteen adults of this new form, laid out beside a series of nearly double that number from various points in Mexico and Central America, are strikingly different; the pale, dull colors of the Lower California bird cannot be matched by any specimen in our series of true *sulcirostris*. The difference is noticeable at a glance but rather difficult to describe. Brewster (1902), in his account of the birds of the Cape Region of Lower California, states that the Groove-billed Ani is not known to occur in central and northern Lower California, and that the colonies which have become established in the Cape region were probably originated by birds which came from western Mexico. However this may be, the isolated colony of Cape St. Lucas has developed into a very distinct form, worthy of recognition.

Griffing Bancroft writes to me, on August 6, 1937, that he is "as nearly satisfied as it is possible to be from negative evidence" that this form of the groove-billed ani is extinct.

Nesting.—Lyman Belding (1883) writes: "The 1st of April I discovered four of these birds in a marsh, in which was a rank growth of *tule*, flags, and reeds. Having shot one of them, and the others were not molested, they remained in the marsh until May 15, or later. A nest found April 29 contained eight eggs. It was fastened to upright reeds, and was composed of coarse weed stalks and mesquite twigs, lined with green leaves. The female, while incubating, was very wary, slipping quietly away from the nest and returning to it very stealthily, below the tops of the reeds."

William Brewster (1902) says that the nest "taken by Mr. Frazar was in a willow about twenty feet above the ground. It is a flat-loose, but withal rather neat structure, formed outwardly of dead twigs and very substantially lined with cottonwood and willow leaves, which look as if they must have been dry when gathered. * * * This nest measures about six inches across the top, and the cavity is nearly an inch in depth. * * * Mr. Frazar met with the Groove-billed Ani only at San José del Cabo, where a flock of about thirty frequented some thick brush about pools of water near the mouth of the river."

Eggs.—The eggs of this race are apparently just like those of the species elsewhere. The measurements of 9 eggs average 31.99 by 24.00 millimeters; the eggs showing the four extremes measure **33.5** by 23, 31.8 by **24.9**, and 30.5 by **23** millimeters.

GEOCOCCYX CALIFORNIANUS (Lesson)

ROADRUNNER

Plates 3–6

HABITS

CONTRIBUTED BY GEORGE MIKSCH SUTTON

Most ornithologists are to some extent acquainted with the roadrunner. They have read about him. They have heard strange stories about him. Perhaps, driving along some road in the Southwestern United States, they have even seen him. But he who really knows the roadrunner has risen morning after morning with the desert sun; thrilled at the brilliance of the desert stars; seen day turned to sudden night by the dust-storm; pulled cactus spines from his shins. He who knows the roadrunner, he who has measured the breadth and the depth of this unique bird personality, has lived with him—not for an hour or so, not for a day, but week after week after week.

First impressions almost invariably give us an inadequate concept of this strange bird. We hear him scuttle through dry leaves ahead of us, catch a glimpse of him as he slips back of a rock, know from the flight of a grasshopper that he has gone a certain way, and that is all. Or, coming upon him suddenly, we surprise him into flight; note his short, rounded wings, long tail, and coarsely streaked plumage; watch him sail down the arroyo; and marvel that with the shutting of his wings and reckless plunge into the thicket he is so instantly lost to view.

Sometimes, traveling in a motorcar, we come upon him perched on a fence post or telegraph pole close to the highway. Now we have opportunity to observe how slim he is, how long his legs, how noticeable his crest. But as we pass he leaps to the ground, swings off through the cactus clumps, and is gone. Once more only a glimpse! Once more only the retreat of a timid desert creature that appears to be half bird, half reptile.

But lie in wait for the roadrunner! Watch him race across the sand, full speed, after a lizard. Watch him put out a wing, change his course, throw up his tail, change his course again, plunge headlong into a clump of cactus, and emerge, whacking his limp victim on the ground. Watch him jerk a slender snake from the grass, fling it into the air, grasp it by the head or

neck, pummel it with his hard mandibles, and gulp it head first. Watch him stalk a grasshopper, slipping quietly forward, making a sudden rush with wings and tail fully spread, frightening the doomed insect into flight, then leaping 3 or 4 feet in air to snatch it flycatcherwise in his long bill. Watch the roadrunner for an hour at his daily business of catching food and you will deem him among the most amazing of all the desert's amazing creatures. Snake-killer indeed! Chaparral cock! Not by sitting quietly on fence posts, not by slipping shyly from the path, has the roadrunner earned for himself these bloodstirring names!

So odd, so even *funny* a creature is the roadrunner that it is natural to caricature him a bit in describing him. This J. L. Sloanaker (1913) has done when he writes:

Of all the birds on our list the Roadrunner is doubtless the most unique; indeed, he is *queer*, and would certainly take first prize in the freak class at the Arizona state fair. He is about two feet in length, with a tail as long as his body, color above brown streaked with black, bare spaces around eyes blue and orange, feathers of head and neck bristle-tipped, eyelids lashed, * * * his whole plumage coarse and harsh. Could you imagine such a looking creature? Try and think of a long striped snake on two legs, a feather duster on his head and another trailing behind; or a tall, slim tramp in a swallow-tailed coat, a black and blue eye, and a head of hair standing straight on end! There you are!

Elliott Coues (1903) describes roadrunners as “singular birds—cuckoos compounded of a chicken and a Magpie.” Mrs. Bailey (1902) considers them among the “most original and entertaining of western birds.” Other writers call them “odd,” “anomalous,” and “unique.” They are.

Throughout much of his range the roadrunner is known as the chaparral cock, or merely the chaparral. He also is called lizard bird, ground cuckoo, cock of the desert, and, as we have stated above, snake killer. The Mexicans call him the *paisano* or the *correo del camino*. The first of these names means *compatriot* or *fellow countryman*; according to some writers it expresses affectionate regard, and is to be freely translated “little friend.” The latter is almost the equivalent of our name roadrunner.

That the roadrunner has at times been known as the *churca* we learn from an interesting note by Elliott Coues (1900) who quotes from an anonymous Franciscan priest the following description, published in 1790: “The *Churca* is a kind of pheasant which has a long bill, dark plumage, a handsome tail and *four feet*. It has these latter facing outward in such

fashion that when it runs it leaves the track of two feet going forward and two going backward.”

Coues himself calls attention to the fact that the word “toes” must replace the word “feet” in the above paragraph if the description is to fit the roadrunner. The error may well have been the translator’s. At any rate, anyone who has observed a roadrunner’s tracks in the sand knows how faithfully these record the zygodactylism of the bird’s foot.

Many a fanciful tale is told of the roadrunner. According to the best known of these, the bird builds a fence of cactus spines about a sleeping rattler, letting the doomed reptile buffet itself to weariness until finally, in desperation, it is impaled on the spines or bites itself to death! According to other stories (some of which probably have a grain of truth in them), the bird will deliberately race the swiftest horse across the plains!

The speed of the roadrunner *is* remarkable. Not when he is flying—a flying roadrunner is as much out of his element as a swimming chicken—but when he is afoot. A. W. Anthony (1892) tells us of a pair of birds that apparently enjoyed being chased by a hound that could never catch them. He says:

At Hatchita [New Mexico] a pair came regularly to one of the mines for water, a small pool having been formed near the shaft, from the pumps. The visit was made at nearly the same hour each forenoon, and was eagerly looked forward to by a foxhound owned by one of the workmen. The dog never failed to give chase as soon as the birds were sighted, and the race was as much enjoyed by the birds as by the dog; they seemed to have no difficulty whatever in keeping well out of danger without taking wing, and usually found time during the chase to stop at the water hole and get their daily drink, after which they quickly disappeared.

H. C. Bryant (1916) quotes from Heermann that the roadrunner “may, however, be overtaken when followed on horseback over the vast open plains,” and Heermann is known to have seen “one captured by a couple of dogs.”

Richard Hunt (1920), who was able to check with a speedometer the actual speed of a roadrunner encountered “en route from Soledad to the Galiban Range” in California, writes: “At the top speed to which we provoked our victim, the famous runner was moving at the tremendous rate of 10 miles an hour on a practically level piece of road.”

H. H. Sheldon (1922a), who also checked the roadrunner's running speed with a speedometer, writes of the incident: "The car gained on the bird until about five yards separated us, and I saw it was running at its utmost speed. I instructed my friend, who was driving, not to press him further, and for fully three hundred yards the bird ran from the big monster in pursuit, the while the speedometer registered exactly fifteen miles per hour. When finally we approached very closely, the bird gave up and flew into a palm, where I plainly saw it, beak agape and apparently very much fatigued from the unusual exertion."

There is no doubt in my own mind that a fully adult roadrunner, can, for short distances, run faster than 15 miles an hour. Athletic directors tell us that an average man (not an athlete) can run 9 yards a second, or about 18 miles an hour. I know I can run as fast as the average man, and I know I have failed many a time to gain on a roadrunner that happened to appear on the road a short distance ahead of me. I distinctly recall catching two *young* roadrunners (with tails 7 or 8 inches long) in a little gully near Fort Worth, Tex., in about the year 1913. I had quite a chase and might never have caught them had they not been forced to run up a steep embankment.

Spring.—We have seen the roadrunner sneak off through the weeds—a frightened bird. We have seen him capture lizards, snakes, scorpions, centipedes, tarantulas—a veritable monster. But not until we have heard him sing do we know the finer side (as Aldrovandus of old would say) of his nature. On fine spring mornings he sings as the sun rises, and he may continue his fervent if somewhat monotonous performance for an hour or more. His favorite songperch is the eastern rim of a mesa where, full in the fresh sunlight, he can see far and wide. If there is no mesa, he chooses a dead tree or a high cactus. Here, directing his bill downward until it almost touches his toes, he begins to coo. *Coo, coo, coo, ooh, ooh, ooh, ooh, ooh*, he calls, pumping out the syllables in a hoarse, throaty voice, his head rising a little with each *coo*, until the bill points upward, the pitch of the song meanwhile dropping gradually lower. So he starts with head low and coo high, and ends vice versa. Cattlemen say that before he begins his song he "lays his beak on the rock."

The pumped-out series of coos we have just described doubtless is the roadrunner's love song. Before giving it he may parade in a prominent place, strutting with head held stiff and high, and wings and tail drooping.

On May 2, 1914, I witnessed what I believe were certain courtship antics of the male before the female not far from Fort Worth, Tex. According to the published account (Sutton, 1922) of this performance, the bird's "wings were spread, and he may have been preening and taking a sun bath, but circumstances * * * led me to think otherwise. Now and then he bowed, and affected a close examination of his feet, only to raise his head again, drop his wings, lift them again and spread his tail. * * * Before I knew it [I] was discovered * * * and then, without wings spread, leaped from the dead branch to the next lower one, whence on outstretched wings he sailed to the ground. I rushed up to where he had been, and was surprised to see two birds scuttling * * * off through the vines."

Nesting.—The nest, which usually is situated in a low tree, thicket, or clump of cactus 3 or 4 to 15 feet from the ground, is a rather compact, though not deeply cupped affair, about a foot in diameter and 6 to 8 inches high, with foundation of sticks and lining of leaves, grass, feathers, mesquite pods, snakeskin, roots, and dry flakes of cattle and horse manure. It is sometimes well hidden, sometimes not. In the Black Mesa country of the far western Oklahoma Panhandle John Semple and I found several nests in small cedar trees that grew on the mesa sides. Here the nests were well hidden, and since cedar trees were numerous our usual method of locating nests was watching the parent birds.

Rarely the nest is built on the ground. Such a nest, containing six eggs, was found by W. W. Brown in Sonora, Mexico, on May 15, 1905. This nest is preserved in the John E. Thayer collection.

A. C. Bent writes me of a nest found by F. C. Willard in Cochise County, Ariz., on April 23, 1916, "7 feet from the ground on a packrat's nest in a dense thicket of hawthorn." This nest was "merely a few leaves, etc., in a slight hollow in the rubbish of the rat's nest." While Mr. Bent was hunting ravens' nests among the abandoned oil derricks in the Kettleman Hills, Calif., his companion, J. R. Pemberton, told him that roadrunners sometimes built their nests in the lower parts of the derricks, using the sticks dropped by the ravens. Griffing Bancroft (1930) describes a nest with complete set of two eggs found in Lower California in the heart of a date palm * * * so well concealed that it could not be seen until much of the foliage had been cut away.

Eggs.—Roadrunner nests contain, as a rule, three to five or six eggs. Occasionally two eggs comprise a complete set; and sets with as many as 12 eggs have been recorded. Where large numbers of eggs are found in one nest it is supposed that more than one female has deposited them. Coues (1903) describes the eggs as “ovate or elliptical, white in ground color with an overlying chalky film which may take a slight yellowish tint, ranging in length from 1.45 to 1.75, averaging 1.55 x 1.20. They are laid at considerable intervals; incubation begins as soon as a few are deposited, and is believed to last 18 days for each egg. The development of the chicks is rapid; perfectly fresh eggs and newly-hatched young may be found together; and by the time the last young are breaking the shell the others may be graded up to half the size of the adult.”

[AUTHOR’S NOTE: The measurements of 55 eggs average 39.2 by 30.1 millimeters; the eggs showing the four extremes measure **44** by 30, 41 by 32, and **34** by 27.5 millimeters.]

William L. and Irene Finley (1915) describe a typical nest in which there were “one fresh egg, one egg just ready to hatch, two featherless, greasy, black young, and two young ones about grown and ready to leave home.” These authors were in a position to watch this nest almost continuously for several days. They did not observe more than one female parent. If, then, the several eggs were laid by more than one female, it may be supposed that the species is to some extent parasitic or promiscuous in this regard, as the yellow-billed and black-billed cuckoos also are thought to be.

Frank L. Burns (1915) also gives the period of incubation as “18 days.” It is supposed that only the female incubates. That incubation sometimes begins with the laying of the first egg rather than “as soon as a few are deposited” (see above quotation from Coues) has been observed by M. French Gilman (1915), who tells us of a nest with four eggs in which “one hatched July 20, the others on the three succeeding days.” In cases of this sort it is natural to suppose that all eggs are laid by the same female.

Eggs are usually laid in April and May. W. E. D. Scott (1886) tells us, however, of several nests found on the San Pedro slope of the Catalina Mountains in Arizona, the earliest of which, discovered March 17, 1885, at an elevation of 3,000 feet, contained two fresh eggs. That two broods of young are sometimes, if not frequently, reared in one season has been

reported by numerous authors, though I believe it has not been proved beyond doubt that fresh eggs found in July are laid by females that have already succeeded in bringing out one brood.

Young.—Newly hatched young are odd, “featherless, greasy, black” creatures with a reptilian appearance. I had a good deal to do with young roadrunners in the vicinity of Fort Worth, Tex., as a boy, so I quote my own description here (1922):

The nestling bird, be it ever so young, has an unmistakable cuckoo-like expression in its face, though its eyes, upon which a good portion of the facial expression depends, are quite different from those of the adult, being of a deep dull brown with a bluish pupil. * * * The eyelashes are small, in fact scarcely apparent. Its whole external appearance is very sombre, and rather dirty-looking, as though the creature had been bathed in some unrefined oil, which had not been properly administered. The white hairs, each of which marks a coming feather, all lie in rows and look as if they had been rudely combed into place. The rather large, pale blue-gray feet are strong in the toes, but very weak at the heel, so that the birds cling to the fingers or the twigs of their nest with some power, but are quite unable to rise. Whenever there were very many young birds in the nest they presented a peculiarly scrambled appearance, due, I believe, to the constant disturbance at feeding time more than to restlessness, for they usually lie quite still. By May 1 [the birds were taken from the nest on April 29] feathers were appearing rapidly on my young birds, first on the top of the head, back, and wings, and then on the belly, tail, and throat. Once the blood-quills had started to burst, development was very rapid. On May 4 the birds were quite well feathered, the tails being one and one-half inches long, and they were quite able to walk unsteadily. It is at this period, or a little before, that the young leave the nest, though there must be innumerable dangers for the rather weak-legged creatures. Several times I have come across young birds able to run well, but still in trees, which leads me to believe that the young may, like young Green Herons, spend a portion of their early active life climbing about from branch to branch.

During the spring and summer of 1914 I reared these two young roadrunners, feeding them uncounted hundreds of grasshoppers, cave crickets, tarantulas, centipedes, scorpions, lizards, snakes, mice, cotton rats, and small birds. They were the most entertaining pets I ever had. Eventually they learned to capture their own food. Concerning this period of their life I have published the following paragraphs (1936):

After three weeks they became sturdy enough to catch part of their own food. With patient coaxing they were taught to pick up grasshoppers tossed to them, and finally to run after and capture crippled insects. Content at first, perforce, with sluggish, wingless nymphs which were abundant, they stole about through the weeds, wings pressed neatly against their slender bodies, snapping up the insects as fast as they could find them. Grasshoppers, often still alive and kicking, they swallowed with a toss of the head and a hollow gulp. Large green or gray cave crickets, which live in piles of boards, or in damp, shadowy places, were especially prized. When a yellow- or coral-winged grasshopper rose noisily from the path, the birds crouched in momentary fear, but soon began to mark the

return to earth of the clackety aeronaut and to steal up behind clumps of grass, intent upon a killing.

Finally they learned to capture the biggest, noisiest, and wariest grasshoppers on the prairies. They would watch a coral-wing in his courtship flight and, running stealthily, wait until the performer dropped to the ground. With a bound over low weeds, a dart across the open, and a final rush with outspread wings and tail, they would frighten their prey into the air, leap nimbly after him, nab him unerringly with their bills, and descend gracefully on outspread wings to beat him to insensibility with a whack or two on a stone.

Once they had learned to capture grasshoppers, their food problem was largely solved and, since they showed no inclination to run away, they were at liberty most of the time. They ran about the yard, playing with each other, or catching insects. In the heat of mid-day they sought the shelter of broad, cool leaves, and sprawled in the sand. Daily, often many times daily, I took them for a walk across the prairie. Following me closely or running at my side, they watched the big world with eyes far keener than my own. Grasshoppers which I frightened from the grass they captured in side expeditions. If I paused near a flat stone, they urged me on with grunts, bit gently at my hands, and raced back and forth in an ecstasy of anticipation.

I entertained misgivings concerning these flat stones. What savage creatures might not they conceal? Could young Road-Runners manage swift-tailed scorpions, sharp-toothed mice, or poisonous spiders? Under the first stone there were scorpions. The Road-Runners hesitated an instant, as if permitting an untried instinct to take possession of their brains, then rushed forward, thrust out their heads, and attacked the scorpions precisely at their tails. Perhaps these venomous tails received more than the usual number of benumbing blows, but the scorpions were swallowed with gusto.

I had not supposed that a Road-Runner would capture and devour a tarantula. One day, however, we paused at the tunnel of one of these big, furred spiders. Somewhat in the spirit of experimentation, and following the method known to all Texas boys, I teased the black Arachnid from her lair by twirling a wisp of grass in her face. She popped out viciously and jumped a good ten inches to one side. With a dash one bird was upon the monster before she had opportunity to leap a second time. A toss of the bird's head and one of the eight legs was gone. Free again, the spider leaped upon her captor. The other bird now entered the combat, snatched up the spider, and flicked off another leg. One by one the legs went down, and finally the two birds pulled apart and gulped the sable torso.

Plumages.—The newly hatched roadrunner's only plumage is coarse, long, white or whitish hairs. These do not by any means cover the dark-skinned body, but they apparently give the bird all the protection it needs. At this stage the light-colored egg tooth is noticeable, the feet are dark colored like the rest of the body, the mouth lining has a peculiarly blotched appearance, and the irides are dull brown.

As blood quills replace the long white hairs, the egg tooth disappears, the legs and feet turn blue-gray, the skin about and back of the eye lightens, and the blotching of the mouth lining becomes less conspicuous. The sprouting feathers now bear at their tips the white hairs of babyhood. Some of these hairs cling to the plumage long after the bird leaves the nest.

When young roadrunners begin to capture their own food they wear a plumage that is much like that of the adult. J. A. Allen (Scott, 1886) describes this plumage thus: "The chief difference in color consists in the broad shaft stripes of the feathers of the neck and breast being less sharply defined in the young than in the adult, and in the brown edgings bordering the shaft stripes being paler."

At this stage the bare skin about the eye becomes pale blue, and the naked patch back of the eye light orange. Too, the eye itself changes, a light-colored ring, which contrasts sharply with the brown or gray-brown of the rest of the iris, forming about the pupil. As the bird becomes older the bare skin of the face brightens. Fully adult males, at the height of the nesting season, are fairly resplendent with their high, steel-blue crest, brilliant eye, and bright orange patch back of the eye.

The plumage worn by the young bird after it loses its natal hairs is apparently the first winter plumage. Judging from such specimens as I have seen, the postnuptial molt is the only complete molt of the adult bird. Whether there is a partial prenuptial molt in young or adult birds I cannot say. The roadrunner's life is so strenuous that he doubtless loses feathers frequently. Spring birds with half-grown tail feathers may therefore not be performing a molt in the usual sense of the word.

Food.—Lizards (including the armored horned "toad"), small snakes, scorpions, tarantulas and other spiders, centipedes and millipedes, mice, cotton rats, ground-inhabiting small birds and their eggs and young, young quail, insects of all sorts, various fruits and seeds, including pricklypears: all these are eaten by the roadrunner; and this considerable list but hints at the rapacity and digestive powers of the gaunt bird. One of the most thorough-going reports on the roadrunner's food habits is that by Dr. Harold C. Bryant (1916). Informing us that animal food makes up slightly over 90 percent of the total food of the species in California, Dr. Bryant says:

Almost any animal, from the smaller rodents down to tiny insects, appears to be relished by this bird. Although the stomachs examined showed no large percentage of vertebrates, other published records show that reptiles sometimes form a large part, if not the entire diet. Even these larger elements of food are usually swallowed whole at one gulp. That the digestive apparatus is powerful is evidenced by the fact that bone, hair, and feathers pass through the digestive tract, and are not thrown back out through the mouth in the form of pellets, as is the case with some hawks and most owls.

A diagram in Dr. Bryant's paper makes it plain that grasshoppers and crickets form a considerable part (36.82 percent) of the roadrunner's food in California. Beetles form 18.2 percent; and seeds and fruits, cutworms and caterpillars, bugs, ants, bees, wasps, scorpions, lizards, mammals, fly larvae, birds, and "miscellaneous" items go to make up the rest.

Game officials are usually opposed to the roadrunner, for the bird is reputed to be an enemy of young quail. Regarding the bird's reputation as a quail destroyer, W. L. McAtee (1931) tells us that "the Road-runner is persecuted almost throughout its range * * * as an alleged destroyer of Quail eggs, and state bounties are even paid for its destruction. Yet the Road-runner never has been known to be a special enemy of Quail," and it doubtless "eats more scorpions, centipedes, and tarantulas, those poisonous nuisances of the Southwest, than it does Quail eggs."

Aldo Leopold (1922) writes of shooting a roadrunner "with a light-colored object in his bill." Examining the spot where the bird had fallen he "found a dead [quail] chick, still limber and warm but unmutilated * * * still in the downy stage, with 1/8-inch pin feathers on the wing—smaller than a domestic chick when hatched."

S. S. Visser (1910) writing of birds found by him in Pima County, Ariz., tells us that roadrunners "have been seen leaving the nest of Gambel's Quails carrying an egg in their beak."

That the roadrunner will occasionally capture birds as large as a mockingbird is apparent from several published accounts including that by Robert S. Woods (1927a) who tells us of "an immature but full-grown mockingbird" captured though not actually killed by a roadrunner near Azusa, Calif.; and that by Dr. W. K. Fisher (1904) who reports that a roadrunner was seen to "remove from a nest a young mockingbird and devour it" in Mission Valley, near San Diego, Calif.

In a letter to A. C. Bent, Mr. Woods (who is quoted above) speaks of a short article written by the late Rev. St. John O'Sullivan, of San Juan Capistrano Mission, "describing a roadrunner's method of killing a swift by lying in wait in a creek-bed near Palm Springs and suddenly springing into the air to knock down one of the birds which passed within its reach." This capturing swifts in air seems perfectly plausible to me, for I recall seeing my captive roadrunners capturing English sparrows in much the same

manner. Walking about with a noncommittal air that was comically suggestive of the chickens that fed nearby, they gradually grew nearer to a sparrow, then with a dash to one side and a tremendous leap snatched the fleeing victim from the air.

A. W. Anthony (1896) tells us of suddenly coming upon a roadrunner that had just finished despatching a wood rat (*Neotoma*). “The bird reluctantly withdrew as I came upon the scene,” he writes, “leaving the rat, which I found to be quite dead.”

My own captive birds caught and killed a cotton rat (*Sigmodon*) that lived in a stone wall near our house in Fort Worth, Tex. Two or three times a day this rat scurried across a gap in this wall, and the birds came to look upon him as a possible meal. At first his speed and considerable size kept the enemy at a safe distance, but their interest sharpened daily, and eventually they formed the habit of loitering near the runway. One day I heard a squeal of terror and ran up in time to see the bewildered animal running this way and that, trying to escape two lightning-quick demons who never really held him, but pinched him, tossed him, dealt him blows, buffeted him, made him weary with fighting for life. Over his limp form the roadrunners had an argument. He was heavy. No sooner would one bird start to swallow than the other would be tugging at the hind foot or tail, and down he would drop. I finally cut the rat in two.

Carroll Dewilton Scott has contributed a note concerning roadrunners that catch half-grown “gophers” in his garden at Pacific Beach, Calif. Describing one of the birds he says: “Finally, after ridiculous gulplings and twistings of his neck he got the gopher down, balanced his tail, and ambled away from another conquest.”

Mrs. Bailey (1928) tells us of two roadrunners observed at Carlsbad, N. Mex., that “snapped their bills and chased each other up into a cottonwood on the bank where there were caterpillar nests. To determine what they had been eating, one was shot and its gizzard was found to contain not only caterpillar skins but a number of large grasshoppers, a large black cricket, beetles, a centipede six inches long, and part of a garter snake a foot long. The rest of the snake was down in the crop and the barely swallowed end up near the bill.”

Regarding the economic status of the roadrunner in California, Dr. Bryant (1916) says: "A preponderance of evidence favors the bird. The destruction of such unquestioned pests as grasshoppers, cutworms, caterpillars, and wireworms and of such rodents as mice is to be desired even if the amount of destruction be relatively small. The taking of this sort of food on wild land is evidence that this bird when feeding in cultivated fields is likely to be distinctly beneficial."

Behavior.—Anyone who has observed the roadrunner closely knows what an entertaining creature it is. Its voracity keeps it on the alert for food. To capture a grasshopper one moment, a race-runner lizard the next, and a tarantula the next requires strength, speed, and prowess. If nymph grasshoppers are numerous, the bird has no trouble in obtaining a meal. But flying grasshoppers are difficult to overtake; lizards escape because of the brittleness of their tails; and tarantulas have burrows into which they can pop when danger threatens.

Not often does one see a wild roadrunner capturing its food. I recall watching one a year or two ago, not far from Packsaddle Lake, a small, artificial body of water in western Oklahoma. I had climbed a sandy mound. Peering through the sagebrush, I saw a roadrunner under a bush not far away, busy feeding. He ran out into the sunlight now and then, but his attention was directed principally to grasshoppers that must have been feeding on the leaves. These he snatched, with nimble leaps upward, in the tip of his bill. To reach the insects that were in the midst of the bush, he scrambled noisily through the twigs, caught a few, then sprang back to the ground to catch those that had fallen or jumped out.

W. E. Allen (1932) gives us a breezy description of the capture of a small bird by a roadrunner that had been running just ahead of him. He says:

Before we had gone as much as a hundred yards, however, his [the roadrunner's] pacing routine was broken by a sudden dash from a slouchy pose and the development of a brown streak across the road which ended in his emergence on a bank beyond a parked automobile. Somewhere beneath the car he had struck a full-fledged young bird (probably a "California Linnet") at full speed * * *. The victim appeared to have been caught by the neck, which probably accounts for the fact that it made no outcry. It must have been badly stunned also by the stroke of the heavy beak because it struggled only feebly.

After becoming satisfied that I was not disposed to interfere, the captor moved on to a point about 10 feet farther away. Here he hammered the hard ground two or three times with the body of his victim, evidently destroying all signs of life. Then he dropped it,

grasped a wing near its base, and with a skilful jerk stripped nearly every feather from it * * *. After stripping the wing he spent three or four minutes in picking at the birdling's body with some hammering and jerking mixed in. Apparently this was for the purpose of getting rid of feathers. At any rate, they were thrown around profusely, and the movements were different from those a little later which seemed to be devoted to mauling and crushing the body into a shapeless mass.

Finally this mass (which seemed to be about as large as the roadrunner's head) was picked up with a kind of tossing motion which landed it in the back of his mouth. The first effort at swallowing, consisting of tossings of the head and spasmodic movements of the jaws and throat, only resulted in getting the mass started into the throat. After a short rest another series of these movements shifted it along to a visible extent but it was not till the fourth series was finished that the food appeared to have been swallowed completely. After this was accomplished the bird turned toward me and slouched into a curious pose of indifference mixed with satisfaction.

A roadrunner's program is full enough with only himself to feed. But when he has a nestful of hungry young he must indeed wear himself ragged catching insects and lizards and snakes. These he brings from near and far, going and coming in such a way as to keep himself hidden. With what satisfaction must he start a foot-long garter snake on its way down the gullet of one of his offspring, knowing that one voice at least will be stilled so long as any of that snake remains to be swallowed!

Many an interesting account has been written of the roadrunner's foraging activities. A. Brazier Howell (1916) tells us of difficulties he had in retrieving small bird specimens before roadrunners stole them. He writes:

While I was out collecting, these abundant birds would often be seen skulking about with eyes open for any opportunity, and it was always necessary, in such cases, to make a dash for a specimen after it was shot. On two occasions a roadrunner darted in and grabbed a bird when I had almost reached it, once hopping two feet in air to nip a sparrow that had lodged in the branches of a bush. At another time I was watching a small flock of sparrows as they busily fed in the brush, when I noted a roadrunner stealing up like a cat, taking advantage of every bit of cover. When at the proper distance, it rushed out and sprang into the air at the retreating sparrows.

J. Eugene Law (1923) tells us of a roadrunner that tried to pull a dead golden-crowned sparrow through the $\frac{3}{4}$ -inch-mesh wire of a sparrow trap. This writer states that he did not actually see the roadrunner kill the sparrow, though an autopsy showed "the entire brain area" to be "dark with blood infusion."

My pet roadrunners did not capture horned "frogs" unless other food was difficult to obtain. They killed and ate these well-armored reptiles, however. A horned lizard, confronted by its ancient foe, would flatten out,

rise high on its legs, and sway back and forth as if about to leap or inflict a dangerous bite. But a roadrunner is not to be bluffed. Grasping his tough victim by the head or back he beat it against a convenient stone. Thirty or forty blows were needed to render it sufficiently quiescent for ingestion. If swallowed while yet alive it had to be coughed up for further battering.

When not engaged in pursuing food a roadrunner may rest, seeking either a cool spot on the ground or the shadowy heart of a tree. In the morning it sometimes takes a sunbath. Henry W. Henshaw (1875) tells us that “it loves to meet the first rays of the rising sun, ascending for this purpose to the top of the mesquite trees, and, standing erect on the topmost branch, loosens its feathers, and appears to catch all the grateful warmth possible, remaining in this attitude for many minutes.” My captive roadrunners took sunbaths every day, spreading their wings and exposing the featherless tracts of their backs.

The roadrunner has a streak of domesticity in his nature. Mrs. Bailey (1922) has given us a delightful account of a remarkably tame though uncaged bird that lived about camp. Concerning this bird (which was known as “Koo”), Mrs. Bailey says:

It was not his potential usefulness as a camp watchman or killer of “varmints” but his ready friendliness and attractive ways which attached us to our rare camp visitor. If we were busy when he came he would call *koo, koo*, and then wait for us to discover him. Sometimes we would look hard before finding him and finally make him out standing on the mesquite slope above us, his feathers puffed out spreading the streaks on his chest till they and his light underparts toned in perfectly with a background of straw-colored ground and dry weed-stalks—completely camouflaging him. It was astonishing to see how such a large, marked bird could disappear in its background. And what a contrast that round, bird-like form made to the grotesque running figure we were familiar with—long neck, slender body, and long tail, one straight line.

J. K. Jensen (1923) tells us of a pair of roadrunners that fed “with the chickens on a ranch near Santa Fe,” that “came regularly for a ‘hand out’ and often went to roost in the poultry house.” This author does not specify what the “hand out” was. Needless to say, it hardly could have been corn or wheat.

Frightened from the nest a roadrunner may scuttle off to remain hidden for some time. Again, it may stay close by, attempting to lure the intruder away. M. French Gilman (1915) describes a mother bird that was “very anxious about the eggs,” that “ran around close to me in a mammalian sort of way, flat on the ground, tail dragging, and head stretched out in front

only about three inches from the soil. She did not look like a bird at all, and though making no fluttering demonstration, her antics were calculated to excite curiosity and distract attention from the nest.”

J. R. Pemberton (1916) describing a “variation of the broken-wing stunt” by a roadrunner, writes:

As I was climbing near the nest the bird hopped to the ground. Immediately it began to squirm, scramble, and drag itself away across an open space and in full view. The bird was simulating a broken leg instead of the conventional broken wing! The bird held its wings closed throughout the demonstration though frequently falling over on its side in its enthusiasm. The whole performance was kept entirely in my view, the bird gradually working away from the tree until it was some 35 feet distant when it immediately ran back to the base of the tree and repeated the whole show. I had been so interested up to now that I had failed to examine the nest which when looked into contained five young probably a week old. When I got to the ground the bird continued its stunt rather more frantically than before and in order to encourage the bird I followed, and was pleased to see it remain highly consistent until I was decoyed to a point well outside the grove. Here the bird suddenly ran away at full speed and in a direction still away from the nest.

Voice.—We have already described the spring song of the roadrunner. Howard Lacey (1911) tells us that the bird “makes a loud chuckling crowing noise * * * and also a cooing noise that might easily be mistaken for the voice of some kind of dove; it also makes a sort of purring sound in its throat, *perrp, perrp, perrp.*”

As for this “purring sound,” I am not at all sure that it is *vocal*. One of the roadrunner’s most characteristic alarm sounds is not a *cry* at all; it is an incisive, clackety noise made by rolling the mandibles together rapidly and sharply. Even young birds just out of the nest can produce this sound, though the softness of the bill muffles the sharpness somewhat.

Young birds in the nest make a buzzing sound when begging for food. Well do I remember a nestful of these “infant dragons” that I found in April 1914. Concerning these I have written (1936): “Accidentally I touched an upturned beak and four great mouths, wobbling uncertainly on scrawny necks, rose in unison. I jerked back my hand—the pink-blotched lining of those mouths had an almost poisonous appearance. From the depths of the small frames came a hoarse, many-toned buzzing which gave the impression that a colony of winged insects had been stirred to anger.”

Enemies.—There is little doubt in my mind that the roadrunner’s worst enemy is man. Man wants quail to shoot. Man sees a roadrunner chasing young quail or finds a young quail in a roadrunner’s stomach and lo,

thumbs turn down, another name goes on the black list, and the roadrunner's doom is sealed. In many a southwestern State there have been chaparral-cock drives and contests, bounty on roadrunners, newspaper stories and editorials defaming the bird. Too, there are those who *eat* roadrunners, or who chase and shoot them for "sport." In populated sections the roadrunner has a hard time. Where man appears the roadrunner all too frequently disappears.

Certain predatory birds and mammals doubtless prey occasionally on the roadrunner, though adult birds usually are swift or wary enough to evade such enemies as coyotes and hawks. Crows and ravens doubtless eat some roadrunner eggs and young. Even the snakes themselves may take a hand in keeping the snake killer tribe from becoming too numerous. Remains of a roadrunner were found in the stomach of a red-tailed hawk collected by Dr. Josselyn Van Tyne in Brewster County, Tex., February 28, 1935 (Van Tyne and Sutton, 1937).

W. L. McAtee (1931) in his timely "Little Essay on Vermin" so justly states the case of roadrunner *versus* mankind that his comments are of special significance here:

The Road-runner is persecuted almost throughout its range * * * as an alleged destroyer of Quail eggs, and state bounties are even paid for its destruction. Yet the Road-runner never has been shown to be a special enemy of Quail, and it cannot eat their eggs except during a brief season. The Road-runner is as nearly omnivorous as any of our birds, eating anything in its habitat that is readily available and swallowable. No doubt it will eat Quail eggs, but it is equally certain that not one meal in a thousand of all the birds at all times consists of Quail eggs.

The Road-runner actually lives up to its repute of killing rattlesnakes; without doubt, it eats more scorpions, centipedes, and tarantulas * * * than it does Quail eggs, and it is a voracious consumer of grasshoppers. It is a unique bird, not only in our fauna but in that of the world, has extremely interesting habits, and in its choice of food in the long run undoubtedly does more good than harm. Its persecution is all but baseless and is thoroughly unjustified.

DISTRIBUTION

Range.—Southwestern United States south to central Mexico.

The range of the roadrunner extends **north** to north-central California (Navarro River, Owens River, and Death Valley); Colorado (Meeker, Cañon City, and Las Animas); and southern Kansas (probably Caldwell and

Arkansas City). **East** to Kansas (Arkansas City); central Oklahoma (Norman); Texas (Fort Worth, Kerrville, and San Antonio); Tamaulipas (Matamoras, Soto La Marina, and Tampico); and Pueblo (San Salvador). **South** to Puebla (San Salvador); Mexico (Tenango); and Jalisco (Zapotlan). **West** to Jalisco (Zapotlan); Baja California (Cape San Lucas, San Cristobal Bay, and Rosario); and California (San Diego, Mentone, Santa Barbara, Sebastopol, and Navarro River).

Casual records.—The roadrunner is not known outside of its normal range, but a remarkable occurrence was the finding of a specimen at Marshall Pass, Colo., at an elevation of about 10,000 feet, on October 12, 1907.

Egg dates.—Arizona: 20 records, April 5 to June 24; 10 records, April 20 to June 3, indicating the height of the season.

California: 73 records, March 4 to July 16; 37 records, March 25 to May 2.

Mexico: 5 records, April 16 to May 16.

Texas: 57 records, March 18 to July 5; 29 records, May 3 to June 1.

COCCYZUS MINOR MAYNARDI Ridgway

MAYNARD'S CUCKOO

HABITS

The mangrove cuckoo (*Coccyzus minor minor*) long remained on the A. O. U. Check-list, including the third edition, based on Audubon's record of a specimen taken on Key West and figured in his Birds of America. Ridgway (1916) examined this specimen and identified it as the Jamaican mangrove cuckoo (*C. minor nesiotes*). Now the 1931 Check-list makes the statement that all Florida records prove to be referable to *C. minor maynardi*, and excludes both of the above races from the list.

The mangrove cuckoo, of which Maynard's is a subspecies, is well named, for all races of the species seem to be confined almost exclusively to the mangroves. The only one I ever saw was encountered on our way to Alligator Lake through the mangrove forests near Cape Sable, Fla. Arthur

H. Howell (1932) collected the only two he ever saw “in a black mangrove swamp near the mouth of Allens River, below Everglade.”

Nesting.—Audubon (1842) says: “The nest is slightly constructed of dry twigs, and is almost flat, nearly resembling that of the Yellow-billed Cuckoo.” Mr. Howell (1932) says that “a set of two fresh eggs (now in the Florida State Museum), which the female was beginning to incubate, was taken at Chokoloskee, June 4, 1903, from a nest 7 feet up in a red mangrove.”

Oscar E. Baynard tells me that he took two sets of two eggs each, on May 25 and 26, 1912, in a dense stand of extra large mangroves in Pinellas County, Fla.

Eggs.—The eggs of Maynard’s cuckoo are practically indistinguishable from those of the yellow-billed cuckoo. The measurements of 20 eggs average 30.77 by 23.18 millimeters; the eggs showing the four extremes measure **33.1** by 22.6, 31.4 by **24.3**, **27.9** by 22, and 30 by **21.6** millimeters.

Young.—Audubon (1842) says that this cuckoo “raises two broods in the season, and feeds its young on insects until they are able to go abroad.”

Plumages.—I have seen no nestlings of this species, but a fully grown juvenal, taken in the Bahamas on June 23, has the upper parts everywhere “buffy brown” to “wood brown,” including the crown, which in the adult is grayer than the back; there are narrow white tips on the secondaries, tertials, and all the wing coverts; there are still narrower white tips on the primaries, which soon wear away; the black space below and behind the eye, so conspicuous in the adult, is lacking; the central tail feathers are paler grayish brown terminally than in the adult, and the lateral tail feathers are pale “wood brown,” instead of black, with whitish, instead of pure white, tips, which are also less clearly defined than in the adult. I have seen young birds in this plumage in July, August, and September. The body plumage is probably molted during the fall, but the juvenal wings and tail are apparently retained until the following spring or summer; I have seen young birds molting wings and tail in February, June, and August, thus assuming a fully adult plumage.

Food.—The food of Maynard’s cuckoo consists mainly of caterpillars, spiders, moths, flies, grasshoppers, and other insects; probably a few small

fruits and wild berries are eaten at times. Mr. Howell (1932) says: "The stomachs of two birds taken at Everglade, Florida, in March, were examined in the Biological Survey; the food in one consisted mainly of hairy caterpillars (Arctiidae), the stomach being well lined with caterpillar spines; the remains of 3 mantids (*Stagmomantis*) composed the remainder. The other stomach contained 4 long-horned grasshoppers, lepidopterous larvae, locustid eggs, mantids, and spiders."

Behavior.—Maynard (1896) says of the habits of this cuckoo in the Bahamas:

They frequent thickets near fields, and often venture into the open grounds to feed, but usually when taken by surprise in such places, quickly retreat to the thickets, into which they glide easily. Once within the cover of the shrubbery, their movements are quite deliberate, but when approached, they will jump from branch to branch, and although not appearing to hasten, will manage to elude their pursuer, and become quickly lost in the foliage.

The flight of this Cuckoo is rapid, the wings being moved quickly, much more so than in the Black or Yellow Billed Cuckoos. They generally move straight forward, without doubling, and when they wish to alight, they do so suddenly without any preliminary lessening of their speed, and as soon as their feet touch the branch the tail is dropped perpendicularly. As a rule, this Cuckoo is rather shy, especially when in open fields, but I once came across one near Mathewstown, Inagua, that was feeding in an old field, that was very tame, allowing me to approach within ten feet of it, as it deliberately searched for food among the remains of partly decayed stubs of trees which stood in the clearing.

Voice.—On this subject Maynard says:

All through the winter Maynard's Cuckoo is rather silent, but as spring approaches they begin to utter their singular cries, and at times, more especially before rain, are quite noisy. The notes may be represented by the syllables "ou, ou, ou, ou, qua, qua, qua, coo, coo, coo."

The "ous" are given very rapidly, with a decided Cuckoo-like intonation.

The "quas" are harsher, more like the notes of the Bahama tree frog, and are not hurriedly given. The last three notes are more Cuckoo-like than any of the others. The first four notes are often omitted, then the harshly and gravely given "quas" begin the song and on occasions these quaint sounds are not followed by any other notes, then it is sometimes difficult to distinguish the notes uttered from some of those uttered by the large Andros Island Cuckoo, *Saurothera andrea*. This varied song is uttered in the early morning with rather more energy than at any other time in the day.

Field marks.—The Maynard's cuckoo might easily be mistaken for a yellow-billed cuckoo, as it doubtless often has been, the tail markings being practically the same. But the underparts are decidedly washed with "pinkish buff" and "cinnamon-buff," though these parts are not so deeply colored as in the mangrove cuckoo. If clearly seen at close range, the grayish crown

and the black area behind and, narrowly, below the eye are good field marks.

DISTRIBUTION

Range.—Northeastern South America, Central America to central Mexico, the West Indies, and southern Florida.

The range of the mangrove cuckoo extends **north** to Nayarit (Tres Marias Islands and San Blas); Tamaulipas (Alta Mira); Yucatan (Izamal, Temax, Chichen-Itza, Cozumel Island, and Mujeres Island); western Cuba (Isle of Pines); Florida (Anclote Keys and Cape Florida); the Bahama Islands (Berry, Eleuthera, and Watling Islands); the Dominican Republic (Monte Cristi, Sosua, and Seibo); Puerto Rico (Desecheo, Culebra, and Vieques Islands); and the Virgin Islands (St. Thomas, Tortola, and Virgin Gorda). **East** to the Virgin Islands (Virgin Gorda); the Leeward and Windward Islands (Antigua, Guadeloupe, Dominica, Martinique, Santa Lucia, St. Vincent, and Grenada); and Trinidad. **South** to Trinidad; northern Venezuela (Aruba Island); Panama (Chiriqui); and Costa Rica (San Jose and Puntarenas). **West** to Costa Rica (Puntarenas, Pozo del Rio Grande, and Pigres); Nicaragua (Greytown and Chinandega); Honduras (Roatan Island and Puerto Caballo); El Salvador (La Libertad); western Guatemala (near Ocos); Oaxaca (Cacoprieto); and Nayarit (Tres Marias Islands).

The range as outlined is for the entire species, which has been separated into several subspecies, only one of which is found in North America. This race, known as Maynard's cuckoo, *C. m. maynardi*, is found in the Bahamas, Cuba, and the southern part of Florida, including the Florida Keys.

Migration.—Apparently only Maynard's cuckoo is migratory, and this only in the Florida part of its range. Early dates of arrival in Florida are: Gator Lake in Monroe County, March 22, and Punta Rossa, March 29. A late date of departure from Key West is September 19.

Egg dates.—Florida and the Keys: 13 records, May 17 to July 10.

COCCYZUS AMERICANUS AMERICANUS (Linnaeus)

YELLOW-BILLED CUCKOO

Plates 6, 7

HABITS

The yellow-billed cuckoo, with its western subspecies, covers practically all the United States and some of southern Canada. It is mainly a bird of the Austral Zone, being much commoner in the Southern States than in the northern portions of its range. In New England it is not so common as the black-billed cuckoo, though in some seasons it seems to be a familiar bird. Originally it was probably a woodland bird, but, like many other species, it has learned to frequent the haunts of man, where it is not molested and where it finds an abundant food supply in our shade trees, orchards, and gardens. Its favorite haunts are still the woodland thickets, where the tree growth is not too heavy, brush-grown lanes, shady roadsides, dense thickets along small streams, and apple orchards in rural districts. In dense, heavy woods it is seldom seen.

Nesting.—Unlike the European cuckoo, both of our North American species usually build their own nests and rear their own young, though they are very poor nest builders and are often careless about laying in each other's nests or the nests of other species. Major Bendire (1895) gives the following very good account of the nesting habits of the yellow-billed cuckoo:

The Yellow-billed Cuckoo is one of the poorest nest builders known to me, and undoubtedly the slovenly manner in which it constructs its nest causes the contents of many to be accidentally destroyed, and this probably accounts to some extent for the many apparent irregularities in their nesting habits. The nests are shallow, frail platforms, composed of small rootlets, sticks, or twigs, few of these being over 4 or 5 inches in length, and among them a few dry leaves and bits of mosses; rags, etc., are occasionally mixed in, and the surface is lined with dry blossoms of the horse-chestnut and other flowering plants, the male aments or catkins of oaks, willows, etc., tufts of grasses, pine and spruce needles, and mosses of different kinds. These materials are loosely placed on the top of the little platform, which is frequently so small that the extremities of the bird project on both sides, and there is scarcely any depression to keep the eggs from rolling out even in only a moderate windstorm, unless one of the parents sits on the nest, and it is therefore not a rare occurrence to find broken eggs lying under the trees or bushes in which the nests are placed. Some of these are so slightly built that the eggs can be readily seen through the bottom. An average nest measures about 5 inches in outer diameter by 1½ inches in depth. They are rarely placed over 20 feet from the ground, generally from 4 to 8 feet upon horizontal limbs of oak, beech, gum, dogwood, hawthorn, mulberry, pine, cedar, fir, apple, orange, fig, and other trees. Thick bushes particularly such as are overrun with wild grape and other vines as well as hedgerows, especially those of osage orange are most frequently

selected for nesting sites. The nests are ordinarily well concealed by the overhanging and surrounding foliage and while usually shy and timid at other times, the Yellow-billed Cuckoo is generally courageous and bold in the defense of its chosen home; the bird on the nest not unfrequently will raise its feathers at right angles from the body and occasionally even fly at the intruder.

Of five Massachusetts nests, on which I have notes, the lowest was only 2 feet above the ground in some bushes, and the highest was 12 feet up in a crotch near the top of an oak sapling in a swampy thicket near a brook. Owen Durfee mentions in his notes a nest 5 feet up in a juniper on the edge of a swamp. The others were at low elevations in thickets along brooks.

A. D. DuBois has sent me his notes on five Illinois nests; one of these was on the end of a branch of an apple tree, 8 feet from the ground, near a country schoolhouse; this nest contained 3 eggs of the cuckoo and a robin's egg. Another was near the end of a branch in an osage-orange hedge, 10 feet up; still another was in an isolated clump of willows, between a field and a pasture, 6 feet from the ground.

But cuckoos do not always nest in such low situations; there are several records of their nesting well up in elm trees. Grant Foreman (1924) tells of a pair that nested on his place in Muskogee, Okla., for one or two years, high up in an elm tree; he says: "The next year after nesting in this inaccessible place, they built their nest in a little elm tree in the parking, in a low limb overhanging the curb on an asphalt street where hundreds of automobiles were passing every day, and here in this exposed, noisy place they raised a brood of young. This year they built their nest in a little hackberry tree in the parking along the side of my lot; but here also the nest was on a low limb overhanging the curb on a paved street, and the ice wagon stopped every morning directly under this nest, which was so low down that the driver might have put his hand in it."

George Finlay Simmons (1915) mentions a nest that he found near Houston, Tex., on the horizontal limb of a young pine near the edge of some woods. He says of it: "The nest was a slight platform about eleven feet up, through which I could see with ease; it was composed of small pine twigs, about an eighth of an inch in diameter and averaging six or eight inches long, and was much more concave than I had expected. This shallow saucer was neatly, though quite thinly lined with a few pine needles, a small quantity of Spanish moss and several tiny buds."

George B. Sennett (1879) says that in the Lower Rio Grande region of Texas “ebony trees near the ranch, mesquites among cactuses, thorny bushes in open chaparral, and open woodland, were favored breeding places.”

Wright and Harper (1913) found a well-made nest in Okefinokee Swamp, in a tupelo tree at the margin of the Suwannee. “It was placed in a cluster of mistletoe on a horizontal branch four feet above the water, and consisted of sticks interwoven with Spanish ‘moss’ (*Tillandsia usneoides*).”

Dr. Harry C. Oberholser (1896) gives the measurements of four nests; the average height of the nests was 4 inches, and the greatest outside diameters averaged 7.63 by 6.25 inches.

Both species of North American cuckoos often lay their eggs in each other’s nests. The eggs of the yellow-billed cuckoo have been found several times in nests of the robin and catbird. H. P. Attwater (1892) writes: “In 1884 I found a Dickcissel’s nest which contained five eggs and one Yellow-billed Cuckoo’s egg. The next year some boys brought me three Black-throated Sparrow’s eggs and one Yellow-billed Cuckoo’s, from the same field, which they said they found all together in one nest.” J. L. Davison (1887) says: “I also found a nest of *Merula migratoria*, taken possession of by *Coccyzus americanus* before it was finished, which was filled nearly full of rootlets; and in this condition the Robin laid one egg and the Cuckoo laid two and commenced incubation, when a Mourning Dove (*Zenaidura macroura*) also occupied it and laid two eggs and commenced incubation with the Cuckoo. I found both birds on the nest at the same time, when I secured nest and eggs. The eggs of the Robin and Cuckoo were slightly incubated; those of the Mourning Dove were fresh.”

Bendire (1895) adds the wood thrush, cedar waxwing, and cardinal to the list of birds that have been imposed upon, and says: “Such instances appear to be much rarer, however, than those in which they interlay with each other, and the majority of these may well be due to accident, their own nest having possibly been capsized, and necessity compelled the bird to deposit its egg elsewhere. Such instances do occur at times with species that can not possibly be charged with parasitic tendencies.”

Marcia B. Clay (1929) thus describes the cuckoo’s method of gathering twigs for her nest:

Flying into an adjacent apple tree containing a considerable quantity of dead material, the Cuckoo landed on a limb, selected a dead twig, and grasping it in her bill bent it back and forth until it snapped from the limb, whereupon she flew with it to her nesting-site in the next tree, arranged this twig and quickly returned for another. As she tugged at a stubborn twig, her back was arched and her long tail curved under or waved about. If a twig resisted too well her attack, the bird desisted at once and tried another. Always she worked rapidly with great energy, attacking a twig as soon as she landed in the tree, never carrying more than one twig at a time, holding it squarely at right angles to her bill and flying rapidly with long tail streaming.

The Cuckoo's concentration in the work, coupled with her indifference to observers, was remarkable. Not once did she descend to the ground for material. Not once did she gather material in the tree in which her nest was located. With two exceptions the twigs were all gathered from the same tree. Working thus off and on for an hour or two at a time, the bird completed the nest. The third night the Cuckoo was sitting on the nest at dusk, but after two days she deserted.

Eggs.—The yellow-billed cuckoo lays ordinarily three or four eggs, sometimes only one and rarely five; as many as six, seven, or even eight eggs have been found in a nest, but these larger numbers may be the product of more than one female. The eggs vary in shape from elliptical-oval to oval, oftener nearer the former, and about equally rounded at both ends. The shell is smooth, but without gloss. Bendire (1895) says that the “color varies from a uniform Nile blue to pale greenish blue when fresh, fading out in time to a pale greenish yellow.” Eggs that I have examined in collections vary in color from “pale glaucous green” to “pale fluorite green.” The measurements of 53 eggs average 30.4 by 23 millimeters; the eggs showing the four extremes measure **34.64** by 23.11, 33.53 by **25.40**, **27.43** by 22.86, and 29.21 by **20.83** millimeters.

Young.—The period of incubation is said to be about 14 days; it is shared to some extent by both sexes, but is probably performed mainly by the female. The eggs are sometimes laid on succeeding days, but oftener at more or less infrequent intervals; the young, therefore, frequently hatch at irregular intervals, and young of different ages are often found in the nest.

Snyder and Logier (1931) say of a brood of young that they examined: “The young were quite active when disturbed. They scrambled about the bush, using the wings and bill for climbing. One young which was brought to our camp demonstrated a remarkable reptile-like behaviour. When it was placed on the table and one reached to pick it up, it erected its somewhat horny plumage and emitted a buzzing hiss like the sound of bees escaping

from a tunnel in dry grass. This performance was certainly unbirdlike in all respects.”

Francis H. Allen writes to me: “I found a young one in an open field on the ground. I was attracted to the spot by its loud rasping cry. It fluttered along when I approached, but it could not fly from that position, in rather long grass, though wings and tail were pretty well fledged. When I picked it up, it pecked at my finger angrily. It seemed as fierce as a young hawk, and its rasping cry was probably calculated to inspire terror in its enemies. I placed the bird on a bough of a Norway spruce, where it took a characteristic cuckoo attitude and seemed much more at home than on the ground.”

Dr. Lawrence H. Walkinshaw has sent me some notes on the weights and development of young yellow-billed cuckoos. One “well-grown” young was weighed for three days in succession before it left the nest, at 6 a. m. each morning. It weighed 28.8 grams the first morning, 31 grams the second, and only 26 grams on the third, August 6. The interesting point is that the loss of weight came with the sudden development of the plumage, of which he says: “When I visited the nest on August 5, at 6 a. m., his feathers resembled the quills of a porcupine, long and bluish, stretched out over his wings and back. At 7 p. m., these quills had all opened and the bird had taken on the resemblance of an adult cuckoo. Correspondingly, the following morning, he had lost 5 grams in weight. He left the nest on August 6.”

At another nest a young bird weighed 25 grams on August 25, 27.6 on the 26th, 32.9 on the 27th, and only 28.9 grams on the 28th; this bird left the nest on August 29, with feathers unsheathed. He says that during the unsheathing process the young bird dressed its feathers continually; “the wings, the tail, the scapulars, the rump, and breast all shared alike, then with the feet he would work about the head and throat. When hungry he would pause and call a low *cuk-cuk-cuk-cur-r-r-r-rrr*. If the parent did not come soon, these calls increased in number. While feeding, his wings would vibrate rapidly, and after the parent left his call was more of contentment, a short *curr*, or a *cuk-curr*. When excreting, he simply backed up to the edge of the nest.”

Plumages.—Bendire (1895) says: “The young when first hatched are repulsive, black, and greasy-looking creatures, nearly naked, and the sprouting quills only add to their general ugliness.” This is a very good description, and the young birds do not improve much in appearance during the period of early growth. The body is well covered with the long, pointed feather sheaths until the young bird is more than half grown. But the sheaths burst, the juvenal plumage appears, and the young bird is well feathered before the time comes to leave the nest.

Dr. A. H. Cordier (1923) describes this process very well as follows:

At the end of seven days the young Cuckoo resembled a porcupine more than a bird. I now cut the limb holding the nest and brought it to the ground. Within three feet of it I then put up the umbrella tent that I might at close range observe minutely the rapid transition of the porcupine-looking object into a fully feathered, beautiful Rain Crow. * * *

The first picture was made at nine o'clock. * * * This shows the young by the unhatched egg; the horny, sheathed feathers were fully two inches long, making the bird look like a porcupine. About ten-thirty the sheaths began to burst, and with each split a fully formed feather was liberated. This process took place with such rapidity that it reminded me of the commotion in a corn popper or a rapidly blooming flower. All the while I was within three feet of the bird, and could see every new feather, as it blossomed, so to speak.

At three p. m., six hours after the first picture was taken, I made another photograph, showing this same bird in the full plumage of a Cuckoo, except the long tail.

In this first plumage the young cuckoo looks very much like the adult, perhaps slightly paler above and with a slight wash of tawny or pale buff on the throat and breast; but the tail is quite different, lacking the conspicuous black and white markings so prominent on the sides of the adult tail; in the young bird the dark spaces in the tail are not black, but dark gray or lighter gray, variable in different individuals or in different feathers in the same individual; the light spaces are not so sharply defined as in the adult and are grayish white instead of pure white.

The juvenal body plumage appears to be molted in fall, from August to October; but the juvenal wings and tail are worn through the first winter at least; I have not been able to detect this plumage in spring birds, so I suppose that a more or less complete molt occurs while the birds are in their winter homes, producing a practically adult plumage before they return in the spring. Adults have a complete molt between July and October, and possibly a more or less complete molt in spring before they arrive here, but winter specimens to show it are lacking.

Food.—Cuckoos are among the most useful of our birds, mainly because of their fondness for caterpillars, which are some of our most injurious insect pests and which constitute the principal food of these birds during their seasons of abundance. Edward H. Forbush (1907) writes:

The Cuckoos are of the greatest service to the farmer, by reason of their well-known fondness for caterpillars, particularly the hairy species. No caterpillars are safe from the Cuckoo. It does not matter how hairy or spiny they are, or how well they may be protected by webs. Often the stomach of the Cuckoo will be found lined with a felted mass of caterpillar hairs, and sometimes its intestines are pierced by the spines of the noxious caterpillars that it has swallowed. Wherever caterpillar outbreaks occur we hear the calls of the Cuckoos. There they stay; there they bring their newly fledged young; and the number of caterpillars they eat is incredible. Professor Beal states that two thousand, seven hundred and seventy-one caterpillars were found in the stomachs of one hundred and twenty-one Cuckoos—an average of more than twenty-one each. Dr. Otto Lugger found several hundred small hairy caterpillars in the stomach of a single bird. The poisonous, spined caterpillars of the Io moth, the almost equally disagreeable caterpillars of the brown-tail moth, and the spiny elm caterpillar, are eaten with avidity.

He says elsewhere (1927):

When, in time, the inside of the bird's stomach becomes so felted with a mass of hairs and spines that it obstructs digestion, the bird can shed the entire stomach-lining, meanwhile growing a new one. * * * Mr. Mosher, a competent observer, watched a Yellow-billed Cuckoo eat 41 gypsy caterpillars in fifteen minutes, and later he saw another consume 47 forest tent caterpillars in six minutes. * * * Dr. Amos W. Butler [1897] says that he has known these Cuckoos to destroy every tent caterpillar in a badly infested orchard and tear up all the nests in half a day. This species frequently feeds on or near the ground, and there gets an enormous number of locusts and other pests. In summer and autumn it feeds to some extent on small wild fruits, such as the raspberry, blackberry and wild grape.

The fall webworm is a destructive pest on certain trees, but few birds will eat it. Dr. Sylvester D. Judd (1902) noted that, on a Maryland farm, "a pair of yellow-billed cuckoos continually extracted them from the webs. The destruction of this insect is an habitual practice with the cuckoo. In a single stomach of the species examined by Professor Beal there were 325 of the larvæ."

Henry C. Denslow writes to me that he fed many hairy caterpillars to a cuckoo that he had in captivity, and says: "Many of these this bird sheared the hairs from by slowly moving them from end to end through its beak by a side-shifting motion of the mandibles. The removed hairs collected in a little bunch and, at the end of the caterpillar, fell to the floor. Most of the hairs were thus shorn from these caterpillars. Other caterpillars were swallowed entire, as I gave them to him, hairs and all."

Walter B. Barrows (1912) says that this cuckoo feeds freely on elderberries and mulberries and that “large quantities of beetles and bugs also are consumed, and both species of cuckoo seem to be very fond of grasshoppers, eating especially such forms as frequent shrubbery and trees, among these the destructive tree crickets (*Oecanthus*). Ten specimens examined by Professor Aughey, in Nebraska, contained 416 locusts and grasshoppers, and 152 other insects.”

Audubon (1842) writes: “In autumn they eat many grapes, and I have seen them supporting themselves by a momentary motion of their wings opposite a bunch, as if selecting the ripest, when they would seize it and return to a branch, repeating their visits in this manner until satiated.”

In addition to those mentioned above, yellow-billed cuckoos have been known to eat many other insects, such as armyworms, ants, wasps, flies, and dragonflies. Several of the earlier ornithologists accused this cuckoo of eating the eggs of other small birds and produced some evidence of the bad habit, but some modern observers seem to think that they do very little, if any, nest robbing. C. J. Maynard (1896) writes:

This species in company with the former [black-billed cuckoo] are the terror of other small birds during the nesting season for they will constantly rob their nests. I have frequently seen a Cuckoo enter a thicket in which a Robin or a Cat Bird had built a home and in a moment the air would resound with the shrill cries of distress given by the parents, causing all the small birds in the immediate vicinity to rush to the spot and as each joins in the outcry, the noise produced is apparently enough to frighten away a bolder bird than a Cuckoo.

But in spite of all this din, the glossy thief nearly always succeeds in accomplishing his purpose and emerges from the thicket, carrying an egg impaled on his beak. He does not always escape unscathed, however, for he is pursued by a motley crowd consisting of Robins, Cat Birds, Thrushes, Warblers, etc. that follow him closely, harassing him on all sides, and some of the more courageous will even assault him with blows from their beaks so that he frequently leaves some of his feathers floating in the wind behind him. As the long and broad tail of the Cuckoo is a prominent object and as it is also a portion of the bird which its enemies can seize with comparative safety to themselves, this member often suffers in these forays, in so much, that by the middle of summer, it is quite difficult to find a Cuckoo of either species which has a full complement of tail feathers.

On the other hand, Major Bendire (1895) says: “I am aware that this species has been accused of destroying the eggs and even of eating the young of smaller birds, but I am strongly inclined to believe that this accusation is unjust, and in my opinion requires more substantial confirmation. I have never yet had any reason to suspect their robbing

smaller birds' nests, and the very fact that they live in apparent harmony with such neighbors, who do not protest against their presence, as they are in the habit of doing should a Blue Jay, Grackle, or Crow come too close to their nests, seems to confirm this view."

But then he goes on to quote from a letter from William Brewster, who says: "While I have never seen either of our Cuckoos destroy the eggs of other birds, nevertheless I think they do it occasionally. One of my reasons for this belief is that many of our small birds, Warblers, Sparrows, etc., show great anxiety whenever the Cuckoos approach their nests, and they pursue and peck at them, when they take wing, behaving toward them, in fact, exactly as they do toward the Crows, Jays, and Grackles, which we know eat eggs whenever they can get a chance. My other reason is that one of my friends once shot a Cuckoo (*C. americanus*, I think it was) whose bill was smeared all over with the fresh yolk of an egg."

Yellow-billed cuckoos sometimes eat tree frogs and other small frogs, and, in the Southern States, an occasional small lizard. Marcia B. Clay (1929) relates the following incident: "For an hour a Cuckoo searched about the dead under limbs of a huge untrimmed apple tree, peering and gliding noiselessly around and around. At last, after long and patient search, it dashed to the ground and began to walk directly toward me through the scant grass and weeds, and only then did I see a frog trying to slip away unseen. The bird followed the frog a rod, pecking its victim and gloating softly *Cuk, Cuk*. Having vanquished its prey, the Cuckoo deftly gathered it into its bill and flew away, the frog's legs sticking out stiff and straight together, exactly like the dead twigs which the Cuckoo carries to its nest."

Behavior.—Mr. Forbush (1927) has described the quiet, retiring behavior of the yellow-billed cuckoo very well as follows:

The cuckoo is a graceful, elegant bird, calm and unperturbed; it slips quietly and rather furtively through its favorite tangles and flies easily from tree to tree in the orchard, keeping for the most part under protection of the leaves, which furnish excellent cover for its bronzy, upper plumage, while the shadows of the foliage tend to conceal the whiteness of its under parts. It has a way also of keeping its back with its greenish satiny reflections toward the intruder in its solitudes, and while holding an attitude of readiness for flight it sits motionless, and its plumage so blends with its leafy environment that it does not ordinarily catch the eye. In the meantime it turns its head and regards the disturber with a cool, reserved, direct gaze, looking back over its shoulder, apparently unafraid and giving no indication of nervousness or even undue curiosity; but if the observer approaches too

closely, the elegant bird slips quietly away, vanishing into some leafy, cool retreat where it may enjoy the silence and solitude, dear to the woodland recluse.

The flight of the cuckoo is rather swift, easy and graceful, exceedingly direct and horizontal, but turning frequently from side to side as it threads its way through the branches of the trees, giving occasional glimpses of its white under parts and the telltale black-and-white markings in its tail; it is stream-lined to perfection and glides noiselessly through the air with its long tail streaming out behind. It is very quiet in its movements in its shady retreats; it seldom perches in a conspicuous place but sits motionless for long periods in the dense foliage, watching, or moves about stealthily in search of its prey. It might easily be overlooked, were it not for its characteristic notes, which lead the observer to look for it.

About its nest it is rather shy, while incubating on its eggs, slipping away cautiously when approached, but when there are young in the nest its behavior is quite different. It then becomes quite solicitous and will often remain on the nest until almost touched, and then perhaps throw itself down to the ground, fluttering and tumbling along, feigning lameness, after the manner of many ground-nesting birds, uttering loud, guttural cries of distress.

Voice.—We hear the voice of the cuckoo much oftener than we see the bird; the well-known sound comes to us, like a wandering voice, from the depths of some shady retreat, but we cannot see the hidden author. We can recognize it easily as the voice of a cuckoo, but it is not always so easy to identify the species by its notes, though some keen observers claim that they can do so. Certain songs are characteristic of each of the two species, but both have a great variety of notes and many notes that are much alike in both. The notes of the yellow-billed cuckoo may be a trifle harsher and a little louder, but they are not always recognizable. The characteristic note of the yellow-billed cuckoo is well described by Charles J. Spiker (1935) as follows: “What may be considered the song of this species is a series of rapid, wooden-sounding syllables resembling the following: *Kuk-kuk-kuk-kuk-kuk-ceaow-ceaow-ceaow-ceaow*; the *kuks* being given rapidly, the *ceaows* more deliberately and with longer intervals.”

Bendire (1895) writes:

One of their commonest notes is a low “noo-coo-coo-coo;” another sounds more like “cow-cow-cow” or “kow-kow-kow,” several times repeated; others resemble the syllables

of “ough, ough, ough,” slowly and softly uttered; some remind me of the “kloop-kloop” of the Bittern; occasionally a note something like the “kiuh-kiuh-kiuh” of the Flicker is also uttered; a low sharp “tou-wity-whit” and “hweet hwee” is also heard during the nesting season. Though ordinarily not what might be called a social bird, I have sometimes during the mating season seen as many as eight in the same tree, and on such occasions they indulge in quite a number of calls, and if the listener can only keep still long enough he has an excellent opportunity to hear a regular Cuckoo concert.

Various other interpretations of the different notes have been given by other writers, but the above quotations cover fairly well the ordinary variations. The song, as given by Mr. Spiker above, is sometimes more prolonged by lengthening the series of *kuks*, with increasing speed of utterance and adding to the series of *ceaows*, with slowly decreasing speed. I believe that the black-billed cuckoo never gives this prolonged song, accelerated during the first half and retarded during the last half; its song is given in more even time, and is generally shorter. The song of the yellow-billed cuckoo is often heard during the night, and its notes are often uttered while flying.

Field marks.—A cuckoo may be easily recognized as a cuckoo by its size, shape, and color—a long, slender bird, longer than a robin, with a long tail, olive-brown above and white below; but the two species look very much alike unless the distinctive markings can be clearly seen. The yellow lower mandible of this species can be seen only at short range. But the rufous in the wing feathers is evident in flight, and the lateral tail feathers are conspicuously black, with large terminal white areas clearly defined. At very close range the yellow eyelids of this species may be seen.

DISTRIBUTION

Range.—Temperate North America, the Caribbean region, South and Central America; casual on Bermuda and accidental in western Europe.

Breeding range.—The breeding range of the yellow-billed cuckoo extends **north** to southern British Columbia (Kamloops); northeastern Oregon (Keenys Ferry); northern Utah (Salt Lake City); northern Colorado (Loveland, Greeley, and Fort Morgan); South Dakota (White River, Yankton, and Sioux Falls); Minnesota (Fosston and St. Paul); Wisconsin (Ladysmith, Waupaca, and New London); northern Michigan (Blaney and Sault Ste. Marie); southern Ontario (Listowel, Rosseau, and Ottawa);

northern New York (Watertown and Plattsburg); and southern Maine (Auburn). From this point the breeding range extends southward along the Atlantic coast to Florida (New Smyrna and Kissimmee); the Bahama Islands (Inagua Island); probably the Dominican Republic (Dajabon); and the Virgin Islands (St. Croix). **South** to the Virgin Islands (St. Croix); Jamaica (Port Henderson); Coahuila (Sabinas River); southern Sonora (Guaymas); and southern Baja California (San Jose del Rancho). **West** to Baja California (San Jose del Rancho and Cerro Prieto); California (Wilmington, Watsonville, Santa Clara, and Redding); Oregon (Salem); Washington (Grays Harbor, Tacoma, and Seattle); and British Columbia (Victoria, Chilliwack, and Kamloops).

The range as outlined is for the entire species, which has been separated into two geographic races. The yellow-billed cuckoo (*C. a. americanus*) occupies the eastern part of the range west to South Dakota, Nebraska, eastern Colorado, and Oklahoma, while the California cuckoo (*C. a. occidentalis*) is found over the rest of the country to the Pacific coast.

Winter range.—The winter home of this species has not been accurately determined, but it extends **north** to northern Colombia (Santa Marta, Bonda, Medellin, and Antioquia); and Venezuela (San Cristobal, Altagracia, the Orinoco River region, and Nericagua). It has been found at this season **east** to southeastern Brazil (São Paulo); Uruguay (Rio Negro); and eastern Argentina (Buenos Aires and Lomas de Zamora). **South**, probably only casually to central Argentina (Lomas de Zamora and Saladillo). **West** to Argentina (Saladillo and La Rioja); Ecuador (Nono, Chimbo, Cumbaya, and Guapulo); and Colombia (Cienaga and Santa Marta).

Spring migration.—Early dates of spring arrival are: Florida—Melrose, March 12; Daytona Beach, April 9; Pensacola, April 13. Alabama—Autaugaville, April 16. Georgia—Savannah, March 24; Kirkwood, April 7. South Carolina—Charleston, April 14. North Carolina—Raleigh, April 25; Weaverville, May 1. Virginia—Lawrenceville, April 23; New Market, April 26. District of Columbia—Washington, April 27. Pennsylvania—Philadelphia, April 20; Beaver, May 6; Renovo, May 11. New Jersey—Morristown, April 7; Elizabeth, May 9. New York—Shelter Island, May 2; Rhinebeck, May 3; Watertown, May 12. Connecticut—Hartford, May 9; Jewett City, May 10. Rhode Island—Providence, May 11. Massachusetts—Boston, May 4; Beverly, May 15. New Hampshire—Milford, May 11.

Vermont—St. Johnsbury, May 27. Maine—Lewiston, May 6; Fryeburg, May 19. Mississippi—Biloxi, April 6; Rodney, April 8; Oakvale, April 10. Arkansas—Helena, April 19; Delight, April 24. Tennessee—Chattanooga, April 10; Knoxville, April 27. Kentucky—Eubank, April 22; Bowling Green, April 24. Missouri—St. Louis, April 28; Kansas City, April 30; Concordia, May 1. Illinois—Rantoul, May 2; Chicago, May 4; Olney, May 5. Indiana—Terre Haute, April 17; Fort Wayne, April 29; Waterloo, May 4. Ohio—Oberlin, April 26; Columbus, April 29; Youngstown, May 7. Michigan—Detroit, May 4; Battle Creek, May 11. Ontario—London, May 8; Guelph, May 12. Iowa—Sioux City, April 30; Grinnell, May 9. Wisconsin—Madison, May 10; Racine, May 10; La Crosse, May 12. Minnesota—Minneapolis, May 6; Winona, May 8. Texas—Fredericksburg, April 1; Kerrville, April 7; San Antonio, April 14. Oklahoma—Skiatook, April 20; Oklahoma City, May 3. Kansas—Ottawa, April 25; Onaga, April 28. Nebraska—Red Cloud, April 29; Lincoln, May 7. South Dakota—Vermillion, May 17. New Mexico—State College, May 25. Arizona—Phoenix, May 1; Tombstone, May 20. Colorado—Longmont, May 20; Denver, May 28. California—Petaluma, April 18; Pico, May 5; Berryessa, May 13. Oregon—Sauvies Island, April 24. Washington—Tacoma, May 3.

Fall migration.—Late dates of fall departure are: California—Murphys, September 1; Vineland, September 22. Colorado—Yuma, September 3; Clear Creek, September 8. Arizona—Tucson, September 8. New Mexico—Mesilla Park, September 7; State College, September 18. North Dakota—Grafton, September 4. South Dakota—Lennox, September 15; Yankton, September 24. Nebraska—Dunbar, September 27; Lincoln, October 5. Kansas—Onaga, October 1. Oklahoma—Copan, September 23; Kenton, September 30. Texas—Bonham, September 25; Swan, September 28; Kerrville, October 19. Minnesota—Minneapolis, October 1; Hastings, October 20. Wisconsin—Madison, September 27; Racine, October 6. Iowa—Grinnell, October 22. Ontario—Galt, October 2; London, October 11; Point Pelee, October 16. Michigan—Detroit, October 8. Ohio—Youngstown, October 12; Columbus, October 19; Oberlin, October 21. Indiana—Sedan, October 13. Illinois—Chicago, October 18; Rantoul, October 24. Missouri—Concordia, October 10. Kentucky—Lexington, October 10. Tennessee—Nashville, October 11. Arkansas—Delight, October 20. Mississippi—Biloxi, October 11. Maine—Fryeburg, September 1. Vermont—St. Johnsbury, September 12. Massachusetts—Marthas

Vineyard, September 24; Lanesborough, September 29; Hadley, October 18. Connecticut—Meriden, October 12; New Haven, October 16; Portland, October 17. New York—Rhinebeck, October 1; New York City, October 12. New Jersey—Morristown, October 9; Elizabeth, October 12. Pennsylvania—Beaver, September 27; Philadelphia, October 17; Renovo, November 12. District of Columbia—Washington, October 13. Virginia—Lawrenceville, October 12. North Carolina—Raleigh, October 17. South Carolina—Mount Pleasant, November 7. Georgia—Savannah, October 19. Florida—Pensacola, November 2.

Casual records.—While the yellow-billed cuckoo may be considered only as a casual visitor to Bermuda, an extraordinary invasion was recorded on October 9, 1849, when thousands of individuals suddenly appeared in all parts of the island; a few more have been subsequently reported. During the period from 1825 to 1921 it was recorded fully a dozen times from England, Scotland, Ireland, and Wales, all these occurrences being in fall and mostly during October. One was taken at Bois de Lessines, Belgium, in October 1874, and one was collected at Turin, Italy, on October 28, 1883. Two occurrences have been recorded from the southern part of France, but there is some question that they were correctly identified.

Egg dates.—Arizona: 13 records, June 28 to August 24; 7 records, July 19 to August 22.

California: 55 records, May 15 to August 20; 28 records, June 17 to July 10, indicating the height of the season.

Florida: 19 records, April 12 to August 25; 10 records, April 16 to May 16.

Illinois: 39 records, May 20 to July 19; 20 records, June 4 to 26.

New York: 23 records, May 24 to August 19; 12 records, June 4 to 11.

Pennsylvania: 13 records, June 6 to July 29.

Texas: 34 records, March 22 to June 30; 17 records, May 6 to June 5.

COCCYZUS AMERICANUS OCCIDENTALIS Ridgway

CALIFORNIA CUCKOO

PLATE 6

HABITS

This western race of our common yellow-billed cuckoo has been separated on very slight average characters, hardly worthy of recognition in nomenclature. I am inclined to agree with Harry S. Swarth (1929), who says:

Between the eastern and western races of the Yellow-billed Cuckoo there is a slight average difference in size, the western bird being the larger and with a somewhat heavier bill. There is a rather wide range of variation in specimens from any one locality, * * * and the largest eastern birds do not fall far short of the maximum measurements of western specimens. Birds from the Pacific coast are the largest, those from central Arizona near the type locality of *occidentalis* (the Santa Rita Mountains) are intermediate in size. The subspecies would have a better claim to recognition if restricted to the Pacific coast. * * * The subspecies is certainly as slightly differentiated as any in our *Check-list*, and I feel that no violence to the facts would result from suppression of the name.

The California cuckoo is nowhere abundant but seems to be generally distributed, in suitable localities, throughout its range from British Columbia to Lower California and other parts of Mexico. In southern California, its favorite haunts seem to be the willow thickets and groves along the beds of streams, or in willow-bottom sloughs, such as the famous Nigger Slough, which formerly existed near Los Angeles. Alfred C. Shelton (1911) describes a favorite haunt in Sonoma County, Calif., as follows: "In the locality of which I write, about five miles southeast of Sebastopol, this stream, known locally as the 'Lagoon', becomes, after some winter storm, a turbulent river, flooding acres upon acres of bottom land. In summer its course is marked by a chain of long, rather narrow ponds, many of which are deep. The banks, and much of the intervening space between these ponds, are covered with a thick growth of willow, small ash and scrub oak, while the whole is tangled together with an undergrowth of poison oak, wild blackberry and various creepers, forming, as it were, an impenetrable jungle, hanging far out over the water."

Spring.—Mr. Shelton (1911) has this to say about the spring movements of the California cuckoo:

Of all migratory birds breeding in this vicinity, the Cuckoo is the last to arrive in the spring, usually appearing during the latter part of May or the first week of June. Upon its arrival, this bird keeps to the higher land, among the oaks and other timber, for a period of two or three weeks before retiring to the willow bottoms to breed. During this period it is wild and shy and difficult to approach. Most active in the early morning, its characteristic

note, a loud, clear "kow-kow-kow," may be heard coming from some tree or group of trees, and perchance an answering "kow-kow-kow," may come from another tree, some distance away. * * *

After the birds retire to the willow bottoms to breed, their entire attitude changes. When watched and studied in the seclusion of their brush grown haunts, while engrossed with the cares of their domestic duties, the Cuckoos cease to be the wild, shy birds of the upland timber. The familiar "kow-kow-kow" is now forsaken for another note, a low guttural note, "kuk-kuk-kuk," always uttered by a brooding bird and is the most common call of the cuckoo during the breeding season.

Courtship.—J. H. Bowles (Dawson and Bowles, 1909) writes: "While standing in an open woodland listening to a pair of Cuckoos calling to each other, I saw the male suddenly fly past with a large green worm in his bill. He flew directly to the female, who was perched in a tree a few yards distant, and for a moment or two they sat motionless a few inches apart looking at each other. The male then hovered lightly over his mate and, settling gently upon her shoulders, gracefully bent over and placed the worm in her bill. It was a pretty and daintily performed piece of love-making."

Nesting.—The nesting habits of the California cuckoo are much like those of its eastern relative. In California its favorite nesting sites are in willow thickets in the river bottoms, or in such swampy lowlands as those referred to above. D. E. Brown writes to me that it is a rare bird in western Washington, and says: "It is found mostly in willow swamps, on the shores of fresh-water lakes, and along streams where the underbrush is thick. I have seldom seen it very far away from fresh water and never in real thick woods. It is a late arrival and does not begin nest building until about the fourth of July. I have found about a dozen occupied nests and only one of these was earlier than the above date; this one was on June 19. All these nests were in willow or wild-rose bushes, except one which was in a spirea bush. All nests were 2 to 8 feet from the ground, or water when built in swamps. The nest is a very frail, small affair composed of twigs loosely put together and lined with finer twigs and sometimes a few leaves. Most of the nests that I have seen have been so very flat and small that they would be exceedingly hard to find if the birds were not on them."

A nest in the Thayer collection, taken near Kirkland, Wash., on July 7, 1909, was found in an open space in a fir forest in low ground, which was dotted with a second growth of fir and some *Osmaronia* and *Spiraea*; it was placed on a branch of a fir on the exposed side of the tree, 9 feet from the

ground; it was made of old fir twigs and lined with fresh fir twigs. I have heard of other Washington nests in fir trees.

Major Bendire (1895) says:

The nests here [Arizona] were placed in willow or mesquite thickets, from 10 to 15 feet from the ground, and they were usually fairly well concealed by the surrounding foliage. *
* * If the California Cuckoo showed the same parasitic habit of occasionally depositing one or more of its eggs in the nests of other birds, as its eastern relatives are now and then known to do, I believe that I should have observed the fact in southern Arizona. Here I found eight of their nests with eggs, and fully five hundred nests of smaller birds, which nested in similar localities among the willow thickets and mesquite bushes, overrun with vines, in the creek bottoms, but not a single instance of parasitism came under my observation.

Wilson C. Hanna (1937) has published an interesting paper on the nesting habits of the California cuckoo in the San Bernardino Valley, Calif., with a photograph of a nest containing the unusual number of seven eggs; he writes:

I have rather complete notes on twenty-four nests that I have examined in the field, six along Warm Creek and eighteen along the Santa Ana River, and with two exceptions all were in willow trees. In one case the nest was 11 feet up in an alder tree next to the trunk, and in the other case 30 feet up in a cottonwood tree on top of a bare limb partly supported by a few twigs and therefore conspicuous. The last mentioned nest was ten feet higher than any other nest I have seen. Six of the nests in willows were either partly supported by or covered with wild grape vines, another nest was well concealed in the center of live mistletoe, while still another was well hidden in poison oak that was growing over the dead willow tree. A few nests were placed next to the trunks of trees, but by far the most common location was well out on a horizontal or leaning limb. The average height above ground or water was less than thirteen feet and two were only four feet up. A good supply of rope and a ladder were necessary for examining some of the nests without disturbing them or the surroundings.

Nests were always loose structures, of coarse twigs for a foundation, sometimes with a little superimposed grape-vine bark, cottonwood bark, or rootlets. In some cases there was no other lining and eggs could be seen through the bottom of the nest; but usually there were fresh or old leaves, bark strips, or willow cotton. In only one nest was there a feather in the lining. Often the nests were much longer in one dimension than the other, in one case four inches wide and twelve inches long.

Eggs.—The California cuckoo lays usually three or four eggs, occasionally only two. These are indistinguishable from those of the eastern yellow-billed cuckoo and average only slightly larger. The measurements of 43 eggs average 31.1 by 23.1 millimeters; the eggs showing the four extremes measure **35.5** by 23.5, 33.5 by **25**, and **27.5** by **21** millimeters.

Food.—Bendire (1895) says that in a brood that he watched, the young were fed “always with a large black cricket (*Anabus simplex* or *purpuratus*) * * *. They picked most of these repulsive-looking creatures from grass stalks and low shrubs on which they were feeding, and although there were numbers of them to be found all around, as well as in camp, they generally went off some little distance to get them.”

Mr. Swarth (1929) says that a female, collected in Arizona, “contained in its stomach two green caterpillars and a lizard 100 millimeters long, the latter swallowed entire and rolled into a coil. This seems a startling diet for a tree-dwelling cuckoo, but there is at least one other instance reported, also from the vicinity of Tucson, of a lizard being taken by one of these birds.”

COCCYZUS ERYTHROPTALMUS (Wilson)

BLACK-BILLED CUCKOO

PLATES 8–11

HABITS

The black-billed cuckoo is not so widely distributed as the yellow-billed, being confined in the breeding season to practically the northern half of the United States and southern Canada east of the Rocky Mountains. Within this range it seems to be commoner northward and rarer southward than the other species, ranging farther north and not so far south.

In appearance and habits our two cuckoos are very much alike, and their haunts are similar; both are often found together, or in similar places, though the black-billed is rather more of a woodland bird and rather more retiring than the yellow-billed. William Brewster (1906) says that in the Cambridge region it is “more given to haunting extensive tracts of dry upland woods and to nesting in wild apple trees, Virginia junipers and barberry bushes in remote rocky pastures such as those which lie scattered along the crest and sides of the high ridge between Arlington and Waverley.”

Courtship.—William Brewster (1937a) writes: “On July 15, 1908, I witnessed the coition of two Black-billed Cuckoos in woods near Bethel, Maine. It took place on a branch only three or four feet above the ground.

Although performed listlessly and intermittently, it was singularly protracted, for the two birds remained together at least four or five minutes, and did not finally separate until disturbed by my approach, when the male flew away and presently sang once. The female stayed quietly on her perch until I got near enough to see that she was a fully adult bird."

Nesting.—Most of the New England nests on which I have data were placed at low elevations, 2 to 4 feet above ground, in various small trees, bushes, or thickets. My first nest was the highest, 10 feet up, in the top of a leaning black birch in a strip of swampy woods and brushy thickets along a small brook. Another was 8 feet from the ground in a dense thicket of shrubbery and blackberry vines. Three nests were in small white pines, 3 to 4 feet up, in rather open spaces overgrown with various shrubs and small trees near the edges of the woods; the nests were placed on horizontal branches, against the trunk, and well concealed among the dense branches. Near Asquam Lake in New Hampshire I saw two nests in thick clumps of mountain-laurel in some dense and heavy deciduous woods, where the land sloped down to the lake; the mountain-laurel grew here in extensive patches, but not very high, seldom over 3 or 4 feet; the large and well-made nests were only 2 and 3 feet from the ground, but fairly well concealed.

Some nests of the black-billed cuckoo are very flimsy affairs, but often they are much more substantially built than the nests of the yellow-billed cuckoo. Owen Durfee describes in his notes a well-made nest that he found in a clump of chinquapin oaks near a road in Rehoboth, Mass. It was made of oak twigs and dry fern stalks, many of the twigs being fresh, with leaves attached; it was lined with dry oak leaves and a few fresh ones. It measured 8 inches in outside diameter and 3 by 3½ inches in inside diameter, being hollowed to a depth of 1 inch. There is a beautiful nest in the Thayer collection, taken from a thicket in Lancaster, Mass., 7 feet from the ground; it has a well-made foundation of coarse twigs, tufts of grass, and burs; and it is profusely lined with the green leaves and the cottony catkins of the poplar. S. F. Rathbun writes to me of an interesting nest that he found: "The nest was a saucer-shaped affair made entirely of the burs from the burdock plant, simply stuck together so as to form a shallow receptacle for the eggs. As a lining for the nest a few dry grasses were used, and the burs with the grasses represented the entire structure. The cuckoo certainly showed

ingenuity when it made this nest, for it could not have been more simply or easily constructed.”

Dr. T. C. Stephens has sent me some fine photographs ([pl. 8](#)) of a black-billed cuckoo’s nest taken at the base of a willow sapling, near the shore of Lake Goodenough, in Union County, S. Dak., and says: “I have observed a rather marked tendency for the black-billed cuckoo in this region to build its nest within a very few inches of the ground or on the ground. This nest in the photograph was several inches above the ground outside the clump, but it might be regarded as a ground nest, because there was quite an accumulation of dirt and weed growth immediately below it. Of course, I have found nests of this species at heights about level with a man’s eyes, also.”

A. Dawes DuBois has sent me the data for eight nests of this cuckoo, as found by him in Tompkins County, N. Y., and Hennepin County, Minn. Two of them were 5 feet above ground, one in the top of a bush covered with grape vines on the steep slope of a ravine and the other in a small larch, in the midst of an extensive willow thicket, half a foot from the trunk of the tree on a branch where two branchlets were attached. One of the lowest two was not over a foot above ground, among weeds 2 feet high; the nest was supported by a small dead branch that had fallen from a tree and lay hidden in the weeds. Others were at intermediate heights in more normal situations. The nest in the small larch tree he describes as made first of some cottony seed pods (ripened willow catkins), then a main structure of woody twigs, and lined with finer twigs, a few grasses and bits of the cottony seed pods; two tufts of dry grass, with roots attached, were at opposite sides of the rim.

Major Bendire (1895) says:

The nests of the Black-billed Cuckoo appear to be slightly better built than those of the Yellow-billed species; the platform is usually constructed of finer twigs, the soft inner bark of cedar, fine rootlets, weed stems, etc., and there is generally more lining. This consists of the aments of oak, white and black ash, and maple, willow catkins, and the flowers of the cudweed or everlasting (*Gnaphalium*), dried leaves, and similar materials. The majority of the nests are placed in rather low situations, mostly not over 6 feet from the ground, on horizontal limbs of bushy evergreens, pines, cedars, and hemlocks, or in deciduous trees and shrubs, such as the box elder, chestnut, thorn apple, and beech trees; also in hedges, briar and kalmia patches, occasionally on old logs, and now and then even on the ground.

A rather high nest was found by P. G. Howes (1908) near Stamford, Conn. It was about 15 feet from the ground, at the extremity of a limb of “a

scrubby apple tree at the foot of a hay-covered field.” He says that, in his experience, the nest “has always been lined with maiden-hair ferns.”

H. W. Flint, of Stamford, Conn., in a letter to Major Bendire (1895), mentions a still higher nest, and says: “I know of one spot in this vicinity where the Black-billed Cuckoo might almost be said to breed in colonies—a sloping hillside near a traveled road. Here I have found seven nests of this species within an hour, none of them placed over 3 feet from the ground. I have also frequently found their nest on a fallen limb, the top of which was resting upon underbrush. As an exception to their low nesting, I once found a nest containing two well-feathered young and two fresh eggs over 18 feet from the ground, placed in the top of a cedar tree, in a dense thicket of other cedars.”

As already mentioned under the preceding species, both cuckoos often lay their eggs in each other’s nests. Thomas McIlwraith (1894) mentions three cases of parasitism on the part of the black-billed cuckoo, as observed by Dr. C. K. Clarke, of Kingston, Ontario; he says:

The first birds Dr. Clarke observed being imposed upon were a pair of chipping sparrows, who raised the young cuckoo at the expense of the family.

Next came a pair of yellow warblers, whose *protégé* soon crowded out the legitimate occupants of the nest. They were raised from the ground and placed within reach, but the big boy required all the attention of the foster-parents, and the others died. During the whole period, the old cuckoo was always to be found flitting about in a restless manner, as if she had some doubt in regard to the ability of the warblers to take care of her child.

The third case was another pair of chipping sparrows, in whose nest the cuckoo was observed sitting, and from which she did not move till the observers almost touched her. The result was the same as in the other cases. The young cuckoo threw the sparrows out as soon as he had strength to do so.

Eggs of the black-billed cuckoo have also been found in nests of the wood pewee, cardinal, cedar waxwing, catbird, and wood thrush. Under the preceding species will be found a note, published by J. L. Davison (1887), describing the finding of a nest occupied by a robin, a yellow-billed cuckoo, and a mourning dove. Bendire (1895) published an almost identical account of such a remarkable occurrence, on the authority of the same observer, but with the black-billed substituted for the yellow-billed cuckoo. It seems hardly likely that such an unusual happening could occur with both species in exactly the same way, and leaves us in doubt as to which species of cuckoo was involved.

Eggs.—The black-billed cuckoo commonly lays 2 or 3 eggs; sets of 4 or 5 may sometimes be the product of a single pair; but the larger numbers that have been found in their nests, 6, 7, or even 8 eggs, were probably laid by two or perhaps three females, as these birds are notoriously careless about laying in each other's nests. Bendire (1895) says: "The eggs of the Black-billed Cuckoo are more nearly oval than elliptical oval, and shorter and rounder than those of the Yellow-billed Cuckoo, and much more deeply colored. Like these, they are unspotted; the shell is thin and fine grained, with little or no gloss. Their color is difficult to describe exactly, varying from Nile blue to pale beryl green, and occasionally the shell shows a decidedly marbled appearance, caused by different shades running into each other. * * * Aside from their deeper color, they are also readily distinguished from eggs of the Yellow-billed Cuckoo by their smaller size."

Eggs that I have seen in collections I should describe as "pale glaucous green," "dull opaline green," or "microline green" and varying from oval to elliptical-oval. Some of these, as well as some of those included in the following measurements, may be yellow-billed cuckoos' eggs, as the two are probably not always recognizable with certainty. The measurements of 54 eggs, presumably of this species, average 27.18 by 20.57 millimeters; the eggs showing the four extremes measure **30.48** by **22.86**, **22.61** by 18.80, and 25.40 by **18.29** millimeters.

Young.—Incubation is shared by both sexes and lasts for about 14 days. As the eggs are often laid at infrequent intervals, it is not unusual to find young birds of different ages, or even eggs and young, in the same nest. Both sexes assist in the care of the young and are devoted parents; even when an egg is hatched by a foster parent, the mother cuckoo does not seem entirely to lose interest in her youngster, as related above. On the other hand, William H. Moore (1902a), of Scotch Lake, New Brunswick, says: "I have known this bird to desert its young when the nest was molested, and after the young died they were covered with leaves by the adults." Cuckoos are careless about removing the cast-off shells, which are often found in the nest after the young have hatched. The young remain in the nest 7 to 9 days after hatching and then become quite precocial. While in the nest the young are fed by their parents in a rather peculiar manner, of which Dr. Thomas S. Roberts (1932) writes:

When the old bird returns, the food, which is very likely to be *live* caterpillars, is concealed in the throat. As a nestling raises its head with open mouth and rapidly vibrating wings, the parent thrusts its bill deeply into the open maw and the young bird grasps securely the smooth bill of the old bird, in which action it is greatly aided by several soft papillae or disks in the roof of the mouth. Then, with a slow, pumping motion, the squirming caterpillars are transferred with some difficulty from one mouth to another. The process is a slow one, the birds being attached a minute or more and the transfer aided, apparently, by a sucking effort on the part of the nestling.

When a young Cuckoo opens its mouth widely there are visible, in the roof of the mouth, a number of large, flat-topped, white papillae or tubercles, arranged symmetrically, the function of which is plainly to make it possible for the nestling to maintain its hold on the parent's bill, which is smooth and tapering. A small finger-tip inserted well down into the open, upturned mouth of a nestling is seized tightly and a sucking motion is distinctly perceptible. [Pl. 10.]

Prof. Francis H. Herrick (1935) has made a thorough study of the home life of the black-billed cuckoo, and I cannot do better than to quote a few of his remarks on the behavior of the young. After describing the peculiar appearance of the newly hatched young, he says:

More remarkable than anything about its appearance, however, is the muscular vigor and endurance the cuckoo displays at this tender age, for it seems to be able to withstand heat, hunger, and general neglect that would be fatal to the young of most wild birds. I soon found by experiment that when barely three hours out of its shell this little cuckoo could hang suspended by a leg or even by a single toe for upwards of a quarter of a minute, and that it could raise itself up until its bill was well over the twig that it grasped with both feet; and in a short time it was able to raise itself on to the twig, or even to draw itself up with the power of one leg, which implies an extraordinary muscular development. * * * Though born blind and essentially naked, the young black-bill is neither deaf nor dumb, and in proportion to its size it is probably the strongest and most enterprising altricial nestling on the North American continent. * * *

Like other nestlings, the cuckoo lies flat, with toes clenched, and holds to its fragile nest with a firm grip. Attempt to remove it, and it is likely to pull its nest to pieces rather than loose its hold; or it may even drag out a fellow-nestling, reminding one of crayfishes or lobsters, in the handling of which one may also get a living chain. Should you succeed in displacing a bird, its claws will rapidly open and close in its desperate search for any object to clasp, for a contact stimulus afforded by any solid body can alone satisfy this strong reflex. * * *

At the age of about six days the nestling cuckoo has reached that peculiar transitory state which we may call the "quill" or feather-tube stage. * * *

Towards the end of this brief and unique stage the behavior of the young bird changes in marked and rapid fashion. It indulges in new attitudes, acquires new call and alarm-notes, shows fear, and begins those preening or combing movements which are to effect a relatively sudden and altogether surprising change in its appearance. * * *

In one instance the first preening action was noticed on the sixth day; thereafter this kind of activity became frequent, and the bird would comb every "quill" within reach, drawing the mandibles over it from base to apex. Then, with apparent suddenness, at the close of the seventh day the transparent horny sheaths began to give way at their base—

instead of wearing off gradually from apex to base in the usual fashion—and were raked off by the mouthful. * * *

Fear may become manifest as early as the sixth day, when a frightened bird will sometimes clear the nest at a bound and, seizing a branch with both feet, hold firmly to it. Should it drop to the ground, it can make off with surprising speed. If captured and held, it will emit loud, explosive squeals, than which nothing seems to arouse its parents to quicker attack or bolder measures. Replace it in the nest, and it spreads its wings, stiffens, and lies flat with every feather-tube on end; and it will repeat these defensive measures as often as it is touched or disturbed. * * *

When from seven to nine days old, with half of its feathers unsheathed, the cuckoo suddenly leaves its nest and enters upon a climbing period, which lasts about a fortnight or until it is able to fly. At one of my observation stations I saw three young birds leave their nest in succession, and the procedure in the case of the oldest one was particularly interesting. This bird, which had been sitting in the bright sunshine, for the day was not uncomfortably warm, of a sudden moved to one side of its platform. After having combed off several mouthfuls of feather-sheaths, it sat upright for some minutes and gazed into its outer world. Then, directing its attention to a small branch and ducking its head as if contemplating flight, with a leap it cleared the nest, and, catching hold of a twig, with both feet, it swung free with acrobatic dexterity. In another moment it had pulled itself up and was comfortably perched. If such a first perch is placed in the shade and the young bird is promptly fed, it may keep to it for a long time; but it can move about, and should it drop to the ground, it can mount to safety again.

The vertical position assumed by young cuckoos, probably as a hiding pose, has been noted by several observers, but the following incident reported to me in a letter from Frederic H. Kennard, is quite unusual; he says: “I had just been investigating a big highbush blueberry bush, looking for a nest, when I discovered, to my surprise, a fledgling black-billed cuckoo, squatting on a twig about 6 feet from the ground. The little bird, which really was not able to fly, was squatting on a limb, just as little birds ordinarily do; his wing feathers were fairly well developed, but his tail was only about a quarter of an inch long. When I parted the branches a trifle, so that we could see him better, and finding out that he was discovered, he promptly assumed an almost perpendicular position, with his neck stretched out almost unbelievably and his bill almost straight in the air; and there he sat, immovable, with his bill in the air like a bittern, only oscillating a trifle when the branch on which he was sitting was disturbed a little by the breeze.

“My youngest son, Jack, being interested in the peculiarities of cuckoos’ feet, attempted to pick him off the limb; the little bird fluttered to the ground, where he picked him up. When we had duly examined and discussed the arrangement of his toes, Jack endeavored to put the little

fellow back exactly where he had been when we first disturbed him. Then, as he endeavored to replace him on the limb, he suddenly went limp and, apparently, passed out in his hand, frightened to death, as I supposed. He was perfectly limp and my impression is that his eyes were closed. Jack finally, in trying to get him to stay on the limb, hung him across the limb by the neck, with his head across one side and his body down the other side. Just then there came a little breeze, the body dropped, and that little bird simply scuttled in under the ferns. It was the most astonishing performance that I ever witnessed, first the stake-driver attitude, as a protective position, and then playing dead.”

Plumages.—Professor Herrick (1935) says of the newly-hatched young cuckoo: “Although most birds emerge from the shell wet with the amniotic fluid, the cuckoo just mentioned came out quite dry. It was two and one-half inches long and weighed less than a quarter of an ounce (or 7.4 grams). Its skin was coal-black, sparsely sprinkled with sharply contrasting snow-white ‘hairs’—in reality the feather-tubes of a rudimentary down which never unfolds. These primitive feather-tubes are later pushed out by those of the juvenal contour feathers and for about a week are borne upon their tips, thus giving them a peculiar flagellate appearance.”

When about six days old, the young cuckoo “bristles like the fretful porcupine in every feather-tract”; these bristles are the feather-tubes of the juvenal plumage, referred to above. At about this age, the young bird begins the “combing” process, by which the sheaths of these feather tubes are removed, as described above, and a marvelous change begins to take place in a remarkably short time, as the sheaths are removed and fall in a shower in and about the nest. Professor Herrick (1935) says: “Thus at one stroke one or more of the juvenal contour feathers are exposed and quickly fluff out in all their shapely proportions. * * * The change actually occupies about twelve hours, and it is really not complete, since the sheaths of the wing- and tail-quills flake off gradually, as in other birds, and those feathers of head and neck out of reach of the ‘comb’ remain sheathed for a considerable time longer.”

By the time that the young bird leaves the nest, at the age of 8 or 9 days, it is in nearly full juvenal plumage; the wings are fairly well grown, but the tail is still very short. In this plumage the soft plumage above is “buffy brown,” each feather tipped with white; the under parts are silvery white,

tinged with pale gray on the belly and with pale buff on the breast and throat. During late summer and early fall, most, if not all, of this juvenal contour plumage is molted and replaced by the first winter plumage, but the flight feathers of the wings and tail are retained until spring. This first winter plumage is much like that of the adult, but it is more brownish on the head and back and more greenish olive on the scapulars than in adults; the throat is more buffy and the upper breast more grayish buff than in the adult; the young bird's tail is quite different, the grayish white tips are smaller and are not bordered inwardly with the dusky space, which is clearly visible in the adult tail. The molts are apparently the same as in the yellow-billed cuckoo; the adult plumage seems to be acquired before the young birds return from their first winter in the south, but we have no specimens showing a spring molt.

Food.—The black-billed cuckoo is just as good a caterpillar destroyer as the yellow-billed; in fact the food habits of the two species are almost identical in all respects. An abundance of caterpillars in a locality is very likely to bring with it an invasion of cuckoos. Frank L. Farley writes to me: "As far as I am aware, the black-billed cuckoo was unknown in central Alberta until the summer of 1923. That year the central portion of the Province was infested with tent caterpillars, which, in 1924–25, assumed plague proportions. Entire bluffs of poplar trees, several acres in extent, were entirely denuded of their leaves, while houses and other buildings were overrun with the pests." In June 1924 the cuckoos began to appear for the first time, birds entirely unknown to the residents. "Although caterpillars gradually disappeared after 1925, cuckoos were reported from widely separated parts of central Alberta, the most northerly one being about 150 miles north of Camrose, which is in latitude 53° N. The presence of cuckoos and caterpillars in the same territory during these years would tend to bear out the claims of other observers, that the insects are particularly relished by these birds, which, in some uncanny manner, are able to locate infested territory far removed from their usual place of residence."

Forbush (1927) says:

In seasons when caterpillars of any species are abundant, cuckoos usually become common in the infested localities. They follow the caterpillars, and where such food is plentiful, the size of their broods seems to increase. During an invasion of forest tent caterpillars in Stoneham, Massachusetts, in May, 1898, Mr. Frank H. Mosher watched one

of these birds that caught and ate 36 of these insects inside of five minutes. He saw another in Malden eat 29, rest a few minutes and then eat 14 more. In July, 1899, he reported a family of these birds in a locality infested with the gipsy moth, and said that they were eating large quantities of gipsy caterpillars. In June, 1895, Mr. Henry Shaw reported great numbers of these cuckoos in Dorchester feeding on the same pests. The late Professor Walter B. Barrows, of Michigan, an extremely conservative ornithologist, is responsible for the statement that in several instances remains of over 100 tent caterpillars have been taken from a single cuckoo's stomach. The Black-billed Cuckoo, because more common than the Yellow-billed, is the species that most commonly attacks this insect in New England orchards. During an invasion of army worms, Professor S. A. Forbes found that 95 per cent of the food of this species consisted of that caterpillar.

F. H. King (1883), writing of the food of the black-billed cuckoo in Wisconsin, says: "Of thirteen specimens examined, nine had eaten caterpillars—among them were eight of the fall web-worms (*Hyphantria textor*), thirty-three of the oak caterpillars (*Dryocampa senatoria*), one of the Io caterpillars (*Saturnia io*), six of the antiopa caterpillars (*Vanessa antiopa*), and one of the caterpillars of the archippus butterfly (*Danais archippus*). One contained five larvae of the large saw-fly (*Cymbex americana*); six, twenty-five grasshoppers; one, a cricket; two, ten beetles; and two, two harvest-men."

Other authors have charged this cuckoo with eating minute mollusks and other small animals, fishes and aquatic larvae, fruits and berries, and even the eggs and young of small birds. On the latter point, Henry D. Minot (1877) says that "they do great mischief in destroying the eggs of other useful birds. Like arrant cowards, as they are, they take opportunities to approach stealthily the nests of many birds, whom they would be afraid to encounter, and then feast on the eggs of the absent parents, after which they hurry away. They are scarcely less destructive in this way than the black snakes, though I have never known them to kill young birds."

Behavior.—The two cuckoos are so much alike in haunts, habits, and behavior that most of what I have said about the yellow-billed would apply equally well to this species. The black-billed is rather more swift on the wing than the other, but it flies in the same graceful manner. It is the same shy recluse of the shady retreats, among the dense foliage of the woods and shade trees, unafraid to frequent the orchards and gardens in search for its food, but shunning any intimacy with human beings; we hear its wandering voice but seldom see more than a fleeting glimpse of its graceful form as it fades away into the shadows.

In defense of its eggs or young the black-billed cuckoo is often quite courageous. Olive Thorne Miller (1892) writes charmingly of her experiences with an incubating pair of these cuckoos; she watched them change places on the nest, and found the female quite confiding, but the male never became reconciled to her presence:

It happened that I arrived when the mother was away, and the head of the household in charge. No sooner did I appear on the path than he flew off the nest with great bustle, thus betraying himself at once; but he did not desert his post of protector. He perched on a branch somewhat higher than my head, and five or six feet away, and began calling, a low "coo-oo." With every cry he opened his mouth very wide, as though to shriek at the top of his voice, and the low cry that came out was so ludicrously inadequate to his apparent effort that it was very droll. In this performance he made fine display of the inside of his mouth and throat, which looked, from where I stood, like black satin. * * *

Finding that his voice did not drive me away, the bird resorted to another method; he tried intimidation. First he threw himself into a most curious attitude, humping his shoulders and opening his tail like a fan, then spreading his wings and resting the upper end of them on his tail, which made at the back a sort of scoop effect. Every time he uttered the cry he lifted wings and tail together, and let them fall slowly back to their natural position. It was the queerest bird performance I ever saw.

On another day, she says: "We had not waited long when the head of the cuckoo family appeared. He saw us instantly, and, I regret to say, was no more reconciled to our presence than he had been on the previous occasion; but he showed his displeasure in a different way. He rushed about in the trees, crying, 'cuck-a-ruck, cuck-a-ruck,' running out even to the tip of slender branches that seemed too slight to bear his weight. When his feelings entirely overcame him he flew away, and though we remained fifteen minutes, no one came to the nest."

E. A. Samuels (1883) writes:

Like the other, the Black-billed Cuckoo is very cowardly, and is quickly driven from the neighborhood of the nest of almost any of the other birds. If a robin, or other bird of equal size, discover one of these, to him pirates, in the vicinity of his nest, he immediately assaults the intruder, with loud outcries, pouncing upon him, and pecking with great ferocity. Others of his neighbors, who are near, join in the attack; the Cuckoo, in retreating, dives into the recesses of a stone wall, or the first secure retreat available; very seldom taking to his wings, as another bird would do, I have known of a cuckoo being driven into a barn by a Blue-bird (*S. sialis*), who sat perching on a fence outside for several minutes, keeping his enemy prisoner; and the latter, when pursued and captured by myself, preferred being my prisoner to facing his enemy outside.

Voice.—One cannot always distinguish with certainty *all* the notes of the black-billed cuckoo from *all* those of the yellow-billed. Many of the notes are much alike in both species. The notes of the black-billed are, as a rule,

softer and more liquid than those of the yellow-billed and not so deep-toned.

The ordinary “song” of the black-billed is preceded by a gurgling note, and the rest of the long song is uttered in regularly measured time, not retarded at the end, as is that of the yellow-billed, and the notes are given in couplets or triplets, one syllable in each set being accented. Dr. Charles W. Townsend (1920a) describes it very well as follows: “The full song may be described as a preliminary harsh clearing of the throat followed by from six to twelve short coughs which in turn are succeeded by the more pleasing doublets and triplets of cows. The Yellow-billed Cuckoo repeats his cows or cowks in regular order without dividing them into sets and they sound as woodeny as if he were striking a plank with a mallet.”

Aretas A. Saunders (1929) says: “The bird has a variety of calls. One consists of a gurgling note followed by single notes in even time, ‘krak-ika kuh kuh kuh kuh kuh kuh kuh kuh’. Another is a series of groups consisting of two to six notes repeated many times, with one of the notes strongly accented, such as ‘*kuka kuka kuka*,’ or ‘*kakukaka kakukaka kakukaka*.’ I have known a bird to repeat such phrases over a hundred times without stopping.”

The notes of the black-billed cuckoo are often given on the nest or while the bird is in flight, and they may be heard at all hours of the day or night. Both cuckoos are said to be more noisy just before a rain, hence the name “rain crow.”

Mr. Brewster (1937a) says that “both species coo in the same subdued, mournful, dovelike tones, but when so engaged, the Yellow-bill always utters only a single note at a time, and then waits at least a second or two before following it with another precisely similar; whereas the cooing notes of the Black-bill are invariably doubled or trebled or quadrupled, or perhaps even quintupled, yet separated from one another within such grouping by scarcely appreciable pauses.”

Gerald H. Thayer (1903) has given an interesting account of what he calls “the mid-summer, mid-night, mid-sky gyrations of the Black-billed Cuckoo, as noted by my father and me for three consecutive seasons in the southwestern corner of New Hampshire”:

Several years before we discovered the nocturnal-flight phenomenon, we began to be puzzled by the extreme frequency of Cuckoo calls on summer nights. * * * They uttered both the *cow-cow* notes and the rolling guttural call; but the guttural was much the commoner of the two, except on dark, foggy nights, when the case was usually reversed. * * * The birds were often so far up as to be only faintly audible when directly overhead, with no obstructions interposed; and this on a still night would seem to mean an elevation of at least a hundred and fifty yards. They sometimes flew lower, however, and on cloudy nights often moved about barely above the tree-tops.

* * * On the evening of July 11—a pitch-dark evening with a thunder-shower lowering,—they were remarkably noisy, both sitting in trees and flying high in air. The seated ones, of which I heard only two, made the *cow-cow* notes, while all the flying ones made the liquid gurgle. I heard this note overhead between thirty and forty times in the course of about three hours, during half of which time I was afoot on the road.

Field marks.—This bird may be recognized as a cuckoo by its size, shape, and general coloration. It can be distinguished from the yellow-billed cuckoo by the absence of the distinguishing marks of the latter, the cinnamon-rufous in the wings, and the conspicuously black-and-white lateral tail feathers.

The wings of the black-billed cuckoo are practically uniform in color with its back; and the lateral tail feathers are dark gray, with inconspicuous, grayish-white, smaller tips, bordered inwardly with a dusky spot. The wholly black bill and the red eyelids can be seen only at short range.

DISTRIBUTION

Range.—Southern Canada and the United States east of the Rocky Mountains, south in winter to northwestern South America. Accidental in Italy, the Azores, and Ireland.

Breeding range.—The breeding range of the black-billed cuckoo extends **north** to southern Saskatchewan (Johnstone Lake, Muscow, and Indian Head); southern Manitoba (Carberry, Portage la Prairie, and Shoal Lake); northern Minnesota (Crooked Lake); southern Ontario (probably Goulais Bay, Toronto, and Stirling); Quebec (Hull, Montreal, probably Quebec, and probably Kamouraska); New Brunswick (Scotch Lake); probably Prince Edward Island (North River); and Nova Scotia (Wolfville and Pictou). From this point the range extends **south** along the seaboard to North Carolina (Raleigh and Winston-Salem); Tennessee (Beersheba Springs and Nashville); northwestern Arkansas (Rogers); eastern Kansas (Lawrence and Clearwater); Nebraska (Red Cloud, Kearney, and Antioch); and

southeastern Wyoming (Wheatland). **West** to eastern Wyoming (Wheatland and Dayton); eastern Montana (Terry); and Saskatchewan (Johnstone Lake).

The species has been detected in summer on several occasions west of its breeding range. Among these are: Colorado (Fort Morgan, Clear Creek, Fort Collins, and Wray); Wyoming (Laramie); Montana (Fort Keogh, Knowlton, and Billings); and western Saskatchewan (Eastend, Skull Creek, Medicine Hat, and Big Stick Lake). On June 23, 1924, a partially completed nest was found near Camrose, Alberta, a range extension that for the time being must be considered unusual (see remarks under "Food"). The species also was recorded at Godbout, Quebec, on August 11, 1885.

Winter range.—Available information indicates that in winter this species is concentrated in northwestern South America: Colombia (Antioquia, Medellin, and Bogota); Ecuador (Guapulo, La Carolina, Daule, Puna Lake, and Lechugal); and northern Peru (Huamachuco).

Spring migration.—Early dates of spring arrival are: Florida—Hastings, April 13; Eau Gallie, April 27; Pensacola, May 2. Alabama—Scottsboro, April 18; Barachias, April 22. Georgia—Atlanta, April 24; Athens, April 27. South Carolina—Frogmore, April 24; Spartanburg, April 29. North Carolina—Raleigh, April 15; Piney Creek, May 4. Virginia—Blacksburg, April 26; Lynchburg, May 8. District of Columbia—Washington, April 30. Maryland—Baltimore, April 20. Pennsylvania—Jeffersonville, April 30; Ridgway, May 6; Doylestown, May 9. New Jersey—Milltown, May 2; Passaic, May 7. New York—Medina, May 3; New York City, May 9; Rochester, May 15. Connecticut—Fairfield, May 1; Hadlyme, May 4; New Haven, May 7. Rhode Island—Block Island, May 6. Massachusetts—Belmont, May 4; Marlboro, May 6; North Amherst, May 12; Danvers, May 12. Vermont—Clarendon, May 12; St. Johnsbury, May 13; Wells River, May 18. New Hampshire—East Westmoreland, May 16; South Hooksett, May 16; Concord, May 21. Maine—South Portland, May 11; Winthrop, May 15; Waterville, May 20. Quebec—Montreal, May 10; Hatley, May 28. New Brunswick—Scotch Lake, May 27. Nova Scotia—Halifax, May 12; Wolfville, May 18. Louisiana—Avery Island, April 12. Mississippi—Bay St. Louis, April 14. Arkansas—Fayetteville, April 30; Broma Towns, May 2. Tennessee—Knoxville, April 12; Tate, April 26. Kentucky—Lexington, May 1; Bowling Green, May 6. Missouri—Jonesburg, May 3; Montgomery City, May 4; Columbia, May 7. Illinois—Rantoul, April 17; Elgin, May 5;

Glen Ellyn, May 7. Indiana—Bloomington, April 26; Richmond, April 27; Vincennes, May 7. Ohio—Oberlin, May 1; Columbia, May 2; Upper Sandusky, May 3. Michigan—Detroit, May 1; Brant, May 3; Sault Ste. Marie, May 11. Ontario—Guelph, April 19; Ottawa, May 7; London, May 9. Iowa—McGregor, April 24; Mason City, April 30; Wall Lake, May 3. Wisconsin—Racine, May 8; Beloit, May 10; Whitewater, May 11. Minnesota—Montevideo, April 13; Anoka County, May 5; Excelsior, May 13. Oklahoma—Tulsa, May 3; Norman, May 5. Kansas—Fort Leavenworth, May 6; Bendena, May 8; Clearwater, May 9. Nebraska—Valentine, May 1; Red Cloud, May 11; Omaha, May 13. South Dakota—Yankton, May 1; Huron, May 13; Forestburg, May 15. North Dakota—Grafton, April 25; Antler, May 3; Jamestown, May 23. Manitoba—Aweme, May 20; Pilot Mound, May 24; Reaburn, May 26. Saskatchewan—Muscow, May 12; Indian Head, May 20.

Fall migration.—Late dates of fall departure are: Manitoba—Margaret, September 3; Aweme, September 14. North Dakota—Fargo, September 13; Cando, September 18; Argusville, September 22. South Dakota—Forestburg, September 6; Sioux Falls, September 22; Lennox, October 14. Kansas—Cimarron, September 2; Osawatometie, September 22. Minnesota—Elk River, September 27; Red Wing, October 2. Wisconsin—New London, September 24; Racine, September 24; Madison, September 26. Iowa—Wall Lake, September 28; Osage, October 10; McGregor, October 20. Ontario—Ottawa, September 16; Point Pelee, October 18. Michigan—Charity Island, September 20; Grand Rapids, September 26, Detroit, October 9. Ohio—Saybrook, October 1; Berlin Center, October 12; Columbus, October 15. Indiana—Indianapolis, October 2; Fort Wayne, October 12; Bicknell, October 18. Illinois—Rantoul, October 4; La Grange, October 7; Glen Ellyn, October 21. Missouri—Columbia, October 4; St. Louis, October 16. Kentucky—Danville, October 5; Bowling Green, October 18. Mississippi—Bay St. Louis, October 11. Nova Scotia—Sable Island, September 25. Quebec—Montreal, September 5. Maine—Orono, September 21; Winthrop, October 20. New Hampshire—Jefferson, October 3; Durham, October 5. Vermont—Wells River, September 14; Woodstock, September 18. Massachusetts—West Groton, September 30; North Truro, October 13; Harvard, October 16. Rhode Island—Providence, October 23. Connecticut—New Haven, September 28; East Portland, October 3; Meriden, October 8. New York—Hyde Park, October 6; New York City, October 10. New

Jersey—Elizabeth, October 3; Sandy Hook, October 5; Milltown, October 24. Pennsylvania—McKeesport, October 11; Jeffersonville, October 13; Pittsburgh, October 16. District of Columbia—Washington, October 28. Virginia—Naruna, October 18. North Carolina—Chapel Hill, October 3; Raleigh, October 10; Hendersonville, October 12. Georgia—Atlanta, October 14; Thomasville, October 21. Alabama—Autauga County, October 16. Florida—College Point, September 28; Pensacola, October 23.

Casual records.—While the black-billed cuckoo does not migrate regularly through the Caribbean region it has been recorded on Dominica (September 30, 1904), Tobago, and Trinidad. Gundlach is alleged to have taken a specimen in May (year?) near Cardenas, Cuba, and there is a somewhat doubtful record from the Isle of Pines on May 11, 1909. Specimens were taken in Bermuda in October 1874 and also in April and May 1875. The Ponta Delgada Museum has an undated specimen taken at San Miguel, Ponta Delgada, Azores; a specimen taken in 1858 near Lucca, Italy, is preserved in the Museum of the University of Pisa; and one was taken on September 25, 1871, at Killead, County Antrim, Ireland.

Egg dates.—Illinois: 13 records, May 7 to July 20; 7 records, June 1 to 26, indicating the height of the season.

Massachusetts: 20 records, May 19 to June 20; 10 records, May 30 to June 10.

Michigan: 14 records, May 25 to September 14; 7 records, June 21 to July 20.

New York: 23 records, May 11 to July 18; 12 records, May 29 to June 9.

CUCULUS OPTATUS OPTATUS Gould

HIMALAYAN CUCKOO

CONTRIBUTED BY EDWARD CHARLES STUART BAKER

HABITS

The Himalayan cuckoo very closely resembles the various races of the common cuckoo (*Cuculus canorus*) in its general habits, but it is, I think, a more secretive bird, keeping closely to tall trees with dense foliage, so that

although one may hear its very distinctive call on many occasions quite close by it is often very difficult to see until it takes to wing. In Kashmir and the northwest Himalayas it is found in summer at all elevations between 5,000 and 10,000 feet and occasionally to some 2,000 feet higher but, never, I believe, above the forest line. In Sikkim it is common in well-forested land between 4,000 and 9,000 feet, while Stevens (1925) records having heard it calling at an elevation of 3,500 feet on May 25, so that it may breed as low down as this in that part of the Himalayas. In Assam it is very common between the same elevations as in Sikkim, there also keeping closely to forest, either deciduous, pine, or evergreen with dense undergrowth. All ornithologists seem to agree as to the nature of the country frequented by this cuckoo. In letters to me A. E. Jones mentions "dense forest," "dense deodar forest" as the breeding haunts of birds whose eggs he has found; B. B. Osmaston (MS.) found it in "open, well-wooded forest." Mackenzie and Hopwood (MS.) took eggs in the nests of *Acanthopneuste davisoni* in the Chin Hills, in "heavy evergreen forest." The only exception to this of which I am aware is an egg taken by T. R. Livesey (MS.) found in Kashmir in a nest of *Emberiza cia stracheyi* in a "well-wooded glade in forest."

At all seasons of the year it keeps almost entirely to branches of high trees, some 40 or 50 feet or more from the ground, descending to the undergrowth only when hunting for nests in which to deposit its eggs or for the actual deposition of the egg. When so employed it is, I believe, always silent and, even in its movements, very quiet and secretive unless it is being harassed by small birds who hunt it just as they do the common cuckoo. I have not seen this bird feeding in bushes or on the ground even when there are numerous caterpillars or a flight of termites to tempt it.

When on migration also this cuckoo seems to keep to forest or to exceptionally well-wooded country, and I have no record of its having been found in the open in India, though obviously it must sometimes pass over such country in its movements from one district to another, more especially in the northwest.

Spring.—In Kashmir and the northwest Himalayas the birds arrive from the lower hills and the plains adjoining them in April, a few apparently in the first fortnight, but the majority not until the last week or so. In Sikkim they arrive a good deal earlier, and Stevens (1925) records them as *seen* at

Gopaldhara in the Rambong Valley about the middle of March “when it ascends to an elevation of 7,000 feet,” while he *heard* them calling at the same place as early as March 12. Farther east in Assam it does not arrive at its breeding quarters in the higher hills until about the first week in April and then only in small numbers. In Burma at elevations of about 4,000 feet and over it breeds about a fortnight earlier and is in full call by the last week in March.

This cuckoo is possibly only a partial migrant in India, leaving the higher hills for the broken foothills, where it may be found more or less throughout the winter, as well as in the plains immediately adjoining them. To the east, however, as I show under the fall migration notes, it is a true migrant. According to La Touche (1931–1936), during the spring migration north the Himalayan cuckoo “appears in China from about the 10th April to the end of May,” and he records three cuckoos taken at Shaweishan on May 1, 16, and 17. These birds were presumably on their way north to eastern Siberia or Manchuria.

La Touche does not think that this bird breeds anywhere in the Chinese Hills, its place being taken by the local breeding race *kelungensis*.

The 38 eggs in my collection were taken as follows: Three in April, 14 in May, 20 in June, none in July, and 1 in August. The earliest date was April 25, 1915, and the latest August 20, 1914. At the same time an egg taken in the Chin Hills on April 30 was on the point of hatching, so, in the Burmese Hills, some eggs must be laid in the middle of that month, birds arriving in their breeding haunts at least a fortnight earlier. Other eggs that have passed through my hands were all taken within these dates so far as I have recorded them, so they may be accepted as confirming my dates for migration.

Courtship.—There is nothing recorded upon this subject but, so far as I know, it differs in no way from that of the Khasia Hills cuckoo, and both sexes seem to be equally promiscuous in their sexual relations.

Nesting.—This cuckoo is parasitic principally on the small warblers of the *Phylloscopus* and *Acanthopneuste* group, which lay white eggs, either immaculate or speckled slightly with various tints of reddish brown or red, or the *Seicercus* group, which lay glossy pure white eggs.

In my own collection I have eggs taken from the nests of 16 species and subspecies of fosterers (see the following list); of these the first 10 may be considered to be normal fosterers and the latter abnormal.

The Himalayan cuckoo does adopt a certain territory but hardly in the restricted sense that *Cuculus canorus* and its various races do. The principal reason for this is the comparative rarity of the birds that she selects as foster parents to her eggs. For instance, although the little *Acanthopneuste* breeding in the Khasia Hills is a common bird, it nowhere breeds in the numbers in which birds of the *Suya* and *Cisticola* genera are found. Perhaps half a dozen pairs may be found in one big forest of many square miles, and to find even these six is very hard work. In the same forest another half dozen birds of the genus *Seicercus* of various species may be found. These little warblers of the two genera *Acanthopneuste* and *Seicercus* make very similar nests, balls of moss, quite green and fresh, which are placed in hollows in banks and generally among the same kind of moss as that of which the nests are built. They are exceedingly hard to find by human beings and, unless the birds are watched on to the nest, are generally located only when the female is disturbed from it by accident and doubtless the cuckoo also finds they are difficult to mark down.

LIST OF FOSTER PARENTS	eggs
<i>Phylloscopus inornatus humei</i>	2
<i>Phylloscopus proregulus simlaae</i>	1
<i>Acanthopneuste reguloides harterti</i>	13
<i>Acanthopneuste reguloides davisoni</i>	3
<i>Acanthopneuste reguloides reguloides</i>	1
<i>Acanthopneuste occipitatis occipitatis</i>	4
<i>Seicercus cantator</i>	3
<i>Seicercus burkii</i>	2
<i>Seicercus xanthoschista xanthoschista</i>	1
<i>Seicercus castaneiceps castaneiceps</i>	2
<i>Ianthia rufilata</i>	1
<i>Niltava sundara sundara</i>	1
<i>Suya crinigera assamica</i>	1
<i>Orthotomus atrogularis nitidus</i>	1
<i>Napothera brevicaudata striata</i>	1
<i>Emberiza cia stracheyi</i>	1

I have only two small series of three eggs, each of which I believe to have been laid by one cuckoo, one of these being taken in 1909 by myself. Of this latter the first two eggs were found on June 16 and 18 in a patch of forest in which we marked down three nests of the Khasia crowned willow warbler, one of which contained no cuckoo's egg. On June 13 we found that the pair of warblers first seen on May 16 had again built close by their original nest and the new one contained a third egg, apparently of the same cuckoo, and three of their own. The nests found on May 16 and 18 were between a quarter and a half of a mile apart.

On another occasion, June 27, 1935, three eggs were taken, one in the nest of the willow warbler and two others in nests of *Seicercus*. These were all in the same forest, but only two close together and the third nearly half a mile away. The eggs, however, appeared obviously to have been laid by the same bird.

Eggs.—The eggs of this species of cuckoo are all of the same type, white eggs sparsely marked with tiny black specks, sometimes confined to the larger end only. Several oviduct eggs have been taken, the first by Brooks from the oviduct of a female he shot in Kashmir on June 17, and three others by Rattray in 1903 on June 10, 15, and 17, respectively. All these were exactly alike and similar to the description given above.

The only two exceptions in coloration I have seen are eggs one of which has a faintly green tinge while the other has an equally faint pink tinge. The first of these was taken from a forktail's (*Enicurus maculatus guttatus*) nest and the second from that of a tailorbird (*Orthotomus atrogularis nitidus*).

In shape the eggs are rather long ellipses, and the shell is thin and rather fragile for a cuckoo's.

Forty-one eggs average in size 20.11 by 14.28 millimeters; maxima **25.3** by **16.2**; minima **19.0** by **13.0** millimeters; the same number average in weight 141.4 milligrams; maximum 178; minimum 117 milligrams.

A large double-yolked egg weighs 222 milligrams, while two very small, almost pygmy eggs, weigh only 105 and 112 milligrams.

The method of the deposition of the eggs is not known, but in many cases they could not possibly be laid direct into the nest. The crowned willow warbler makes a nest in a hole in among the roots of trees with so

small an entrance and so far in that the cuckoo could not possibly get in to lay her egg nor could she eject it from the cloaca sufficiently far to reach the egg chamber and, if she did, it would certainly break. Nests of the genus *Seicercus* and of *Acanthopneuste reguloides* are so shaped and situated that though the cuckoo could not enter them she could very often eject the eggs into them without much difficulty by clinging to their tops or sides during the operation.

Young.—Nothing is recorded, as to incubation, but I believe it to take only 10 or 11 days, i. e., much the same period as for the eggs of the small warblers the cuckoo selects to cuckold.

Ejection of young undoubtedly takes place in the same manner as that by which it is effected by the common cuckoo, the young cuckoo hoisting the other occupants of the nest, either eggs or young birds, onto its shoulder and then pushing them over the edge of the nest. I have seen several young birds from two days to a week old in the nests of various warblers, and they have always been alone. The young cuckoo has the same dorsal interscapular arrangement as that found in the nestling *canorus* to assist it in carrying out the fratricide.

Plumages.—Male: Similar to the male of *Cuculus c. canorus* and *C. c. bakeri* but less dark on the upper parts than the latter and always to be distinguished from all races of the *canorus* group by having the edge of the wing pure white and not barred. On the lower surface the white bars are broader and the black bars in consequence wider apart, while they are blacker and bolder in appearance.

Female: Similar to the male but generally with a more rufous tinge on the breast and abdomen; the under tail coverts are often pale fulvous with black crossbars.

The female has a hepatic phase like that of *C. c. canorus*.

Juvenile: In first plumage blackish brown above, the feathers all broadly edged with white; the wing quills are barred with rufous on the outer webs; the chin, throat, and breast are blackish, the feathers narrowly fringed with white and the rest of the lower parts white, or faintly fulvous-white, heavily barred with strong bands of black.

In intermediate stage between juvenile and adult the plumage is slaty, the feathers very narrowly edged with white; below, the chin, throat, and upper breast are blackish, the feathers broadly fringed with white and the remainder of the lower parts like the adult.

Hepatic young are like the hepatic female adult but less richly colored and more heavily banded below with blackish, especially on the chin, throat, and breast. Some specimens of young birds in the British Museum collection seem to be changing from a juvenile hepatic plumage into a normal adult plumage.

The white nuchal spot is very seldom seen in the young of this cuckoo but is occasionally present.

Measurements.—Wing 208 to 226; tail 151 to 176; tarsus about 20 or 21; culmen 20 to 22 millimeters. The female is little, if any, smaller than the male in Indian birds, but of Chinese birds La Touche gives the following measurements: Male, 23 (2 examples), wing 200 to 226; female, 22 (1 example), 190–200; culmen, male, 20–22; female, 19 millimeters.

Colors of soft parts.—Iris yellow or “grey with brown inner circle” (La Touche); bill dark horny green, the base of the upper, and most of the lower, mandible yellowish horny, the gape still more yellow; legs and feet wax-yellow to rather bright yellow.

Food.—This also is the same in character as that of *Cuculus canorus*, but *C. optatus* also devours more hard-bodied insects such as Cicadae and many small and some quite large beetles, while it feeds rather more exclusively on food obtained near the tops of high trees. I have never seen it feeding on the ground, nor have I noticed it in scrub or bushes except when it was probably hunting for a nest in which to deposit its egg.

Behavior.—I do not think that this bird could be distinguished from *Cuculus canorus*, except by its voice, by any field naturalist until the bird was actually in his hand. In flight, perching position, and general action I have been able to discern no difference of any kind between the two birds unless it is that when the two were seen together *optatus* looks a smaller, slighter bird than *canorus* and may fly a trifle faster with quicker wing beats.

Both sexes have the habit, as *canorus* has, of sitting, almost motionless, for a very long time in one position, possibly in the case of the female while she is watching certain birds and waiting for them to give away the site of their nest, or when she is waiting for the precise moment at which to fly down and place her egg in the nest of the foster parent selected to receive it.

Voice.—The call of the male consists of four notes, two rapid, then a pause, and then two more, all of the same cadence, sounding like *hoo-hoo hoo-hoo*. When one is close to the bird a fifth note can be heard preceding these, much higher pitched and far less resounding so that at a little distance only the four notes are heard. The call is a typical cuckoo note, and hearing it one would expect to find that a cuckoo had uttered it. There is also a sweet trilling note that is rarely heard, and I cannot say whether it is uttered by one sex or both, but I suspect it to correspond to the bubbling note uttered by the female *canorus*. Although occasionally this cuckoo calls on moonlight nights it does not do so with anything like the perseverance of the common cuckoo, nor is it so incessantly vocal during the day.

Enemies.—The same as those of the common cuckoo, but the young do not suffer so much from exposure by falling out of nests too small or too weak to hold them. In the case, however, of eggs laid in holes occupied by the nests of the crowned willow warbler the young birds have to vacate them at a very early stage or they would be unable to do so when full grown and would be incarcerated in them for life. As these cuckoos select ground nests so largely in which to deposit their eggs, these and the young birds when hatched do not suffer so greatly from the crows and magpies who hunt the hedges and bushes, but, on the other hand, they are easier for snakes and lizards to get at.

Fall.—Migration from the breeding grounds in northwest India apparently commences in the end of August and continues to the end of September or early October. From northeast India they would seem to migrate later by about a month, birds continuing to call up to the beginning of August, while in 1914 an egg was taken as late as August 20. It is a curious fact that in the continent of India birds do not seem to move far south and have only been recorded in winter from many places in the plains of the northwest and Punjab and from Lucknow, Jodhpore, Fategarh, Bihar, Dibrugarh and Cachar, in the United Provinces, Bengal, and Assam. Farther east it migrates much farther south and has been recorded from all over

Burma to the extreme south, and from the Andamans and Nicobars, while it also ranges throughout the Malay Peninsula and the Austro-Malaysian islands to New Guinea and Australia. Birds found in winter east of Burma are certainly those that breed north of China and probably migrate, more or less, due north and south, and I am inclined to think it is possible that birds from the Himalayas migrate almost entirely to the east in India and then southeast through Burma and the Malay States. Otherwise it seems incredible that no specimens should ever be obtained, in south India or even central India, of a migrant which in and from China wanders so very far south in the winter months.

DISTRIBUTION

The Himalyan cuckoo is found throughout the Himalayas and central Asia from the extreme west of India and Baluchistan, while in central Siberia it extends from Dauria and Lake Baikal to the extreme east of the northern Chinese mountains and, possibly, to Japan. In China La Touche (1931) summarizes its distribution as follows: "China generally, Szechuen, Kwangsi, Fohkien, Lower Yangtse, Shaweishan, Shantung, Chihli (migrant)." To this he adds, "On the whole this Cuckoo is not common in North China, and very few have been noted by Dr. Wilder," and, again, noting on my distribution of the summer range he adds, "Corea and Manchuria should probably be added."

It certainly occurs and breeds in all the higher hills in Burma; quite commonly in the Chin Hills and rarely to the east in the Ruby Mines district and the Shan States.

Its appearance on the American list is due to a specimen obtained on St. Paul Island, Pribilof Islands, Alaska, on July 4, 1890 (W. Palmer, 1894).

CUCULUS CANORUS BAKERI Hartert

KHASIA HILLS CUCKOO

CONTRIBUTED BY EDWARD CHARLES STUART BAKER

HABITS

This most interesting race of the common European cuckoo was given a name in 1912 by Dr. Hartert (1912), no museum possessing sufficient material until 1911 to substantiate the differences that both Dr. Hartert and I believed to distinguish this subspecies from those others already accepted.

The Khasia Hills cuckoo is a race that breeds at considerably lower elevations than do either *C. c. canorus* or *C. c. telephonus* and, speaking generally, haunts during the breeding season more densely forested areas in much hotter and far more humid climates. Thus in the Khasia Hills of Assam, where it is extraordinarily common, it may be found in all the higher ranges at and about Sherraponji between 3,500 and 6,500 feet frequenting forest that is always lush and green, with luxuriant undergrowth and with an almost impenetrable tangle of creepers, beautiful orchids, ferns, and parasites growing on every tree. Here the rainfall averages about 550 inches in the year, while as much as 72 inches have fallen in 24 hours and over 700 inches in the 12 months. They are, however, equally numerous, between 4,000 and 6,000 feet around Shillong, where the rainfall is less than 150 inches in the year and where, in many parts, pine woods take the place of the wet evergreen forests.

Although on the whole this race of the cuckoo certainly frequents denser and thicker forest than do the other forms, individuals vary greatly in their tastes and those parasitic on pipits and other species of birds, which breed in the open grass plains, also haunt the same kind of country as that in which these birds breed. At the same time even in these places they always select those that contain a certain number of trees on which they can perch and from which they can survey the surrounding area and watch the fosterers to their nests. In the Khasia Hills many cuckoos are parasitic on the common Indian and Blyth's pipits, which are to be found nesting on the wide stretches of open grassland above Shillong, the former on ridges and plateaus up to some 5,500 feet and the latter on those above that altitude. Intermediate in habits between the evergreen forest cuckoos and the open country cuckoos there are others that haunt the more open pine forests, especially those that have brooks and small rivers running through them, or are broken up by ravines in which small deciduous trees, bushes, and brambles grow freely to the exclusion of the somber pine trees.

In Burma this cuckoo is commoner in the Ruby Mines district and in the Southern Shan States than anywhere else. T. R. Livesey, who has had a

wonderful experience with them in the latter country, informs me that in these districts they frequent both the open deciduous forests as well as the open country of scrub and grass land around towns and villages, even depositing their eggs in the nests of birds breeding in and close to the gardens of houses occupied by Europeans or by Shans.

After the breeding season this cuckoo may be found in almost any kind of country that is fairly well-wooded, while in the plains of Bengal and Lower Assam it seems to have a special liking for mango orchards and patches of jungles in tea plantations.

Spring.—In Assam those individuals that have migrated into the plains and lower hills the preceding autumn return to their breeding quarters in the end of March or early April, their numbers increasing rapidly up to the end of that month, when their mellow call may be heard almost continually in every direction at all elevations between 3,500 and 6,500 feet. Exact dates of arrival are recorded nowhere that I can trace, but I have eggs taken on April 14, odd eggs taken in the third week in April, and many after the 25th of that month.

In the Chin Hills, between Assam and the Ruby Mines district and Shan States, most cuckoos commence breeding about the middle of April, arriving in their breeding haunts nearly a month earlier. Here they are common at about 4,000 feet upward and haunt much the same kinds of country as they do in Assam, certainly breeding up to 7,000 feet, or a little over, as eggs have been taken on Mount Victoria and the hills above Fort White.

In the Shan States Mr. Livesey informs me that few if any birds remain above 4,000 feet in winter, in the end of August and early September the great majority moving into the lower hills and adjoining plains. The first birds returning to the higher hills put in an appearance early in March and commence to lay the last of that month, March 27 being the earliest date on which he has taken eggs and July 1 the latest.

The length of the breeding season is much more extended than is that of the European cuckoo (*C. c. canorus*) or the central Asian bird (*C. c. telephonus*), doubtless owing to the same factors of cold, food supply, etc., that compel the more northern birds of all families to compress their breeding arrangements into a far shorter time than that occupied by the

southern birds, for whom temperature, food supply, and other factors are more or less favorable almost throughout the year.

The individual cuckoo, however, is governed in her time of egg-laying by the habits of the foster parent that she selects to bring up her young, while all cuckoos, as well as the foster parents, are affected more or less in their spring migration and breeding time by the late or early arrival of the rains and the consequent shortage or abundance of insect life. In Assam, where the rainy season is long and the rainfall heavy, eggs may be found in some numbers from April 20 to the middle of June and, then, in decreasing numbers, to the end of June, after which they further decrease in number until, after the middle of July, few will be taken, though my latest recorded date for a cuckoo's egg is August 24. In Burma, as already noted, birds commence laying about a month earlier, and few eggs are found after early June.

The 1,366 eggs in my collection were taken as follows: One in March (taken in Burma), 104 in April, 608 in May, 527 in June, 114 in July, and 12 in August. It may, I think, be accepted that migrating birds arrive on their breeding ground some two to three weeks before commencing to lay and leave again about one to two months after the deposition of their last eggs.

In Burma the peak of laying is reached in early May; in Assam not until the end of that month.

Courtship.—I can find nothing recorded as to the conduct of courtship between males and females during the breeding season and have, therefore, only my own observations to guide me. There appears to be no true courtship, and I have no doubt that, except in very rare instances, cuckoos do not pair, though in certain areas, where the bird is uncommon, a male and female may possibly mate and remain together throughout the breeding season. This is probably the case in North Cachar, where the bird is rare, breeding in deeper forest than I have seen anywhere else, and where I have seldom heard more than one male calling in any one particular area or found more than one type of egg deposited therein. In the Khasia Hills, in which district the birds swarm in incredible numbers, both sexes were undoubtedly promiscuous in their sexual relations. I have personally seen a female accept the attentions of one male and then fly off to a tree close by and accept another pursuing male within a few minutes. On this occasion

several, I think four, males were pursuing the one female, who flew slowly from one tree to another on a roadside. Settling on one of these she crouched on a large bough, half lying there with both wings hanging slightly lowered and quivering, while her head was held quite low, her whole attitude expressing invitation. The first male, of those following her, at once accepted this after which he settled on a branch close by while the female flew away to a tree about 50 paces distant followed by the other three males, one of whom was again accepted by the female who, after the completion of the act flew off accompanied by the remaining two males to a tree beyond my view.

Among the Khasias, certain classes of whom are very intelligent observers, it appears to be well known that both sexes of cuckoo exercise no discretion in their love affairs.

Nesting.—The female cuckoo is, of course, parasitic in her habits, and I have personally taken eggs in the nests of no less than 103 species and subspecies of fosterer (see following list), the very great majority of which are, however, undoubtedly abnormal or casual only, for I feel satisfied that when the usual foster parent has no nest available the cuckoo will eventually deposit her egg in that of any other bird that may be convenient. Repeated instances have come within my own experience in which a cuckoo normally parasitic on one bird deposits one or more of a series of eggs in a nest utterly unlike those of the usual fosterer, merely because she has already exhausted all the nests known to her of the latter.

LIST OF FOSTER PARENTS	eggs
<i>Suya crinigera</i> (subspecies 6)	215
<i>Cisticola jundicis cursitans</i>	276
<i>Cisticola exilis tyleri</i> (generally with <i>C. j. c.</i> series)	7
<i>Tribura luteiventris</i>	32
<i>Orthotomus</i> (species and subspecies 2)	27
<i>Acanthopneuste reguloides harterti</i>	9
<i>Megalurus palustris</i>	7
Miscellaneous warblers (abnormal fosterers, 9)	25
<i>Anthus</i> (various species and subspecies, 8)	199
<i>Rhodophila ferrea haringtoni</i>	18
<i>Saxicola caprata burmanica</i>	55
<i>Rhyacornis fuliginosus</i>	25

Total Fosterers: Normal 23, abnormal 80 (= 103).

LIST OF FOSTER PARENTS	eggs
<i>Enicurus</i> (species 2)	25
<i>Petrophila (Monticola)</i> (species 2)	21
Turdidae (abnormal fosterers, species 8)	15
<i>Cyanosylvia leucura</i>	13
<i>Eumyias thalassina thalassina</i>	63
<i>Niltava</i> (species 3)	43
Muscicapidae (abnormal fosterers, species 12)	22
<i>Lanius nigriceps nigriceps</i>	35
Laniidae (abnormal fosterers, species 2)	2
Troglodytidae (species 3)	21
<i>Mesia agentauris</i>	46
<i>Leiothrix calipyga</i>	33
Timaliidae (abnormal fosterers, species 32)	92
Other abnormal fosterers, <i>Passer</i> , <i>Sitta</i> , <i>Parus</i> , etc. (7 species)	8
Total Fosterers: Normal 23, abnormal 80 (= 103).	

Like the English cuckoo, as Chance (1922) has proved, the Khasia Hills female cuckoo sometimes adopts a certain area of country as its own breeding territory into which it will allow no other female, parasitic on the same foster parent, to enter, though it seems to have no objection to the entry of any number of males or of another female parasitic on a different species. Thus in a certain area close to the house in which I lived in Shillong I found three cuckoos breeding; one depositing its eggs in the nests of the little fantail warbler (*Cisticola juncidis cursitans*), the second in the nests of the verditer flycatcher (*Eumyias thalassina thalassina*), and the third in the nests of the black-headed shrike (*Lanius nigriceps nigriceps*). On another occasion I took eggs of three different cuckoos from nests of *Cisticola*, *Niltava*, and *Leiothrix* in the same strip of jungle and not 50 yards from one another. Yet another instance of shared territory was a grass-covered hillside occupied by two female cuckoos, the one parasitic on the same little fantail warbler and the other on pipits (species various). Other similar instances were quite common in this district.

My personal experience confirms what those interested in cuckoo-work have for some time maintained in regard to cuckoos being consistently parasitic on one selected foster parent. A good example of this constancy is a series, collected for me by a family of four men trained by myself in cuckoo-work in the Khasia Hills. From 1925 to 1935 these men obtained for me no less than 132 eggs of one cuckoo parasitic on the fantail warbler,

all collected in the same area of grass and scrub surrounded by pine forest. Of these 132 eggs only 4 were taken from nests other than those of the little fantail and of these 4, 3 were placed in the nests of another little warbler (*Orthotomus sutorius patia*) and the other in the nest of the Assam brown hill warbler (*Suya crinigera assamica*), both, of course, in the same territory.

In addition to the above I have many series of eggs, numbering 3 to 15, from individual cuckoos, occasionally taken in two or more consecutive years, all placed in nests of the same warbler. Again I have other series of cuckoos' eggs, some taken from nests of *Suya* (various species); others from those of *Eumyias*, pipits (various species), the silver-eared mesia (*Mesia a. argentauris*), the Pekin robin (*Leiothrix lutea calipyga*), and other fosterers. As a rule the cuckoo also shows equal constancy to its chosen breeding area, returning year after year to it and leaving it only when the supply of fosterers is exhausted or the nature of the country has been altered by man or some other agency. Where the normal foster parent is exceptionally common, such as are the warblers of the genera *Cisticola* and *Suya*, the cuckoo often has ample nests in a compact area to act as hosts to her whole series of eggs, and we have found as many as 18 eggs laid by the same individual within such an area. Where, however, the fosterers such as pipits or mesia are less numerous or are breeding in comparatively small areas, some 4 to 10 eggs are laid in the one area and then, when the available nests have all been made use of, the cuckoo removes to another area, generally close by, eventually often returning to the original one later in the season and again cuckolding other nests of the same fosterer.

There can, I think, be no doubt that a cuckoo brought up in a nest of a certain fosterer will deposit its own eggs in nests of the same bird. This theory seems now to be one generally accepted, and certainly my experience goes to confirm this, and in my own collection I have eggs of a very definite type taken for over 30 years in the nests of the same species of fosterer in the same district.

Eggs.—The eggs of the Khasia Hills cuckoo are of very many and beautiful types of color and character of markings, while the eggs of each individual cuckoo, though varying slightly *inter se*, do not show so great a variation as is found in the eggs of single clutches laid by many species of Passeres or other orders.

Among the most common types of eggs are, naturally, those that have been evolved to assimilate with the eggs of the birds most frequently employed as foster parents. Among these may be mentioned the following:

An egg with a white ground lightly marked, chiefly at the larger end, with specks, spots, or small blotches of red, reddish brown, or brown. These eggs are excellent counterparts in all but size of the eggs of the fantail warbler, in whose tiny nest they are usually deposited.

The next commonest type has the ground color more or less distinctly tinged with reddish and is rather more profusely marked with larger blotches of various shades of red-brown. These are normally deposited in the nests of various species of *Suya*, or brown hill warbler, the most common types of whose eggs agree well with those of the cuckoo.

A third and common type of cuckoo's egg has the ground color a beautiful salmon-pink or buff-pink and has the surface freckled with deeper reddish; in some cases so finely and thickly that the eggs appear at first glance to be unicolored; in other cases more or less boldly, though sparsely, covered with reddish brown and underlying faint marks of gray or pale purple. These eggs agree well with various types of eggs laid by the verditer flycatcher and, even still more so, with those of the beautiful niltava (*Niltava sundara*), while it is in the nests of these birds we find them deposited. Cuckoos, however, that normally deposit their eggs in the nest of these birds seem regularly also to cuckold the large niltava (*N. grandis*) and the white-tailed chat (*Muscisylvia notodela*), which lay the same colored eggs and make similar cup-shaped nests of living moss, built in exactly similar positions in holes in banks and among boulders.

Another beautiful type is bright pale blue, often immaculate but sometimes faintly flecked with primary reddish and secondary gray blotches. In the Khasia Hills these are almost invariably deposited in the nests of the silver-eared *Mesia* or the red-billed *Leiothrix* (*L. lutea calypga*), two species that lay exactly the same type of egg to that of the cuckoo, though more boldly blotched, while the two species also make similar nests, which they place in somewhat similar positions in bushes, etc.

In Burma, more especially in the Ruby Mines district and in the Shan States, we have two dominant types of eggs: One blue, much darker in tint than those referred to as being deposited in *Leiothrix* nests, laid with the

similar eggs of the Burmese dark gray bushchat (*Rhodophila ferrea haringtoni*), and the other having a pale pink ground, freely blotched all over with reddish, deposited in the nests of the Burmese stonechat (*Saxicola caprata burmanica*), which also lays eggs of this color and character. In connection with these two types of egg an interesting state of affairs has now been arrived at in parts of the Shan States. Thirty years ago the bushchat was extremely common in certain districts in hills between 4,000 and 5,000 feet, and the great majority of cuckoos found there were those laying blue eggs. Cultivation has now wiped out the scrub and bush jungle, beloved by the bushchat, and fields of rice, gardens, and the vegetation surrounding villages have taken its place. With this change in the character of the jungle growth has also come a change in the birds frequenting it, the bushchat has almost disappeared, and the little stonechat has taken its place, breeding everywhere in gardens, village grounds, and cultivated fields. The elimination of the bushchat, although so recent, has already gone far to eliminate also the cuckoo that lays blue eggs, while the one that lays eggs like those of the stonechat has become much more numerous and has become the common form. Even now, however, an occasional blue egg of a cuckoo will be found in the stonechats' nests, the latter similar in every respect to the nests of the bushchat and therefore cuckolded, *faute de mieux*, by the cuckoo.

It is impossible here to deal with the problem of the evolution of the various types of cuckoos' eggs, but the facts recorded above seem to go far toward proving that cuckoo eggs, to assimilate with those of their fosterers, have been evolved by discrimination among the foster parents leading to the slow but sure destruction of the unfit, i. e., Darwin's doctrine of the survival of the fittest in its crudest form.

In shape the eggs are rather broader ovals than those of the European cuckoo, and they also average larger and heavier. It is indeed much easier to separate the various subspecies of cuckoo by the eggs they lay than by the plumage of the birds that lay them.

The number of eggs laid by the Khasia Hills cuckoo probably varies somewhat individually, though I believe it generally to be 12 to 20. It is, however, very difficult to decide this definitely, as it is impossible to be satisfied that every nest of the selected fosterer in any given area has without doubt been marked down. It was not until 1907 that I concentrated

on the attempt to solve any one of the numerous cuckoo problems, and it was some years after this before satisfactory evidence had accumulated on this particular point. My different series of eggs of individual cuckoos vary greatly in number, being, I believe, governed entirely by the number of nests of the foster parent available in the area searched over. In 1908, on May 25, in one small grass glade surrounded by pine forest I found five nests of *Cisticola*, three containing young cuckoos or eggs of the cuckoo, and I am practically sure that I missed no *Cisticola* or *Suya* nests. One of these nests had in it a young cuckoo about 2 days old, another nest had a young cuckoo just hatching, while a third had a slightly incubated cuckoo's egg. The two other nests of the *Cisticola* were unfinished and empty, but on June 1 one of these contained a slightly incubated cuckoo's egg and three of the foster parent, while, finally, on June 4 the last nest contained a fresh egg of the cuckoo and four of the warbler. As the egg from which the older of the nestling cuckoos had been hatched must have been laid about May 13, while the last egg was laid on June 3 or 4, we have only five eggs laid in 21 days, whereas we know now that the larger cuckoos lay every second day, so the five eggs and the two young found cannot possibly include all then laid. A series of 14 eggs, all found between May 19 and June 10, 1910, in a similar but much larger area, were probably laid between May 16 and June 10 and represent a complete series laid every alternate day. In another small series of five eggs of one and the same cuckoo found in a narrow strip of grassland three eggs were deposited in *Cisticola*'s nests on May 16, 18, and 20, this exhausting all the nests then available. After this she apparently departed, but on July 5 and 7 the same cuckoo returned and placed two more eggs, one each in two new nests of *Cisticolas*. Another series of 14 eggs, of which 12 were placed in nests of *Cisticola jundicis cursitans*, one in a nest of *C. exilis tytleri*, and one in a nest of *Suya atrogularis khasiana*, were laid in two periods: The first from May 15 to 23 occupying each of the five nests of the fantail warbler available between these dates, and, then, from June 1 to 18 in all the nests then available in that particular area. Finally a series of eggs of one cuckoo that were taken from 1925 to 1935 consisted of the following numbers: 6, 8, 14, 9, 10, 12, 15, 18, 14, 11, 15. In 1925 and 1926 woodcutters were working in the pinewoods surrounding the open patch in which the fantail warblers were breeding and doubtless drove these little birds away and so deprived the cuckoo of foster parents during the latter period of her laying, or probably larger series would have been

obtained in that particular patch of grassland. The various series prove satisfactorily that eggs are normally deposited every alternate day and, also, that there is no interval in the laying period dividing it into two.

The weights of 1,368 eggs of the Khasia Hills cuckoo are as follows: Average weight 231; maximum 307; minimum 153 milligrams; this latter is, however, an abnormally small light egg and very few will be found less than 180 milligrams.

The same number of eggs measure: Average 23.76 by 17.43; maxima, 28.5 by 18.0 and 27.1 by 20.0; minima 20.9 by 16.3 and 23.4 by 15.0 millimeters.

I should, perhaps, not omit to say that though the generally accepted idea that cuckoos' eggs can be distinguished by their weight is in most cases correct, it is not always so. For instance, I have 37 eggs of this cuckoo taken in shrikes' nests, and a comparison of weights and measures of the eggs of the two species is as follows:

<i>C. c. bakeri</i> , 37 eggs	<i>Lanius n. nigriceps</i> , 100 eggs
Average size: 23.56 by 17.50 mm.	23.60 by 17.9 mm.
Average weight: 232 mg.	215 mg.
Maximum weight: 266 mg.	249 mg.
Minimum weight: 190 mg.	185 mg.

The comparative weight of the eggs of different species of birds varies greatly. Thus *Anthus* eggs are normally very light while the eggs of *Passer* are very heavy, two of the former about equaling three of the latter of the same size and, in a few cases, otherwise indistinguishable in coloring, shape, etc. Hoopoes' and spine-tailed swifts' eggs are very heavy, far more so in comparison with their size than those of cuckoos.

This shows that weight alone in some cases does not suffice to distinguish cuckoos' eggs from their fosterers' eggs, and other comparisons of cuckoos' eggs with those of other species could be quoted to confirm this. The hard *gritty* shell with fine pits at wide intervals is a further good distinguishing feature of cuckoos' eggs, while when blowing it is noticeable that most cuckoos' eggs have the yolks tinged with flesh color and very pale, while the white is rather more opaque, like the white of a duck's egg when compared with that of a fowl.

In Europe it is very rare to find two eggs of cuckoos in a nest laid by the same female; in India this is not so rare, and I have on three occasions found three eggs of the same cuckoo in one nest.

It is not possible here to go fully into the fascinating subject of the method of deposition of eggs, but briefly it may be said that it has been fully proved that in many cases cuckoos lay their eggs directly into open nests, an act that has been witnessed by myself, Whitehead, and others in India and proved by Chance (1922) to be the case in England.

There are, however, any number of eggs deposited by cuckoos in nests into which it is utterly impossible for the cuckoo to gain an entrance. Into many of these the cuckoo projects her egg from the cloaca by pressing herself up against the entrance to the nest and ejecting her egg with sufficient force to propel it the 2 or 3 inches that may be necessary for it to reach the nest (Livesey, 1936; A. E. Jones, 1937).

On the other hand, there are many nests to which this method also would not apply, such as nests of small birds, more than 2 or 3 inches inside holes of various character, in some cases a corner having to be turned before the nest is reached. Into these I believe the cuckoo places her egg with her bill, and there is some evidence to support this which I hope to give in my proposed book on "Cuckoo Problems."

Young.—Incubation, I think, takes usually 12 or 13 days, rarely only 11 days but occasionally extending to 14. The period the nestling remains in the nest is 4 to 6 weeks, but in many cases the nest is far too small to retain the young cuckoo until it is full grown. Thus when the eggs are deposited in the nests of birds such as *Cisticola* and, to a lesser degree, *Suya*, the young bird when a quarter grown fills the small egg-shaped nest, sitting in it with its head projecting from the entrance at the top side. Gradually, as the young cuckoo grows, the nest is expanded until it looks like basketwork around it, which finally bursts, depositing him or her on the ground. This generally occurs when the cuckoo has fair feathering and is about half grown or a little later.

The young Khasia Hills cuckoo ejects the fosterer's eggs or young from the nest in the same way as its English cousin does, possessing the same curious interscapular pit to assist it in doing so. This structural aid to ejection is found in all such genera as *Cuculus*, *Cacomantis*, *Penthoceryx*,

and others that eject their foster brothers and sisters, but not in the young of *Clamator*, *Eudynamis*, and those cuckoos that do not commit such murders. In the cuckoos that possess it, the pit soon fills in and young cuckoos lose the impulse to eject after a very short time, sometimes within 4 days and almost invariably within a week of being hatched.

Plumages.—Male: Whole upper plumage and wing coverts a dark slaty-gray or blackish slate, decidedly darker than the same parts in the European cuckoo; the lower back, rump, and upper tail coverts are a purer and somewhat lighter gray; the wing quills and concealed portions of greater coverts more brown, the quills slightly glossed and barred with white on the inner webs of the outer primaries, turning to rufous on the inner primaries; tail ashy black tipped with white and with white notches along the sides of the shafts, the white increasing in extent on the lateral tail feathers; chin, throat, sides of head and neck, and the upper breast ashy gray, not so dark as the back; remainder of lower parts, axillaries, and under wing coverts white with rather irregular bands of black, broader and farther apart than in *C. c. canorus*. Under tail coverts the same but with the dark bars still farther apart.

Female: Differs in having a rufous tinge on the upper breast and sometimes on the throat and sides of the neck.

Nestling: Naked when hatched.

Juvenile: First plumage; whole plumage brownish gray or slate, obsoletely barred with buffish white; a patch of white on the nape or hind neck; whole lower plumage barred white or rufescent white and dark brown, very heavily on the chin, throat, and breast, and less so on the under tail coverts.

The young male after the first molt is like the adult but nearly always retains traces of the juvenile barring, more especially so on the wings.

Hepatic females have the whole upper parts barred chestnut and blackish slate or blackish brown; the lower plumage has the chin, throat, and breast barred pale chestnut and blackish and generally with a strong rufous tinge on the breast and abdomen.

Young hepatic females are duller in color than the adults and have the feathers of the upper part fringed with white.

Colors of soft parts.—Iris pale to deep yellow, sometimes brownish in young birds of the year; bill dark horny brown, or very dark horny green, paler and yellowish at the base and on the commissure and orange-yellow on the gape; legs and feet wax-yellow.

Measurements.—Wing 220 to 227; tail 155 to 178; tarsus 18 (La Touche) to 19; culmen from feathers of forehead 20 to 22 millimeters (24 La Touche).

Food.—The principal food of this, as of other cuckoos, consists of caterpillars, pupae, chrysalides, and soft insects of any kind. During flights of termites cuckoos may be seen both catching them on the wing and eating them on the ground as they emerge from it. I have also taken Cicadae from their stomachs, and occasionally quite hard beetles of considerable size.

The actions of this bird when attempting, and indeed succeeding in, the catching of termites in flight are very clumsy and labored and their progress on the ground very slow, similar to the progress of certain woodpeckers when hunting a lawn for ants and other prey.

As a rule they feed in trees at some height from the ground, but any plague of caterpillars will tempt them down to quite low undergrowth or even onto low grass in the open, and I once saw four cuckoos all feeding on small green caterpillars on the ground in an open glade in pine forest.

Behavior.—Normally cuckoos are seldom found on any but high trees, often resting in one position for a long time on some lofty branch and then flying to another tree with strong, easy beats of the wing, at a considerable speed. In the breeding season the females undoubtedly mark down the nests in which they intend to deposit their eggs, and they may be seen perched in a tree, watching their victims until these latter give away the position of the nest. While thus engaged the patience of the cuckoo seems inexhaustible, and it will sit for hours in one position, hardly moving, yet obviously watching the intended fosterers, which may be loath to return to their nests though they may come and perch on the same tree as that occupied by the cuckoo, sometimes within a few feet of her. Apparently they not only mark down nests for immediate victimization, but others to be made use of when later eggs are to be laid. At other times they seem to be able to ascertain the approximate, yet not the exact, position of a nest. (Livesey MS.) gives me a most interesting example of this. He writes:

“Yesterday, May 19, 1937, I was out for a walk with my wife about 5 p. m. at Taungyi, which has an elevation of some 5,000 feet. Close to my cottage the dogs chased a bird off a ploughed field, which I recognized as a cuckoo. The cuckoo returned with two chats after her and perched on some bamboo rails, so, suspecting that the chats had a nest somewhere near in which the cuckoo was going to lay, I sat down to wait and watch with my glasses. Back came the cuckoo and flew low over the field almost settling on a place some 50 yards in front of us. Twice she flew backward and forward, chased by the chats, finally settling where I expected the nest to be. I could see her very plainly through the glasses as with throat feathers puffed out and crest feathers sometimes raised she jumped clumsily from clod to clod searching for the nest, straining her neck up and looking everywhere. The chats were now mobbing her furiously and in retaliation she only opened her beak and made faces at them. She did not appear to know where the nest was and began a systematic search for it in an area about 5 by 3 yards. The clods in the field were very large and lumpy with all sorts of holes which might have held a nest, and from time to time she disappeared from sight as she hopped into the various depressions, going backward and forward in the most persistent manner, mobbed all the time by the chats.

“She was, I think, a full 10 minutes searching for the nest, but, at last she dipped out of sight and a flutter of her wings suggested she had found the nest and was in the act of laying. She was out of my sight for about 4 seconds and then sped away in a great hurry.

“We climbed the fence and walking up to the place whence she had flown found a very well-concealed chats’ nest in which by stooping down to the ground I could see that the nest contained two eggs, the cuckoo’s and one of the chat’s; by using the tips of my two fingers I was just able to draw the eggs out, one at a time with considerable difficulty.”

Voice.—The call of the male during the breeding season is exactly the same as that of the European cuckoo, but in winter it also has a single note sounding like *chuck* softly repeated two or three times. At the beginning of the season the bisyllabic note, from which the bird derives its name, is not perfect and the imperfections cover quite a wide range of variations. Sometimes the note is single instead of double; often it is preceded by a rather hoarse note of the same character as the call, while sometimes the

cuck-oo is followed by another hoarser note. The female has the summer bubbling note of its cousin and also certain chuckling notes, very seldom uttered, while in winter it gives vent to the same soft *chucks* as the male. Whether the female ever calls *cuckoo* is disputed, but, personally, I am fairly certain that she does, although it may be but seldom. The bubbling note is the call to the male and is also repeated after she has met her mate, but it is then, I think, lower and softer, perhaps an expression of satisfaction.

During the daytime the calling of the male is almost continuous, but that of the female far less so as the former calls in the presence of a female, while the latter does not “bubble” before the male except at the moment she takes to flight and invites him to follow her.

On moonlight nights the male often calls with as much persistency as during the day, but I have never heard it calling on dark nights until dawn is advanced.

Enemies.—Cuckoos have the usual enemies of all bird life, vermin of every kind, which during the breeding season hunt for and devour all the eggs and young they can find. All the civet-cat tribe, snakes, lizards, and iguanas are inveterate thieves of eggs and young, while even more destructive than these are the birds of the crow and magpie tribe, which systematically hunt out the nests and devour their contents. I have often watched the birds of the genus *Dendrocitta* and *Cissa* beating over an area in search of nests, quartering it with the care and energy of a spaniel after game. The young birds suffer also very greatly from being too large for the nests in which they are hatched. Half grown they fall out on to the ground, and many, which escape death from vermin, are killed by exposure to heavy rain.

The older birds seem to have no special enemies, their swift flight and comparatively large size saving them from the sudden death so often the fate of smaller, slower birds. At the same time birds of prey undoubtedly attack and kill them just as they would any other bird of similar size and their superficial resemblance—in human eyes—to a sparrow hawk, would certainly not deceive their would-be destroyers even if these were sparrow hawks. Nor do the small birds attack cuckoos because they *believe* them to

be hawks, but because they *know* them to be cuckoos and, in their own way as objectionable as hawks.

Fall.—The Khasia Hills cuckoo is far more sedentary in character than its nearest relations and, possibly, is originally a sedentary race from which the migratory forms have sprung. I have seen the bird in the Khasia Hills in every month of the year except February and, as it is silent in the winter months and does not call attention to its presence, it is probably even more numerous in its breeding range at this time than has hitherto been supposed, while some individuals may be resident in the same locality all the year round. At the same time migration does take place in some degree and this dark race has been found in winter, certainly, in Bengal, more especially in the eastern districts on the Bay, in Orissa, once by Annandale (MS.), while it extends through Burma, south to Prome (Mackenzie and Hopwood, MS.); south Siam (E. G. Herbert, MS.) and finally, almost certainly, to southwestern China. I have no proof that in the Indian Empire adult cuckoos migrate any earlier than the birds of the year. About September the pleasant call, which has been heard continually up to the end of July and casually up to the end of August, ceases entirely, and the birds are also far less frequently seen on the higher ranges of hills and, by October, nearly all the birds, old and young, have left these and have taken to the lower hills and the broken country at their bases, thence slowly and gradually extending into the plains in the districts already mentioned. It is also possible, of course, that this cuckoo may range farther south in winter than stated above, as records of *Cuculus canorus* (subspecies?) have been recorded from Madras and from the islands of the Austro-Malayan region, between October and March, some of which almost certainly refer to this form. At present, from the evidence at my disposal, I can neither substantiate nor refute this suggestion, as it is quite impossible to recognize one subspecies from another in the field unless a particularly bright light shows up the comparative slaty darkness of this bird, to an observer with some experience of cuckoos.

DISTRIBUTION

Exact details as to the distribution of the Khasia Hills cuckoo are still wanting as so many records of cuckoos refer merely to the species, *Cuculus*

canorus, while the subspecies is not given or, indeed, in many cases distinguished. The breeding area has been *proved* to extend throughout the hills of Assam and Burma as far south as Karen-nee and as far east as Yunnan, while Bangs and Peters (1928) consider it is this form that is found and breeds in eastern Tibet and Szechwan. To the west the breeding cuckoo in the Bhutan Hills is undoubtedly of this race, but how much farther west it may extend is not known though, almost certainly, it may be found breeding in the lower Himalayas below Sikkim. Stevens informs me that he believes it does.

A fine series of breeding specimens in the Stevens collection, collected by him in the Sikkim Hills, indicates that the range of *bakeri* is below 7,000 or 8,000 feet, and that above that, up to 12,000 feet, *telephonus* is the breeding form.

In the cold weather it extends to the countries mentioned above in this article as being visited on migration. To these may be added that it occurs in Siam, as far south as Bangkok and the Siamese peninsula west of Tenasserim. Finally, extraordinary as it may seem, it has been recorded by Friedmann (Friedmann and Riley, 1931) as having once been obtained in St. Lawrence Island, Bering Sea, an occurrence that entitles it to a place in the American avifauna. This specimen was originally described as belonging to the central and northeastern Asiatic race *telephonus*, which one might expect would occur at long intervals in Alaska.

Order TROGONIFORMES

Family TROGONIDAE: Trogons

TROGON AMBIGUUS AMBIGUUS Gould

COPPERY-TAILED TROGON

HABITS

This gorgeous Mexican species brings color from the Tropics, all too rarely, across our borders in extreme southern Texas and southern Arizona. Ever since Lieutenant Benson shot an immature male in the Huachuca Mountains on August 24, 1885, it has been known to occur there and in other neighboring localities in Arizona as one of our rarest birds. Specimens have been taken there under circumstances that would indicate that sometime its nest will be found within our borders. An adult female was shot by F. H. Fowler (1903) in the Huachucas during August 1892. Major Bendire (1895) writes: "Another adult female, which evidently had a nest close by, was obtained by Dr. Edgar A. Mearns, United States Army, on June 23, 1892, on the east side of the San Luis Mountains, close to the Mexican boundary line. The long tail feathers in this specimen are much worn and abraded, and look as if the bird had passed considerable time in very limited quarters. Its mate was also seen, but not secured. Judging from the character of the country this species inhabits in southern Arizona, that is pine forest regions, it is probably only a straggler, in the lower Rio Grande Valley in Texas, and does not breed there."

Mrs. Florence M. Bailey (1923) reports that A. B. Howell discovered a pair of these trogons in the Santa Rita Mountains in southern Arizona in 1918, of which he writes: "While wrapping two birds which I had shot at 6,000 feet in a canyon, on August 4, I looked up and saw a pair of these birds watching me from live oak branches at perhaps a hundred yards. I had an unobstructed view of their bright underparts and characteristic form and

flight, and identification was sure. They were very ‘wise,’ and as I carefully approached, they as slowly receded, flying from oak to oak until they separated and I lost them in the denser growth. The trees were almost entirely live oaks here with a very occasional pine.”

Herbert W. Brandt has sent me the following notes on the status of this beautiful bird on the western slopes of the Huachuca Mountains, Ariz.: “The turkeylike call of this rare, semitropical visitor is a common bird note in Sunnyside Canyon and in the lower reaches of Bear Canyon. There are at least three pairs of birds in each of these valleys; and in the morning they noisily call back and forth to one another. These valley floors are usually densely wooded and would make this bird difficult to study were it not for its inquisitive nature, for it is usually easily lured by the ‘squeech.’ One lavishly garbed male and his more modest mate repeatedly allowed me to walk up to within about 20 feet of them before they would fly a short distance and then allow me to approach them again. Each time they both called their hen-turkey-like notes, *kum-kum-kum*, ever answering my squeeches of a like count and inflection. That this bird breeds in the vicinity there is little doubt, but we did not spend time seeking its home, as Arizona has wisely put it on the permanently protected list, and in consequence this mountain-loving species is becoming common again in its densely tangled retreats.”

A. J. van Rossem (1936) adds the following news:

Regardless of its status in former years, this trogon may now be counted a fairly common summer visitant in the Santa Ritas. Possibly it has always been more numerous than was supposed, for one of the rangers, who has been stationed for many years in the Santa Ritas, knew the bird well and told me of having seen as many as five or six feeding together at a single patch of manzanita. At any rate there were several pairs in Madera Cañon in the summers of 1931 and 1932. * * *

On June 27 [1931], Mr. Gorsuch and I saw or heard eight birds between the forks of the cañon at 6000 feet, and Littleshot Cabin at 7000. On that date a fully adult male was collected by Mr. Gorsuch for the museum at the University of Arizona. On June 28, a very young trogon, about two-thirds grown and evidently just out of the nest, was shot, quite unintentionally, in a patch of oaks at 6000 feet.

At least two pairs were noted on May 30, 1931; “the association in which they were noted was the oak-sycamore growth near the juncture of Upper Sonoran and Transition. Two males (both of which presumably had mates) were heard in the left (north) fork—one at 7000, the other at 8000

feet altitude. These altitudes are in the pine-oak association in the Transition Zone.”

Col. A. J. Grayson (Lawrence, 1874) says that, in western Mexico, “it is to be met with only in the dark forests of the *tierra caliente*.”

Nesting.—Colonel Grayson says that “it breeds in the hollows of trees like the parrots.” According to Mrs. Bailey (1928), the nest is “reported in cavities in large trees, generally in large deserted woodpecker holes, but also in holes in banks.” There are ten sets of eggs of the coppery-tailed trogon in the Thayer collection in Cambridge, all collected by, or for, Frank B. Armstrong near Ciudad Victoria, Tamaulipas, Mexico, between March 29 and April 27, 1908. If these dates are all correctly recorded, these trogons must breed very plentifully in that region, or Mr. Armstrong’s collectors must have been very industrious. There are a number of sets in other collections from the same locality, all taken by the same collectors. These eggs were all taken from nests in holes in trees, apparently natural cavities; the holes were at various heights, ranging from 12 to 40 feet above the ground; some of the trees were in a river bottom and others in “big woods near town.” The nests were made of various materials, such as hay, straw, trash, moss, wool, down, feathers, vines, and thistle-down.

Eggs.—The coppery-tailed trogon lays ordinarily three or four eggs but probably sometimes only two. These are rounded-ovate to nearly oval in shape; and the shell is smooth but not glossy. The color is dull white or faintly bluish white and entirely unmarked. The measurements of 55 eggs average 28.50 by 23.18 millimeters; the eggs showing the four extremes measure **30.7** by 24.6, 29 by **25**, **26** by 22.8, and 29.5 by **22.1** millimeters.

Plumages.—I have not seen any nestlings or very young birds. Ridgway (1911) gives very full and accurate descriptions of all the known plumages of both sexes of this trogon; but his accounts are too long to be quoted in full here, so I shall mention only the most conspicuous features of the different plumages, by which the reader may recognize them.

In the juvenal plumage, in July, the sexes are alike, or nearly so, and closely resemble, on the upper parts, the adult female, except that the central pair of tail feathers have very narrow black tips, instead of broad ones; the next three pairs of rectrices are black, and the two lateral pairs are mostly white, barred with black, except for a large terminal white area; the

lesser, median, and to a lesser degree the greater wing coverts are tipped with a large spot of pale buff or buffy white, bordered with black; chin and throat grayish brown above a quite distinct white pectoral band; below this band the under parts are indistinctly barred, or mottled, with grayish brown and grayish white.

This unadulterated juvenal plumage is apparently worn through the first summer and early fall; I have seen it in its purity in birds collected at various dates between July 23 and September 20; but, on the other hand, some specimens show the beginning of a molt before the end of August. During all the remainder of their first year, young birds show more or less continuous progress toward maturity by a gradual and irregular molt. At an early age, between August and November, young males begin to show metallic green feathers in the back and throat, and metallic blue feathers in the rump and upper tail coverts; during winter and spring these metallic colors gradually increase; and on the under parts, below the white band, there is a gradual decrease in the brown and white and a corresponding increase in the "geranium red" of the adult plumage. At the same time, young females are acquiring more and more of the "peach red" of the adult female on the posterior under parts.

These transition plumages may be seen, in the series I have studied, in birds collected in November, December, February, March, April, May, June, and July, during all of which time the juvenal wings and tail are retained. From this I infer that the annual molt occurs in summer and fall and that young birds do not acquire the fully adult plumage until they are at least 15 months old, or perhaps much older.

Food.—Some coppery-tailed trogons that E. C. Jacot collected for me were feeding on wild grapes. Dr. A. K. Fisher wrote to Major Bendire (1895) that "a rancher who raises fruit in Ramsay Canyon stated that the species visited the gardens in considerable numbers, especially during the period when cherries were ripe." Major Bendire (1895) says of other members of the trogon family: "Their food consists of fruit, grasshoppers, and other insects, and in their actions while catching the latter they are said to resemble a Flycatcher, starting and returning from a perch like these birds, and often sitting for hours in the same place."

Cottam and Knappen (1939) state that a bird, collected by Dr. Fisher in the Huachuca Mountains in June, "had fed exclusively on the adults and larvae of lepidopterous insects." They examined the stomach of another bird, collected in October in Mexico, that contained 68 percent insects and 32 percent fruits. The insect food included one grasshopper nymph, long-horned grasshopper eggs, three Mantidae, three stink bugs, other Heteroptera, one leaf beetle, one very large larva of a hawk moth, larvae of undetermined Lepidoptera, and two sawfly larvae. The vegetable food consisted of fruits of cut-leaved cissus, fruit of red pepper, and undetermined plant fiber.

Behavior.—F. H. Fowler (1903) writes:

On June 9, 1892, my father and I accompanied Dr. A. K. Fisher to Garden Canyon seven miles south of the post. We reached the canyon and were riding up the narrow trail bordered with pines and live oaks, when suddenly a beautiful male trogon flew across the path just ahead of us, and perched on a live oak bush on the other side of the small stream which flows through the canyon. The Doctor tried to approach it, but the noise caused by his passage through the thick brush and over the sliding rocks on the hill side alarmed the bird, which from the first had seemed a trifle uneasy, and it was soon lost to view among the trees down the canyon. Higher up among the pines, on the same day, we heard the calls of another which sounded much like those of a hen turkey. While we were eating lunch on the way down, we heard still another calling from the hillside above us, and the Doctor, who found it perched on the lower limb of a pine after a short search, watched its actions for a few moments and then shot at it. It sat erect, the tail hanging straight down, and when uttering the call threw its head back until its beak pointed nearly straight up.

On August 14 of the same year I again found the trogon in Garden Canyon, this time higher up however at the Picture Rocks. A beautiful pair flew up from a fallen pine to the lower limb of a tree, and sat there quietly watching me. I dismounted and fired a reduced charge at the male, but the only effect was that he flew off through the trees unhurt, while the female flew up to a small tree on the hill, where she sat, looking at me until I loaded my gun, when I shot her. At the second shot the male flew up the canyon his beautiful carmine breast gleaming in the sunlight like a streak of flame. Both birds sat nearly erect when at rest, with their long tails hanging nearly straight down. Their flight was nearly like the slow flight of a magpie, until startled, when they flew like a dove and nearly as fast.

Voice.—Mr. Jacot says, in his letter, that the young birds were silent but that the adults had many notes, one of which was "almost like the chattering of our gray squirrel." A note referred to above was like that of a hen turkey. Dr. William Beebe (1905) says: "The call of the trogon, uttered especially toward evening when it came to drink, was a soft series of melodious notes, reminding one somewhat of the content-call of a hen with chickens. Regularly at dusk two of these birds went to roost in a dense tangle of wild

clematis.” Mr. van Rossem (1936) refers to the note of the male trogon as a “loud, hoarse, ‘kó-a-kó-a-kó-a’.” Evidently the bird has a variety of notes.

Field marks.—The shape and posture, referred to above, as well as its brilliant colors, would mark this beautiful bird definitely as a trogon. The only other trogon likely to be met with anywhere within the range of this species is the Mexican trogon, found on the highlands of Mexico. The two species can be recognized in life by the color patterns of the outer tail feathers; in *ambiguus* these feathers are largely white, barred or vermiculated with black; in *mexicanus* they are largely black, broadly tipped with white.

DISTRIBUTION

Range.—Southern Arizona, south to central Mexico; accidental in the lower Rio Grande Valley of Texas.

The range of the coppery-tailed trogon extends **north** casually to Arizona (possibly Santa Catalina Mountains, Huachuca Mountains, and possibly Tombstone); northern Chihuahua (San Luis Mountains and the Sierra de la Campana); Nuevo Leon (Monterey and Montemorelos); and central Tamaulipas (Ciudad Victoria and Soto la Marina). **East** to Tamaulipas (Soto la Marina, Xicotencatl, Altamira, and Tampico); central Veracruz (Orizaba); and central Oaxaca (Talca and Juchatengo). **South** to Oaxaca (Juchatengo); Oaxaca (Omilteme and Amula); Michoacan (Tancitaro); and Jalisco (Zapotlan). **West** to Jalisco (Zapotlan, Ameca, and San Marcos); Nayarit (Mazatlan, Tres Marias Islands, and Mexcatitlan); Sinaloa (Escuinapa, El Limon, Angostura, and San Javier); eastern Sonora (Alamos, Chinobampo, and Guiracoba); and southeastern Arizona (Huachuca Mountains and possibly the Santa Catalina Mountains).

Casual records.—Although this bird can only be considered as rare anywhere in the United States, it appears to be fairly regular in the Huachuca Mountains, Ariz. During the summer of 1877 a trogon was killed near Ringgold Barracks and another at Las Cuevas, Tex. Although not preserved they were fully described, by the persons who shot them, to Dr. James C. Merrill, and were undoubtedly this species.

Egg dates.—Mexico: 23 records, March 29 to June 30; 12 records, April 15 to May 13, indicating the height of the season.

Order CORACIIFORMES

Family ALCEDINIDAE: Kingfishers

MEGACERYLE ALCYON ALCYON (Linnaeus)

EASTERN BELTED KINGFISHER

[PLATES 12–14](#)

HABITS

Our North American representative of the large and interesting kingfisher family is not so gaudily colored as some of the foreign species and is intermediate in size between the largest and the smallest members of the family, but it is an interesting bird, striking in appearance and voice, and unique in form. Its long, heavy bill and its large head with its prominent crest, contrasting with its diminutive feet and its short tail, seem entirely out of balance and give it a top-heavy appearance. But its peculiar proportions and structure are beautifully adapted for the life it leads; its large beak and head form an effective spearhead for use in its deep plunges, and they are well built to stand the shocks of frequent diving. Unlike the osprey, it does not need to use its feet in fishing; but the short legs and shovellike feet are most useful in shoveling the loose soil from its nesting burrows, after it has been loosened by the powerful beak.

The belted kingfisher, as a species, covers nearly all the North American Continent, breeding from northern Alaska and central Labrador southward to the southern border of the United States. Being essentially a fish-eating bird, its haunts are naturally near large or small bodies of water. It is common on the seacoast and estuaries, where it may be seen perched on some stake or pier, watching for its prey; or along the shore of a lake or pond, its favorite outlook may be the branch of a tree overhanging the water; I believe that it prefers to perch on a dead or leafless branch, where its view is unobstructed. Trout brooks, especially swift and rocky mountain streams, are favorite resorts, where its loud, rattling cry is often heard, as it

flies up and down, patrolling its chosen fishing ground and driving away any intruders of its own species; it prefers to play the role of the lone fisherman.

Courtship.—Very little seems to be known about the kingfisher's courtship. Laurence B. Potter says in his notes: "Sometimes I have watched as many as five or six high up in the air, tumbling and wheeling about, uttering their harsh rattle; they appear to be doing it merely for the joy of flying, or it may be their courtship antics." Francis H. Allen writes to me: "From courting birds—a group of them—I have heard a mewling note uttered in rapid succession, almost if not quite as loud as the familiar rattle of the species. These same birds—or two of them at least—also kept up a continual, prolonged rattle."

Nesting.—The nest of the belted kingfisher is almost invariably in a burrow in a sandy, clay, or gravelly bank, excavated by the birds themselves. The site chosen is preferably near water and as near the favorite fishing grounds of the birds as a suitable bank can be found. But such banks are not always to be found in the most convenient places, so the birds are forced to nest in any bank they can find, often at a long distance from any water, such as the embankment of a railroad cut, the cliff of a sand dune, or a bank by a roadside where sand or gravel has been taken out for grading. On Cape Cod, the sandpits made while sanding cranberry bogs are favorite sites. The burrow may be at any height from the base of the cliff, depending on the height of the cliff, but it is usually not more than 2 or 3 feet from the top, though Major Bendire (1895) says that it is sometimes as much as 20 feet below the top of the cliff. The burrow extends inward, sloping slightly upward, for varying distances, usually from 3 to 6 feet, but sometimes as much as 10 or even 15 feet; as kingfishers sometimes use the same burrow for several years in succession, it may be that the deepest burrows are the oldest and have been extended from year to year to provide a fresh, clean nest. The burrow is usually straight, or nearly so, but often it curves somewhat, or makes a more or less abrupt turn to the right or left. One that I dug out ran straight in for 3 feet, made an abrupt turn to the left, and then made a reverse curve, so that the nest was only about 2 feet from the face of the cliff. The entrance and the tunnel itself are not quite circular, being usually about 3½ to 4 inches wide and 3 to 3½ inches high; an occupied burrow can generally be recognized by the footmarks of the bird, a central

ridge with a furrow on each side of it, made by the bird as it enters or leaves the nest. The nest is placed in an enlarged chamber, which may be directly at the end of the tunnel, or a little to one side of it, and usually a little above the level of the tunnel. The chamber varies considerably in size and shape but is approximately circular and dome-shaped; it is usually 10 to 12 inches in diameter and 6 to 7 inches in height. Often the eggs are laid on the bare sand or gravel which probably indicates that the nest is a new one, or that the eggs are fresh; oftener, perhaps, the nesting chamber is lined with bits of clean, white fish bones, fish scales, or fragments of the shells of crustaceans; these, I believe, are the remains of ejected pellets and indicate that the nest has been previously occupied by young birds or that the female has been fed on the nest for some time; there is no evidence to indicate that the birds ever bring in such material intentionally.

Bendire (1895) writes:

The time required to dig out a burrow depends largely on the nature of the soil to be removed, taking sometimes two or three weeks, but generally much less. I have personally seen an instance where a pair of these birds excavated a new burrow in a rather friable clay bank near Fort Lapwai, Idaho, to a depth of 5 feet (estimated measurement) in a little over three days. How they managed to dig so rapidly, considering their short and weak-looking feet, with which they must remove the greater part of the material, has always been a mystery to me, and I would not believe them capable of accomplishing such an amount of work had I not seen it done. When not disturbed the same nesting site is resorted to from year to year. Sometimes the male burrows an additional hole near the occupied nesting site, usually not over 3 feet deep, to which it retires to feed and to pass the night.

Dr. Thomas S. Roberts (1932) published the following account on the authority of Miss Frances Densmore, of Red Wing, Minn.:

On April 25, 1928, I found a pair of Kingfishers digging their tunnel at the top of a high cut about a foot below the surface, just where the black loam met the under sand, some one hundred feet, or thereabouts, above the water. They both dug, taking turn and turn about, except when she thought he hadn't stayed in long enough and sent him back. After watching them for an hour or more I formed a theory as to how they managed it. One would go in and work for two or three minutes and then push the dirt ahead of it to the entrance and fly out over it. No dirt ever came out with the bird that had been digging, but when the other went in there was a veritable fountain spurting out for nearly a minute after it entered. Then this subsided and more digging was done by the bird that had cleared the hole. They kept very close to their schedule of two or three minutes each. On this day the dirt they brought out was sand, but on the 27th it was black loam from above, and I decided that they had got back to their "sitting room." On May 1 they weren't working and, as both were in and out at the same time, I judged that there was room to turn and that they would call it done.

Dr. A. K. Fisher wrote to Major Bendire (1895) as follows:

On June 6, 1882, the writer found two nests of the Kingfisher in the side of a railroad cut near Croton Lake, Westchester County, New York. The burrows were placed in a bank not over 7 feet above the roadbed and within 18 inches of the top. That of the first one ran in about 7 feet and turned to the right as it entered the nesting chamber. The seven fresh eggs were placed in a nest of coarse grass, which, although rather scanty, covered the floor of the cavity on all sides. The burrow of the second one extended in about 4½ feet, and, like the other previously mentioned, turned toward the right as the expanded nesting cavity was reached. The nest, which was quite elaborate, was composed wholly of fish scales and bones, arranged in a compact, saucer-shaped mass. The writer made a tunnel from the top of the bank so as to intercept the burrow as it entered the nesting cavity. Viewed through this hole, the nest was a beautiful affair. The scales, which looked as if made of frosted silver, formed a delicate setting for the six pure-white eggs lying in the center, and by the projected light made a most effective picture. On two occasions, near Sing Sing, New York, the writer found the Kingfisher and Rough-winged Swallow using burrows having a common entrance. It is probable in each case that the swallow had commenced its diverging burrow after the larger bird completed its work.

A few cases have been recorded of the kingfisher nesting in other cavities, where suitable sandbanks were not available. Mr. Forbush (1927) says: “Mr. Herbert F. Moulton of Ware, Massachusetts, tells me that he found a kingfisher’s nest in a plowed field on a hillside. The entrance was made in a ‘dead furrow.’” Arthur H. Howell (1932) says: “Baynard (1913) writes that in Alachua County [Florida] the Kingfisher nests early in April in holes in dead trees or stubs over water. He states, also (verbally), that he once found a nest at Clearwater, 4 feet above water in a leaning stub, the entrance hole being on the under side; this nest contained 4 eggs on May 6.”

Beyer, Allison, and Kopman (1908), referring to Louisiana, say:

The character of the nest varies greatly with different conditions of soil. On the coast it is content with such elevations as can be found on the shores, and the burrow is sometimes scarcely more than a pocket in the clayey banks; in the upper districts, the site is often far from water, and the soft, coarse-grained soil renders easy the excavation of a burrow five or six feet deep, enlarged at the end, and often partly lined with leaves and pine straw; and finally, a unique condition exists in the extensive gum-swamps in the lake region of the southeast, where the land—always submerged—is perfectly flat, and nothing stands above water except innumerable trees and stumps of *Nyssa*; the nest is placed in the top of a decaying stump, with no attempt at excavation.

But tree-nesting is not wholly confined to the southern swamps, for Dr. George M. Sutton (1928) writes:

On May 27th, 1927, while observing Chimney Swifts at Bethany, Brooke County, West Virginia, I saw a Kingfisher fly rapidly across an open field from a near-by deep pool in Buffalo Creek. Wondering that it should thus cross overland, I watched it as it flew to a large, dead sycamore, not far from me, and disappeared in a hole at the end of a short, thick, horizontal stub. Upon going to the tree I heard the buzzing cries of the young birds. Shortly thereafter the male parent flew away as the female came in. The nest was located

about ten feet from the ground in a large cavity near the juncture of the bough and the main trunk. The young birds were lying about seven feet from the entrance. The cavity was almost as dark as a bank burrow would have been. It is odd that the Kingfishers chose such a site for their nest, since earthen banks admirably suited to their needs were available along the creek.

Eggs.—The number of eggs laid by the eastern belted kingfisher varies ordinarily from five to eight, the commonest numbers being 6 and 7; on rare occasions as many as 11 or 14 eggs or young have been found in a nest. If the first set of eggs is taken, the birds will dig another burrow, often within a few feet of the first, and lay a second set; sometimes a third, or even a fourth, attempt will be made. But only one brood is raised in a season. The eggs are short-ovate or rounded-ovate in shape; the shell is smooth and rather glossy; they are pure white in color. The measurements of 54 eggs average 33.9 by 26.7 millimeters, the eggs showing the four extremes measure **36.8** by **27.9**, **30.8** by 26.4, and 33 by **25.4** millimeters.

Young.—The incubation period is said to be about 23 or 24 days. Bendire (1895) says:

The male does not assist in incubation, but supplies his mate with food while so engaged, and she rarely leaves the nest after the first egg has been laid; at any rate I have invariably found the bird at home if there were any eggs in the nest. Incubation lasts about sixteen days. The young when first hatched are blind, perfectly naked, helpless, and, in a word, very unprepossessing. They scarcely look like birds while crawling about in the nest, where they remain several weeks, their growth being very slow. The excrement of the young is promptly removed and the burrow is kept rather clean. They utter a low, puffing sound when disturbed, and frequently vary considerably in size, as if incubation, in some instances at least, began with the first egg laid. The young, even after they have left the nest for some time, require the attendance of their parents before they are able to secure subsistence for themselves.

I believe that Bendire's statement above, that the male does not incubate, is incorrect; perhaps he may not do so regularly, or to the same extent that the female does, but several observers have reported finding the male on the eggs, or at least in the nest. The young remain in the nest for about 4 weeks or more, and do not leave it until they are able to fly.

William L. Bailey (1900) made an interesting study of a family of young kingfishers, by digging a hole in the rear of the nesting cavity on four different occasions, taking photographs of the young at four different ages, and filling up the hole each time, so as not to disturb the birds too much. He says that when the young were about two days old they "were not only found wrapped together in the nest, but the moment they were put on the

ground, one at a time, though their eyes were still sealed, they immediately covered one another with their wings and wide bills, making such a tight ball that when one shifted a leg, the whole mass would move like a single bird. This is a most sensible method of keeping warm, since the mother bird's legs are so short that she could not stand over them, but as they are protected from the wind and weather they have no need of her. Their appearance is comical in the extreme, and all out of proportion. This clinging to one another is apparently kept up for at least ten days, for a week later, when nine days old, they were found in exactly a similar position."

On his last visit, when the young birds were 23 days old, he made an interesting discovery, of which he says: "Taking the precaution to stop the hole with a good-sized stone, I proceeded to my digging for the last time on the top of the bank. This time I found the chamber had been moved, and I had some difficulty in locating it about a foot higher up and about the same distance to one side. The old birds had evidently discovered my imperfectly closed back door, and either mistrusted its security, or else a heavy rain had soaked down into the loosened earth and caused them to make alterations. They had completely closed up the old chamber and packed it tightly with earth and disgorged fish bones."

Mrs. Florence M. Bailey (1928) makes the following apt quotation from the writings of Professor Herrick:

From the time of birth the young lie huddled in a cluster in their dark underground chamber.... As they grow in size and strength the monotony of sitting still, often with legs and wings interlocked, must become very great, and ... they soon begin to bite and tease one another like young puppies. Should one be hard pressed, the only way of escape lies along the narrow passage, which they naturally traverse head first; but the instinct to return to the warm family cluster is strong, and to do this they are obliged to walk backwards. Again when the rattle of the *alma mater* announcing the capture of another fish is heard, each struggles to get down the narrow passage-way first, but when the parent enters the hole she hustles them all back.

The young are fed by their two parents while they remain in the nest, and for some time afterward while they are learning to fish for themselves. Mrs. Irene G. Wheelock (1905) made the following observations:

By care in concealment we were able to discover that the adult came to the nest on the first day with no visible supply of food in the bill but with a gullet conspicuously swollen. We had previously excavated the nest from the rear making a false back to it so that it would be protected from the weather and at the same time open easily. As soon as feeding was completed and the adult out of sight, we opened the nest at the false back, took out the

young, then one day old, and examined the crops. They contained a dark gray, oily mass, nearly fluid and very ill smelling, but with no bones or scales in it. If fish they were very small and digested. Returning the young fishers to the tunnel, we closed it. Two days later the experiment was repeated with the same results. Four days later, or the seventh day after hatching, we examined again. This time one of the nestlings had swallowed several small fish about one and one half inches long and the others were still hungry. As yet we had not seen either of the adults bring visible food and the most frequent feedings had been forty minutes apart, I believe all by regurgitation. No record was kept from the seventh to the fourteenth day when an examination was made for the third time. We now found the young showing well developed pin feathers, and there were traces of disgorged fish bones and scales in the nest which had not been there before. The crops examined showed fish only slightly digested and regurgitative feeding had evidently given place wholly or in part to fresh food. On this day one of the adults brought several fish, possibly four inches long to the nest in different journeys. Examinations made on the twenty-first day revealed the same food conditions as the fourteenth. The pile of fish bones and scales was a trifle larger but was partially buried in the earth. There was surprisingly little of this debris in the nest or tunnel but the ground seemed to be saturated with fishy oil. On the twenty-eighth day the young kingfishers resented being examined or photographed, and made good their escape when taken from the nest.

Henry R. Carey (1909) writes:

The food brought to the nest-hole consists of various kinds of small fish. It not infrequently happens that one of these fish is too large to be carried by the parent bird into the narrow passage; it is then dropped upon the sand and is allowed to rot. * * * I once found a common salt-water flounder, four and one-half inches long and proportionately wide, which, being rather unwieldy for the parent bird to handle, had been left in this way. Another time I found a young Sculpin (*Callionymus aeneus*) in the same condition, and, yet again, a live minnow, which, in spite of a great patch on its side devoid of scales, was finally freed in perfect health. * * *

The young birds leave the earth about July 25 [in New Hampshire]. They are a sombre-looking lot, as for several days they sit tamely about the wharfs or venture on short, erratic flights, which makes one feel that they have not yet got used to the light after their long imprisonment underground. It is at this time that both parents and young, somewhat crowded in the vicinity of the home nest by their sudden increase in population, begin to seek out new fishing-stubs, or to use old ones for the first time in the year. When the young are able to care for themselves, the old birds leave them and lead once more the single life which they seem to enjoy most.

At this time of year, frequent quarrels occur among them, mostly about the best fishing spots, and now that strange, whining note, which Herrick describes as resembling the grating of two tree boughs in the wind, is often heard. It appears to be a note of anger; I have heard it when one bird, wanting the perch of another, hovered menacingly over him. Once I saw two birds dive simultaneously for the same spot in the water, the same note escaping them as each reluctantly swerved aside.

Floyd Bralliar (1922) writes interestingly of how the young learn to catch their own fish:

The young birds did not remain in the neighborhood of the nest more than a few days, but those few days were busy ones, for in that brief time the mother was teaching her

children how to earn a living. She would perch by their side on an overhanging limb and patiently wait for the glimmer of a fish below. The first day or two she usually caught the fish, beat it into partial insensibility and then dropped it again into the water. The young were persistent in their plea for food, but the mother was as insistent that they catch their living if they got any. There was very little current where they hunted, and a fish did not float out of sight quickly. The young birds would crane their necks and look hungrily at the fish below until finally one more hungry or more bold than the rest would make a dive for it. At first the aim was not good, and the bird would miss even a dead fish more often than he succeeded in catching it. Usually, however, he fluttered about the surface of the water until he got his fish, even tho he had missed it when he made his plunge. * * * During the first few days when the young birds became too hungry, the mother would occasionally relent and feed them, but before the week was over, no matter how hungry they became, no food was coming until they caught it. Within ten days the young birds were catching live fish instead of half dead ones.

Then a young bird would catch his fish, carry it to his perch, whack it over the limb a few times, toss it in the air, catch it by the head as it came down and swallow it with as much skill as his mother. As soon as she was convinced of the skill of each of her brood, she forsook them entirely. I do not know whether she ultimately drove them from the neighborhood or whether they left voluntarily, but when July was past only the old birds were to be seen in the neighborhood.

Henry Mousley (1938), who made a prolonged study of a family of belted kingfishers, estimated the period of incubation to be about 24 days. His report says, in conclusion:

On summing up I find forty-two hours were spent with the birds (May 11-July 24), during which time the young were fed one hundred times, or at an average rate of once in every 25.2 minutes. Of course there were periods when the feeding was much faster, as for instance, once in every 8, 9, 13, 20, and 21 minutes respectively. Sometimes the parents were absent from the nest for long periods of time, such as, 150, 120, 105, 97, 93, 90, 85, 75, 70, and 60 minutes at a time, when of course the young were without food. It was after these long spells that the more rapid feedings generally took place. As already remarked, the male seemed to pay the most attention to this part of the business, for I find of those times when I was perfectly sure of the sex of the parent, the male fed twenty-eight times to his partner's fourteen, or just double. It was the male parent which was the last seen at the nest previous to the departure of the one surviving young—a male. The food for the most part consisted of small fish, crawfish, minnows, tadpoles, and probably beetles.

Plumages.—The young kingfisher is hatched naked, blind, and helpless, a shapeless mass of reddish flesh, looking much like a very young puppy with a huge conical bill. Its eyes do not open for about two weeks. Within a week, or less, the pinfeathers or feather sheaths appear, and soon the young bird bristles in all the feather tracts with quills like a young porcupine, which grow to varying lengths and show, before they burst, the pectoral band and the general color pattern of the adult. When the young bird is 17 or 18 days old, a remarkable change takes places within the short space of

24 hours, for the sheaths rapidly burst and the juvenal plumage blossoms out all over the body.

In this plumage the young bird looks strikingly like the adult with only minor differences, and the sexes are alike. Young birds of both sexes, when fully fledged, have the pectoral band more or less heavily tipped or mixed with cinnamon, rufous or dull brown; this usually consists of narrow edgings in young males, but in young females many feathers are half or more brown; the rufous band of the female adult is only partially shown in the young female, mainly on the flanks, and it shows to some extent in nearly all young males, some having nearly as much as young females; the crest is darker than in adults, there is more white in the wing coverts, the white tips of the secondaries are more extensive, and the central tail feathers are spotted, as in the adult female. This juvenal plumage is also a first winter plumage, for it is worn without much change until spring. Young birds have a first, prenuptial molt early in spring, which involves most of the body plumage, perhaps all of it, the tail, and apparently the wings also; this takes place between February and April. Adults have a complete postnuptial molt in August, September, and October.

Mr. Brewster (1937a) made some observations on the roosting habits of kingfishers at Umbagog Lake, Maine, of which he says: "Every evening, a little after sunset, two or three Kingfishers resort to Pine Point to spend the night. They fly directly into the forest and go to roost among the densest foliage, often that of spruces and arbor vitae, growing anywhere from four to ten rods back from the nearest shore." One that he watched "flew up into a tall, slender Paper Birch, and alighted near the end of a long branch, about thirty feet above the ground. * * * As darkness gathered, the bird settled lower and lower on the branch, and drew in its neck without, however, burying its head in its plumage, for I could still see its bill pointing outward over the breast. At 9 P. M. I went under the tree, and by the light of a lantern dimly made out the Kingfisher, crouching in the same attitude, in exactly the same place."

Food.—The kingfisher is a fish eater and an expert fisherman, well deserving its name. It evidently prefers fish to any other food and would probably live on fish exclusively, if it were always able to secure all it needed. It catches mainly small fish, preferably not over 6 inches long, and mostly those species that are of little use, or even harmful, economically as

far as human interests are concerned. The kingfisher has been condemned by trout fisherman as a great destroyer of trout, but in a large measure unjustly. Kingfishers undoubtedly visit trout hatcheries frequently and can easily catch plenty of trout in the open pools, where the trout have no place to hide and where they are congregated in large numbers; they can do considerable damage in such places, but the trout can be easily protected by placing wire screens over the pools. In the trout streams wild trout are not so easily caught, for, as every trout fisherman knows, the trout are seldom seen in the open places except when darting swiftly across them, but spend their time hiding under overhanging banks, or under logs or stones, and only dashing out occasionally to capture their prey. On the other hand, chubs, dace, suckers, and sometimes sculpins are very common in most trout streams, generally in larger numbers than the trout; they frequent the open spaces, are much slower in their movements than the swiftly moving trout, and consequently are more easily caught. The records show that these neutral or harmful fishes make up the greater part, or nearly all, of the kingfisher's catch in trout streams. Chub and sculpins are very destructive to small trout fry; and suckers eat quantities of trout spawn; consequently the kingfishers are really doing the trout fisherman a favor by reducing the numbers of these fishes in the streams and should be protected rather than persecuted.

The kingfisher must of necessity do its fishing in clear water, so is seldom seen on muddy streams or ponds, or on those that are choked or overgrown with thick vegetation, such as is often found along trout streams. It usually perches on some stake, snag, or pier standing in the water, or on some bare overhanging branch, where it can watch patiently for some passing fish. Favorite perches of this sort in good fishing spots are resorted to regularly and rival kingfishers are driven away, for the kingfisher is a solitary bird except during the nesting season or while training its young. From such a perch the bird may dive obliquely into the water to seize the fish in its powerful bill; or, rising 30 or 40 feet into the air and scanning the water below it in more active pursuit of its prey, it may stop and hover for a few seconds, with rapidly vibrating wings, and then make a straight or spiral dive directly downward, disappearing beneath the water sometimes for several seconds. It is not always successful in its plunges, as the fish may move and cause a sudden change of direction in the bird's rapid dive either above or below the surface; but it is not easily discouraged and is

always ready to try again. Having secured the fish, it flies with it to some favorite perch, where it beats the fish into insensibility, tosses it into the air, or otherwise adjusts it, so that it can swallow it head first and thus avoid any injury to its throat from the sharp spiny fins. Sometimes the fish is too large to be swallowed completely, in which case the tail must be left protruding from the mouth until the rapid process of digestion enables the bird to gradually work it down. Manly Hardy wrote to Major Bendire (1895) of such a case, and says: "I shot a Kingfisher last spring that had swallowed a pickerel considerably longer than the bird from the end of the bill to the tip of the tail, and the tail of the fish protruding from the throat, while the head was partly doubled back, causing a large protuberance near the vent."

Where fish are not readily obtainable, especially in the arid regions of the Southwest where the streams largely disappear during the dry season, the kingfisher seems able to make a good living on various other kinds of food. The list includes crabs, crayfishes, mussels, lizards, frogs, toads, small snakes, turtles, grasshoppers, locusts, rickets, salamanders, newts, butterflies, moths, beetles and other insects, young birds, mice, and even berries. On the seacoast, it has been known to feed on clams and oysters, which sometimes results disastrously for the bird. H. C. Hopkins (1892) reports finding a kingfisher with "its bill held fast between the shells of an oyster." The bird had evidently inserted its bill into the open shell of the oyster, which had closed upon it; "the tongue [was] quite black from non-circulation of the blood, which showed that it must have been held prisoner for some time." The bird would probably have drowned at the next high tide. Dr. B. H. Warren (1890) reports that a friend of his once caught one of these birds on a hook and line while fishing with a live minnow for bait. He also says: "One day B. M. Everhart found a kingfisher lying on the bank of a small stream. On making an investigation, Mr. Everhart ascertained that the bird was unable to fly, as its bill was tightly clasped in the grasp of a large fresh-water mussel. I have heard of several instances where kingfishers have been captured under similar circumstances, which could naturally lead one to suppose that they feed to a limited degree on the flesh of these bivalves."

In the Bermudas, kingfishers are said to feed on squids. Walter B. Barrows (1912) quotes Professor Aughey, of Nebraska, as follows: "One

that was sent to me to identify in September, 1874, had 18 locusts, in addition to portions of some fish, in its stomach. One that I opened in September, 1876, had mingled at least 14 locusts with its fish diet.” Bendire (1895) caught a kingfisher in a trap baited with a mouse, and believed “that not a few mice, and possibly small birds also, are caught by them during their nocturnal rambles, and they are certainly fully as active throughout the night as in the daytime.”

Kingfishers disgorge as pellets the indigestible portions of their food, such as fish bones and scales, the shells of crustaceans and the seeds of berries; the bones and scales found in the nests are the remains of such pellets. Henry K. Carey (1909) writes:

Only once have I seen a pellet of fishbones and scales being disgorged from the bird’s beak, as he sat on his hunting perch. These pellets are found wherever the birds are accustomed to sit for any length of time. I once found one completely composed of various parts of the shell of a small crab. Only a few days later I had the pleasure of seeing a crab actually caught. The bird captured him by diving in the usual way and took him to a low rock where he proceeded to bang him just as he would have done to a minnow. During this process the crab, which measured an inch and a half sideways across the shell, lost several legs and was dropped upon the rock, from which by a considerable effort he managed to fall by scrambling to the edge with his remaining legs. The bird, perhaps seeing that he was rather a large morsel to swallow whole, then forgot him completely and went on with his fishing.

Ora W. Knight (1908) has seen kingfishers chase and capture moths and butterflies, taking them on the wing. Dr. Thomas S. Roberts (1932) says that they have been known to eat young sparrows; also that “crawfish are pounded and crushed before swallowing, and fish that are too large may be divided into pieces by the powerful bill. Miss Densmore, of Red Wing, once saw a Kingfisher that was making unsuccessful attempts to handle a small turtle. This was thoroughly pounded and variously manipulated but had to be discarded in the end.”

Dr. Charles W. Townsend (1918) writes: “Early in August, 1917, Mr. John Hair, gamekeeper of Mr. R. T. Crane at Ipswich, missed six of a four days old brood of Bob-whites. He had seen a Kingfisher nearby and later the same day saw it perched on the gable end of the little house where the Bob-whites had been hatched, and from there pounce on the young birds as they ran in and out. He shot the Kingfisher, and, on opening the bird, a female, found the legs and feathers of the young Bob-whites in its crop.”

When hard pressed for animal food the kingfisher has been known to eat wild cherries and probably other wild fruits. Dr. Elliott Coues (1878) published the following note from Mrs. Mary Treat about a kingfisher that fished regularly from a private wharf in front of her house in Florida: "When the water is so rough that it is difficult for him to procure fish, instead of seeking some sequestered pool he remains at his usual post, occasionally making an ineffectual effort to secure his customary prey, until, nearly starved, he resorts to a sour-gum tree (*Nyssa aquatica*, L.) in the vicinity, and greedily devours the berries. Returning to his post, he soon ejects a pellet of the large seeds and skins of the fruit."

Behavior.—The belted kingfisher is a striking and picturesque feature in the landscape whether in action or at rest. The mountain trout stream would lose much of its charm without the rattling call of the lone fisherman and the flash of his broad, blue wings, as he follows the course of the stream, flying well below the treetops until he glides upward to alight on his accustomed perch, there to tilt his short tail nervously up and down and raise and lower his long crest a few times, as he sounds his battle cry again. At the seashore, too, his trim, unique figure and his conspicuous color pattern, as he perches day after day on his favorite stake or goes rattling along the shore, add color and a tinge of wildness to the scene.

The flight of the kingfisher is strong, swift, and graceful, usually low, but high above the treetops when traveling; often there are five or six rapid strokes followed by a long glide on half-closed wings. Mr. Carey (1909) writes of it:

The Kingfisher's flight is remarkable for its beauty. How easily those long wings carry him about, as he skims so close over the water that their tips are sometimes wetted, or, as he hovers, his body appearing absolutely motionless, in that wonderful way which few birds can equal, for indefinite periods of time. Sometimes, especially in water half a foot or less in depth, he dives while flying nearly parallel to its surface. Sometimes, in his journeys from perch to perch when fish are plentiful, he dips again and again into the water in this way, reminding one of the Swallow as he gracefully touches the water here and there in his flight over the mill-pond. Again, he drops like a falling stone in a nearly perpendicular line upon his fishy prey.

Again he writes, referring to times of keen competition over good fishing grounds:

On such occasions one bird is often angrily pursued by another. These pursuits are most reckless and enduring in character. One sees the two birds swirl by like two blue flashes of light, to disappear in an instant of time on perfectly controlled wings perhaps far away in

the pine woods, almost grazing the tough trunk of some mighty tree, or heading straight for a sheer cliff and rising fifteen feet or more to clear it when it seems that they must be dashed to pieces on the rock. I once saw a Kingfisher, hard-pressed in such a pursuit, adopt a clever means of escape. His pursuer was close upon him—about five feet behind. On they came down the creek, neither bird seeming to gain upon the other. Both were flying at top speed low over the water. Suddenly there was a splash, and the foremost Kingfisher disappeared under the water. The bird behind swept on and lit on a nearby stub, not attempting to renew the chase when his enemy reappeared.

The above was probably a case of one bird defending its territory against the invasion of it by another kingfisher. Frederick C. Lincoln (1924) saw a striking example of this during his field work in the marshes of the Illinois River; he says: “During the period of greatest abundance, practically every channel had its quota of birds, each of which appeared to patrol or to hold dominion over a certain well-defined section.” As many as 8 or 10 birds were encountered, each always confined to its own limited section. One of the boatmen told him that “he took much pleasure in informing club members and others who might be with him in the club launch, just how far the kingfisher then in sight would go and where the next one would be met.”

William Brewster (1937a) watched a kingfisher at Lake Umbagog that “plunged into the water, striking a fish so large, that he had to let it go after a brief struggle, during which he failed to bring it to the surface, although evidently trying his best to do so.”

Voice.—There is not much beauty in the voice of the kingfisher, but the loud rattling call always produces a thrill in the listener; it is a wild, weird, wilderness call that enlivens the solitudes and punctuates the stillness of lonely shores or forest streams; it seems to fit in well with the active vigor of this aggressive guardian of his domain, as a warning to his rivals. It consists of a series of harsh, wooden, rattling notes of great carrying power. It has been likened to the sound made by an old-fashioned policeman’s or watchman’s rattle, a very good description for those of us who are old enough to remember such out-of-date sounds; but it may remind the younger generation of the sound made by certain noise-making instruments used at get-together dinners, political rallies, or other joyous gatherings.

It is not easily expressed in syllables, but Mr. Bralliar (1922) has written it fairly well as *rickety*, *crick*, *crick*, *crick*. The call varies some under different circumstances, sometimes being quite soft and low, as if in a

conversational tone with its mate. The courtship note, referred to above by Mr. Allen, was “a mewling note uttered in rapid succession, almost if not quite as loud as the familiar rattle.” Then there is the whining note, referred to by Mr. Carey above, “resembling the grating of two tree boughs in the wind,” which seems to be a note of anger while quarreling over fishing rights. Brewster (1937a) says that “on such occasions they often utter a harsh *cah-car-car-car*, quite unlike the usual volley of watchman’s rattle-like notes.”

Field marks.—The kingfisher could hardly be mistaken for anything else. Its shape is distinctive, its large head and crest and its long, heavy bill are all out of proportion to its small body, short tail, and tiny feet. As it flies its great blue wings, its white collar, and its banded breast are unique. Its loud, rattling note proclaims it beyond doubt.

Enemies.—The most serious enemies of the kingfisher are the selfish fisherman, who wants all the fish for himself and begrudges the poor bird an honest living, and the proprietor of a trout hatchery, who is unwilling to go to the trouble and expense of screening his pools to protect his fish. The former shoots every kingfisher he can with misguided satisfaction; the latter either shoots or traps any that visit his pools. A small, unbaited, steel trap is set and fastened to the top of a stake or post near the bird’s favorite fishing pool; if the trap is so set that the pan is at the highest point, the bird is almost sure to alight on it and is caught. Hundreds of kingfishers are caught and killed in this way along private trout streams, or about trout hatcheries, every year.

The natural enemies of the kingfisher are of no great menace to its welfare. The Cooper’s and the sharp-shinned hawks often pursue it, perhaps largely for sport; under the accounts of these two hawks, in a previous volume, will be found references to these attacks and the successful attempts of the kingfisher to escape by diving; it even seems as if the kingfisher enjoyed the sport, judged by its derisive “laughter” at the defeat of the hawk.

The remains of a kingfisher have been found in the stomach of a red-tailed hawk; the former must have been caught unaware, for the hawk is no match for it in flight. The kingbird sometimes makes life miserable for the kingfisher; Fred T. Jencks (1881) writes:

The Kingfisher had poised himself several times to look for fish, and was just moving to do so again as the Kingbird approached and attacked him. The Kingfisher is not a troublesome bird, and always minds his own business. He was entirely unprepared, and acted as though he could not believe that the other had any evil intentions, for he tried to poise again. The second attack seemed to undeceive him, and show him his enemy was in earnest. He vaulted and turned, vainly endeavoring to rid himself of his persecutor. He soon saw he could not save himself by flight and tried diving. As soon, however, as he appeared at the surface he attempted to fly, but the Kingbird, keeping up an incessant twittering, forced him to dive again. Two or three times this was repeated, both birds making considerable noise, until the Kingfisher seemed convinced that escape in that direction was impossible, so he sat like a duck upon the surface, and as his persecutor would swoop at him he would go under. This lasted for some little time, until even the Kingbird seemed wearied and flew away.

Snakes and perhaps skunks or minks may crawl into the nesting holes, while the parent birds are away, and destroy the eggs or young; but it would seem that the formidable beak of the kingfisher, if at home, would prove to be an effective weapon of defense. H. H. Bailey (1907) says: "While digging out some Kingfishers' nests this season I was surprised to find a dead bird in about every fourth or fifth hole. This I was at loss to account for, as the birds showed no signs of combat or disease, while the plumage was not even disarranged. The bodies, though, seemed to be dried up, with no signs of blood in them, so I presumed that something had crawled into the holes and sucked the blood from them, leaving the carcass intact. This surmise proved correct, as the last hole I dug out contained a large black snake, and a dead kingfisher still warm."

I quote the above for what it is worth, but cannot agree with Mr. Bailey's conclusion. I have never heard that the black snake is a blood-sucker and doubt if it would attack a bird as big as a kingfisher. If such well-known blood-suckers as minks or weasels had attacked the bird, there would have been evidences of a struggle. I believe that the snake was looking for eggs. The kingfishers may have died from an epidemic of roundworms or ringworms, which have often proved fatal to these birds.

Frederic H. Kennard told me that while he was fishing on Grand Lake, Maine, he and his daughter heard a splash behind them and their guide saw a kingfisher dive, disappear beneath the surface and not come up. Although they all watched for some time, the bird never appeared. They paddled over to the place where the kingfisher dived but could see no trace of the bird in the water or in the air. They suspected that some of the large fish, pickerel, salmon, or togue, that abound in that lake, may have caught the kingfisher.

Arthur W. Brockway tells me of a nest that was dug into by a skunk and the young devoured, the excavation being made from above the nest.

Winter.—The belted kingfisher is a hardy bird, and remains as far north in fall and winter as it can find open water in which to catch a fair supply of fish. A few kingfishers remain all winter, especially during mild winters, in southern New England, frequenting to some extent the open inland streams, but more regularly along the southern coast from Cape Cod to Long Island Sound.

There, Mr. Forbush (1927) says, “they go into winter quarters in December, especially about river mouths where at that time the little frost-fish come in, and there they remain, unless extreme cold locks rivers and shores in ice.”

They are occasionally seen in winter in the northern tier of States. Dr. L. H. Walkinshaw writes to me from Battle Creek, Mich.: “Along open stretches of water, a few kingfishers can be found during the entire winter. I have several dates for southern Michigan for December, January and February. I have watched them on zero, or near zero, days dive from some dead branch after minnows in the open stream.”

Mr. Skinner says in his notes from Yellowstone Park: “Ordinarily this is a migratory bird, but here a few remain all winter along the streams kept open by hot water. During cold winds and storms, they are often seen in the most protected places, but with feathers all fluffed out. The winter birds noticed have been males, the females not appearing until March 17. In winter, I believe they roost in the tops of the thick cedars in the Gardiner Canyon, but I have no data about the other localities.”

Most of the kingfishers migrate to the more southern States during the late fall, where they find open water and plenty of fish. There they establish regular fishing stations and live their solitary lives, each in its own territory. We often saw one in Florida perched day after day in practically the same spot, presumably the same bird in each case.

DISTRIBUTION

Range.—North America south to northern South America; accidental in the Azores, Ireland, and the Netherlands.

Breeding range.—The breeding range of the belted kingfisher extends **north** to central Alaska (Kowak River, Fairbanks, and probably Fort Yukon); Mackenzie (Fort McPherson, Fort Wrigley, Fort Providence, and Fort Smith); central Saskatchewan (Knee Lake and Pelican Lake); central Manitoba (Oxford Lake); Ontario (Rossport and North Abitibi River); Quebec (Lake Mistassini, Godbout, and Romaine River); and east-central Labrador (Grand Falls). From this northeastern station the range extends southward through Newfoundland and along the coasts of Nova Scotia and the Eastern United States to central Florida (Micanopy and Clearwater). **South** to Florida (Clearwater, probably St. Marks, and Chipley); southern Louisiana (Bird Island and Bayou Sara); Texas (Giddings, Corpus Christi, Kerrville, and Pecos); southern New Mexico (Carlsbad and Chloride); and southern California (Escondido). The western limits extend northward through California, Oregon, Washington, and British Columbia (sometimes along the coast) to Alaska (Sitka, Hope, Mount McKinley, and Kowak River).

There are a few records north and west of the range as outlined that may possibly represent occasional nesting. One was seen August 5 and 6, 1915, at a point 100 miles above Bethel, Alaska, and another was seen on July 6, 1917, at Iditerod. MacFarlane (1891) reported seeing several birds in the Fort Anderson region of Mackenzie, but he states that no nests were found. An adult female was taken at Churchill, Manitoba, previous to 1845; one was seen at Fort Du Brochet on September 22, 1920, while the species also has been reported from York Factory.

Winter range.—In winter the species is found **north** with fair regularity to southeastern Alaska (Sitka and Wrangell); southeastern British Columbia (Okanagan Landing); Wyoming (Yellowstone Park and rarely Wheatland); rarely central Missouri (Marionville); rarely central Indiana (Crawfordsville and Richmond); Ohio (Canton); and southern New Jersey (Cape May). From this point the eastern boundary of the winter range extends **south** along the coast to southern Florida (Royal Palm Hammock and Key West); the Bahama Islands (Nassau); the Dominican Republic (Sanchez); Puerto Rico (Fortuna); the Lesser Antilles (Anguilla, Antigua, St. Lucia, Carriacou Island, and Los Testigos); and northeastern Venezuela (Cariaco). South to Venezuela (Cariaco, Guarico, and Apure); rarely northern Colombia (Santa Marta); Panama (Barro Colorado Island); and Costa Rica (San Jose). **West**

to Costa Rica (San Jose and Rio Frio) and northward along the west coast of Nicaragua, Guatemala, Mexico, the United States, and British Columbia, to southeastern Alaska (Craig, Ketchikan, and Sitka). Occasionally kingfishers will be recorded in winter from points in the northern United States (Montana, Iowa, Wisconsin, Michigan, New York, Vermont, and Massachusetts), as well as from northern Ontario (London, Guelph, and Toronto). It also is a winter resident on Bermuda.

The range as outlined is for the entire species, of which two subspecies are currently recognized. The eastern belted kingfisher (*Megaceryle alcyon alcyon*) occupies all the range east of the Rocky Mountains and north to Quebec and Mackenzie; the western belted kingfisher (*M. a. caurina*) is found west of the Rocky Mountains and north to Alaska and Yukon.

Spring migration.—Early dates of spring arrival are: Pennsylvania—Renovo, February 22; Oil City, March 3; Beaver, March 22. New York—Ballston Spa, March 19; Rochester, March 21; Watertown, March 27. Connecticut—Hartford, March 16. Rhode Island—Providence, February 23. Massachusetts—Boston, March 19; Harvard, March 25. Vermont—St. Johnsbury, March 26; Rutland, April 4; Wells River, April 12. New Hampshire—Tilton, March 28. Maine—Orono, April 6; Portland, April 7; Machias, April 18. New Brunswick—Scotch Lake, April 22. Nova Scotia—Picton, April 6. Prince Edward Island—Alberton, April 17. Quebec—Montreal, April 21; Quebec City, April 24; Godbout, May 5. Missouri—St. Louis, February 25; Kansas City, March 10; Concordia, March 17. Illinois—Chicago, March 11; Odin, March 13. Northern Ohio—Oberlin, February 13. Michigan—Vicksburg, March 8; Detroit, March 20; Sault Ste. Marie, April 11. Ontario—Toronto, March 6; Ottawa, April 5. Iowa—Iowa City, March 7; Sioux City, March 18. Wisconsin—La Crosse, March 11; Madison, March 16; Stevens Point, April 1. Minnesota—St. Cloud, March 23; Minneapolis, March 24. Kansas—Hays, February 7. Nebraska—Valentine, March 3. South Dakota—Vermillion, March 22; Dell Rapids, March 23. North Dakota—Charlson, April 5. Manitoba—Margaret, April 1; Aweme, April 11; Reaburn, April 22. Saskatchewan—Indian Head, April 22; Eastend, April 23. Idaho—Rathdrum, April 9. Montana—Bozeman, March 26; Great Falls, April 13, Alberta—Banff, April 25; Stony Plain, May 12. Northwestern Alaska—Kowak River, May 21.

Fall migration.—Late dates of fall departure are: Alberta—Lac la Biche, September 27; Banff, September 30; Camrose, October 24. Montana—Great Falls, October 1; Big Sandy, October 9; Fortine, October 31 (winters rarely). Saskatchewan—Indian Head, September 20; Eastend, October 5. Manitoba—Killarney, October 15; Aweme, October 25. North Dakota—Rice Lake, October 2; Kindred, October 22. South Dakota—Yankton, October 16; Sioux Falls, November 11. Nebraska—Valentine, October 24; Lincoln, October 30; Red Cloud, November 10. Kansas (sometimes winters)—Hays, November 21; Wallace, November 24; Harper, November 30. Minnesota—Lanesboro, November 16 (sometimes winters); Minneapolis, November 25. Wisconsin—Madison, November 12. Illinois—Rantoul, October 27; Chicago, November 20. Ontario—Toronto, November 3; Ottawa, November 26. Michigan, Sault Ste. Marie, October 16; Detroit, December 4. Quebec—Montreal, October 23. Prince Edward Island—Alberton, October 10. New Brunswick—Scotch Lake, November 5. Maine—Portland, October 13; Phillips, October 17. Vermont—St. Johnsbury, November 9; Wells River, November 23. Massachusetts—Boston, December 13. Rhode Island—Providence, December 2. Connecticut—Hartford, December 26. New York—Watertown, October 31; Rochester, November 13. Northern New Jersey—Elizabeth, December 15. Pennsylvania—Beaver, November 28; Renovo, November 30.

Further insight into the migratory flights of individual kingfishers is provided by banding records. One banded as a nestling at Noblesville, Ind., on June 21, 1924, was killed in the Naches River Valley, Tex., on November 19, 1924, while another, banded on June 30, 1937, at Waukesha, Wis., was retaken at Society Hill, S. C., on November 22, 1937.

Casual records.—At least a part of the reported occurrences of the belted kingfisher in the Old World are unsatisfactory. Two cases, wherein specimens were alleged to have been taken in Meath and Wicklow, Ireland, in the autumn of 1845 are now believed to be based upon fraud. A specimen is supposed to have been taken on the island of Flores in the Azores, but neither its disposition nor the details of collection are known. A male bird was taken, however, in Holland on December 17, 1899, and another was collected on Westmann Island, off the south coast of Iceland, in September 1901.

Egg dates.—California: 16 records, April 7 to June 24; 8 records, April 21 to May 17, indicating the height of the season.

Illinois: 5 records, May 10 to June 8.

Massachusetts: 8 records, May 11 to June 6.

New York: 27 records, April 10 to July 15; 14 records, May 10 to 29.

Ontario: 7 records, May 24 to June 28.

MEGACERYLE ALCYON CAURINA (Grinnell)

WESTERN BELTED KINGFISHER

PLATE 15

HABITS

It has long been known that the kingfishers of the Pacific coast are appreciably larger than eastern birds, but Dr. Joseph Grinnell (1910) was the first to give the western race a name. He characterizes it as “similar to the *Ceryle alcyon* of eastern and southern North America, but size throughout greater, especially measurements of flight-feathers,” and goes on to say that “the secondary wing quills are *proportionally* longer in the northwestern birds. This means that in addition to its greater expanse of wing and generally larger size, the wing of *caurina* is *broad*er. In the closed wing this difference presents itself conspicuously in the interval between the end of the longest secondary and the tip of the longest primary. In the northwestern birds this interval averages only 27.3 mm., while in the eastern birds it averages 33.7 mm. This is in spite of the larger size of the former. The ratio to total wing length in the two cases is 17 and 22 per cent., respectively.”

The nesting habits of the western belted kingfisher seem to be very similar to those of the eastern bird; I find no record of its nesting anywhere but in sand, clay, or gravel banks; such banks are often much higher than those in the East, so the nesting burrow is often at a considerable height from the base, but usually near the top of the cliff.

In all other respects its habits are not materially different from those of the species elsewhere. Grinnell and Storer (1924) say: “This bird’s wing-

beat is characteristic, three quick beats followed by two executed in a more leisurely manner, like this: one, two, three; four; five.

“A Western Belted Kingfisher watched by Mr. Walter P. Taylor came to a perch on a bare limb overhanging some rapids in the river, and sat there motionless. The outline of the bird’s body at once became indistinguishable from the light and shade of its background; in other words it was obliterated because of the disruptive pattern of its coloration, white and slate areas alternating. If the fishes in the water beneath got the same impression as did the human observer, the kingfisher must have become invisible to them, remaining so until the moment of its headlong plunge in their pursuit.”

The eggs of the western belted kingfisher are similar to those of the eastern bird and average only slightly larger. The measurements of 38 eggs average 34.78 by 26.89 millimeters; the eggs showing the four extremes measure **36.6** by 27.4, 35.7 by **28**, **31.6** by 27, and 33 by **25.4** millimeters.

MEGACERYLE TORQUATA TORQUATA (Linnaeus).

RINGED KINGFISHER

PLATE 17

HABITS

This handsome bird is the largest of the three species of kingfisher found on the American Continent. Its claim to a place on our list is based on the capture of a single specimen, an adult female, by George B. Benner, on June 2, 1888, about a mile below Laredo, Tex., on the United States side of the Rio Grande. The specimen was presented to the Academy of Natural Sciences, in Philadelphia, of which Dr. Witmer Stone (1894) says: “It was sitting on some old roots which had been washed up into a heap by the current of the river, and was shot immediately, so that he did not see it fly or hear its call. Mr. Benner further states that he never saw one of these birds in the vicinity either before or since. Upon the strength of the evidence just given this species seems entitled to a place in the fauna of the United States, along with several other tropical birds which occasionally reach the Rio Grande valley.”

Col. A. J. Grayson, in his notes from western Mexico, sent to George N. Lawrence (1874), says:

I have seen the largest kingfisher only near the sea coast, in the vicinity of Mazatlan River, but not on that stream. They seem to prefer the stagnant pools and lagoons, whose waters are murky and densely shaded with overhanging trees; here upon some dried branch it sits quietly watching the opaque water for whatever finny creature may make its appearance upon the surface, when if not too large, it instantly darts or plunges headlong upon it; after securing the prey in its powerful bill, it bears it to the perch, and beating it a few times upon the perch swallows it entire. Small fishes constitute almost its principal food, but frogs and small water reptiles are often struck and devoured by it. This species does not seem to be so wary as its near congener, the Belted. I have approached it quite near, in order to observe its habits, and it appeared to be very little concerned at my presence. In examining the stomach of one shot by me, I found it crammed with the small fry peculiar to muddy pools, among which was a mud catfish of considerable size. It doubtless breeds in holes scratched in sand cliffs, like the other members of this family, but I have never encountered the nest.

Dickey and van Rossem (1938) say that in El Salvador—

the ringed kingfisher is of general distribution along the coast where it shows decided preference for mangrove lagoons. Locally, the species is almost equally common on fresh water wherever there is a plentiful supply of small fish. It is notable, though, that the species attains its upward limit at Lake Guija at 1,450 feet, and seems to be absent from the higher lakes such as Ilopango. It is customarily solitary, although local conditions such as a very favorable stream may result in a number being found at one point. A great deal of territory may be covered by individual birds, for they seem to have regular routes along lake borders and rivers with lookout perches at intervals of every few hundred yards. In activity this species is far ahead of any of the other resident kingfishers and evidently prefers to range widely for its prey rather than to stake out a limited, private preserve.

Alexander F. Skutch tells me that in the Caribbean lowlands of Guatemala and Honduras they are partial to the larger, more open streams and lagoons, and avoid the narrow, tree-shaded waterways that the smaller species of kingfisher sometimes frequent.

Nesting.—Major Bendire (1895) quotes Dr. Herman Burmeister (1856) as saying that “it nests in perpendicular banks, occasionally quite a distance from water, in burrows from 5 to 6 feet deep, and lays two white eggs.”

There are two sets of eggs, one of five and one of six eggs, in the Thayer collection, taken for Frank B. Armstrong, on March 21 and 30, 1910, near Ciudad Victoria, Tamaulipas, Mexico. One of the nests is described as a hole in the bank of a river, 8 feet deep and 10 feet above water.

Dickey and van Rossem (1938) state that “at San Sebastián in late July, 1912, nest holes were occasionally noted in vertical sandy banks. In size

these holes were on an average four inches wide and three inches high at the entrance. Those investigated went straight back into the bank, in one case as far as six feet. This particular burrow then may have made a turn, but it was not dug out.”

Alexander F. Skutch has very kindly lent me his unpublished manuscript for a proposed work on the birds of the Caribbean lowlands of Central America, which contains much interesting information on this species, from which I shall quote freely. He says that, in that region, these kingfishers “begin the excavation of their burrows in February, if not earlier. Their nesting is of necessity limited to the drier season when the rivers are lowest, which in this region extends from February to the end of May. The flood waters of June often undermine and eat away the banks in which they nest, if they do not actually rise high enough to flood their tunnels. Since they begin the excavation of their burrows early and raise a single brood each year, they proceed with their task in a leisurely fashion, and one may watch long and in vain for them to return to their work.

“One morning toward the end of February I concealed myself in a blind before a burrow in a bank of sandy loam beside the Rio Morja, near the boundary between Guatemala and Honduras. The tunnel was already well advanced and went far beneath the banana plants at the top of the bank, which had been freshly cut when the river ate into the plantation at the last high water. Although I began the watch early in the morning, it was 11 o’clock before I heard the measured *kleck kleck kleck*, and turning saw one of the pair approaching from upstream. It soon entered the burrow and remained within several minutes, appearing not to notice the rough-winged swallow that fluttered before the tunnel and several times rested in the entrance while it was busy inside. On emerging, the kingfisher perched atop a banana leaf and kept up a running conversation in low rattles with its mate, out of sight around the bend up-river. In five minutes it flew upstream to join the other, calling with a loud *kleck kleck kleck*.

“Soon the pair returned together and began to work in earnest. Each time one entered the burrow there was a jet of earth thrown out behind. As the bird moved inward the jet fell short of the entrance until it could be seen no more. Doubtless the bird continued to kick the earth back until it reached the head of the excavation, and so the material loosened by the bill was gradually pushed out of the tunnel. The kingfishers invariably emerged

head first, indicating that the burrow had reached its final length and had begun to widen at the far end into the nesting chamber. The bird inside called in low *klecks*, which were answered by the mate perched on a rusty tram rail, washed out by the flood, which leaned against the bank just below the burrow. They seemed to encourage each other in their dark subterranean labors. Both sexes shared equally in the toil, and as soon as one emerged and flew up beside the other on the rail, the latter went into the burrow, throwing out a jet of earth as he disappeared into the darkness. Four or five minutes was the usual period spent in the earthwork. On one occasion both were together in the burrow for a few minutes. Just after noon, while one bird was working inside, its mate became tired of waiting on the rail and flew upstream. The other, when it emerged and found itself alone, followed in this direction. Although I remained until the middle of the afternoon, the birds did not return, having worked less than an hour that day.

“I waited almost a month before daring to open the nest. I probed the length of the burrow with a slender vine, repeating the measurement several times to make sure I had reached the back. I found the burrow to be 7 feet 3 inches long, measured back this distance from the top of the bank, and tried to calculate the position of the nesting chamber from the direction of the portion of the tunnel I could see from the front. Experience with Amazon kingfishers and motmots had taught me that by far the safest way to open a burrow is to dig down behind it and make an opening in the back of the nesting chamber just large enough to reach the eggs, afterward closing it with a stone or a board and carefully covering over the excavation.

“As I began to dig almost above it, the incubating bird, who had stood its ground in face of the thrusts with the vine, flew out, uttered a few *klecks* and headed upstream, where it perched on a giant cane leaning over the current and soon plunged for a fish. I now saw for the first time that it was the male. Like other burrow-nesting birds under the same circumstances, it seemed rather unconcerned, and this despite the fact that I afterward found it the most devoted of parents.”

Because he “had faith that the activities of birds, including their periods on the nest during incubation, are rhythmic and more or less constant for the species, rather than irregular and arbitrary,” Mr. Skutch was determined to learn how these kingfishers arranged their shifts on the nest. In his efforts to solve this problem, he was at “first baffled, next challenged, and finally

surprised.” For many weary hours, during nearly two weeks, and at various hours from dawn to sunset, he patiently watched that hole in the bank, mostly with little or no results. The use of what he called a “silent monitor,” a small stick stuck upright and loosely in the entrance of the burrow, enabled him to tell that a bird had either entered or left the nest, but did not tell him the hour or the sex of the bird. His account of how he accomplished this and finally learned the secret is too long a story to be told here. But he did finally discover “that there is a single nest relief each day, early in the morning. This is comparatively easy to observe, for the bird coming to take its place in the burrow usually flies downstream sounding his powerful, metallic *kleck kleck* at measured intervals, and so heralds his own arrival.” After three or four false starts he finally gathers courage to enter the burrow; and, after a minute and a half or two minutes, his mate launches forth and flies off upstream. The times at which the relief took place varied from 7:05 to 10:01 a. m., most of the shifts being made between seven and nine o’clock. He says, further: “After I learned what precautions were necessary in order to determine the sexes of the birds, I found that on some mornings it was the male who entered and the female who departed, while on other mornings the reverse was true. There was a regular alternation, the male entering one morning and the female the next.

“Each afternoon the incubating partner took a single recess, for food or exercise, from its long 24-hour turn on the eggs. It emerged suddenly and without warning, at some time between 1 and 4 o’clock, flew upstream to the feeding ground, leaving the nest unoccupied, and returned in half an hour to an hour. On returning, it flew downstream low above the water and entered the nest directly, without perching or calling, in a manner very different from the morning entry, since there was no mate on the nest to be advised of its arrival. Then it remained until relieved by the mate the following morning.

“Few birds incubate so continuously as the ringed kingfishers. One day the female took her afternoon recess early, and on returning remained on the nest more than 19 hours, for her mate was very late in relieving her the next morning, and did not appear until 10 o’clock. The usual period between their return in the afternoon and their relief the following morning is 16 or 17 hours.”

Mr. Skutch found another nest of this kingfisher “400 feet downstream in the same bank. I opened it only two days before the eggs hatched. The bird on the nest, with a degree of attachment I have rarely seen equaled, remained bravely in the tunnel while I probed its length, dug in the rear, took out the eggs for measurement, fitted a stone in the aperture I had made, and tamped the soil above it. All this occupied well over an hour. This burrow was 7 feet 9 inches in length and the nesting chamber, 22 inches below the surface, contained, like the last, four white eggs. The territories of these two pairs extended in opposite directions from their burrows, the birds from the upper nest always entering it from upstream and returning thither when relieved, those in the lower nest fishing in the river below it.”

Eggs.—The ringed kingfisher apparently lays three to six eggs to a set; probably four and five are the commonest numbers. What few eggs I have seen are pure white; they vary in shape from ovate to short-ovate; and the shells are smooth and quite glossy. The measurements of 19 eggs average 43.72 by 34.52 millimeters; the eggs showing the four extremes measure **46.8** by 35, 43.7 by **35.6**, and **39.7** by **32.5** millimeters.

Young.—On the morning of March 28, the female arrived, at the first burrow examined by Mr. Skutch, at sunrise, an unusually early hour for her to appear, and seemed to be excited. He opened the burrow and found that two of the eggs had hatched. The nestlings were “pink skinned and were without the least vestige of feathers. They could already stand upright, supporting themselves on their feet and belly, and attempted to walk, which they did in a weak and tottering fashion.” Three of the eggs in the other nest hatched the same day; as he removed the stone from the rear, “a sizzling noise arose from the earth,” and two of the nestlings retreated into the tunnel with their mother, where they could not be reached.

Soon after the young in the first nest had hatched, some malicious person dug into the burrow at both ends but could not reach the young in the tunnel. Three of these nestlings eventually died, but the parents continued to feed and brood the fourth in the exposed and gaping burrow. Then, Mr. Skutch continues: “Stuffed with whole fish to the bursting point, the single nestling grew at a tremendous rate. Its eyes opened by the tenth day, when it uttered a high-pitched, trilling sound in response to its parents’ rattle. It was 14 days before the upper mandible caught up to the lower in length. The young kingfisher was beginning to defend itself with energy

and bit hard with its great mandibles whenever I picked it up. A few days later it squealed and fought like a fury. When 4 weeks old it was fully feathered except for the naked belly, which rested upon the sandy floor of the foul burrow. To its biting and squealing it now added an alarm rattle almost as loud as that of the parents, which at length drew one of them, who answered in kind. Still it did not attempt to flutter, and when I placed it on the ground it could do no more than hop along with outstretched wings. It remained in the burrow a full week after it was completely feathered, and finally departed at the age of 34 or 35 days.

“Although this nestling was one of the most vociferous I ever encountered, its struggles and cries were very mild in comparison with those of the single youngster who survived in the burrow downstream.” When about 4 weeks old, it was almost as big as its parents. When he attempted to reach it from the back of the nest, “it fled through the tunnel and jumped into the river, where it spread its wings, turned upstream, and flapped its way slowly against the current. When it encountered obstacles of stranded brush it hooked its bill over them and scrambled across. Thus it led me a merry chase, wading in the muddy shallows, until a fallen banana plant stopped its wayward progress and I seized it. Unlike its neighbor of the same age upstream, it had not become accustomed to being taken in hands; its deafening screams and fierce attempts to bite made, for duration and intensity, the best efforts of the other pale to insignificance. It was ten agonized minutes before it became reconciled to me, and I let it perch on my hand until its feathers dried before returning it to the burrow. This bird left its burrow between its thirty-fifth and thirty-seventh day.”

Plumages.—I have seen no very young birds. In immature birds, probably birds of the first year, the sexes are distinguishable. In the young male, the jugular area is dull gray washed with “cinnamon,” and the under tail coverts are pale cinnamon, both of which parts are pure white in the adult male, though some adults have the under tail coverts barred with bluish gray; the brown areas on the under parts are paler than in the adult, “ochraceous-tawny” to “cinnamon,” more or less mixed with white; these parts in the adult are rich browns, “Sanford’s brown” to “cinnamon-rufous”; the upper parts and wing coverts are dotted with small white spots, and many of the feathers have shaft streaks or median wedges of black; these parts are clear, grayish blue, or “delft blue,” in adults, without streaks;

young birds also have broad, median, black streaks in the crest. Young females are similar to the young males but may be readily recognized by the broad pectoral band, which is never present in the male at any age; in the adult female this is clear bluish gray, but in the young bird it is broadly edged with rufous, or dull brown, and sometimes mainly rufous; in the young female the under wing coverts are wholly "cinnamon-rufous," whereas in the young male these are partly white.

I have not seen enough material to determine how long these immature plumages are worn, or to learn much about the molts.

Food.—Mr. Skutch (MS.) says on this subject: "The ringed kingfisher's diet is rather monotonous. They live almost entirely, if not exclusively on fish, often of large size, which they catch in the regular manner of kingfishers, but being larger birds their plunges are more spectacular than those of the others."

One day "a female flew into a balsa tree, growing beside a small stream, with a fish fully half as long as herself dangling crosswise in her bill. For more than two and a half hours by the watch she held it thus, changing her position only from one branch to another of the same tree. I was at the time watching a green kingfisher's nest, and I could keep her in sight without additional effort. When at length I was ready to leave, she had begun to beat her fish against the limb, although it must have been dead long since. After this exhibition of stolidity, I no longer felt sorry for the ringed kingfishers because the customs of their race obliged them to sit on the nest for such long periods."

Behavior.—Dr. Charles W. Richmond (1893) found this species very common in Nicaragua and Costa Rica, and he writes of its habits:

One morning a pair of these birds went through a very curious performance. Attention was first called to them by their loud rattling cry, which was kept up almost constantly as they circled and gyrated about over the water, occasionally dropping—not diving—into the water, and sinking below the surface for a moment. This maneuvering lasted some minutes, after which both birds flew upstream uttering their ordinary note.

Two or three individuals were in the habit of passing the night at some point on the creek back of the "I. P." plantation, and came over just about dusk every evening. I noticed them for several months, and was struck with the regularity of their coming, and the course taken by each on its way to the roost. The birds could be heard a considerable distance away, just before dusk, uttering their loud single "chuck" at every few beats of the wings. They appeared to come from their feeding grounds, often passing over the plantation opposite, probably to cut off a bend in the river. One of the birds invariably passed close to

the corner of the laborers' quarters, though at considerable height, and the other near a trumpet tree some distance away. The third bird was only a casual visitor. At times the birds came together, but usually there was an interval of several minutes. Their routes met at a turn of the creek a few rods back of the house, where they usually sounded their rattling notes and dropped down close to the water, which they followed to the roost. This was a huge spreading tree, covered with parasitic plants and numerous vines, which hung in loops and festoons from the limbs. On one occasion I shot at one of the birds as it came clucking overhead, and caused it to drop several small fish. A female nearly ready to deposit eggs was shot October 9.

Referring to the behavior of ringed kingfishers, Mr. Skutch (MS.) writes: "Watercourses are their highways, and, like men, they are frequently reluctant to leave them. One day, ascending in a motor launch the Toloa Creek in Honduras, we drove a ringed kingfisher before us for possibly a mile. Each time the boat approached he would leave his perch, fly a few hundred feet ahead, and finally alight in a branch overhanging the stream. Here he would wait until the launch was almost opposite him, then fly ahead of it a few hundred feet more. Only after this procedure had been repeated many times did he finally double over the bank and return downstream."

He noticed, while studying the program of nest relief, that these kingfishers "showed a certain amount of formalism in their natures." One morning while he was on watch, "at 7:30 the male emerged from the burrow without warning [and without waiting to be relieved by the female in the usual way] and flew upstream as usual. I was surprised at this unexpected behavior, but probably no more than his mate. Five minutes later she appeared from the direction of his departure and perched in a trumpet tree growing on the bank a few rods from the burrow, where she often rested before going to relieve the male on the nest. Soon he reappeared and the female, who had not moved, greeted him with a rapid, low rattle, which was evidently a scolding. A minute later he flew upstream again. I thought now the female would certainly enter, it was her day upon the nest, but such an unwelcomed entry would have been a breach in formality, beneath the dignity of a well-bred kingfisher matron. She delayed another minute, as though considering what course to follow, then flew off after the delinquent. Soon they both returned, but separately, and there was a rather lengthy conversation between the pair as they perched on the banana leaves near the entrance of the burrow. Finally, after flying back and forth several times before it, the male entered, rather sheepishly we may suppose, while his mate continued to *kleck* from her perch in a low voice.

“Mrs. kingfisher had won her point. She delayed a seemly interval for him to compose himself for her reception, then entered herself with the usual warning. I shall probably never know what passes between them in the fraction of a minute they must be together in the nesting chamber, but whatever form of greeting they indulge in, we may well suppose it was not as cordial as usual. The male came out in record time, about one minute after his mate’s entry, and turned upstream, klecking loudly, a free bird at last.”

A change in the behavior of the two males was noted after the young had hatched in the two nests, of which Mr. Skutch (MS.) writes: “The males of both nests, both of whom happened to be free that day, behaved differently than on previous mornings. After being relieved, they had gone off as usual to their respective territories up and down the stream; but instead of remaining there, as they had always done before the eggs hatched, they soon returned to perch at no great distance from their nests, and loudly protested my presence. In their excitement, both perched at the same time in the trumpet tree which grew on the bank of the river between the two nests, and was apparently the boundary between their territories. They stood side by side on a branch, their beautiful, white-barred, slate-colored wings spread until they almost touched. One raised his crest, but the other laid his flat, and, with open bills and angry *klecks*, each defied the other to cross the accepted frontier.”

Voice.—The same patient observer says: “Now that there were nestlings, I heard an utterance from the kingfishers that I had never heard before. It was really not so much a new note as a different manner of using the old familiar one, for their entire vocabulary consists of a sound that to our ears is suggested by the syllable *kleck*, but they employ their single word in a great variety of ways to express different meanings and emotions. A single loud *kleck*, uttered at measured intervals, punctuates their flight; a softer, rapidly repeated *kleck* is the signal that a bird wishes to relieve his mate on the nest; and now there were nestlings to guard they expressed their anxiety by a very loud, rapid, mechanical klecking, continued with momentary pauses so long as danger seemed to threaten. This harsh, deafening rattle was uttered while the bird perched with the bill open, the mandibles held motionless, and the tail vibrating rapidly up and down. I never heard another bird make more noise when its nestlings seemed to be in danger,

nor give more evident signs of distress, yet, in common with other birds which nest underground, they never darted at me nor made any demonstration. They merely perched in full sight and rattled interminably.”

DISTRIBUTION

Range.—Mexico, the Lesser Antilles, Central and South America; casual in the lower Rio Grande Valley, Texas; nonmigratory.

The range of the ringed kingfisher extends **north** to Nayarit (Tres Marias Islands and San Blas); Tamaulipas (Rio Cruz and Tampico); Honduras (Tolosa and Lancetilla); northern Colombia (Cartagena, Sabanilla, and Santa Marta); northern Venezuela (Lake Valencia); the Lesser Antilles (Goyave); British Guiana (Georgetown and Blairmont); Surinam (Paramaribo); and northern Brazil (Santarem and Capim River). **East** to Brazil (Capim River and Cantagalla); Uruguay (Rio Negro); and Argentina (Villegas, Puerto Santa Elena, and Tierra del Fuego). **South** to southern Argentina (Tierra del Fuego); and southern Chile (Chonos Archipelago). **West** to Chile (Chonos Archipelago and Chiloe Island); western Peru (Huachos and Lima); western Ecuador (Tumbez, Bucay, Babahoyo, and Vinces); western Colombia (Cali and Honda); Panama (San Miguel and Almirante Bay); Costa Rica (Pozo Azul and Bolson); western Guatemala (Rio Morja); Jalisco (Las Penas Island); and Nayarit (Tres Marias Islands).

Casual records.—The only United States record of this species is a specimen in the Academy of Natural Sciences of Philadelphia that was collected on the American side of the Rio Grande, near Laredo, Texas, on June 2, 1888.

Egg dates.—Guatemala: March.

Mexico: March 21 and 30.

Peru: May 29.

CHLOROCERYLE AMERICANA SEPTENTRIONALIS (Sharpe)

TEXAS KINGFISHER

PLATE 16

HABITS

I made the acquaintance of this pretty little kingfisher when I made a short visit to Cameron County, Tex., as the guest of George Finlay Simmons, in May 1923. This is a most interesting bird country, rich in the number of Mexican species that reach their northern limits here and the only region in which some of them can be found within the limits of the United States. About Brownsville the chaparral, the open prairies, the tree claims, the ponds, and the swamps were all teeming with bird life of many species; but perhaps the most interesting of all were the dense forests along the resacas or stagnant watercourses, the old beds of rivers; these often contained large trees, mesquite, huisache, ebony, palms, etc., with a thick undergrowth of many shrubs and small trees, such as granjena, persimmons, coffee bean, and bush morning-glory. Here we found the characteristic birds of the region in abundance, such as the chachalaca, the red-billed pigeon, the noisy derby flycatcher, the brilliant green jay, and Audubon's oriole. Here, too, I was delighted to see my first Texas kingfisher, as it sat on a dead fallen tree over the water and then went flying away upstream, uttering its rattling twitter, suggestive of but different from that of the belted kingfisher.

Mr. Simmons (1925) says that it is "resident along clearer mountain streams of southwest-central Texas, from Comal County southward. * * * Occurrence depends largely on its habitat, the bird requiring the clearest of waters, particularly the crystal-clear rivers and brooks of the central Texas hill area; dislikes water the least bit muddy; larger, clearer streams and rarely smaller ones," as well as "shady little brooks," are mentioned as its favorite haunts.

Dickey and van Rossem (1938) record this kingfisher as a "common resident throughout the arid Lower Tropical Zone on all freshwater lakes, streams, and marshes below 2,300 feet, and also coast-wise in the mangrove belt," in El Salvador. Of its haunts they say: "A favorite environment is along small, rocky streams of the uplands, and in such places the population averages about a pair to the mile. When the young of the year are on the wing, this average is considerably increased for a time. Probably the section of stream inhabited by any individual pair has pretty definite limits, for although individual birds or pairs show no hesitancy in keeping well ahead of a person for a time, sooner or later they will make every effort to break

back along the route to the places from where they were first started. Salt water is apparently not greatly to their liking, and in the mangrove lagoons they were decidedly uncommon. Scarcity of suitable nesting sites may, however, be in part responsible for this condition.”

Alexander F. Skutch found this kingfisher on small streams as high as 7,000 feet above sea level in the highlands of Guatemala, and writes (MS.) of its haunts: “While toiling over the rocky bed of some narrow torrent rushing down a mountain valley, where the huge trees arch overhead and shut out the sky, I have often heard a pleasant *cheep* and turned to watch the solitary figure of a green kingfisher fly swiftly past, low above the water and following all the twistings of the channel, until lost in the depth of the forest. While his larger relatives require deeper water and a longer drop, he is often content to plunge from the top of a boulder projecting a foot or so above a shallow channel, and fishes on the smaller streams from which they are absent; but he joins them on the broader and more sluggish waterways. He rarely hovers above the water in the manner of the larger kingfishers.”

Nesting.—The first account of the nesting of the Texas kingfisher is by William Brewster (1879), who writes: “This beautiful little Kingfisher was found by Mr. Werner in comparative abundance at several points in Comal County, notably about some of the springs that empty into the Guadalupe River. A set of six eggs, taken April 25, 1878, was authenticated by the capture of both parent birds, the female being caught on the nest. * * * The nesting cavity was in a sandy bank near the water’s edge. The eggs were laid on the bare sand, no fish bones or other extraneous material being near. The entrance was not quite 1¾ inches in diameter, and the hole extended inward from the face of the bank about 3½ feet.”

Bendire (1895) says:

The nests of many of these little Kingfishers are yearly destroyed by high water flooding their burrows, caused by heavy rains and cloud-bursts, which are more or less prevalent in southern and western Texas. It is not uncommon on both the Medina and San Antonio rivers, and a nesting site on the last-mentioned stream found by Mr. C. H. Kearny, in the spring of 1892, containing six fresh eggs, is described by him as being located in a bank about 15 feet high and about 5 feet above the water level. The nesting chamber, which was slightly larger than the tunnel leading to it, was placed about 2 feet from the mouth of the hole. There was no nest proper, but a few fish bones and scales were scattered about the eggs. In the same bank a number of Bank Swallows (*Clivicola riparia*) had taken up temporary homes, and one of their holes was located within a foot of that of the

Kingfishers. They are devoted parents, and these birds will usually allow themselves to be caught rather than forsake their eggs.

There are four sets of eggs of this kingfisher in the Thayer collection and one in the writer's collection, all taken by, or for, Frank B. Armstrong, near Ciudad Victoria, Tamaulipas, Mexico, between March 12 and 28, 1908. The nests were all in holes in banks, about 3 feet deep and 6 to 8 feet above the water.

Mr. Skutch says (MS.) that this kingfisher, which he calls the green kingfisher, is one of the five species of birds that nest most commonly in the banks of the streams in the Caribbean lowlands of Central America. "The burrows of the ringed kingfisher are distinguished at once by the large diameter of the tunnel, 6 inches in width. Next in size are those of the Amazon kingfisher, $3\frac{3}{4}$ inches wide. Then come those of the turquoise-browed motmot, about $3\frac{1}{2}$ inches in width; and finally those of the green kingfisher, only 2 or $2\frac{1}{4}$ inches in horizontal diameter.

"While the burrows of the larger kingfishers and the motmot are placed in plain sight in the bare and exposed banks, so that he who runs may see them, those of the green kingfisher I have found concealed by the fringe of vines and dead vegetation draping the top of the bank, or else behind exposed roots, and I discovered them only by seeing a bird enter or leave. Theirs were the last of the three kingfishers' nests I encountered, and I found only two, late in the season, although the species is equally abundant with the Amazon kingfisher. As befits the smaller bird, their burrows are far shorter than those of the larger species. The two I measured were only 22 and 25 inches in length.

"One morning at the end of April, I sat down to eat my breakfast upon a fallen log beside the Quebrada de Arena, a little brook so narrow that one can easily jump across it, flowing through a pasture grown up with low bushes and thorny vine tangles. Presently a male green kingfisher flew downstream, perched on a branch ahead of me, *ticked* a great deal and seemed excited about my presence. The bird's persistence in remaining in that stretch of the river and his evident excitement renewed my conviction [on a previous visit] that his burrow was not far distant. I removed my shoes and waded up and down, examining every likely bit of bank, while the bird flew back and forth to keep out of my way. I discovered only old burrows whose lack of fresh foot-furrows proclaimed clearly enough they were not

in use. Quite baffled, I paused beneath the shelter of some overhanging bushes to watch the bird, and in a few minutes, after calling again *tick tick tick*, he flew up beneath the exposed roots of a dead stump, half washed out, and disappeared. This was almost the exact spot whence I had seen him or his mate emerge three weeks before, but the entrance to the burrow was so well concealed by the overhang of the top of the bank, the projecting roots and the vines which draped over them, that it had completely escaped me.

“I lost no time in opening the burrow. The male flew out with the second push I gave the machete which I used to dig, for the tunnel sloped upward so sharply that the nesting chamber was less than three inches below the surface of the ground, and I broke through into the rear of it before I supposed I had well started to dig. There were three white eggs, well advanced in incubation, to judge from their opacity. The male fluttered several times in front of the burrow, eager to enter before it had been closed again. I fitted a stone over the small aperture I had made, covered it with earth, and placed logs across the roof of the nesting chamber, to prevent the mules’ stepping upon it and possibly breaking through. The birds continued to incubate.

“The pair arranged their turns on the eggs in much the same manner as the Amazon kingfishers. The female spent each night on the nest. Soon after 6 a. m. the male flew downstream, low above the water, uttering at intervals the high-pitched *cheep*, which is his flight call. He perched on one of the roots of the old stump projecting in front of the nest and called *tick tick tick* in a low voice, which his mate heard in the burrow. She came forth, greeting him with a single *cheep*, as she flew swiftly past and turned down the brook to her feeding grounds. Her behavior was rather erratic.

“One morning she could not await his arrival, although he was hardly late. She popped out of the nest without warning and flew off, but a minute later the pair returned together. The male went to the root in front of the burrow, *ticked* just as much as he was accustomed to do to call off his mate, although he could certainly see she was not inside, then entered the empty burrow. The following morning she acted in quite another manner. Just after 6, on a cloudy morning after a night of hard rain, the male flew downstream, perched in front of the burrow, and *ticked* for her to come forth, but she paid no heed to his repeated calls. He flew a few rods downstream, then returned to call *tick tick tick* again. Still no response, so

he flew away out of sight. Ten minutes later he reappeared and perched again on the root in front of the burrow, where he called at intervals for two minutes before at length she darted forth. Then he entered for the morning. The male incubated until the female returned from her breakfast to relieve him. One morning she left him on the nest less than two hours, but the next it was nearly three. She covered the eggs for the remainder of the morning. The male was chiefly responsible for keeping them warm during the afternoon, until his mate called him from the nest and entered for the night at some time between 5 and 6 o'clock."

Eggs.—The Texas kingfisher lays three to six eggs, but five seems to be the commonest number. These vary in shape from oval to elliptical-oval; the shell is smooth and thin; some show little or no gloss and others are quite glossy; the color is pure white. The measurements of 64 eggs average 24.36 by 19.23 millimeters; the eggs showing the four extreme measures **26.4** by 20.3, 24.1 by **20.7**, **22.3** by 19.3, and 23.8 by **17.5** millimeters.

Young.—The second nest that Mr. Skutch found contained five "pink-skinned, blind, and totally naked nestlings," which "like those of the larger species, had undershot bills and heel callosities." He visited this nest 18 days later and found that the young, which were not more than 25 days old, "could flutter just a few feet. One flew into the river, where she spread her wings on the surface and headed for the shore. I threw her into the shallow water again, and again she turned unerringly toward the marginal rocks, beating her wings on the surface until she gained a footing. The following morning I placed them on the shore for a photograph, but two found their wings and easily traversed the 50-foot channel, flying low above the surface. The power of flight had come to them almost overnight.

"One evening early in June, after the sun had fallen behind the bordering fringe of willow trees, I was resting on a log stranded on the flood plain of the river, when a young green kingfisher flew upstream, calling *cheep* at intervals, and perched on a pile of brushwood almost in front of the burrow in which it was hatched. Presently its father came flying downstream, with a small minnow in his bill, and perched on the same pile of brushwood, not far from the other. The young bird came toward him, as if to receive the fish, but the other raised his wings above his back to fend it. The youngster took this as a hint to remain aloof and perched at a little distance. Not satisfied with the interval that remained between them, the male darted

at the young bird, which retreated a few feet. Several times it started to approach its father, but each time was warned to remain away by the spread wings, a very picturesque attitude. Several times, too, the bird with the fish drove at the applicant for it and finally, still holding the fish in his bill, chased it down the stream and out of sight. The young kingfisher had been out of the nest 29 days and must now at least learn to dive for its own fish.”

Plumages.—The young kingfishers are hatched naked and blind, as described above, but the juvenal plumage is acquired before the young leave the nest. A young male, taken on August 21, fully grown and fully fledged in juvenal plumage, is much like the adult female; the rich brown pectoral band of the adult male is only faintly indicated but is replaced by a band of greenish-black spots; and there is more black spotting on the flanks than in the adult male. Apparently there is a gradual change toward maturity during the first year, for October, February, and April birds show a gradual increase in the rufous band.

The youngest female I have seen, taken on June 8, is much like the adult female, but the pectoral band of greenish black spots is lacking, or nearly so, the under parts being nearly immaculate. These spots, which are more in the form of broad streaks than in transverse spots, increase with age to form one complete and one nearly complete band in the oldest birds. The observations made by Mr. Skutch on the brood of young that he studied agree substantially with the above.

I have no data on the molts, but Mr. van Rossem (1938) says that “the annual molt * * * takes place in the fall, but the definite time is not known.”

Food.—The food mentioned by Mr. Skutch (MS.) consisted of small minnows. Mr. Simmons (1925) says that this kingfisher is a “business-like little fisherman, perching atop a stick or stake in the water or on a low branch overhanging low water”; it “frequently flies back and forth over the water, hunting for small fish.” It is “often driven off feeding-grounds by the larger Belted Kingfisher, with which it is sometimes found.”

Voice.—Mr. Simmons (1925) says that the voice of the Texas kingfisher is a “rather sharp, rattling twitter, uttered on the wing; quite different from and shriller than the loud, harsh rattle of the Belted Kingfisher.” Mr. Skutch refers to the flight note as *cheep*, and the call or alarm note as *tick tick tick*.

Field marks.—This is such a small kingfisher that it could hardly be mistaken for anything else within its range. Its upper parts are dark, glossy green, and spotted with white, and it has no occipital crest. The under parts are white, with a rufous pectoral band on the male and a ring of black spots across the breast of the female.

Enemies.—The first set of eggs that Mr. Skutch found failed to hatch, as they were destroyed by ants. He writes (MS.): “Opening the burrow, I found it swarming with myriads of small, amber ‘fire ants,’ a scourge to man and beast alike. Invading the nest, they had worried the birds until they fidgeted on their eggs and cracked them; then they had worked into the cracks and begun to eat the embryos. I had cleaned them out the previous evening, but all to no avail. The nest was completely ruined. That same morning they had attacked and killed three young woodpeckers in their nest in a dead stub standing a few paces from the kingfishers’ burrow. In the humid coastal regions, ants are one of the principal enemies, if not actually the chief enemy, of nesting birds. I have found more eggs and nestlings destroyed by them than by all other known agents combined.”

DISTRIBUTION

Range.—Southern Texas, Mexico, and Central and South America; casual in southern Arizona; apparently a nonmigratory species.

The range of the green kingfisher extends **north** to Sinaloa (Mazatlan); Durango (Rio Sestin); southern Texas (Turtle Creek and New Braunfels); Quintana Roo (Xcopan); eastern Nicaragua (Pis Pis River); northern Colombia (Santa Marta and Bonda); northern Venezuela (La Guaira); British Guiana (Potaro Landing and Bartica); Surinam (Paramaribo); and northern Brazil (Quixada). **East** to Brazil (Quixada, Rio Taquarussu, and Goyaz); and Uruguay (Santa Elena). **South** to southern Uruguay (Santa Elena); and central Argentina (Santa Elena, San Jose, and Tucuman). **West** to northwestern Argentina (Tucuman); Bolivia (San Jose); Peru (La Merced and the Ucayale River); Ecuador (Vinces); western Colombia (Tumaco, Cali, and Rio Frio); Panama (Sapo Mountains, Gatun, and the Chagres River); western Guatemala (Duenas, and Lake Atitlan); Oaxaca (Juchatengo); Nayarit (San Blas); and Sinaloa (Escuinapa, and Mazatlan).

The subspecies, known as *Chloroceryle americanus septentrionalis*, is the only form to enter the United States. It ranges southward from Texas to Yucatan.

Casual records.—A specimen was taken at Decatur, Tex., north of the normal range in this State, on January 3, 1889. Dr. Elliott Coues reported seeing this species in September 1865 at points on the Colorado River, Ariz., between Forts Mohave and Yuma, and one was taken on the San Pedro River near Fairbanks on February 13, 1910. One was collected on September 8, 1893, at Cajon Bonito Creek, Sonora, a few miles south of the New Mexico line. One was taken on the Santa Cruz River, Ariz., on October 1, 1938.

Egg dates.—Mexico: 24 records, March 5 to June 13; 12 records, March 12 to April 19, indicating the height of the season.

Texas: 4 records, April 11 to June 15.

Order CAPRIMULGIFORMES

Family CAPRIMULGIDAE: Goatsuckers

ANTROSTOMUS CAROLINENSIS (Gmelin)

CHUCK-WILL'S-WIDOW

PLATES 18–20

HABITS

CONTRIBUTED BY ALEXANDER SPRUNT, JR.

Dusk falls gently over the salt marshes, which reach out from a shoreline where moss-bannered live oaks and stately pines rustle softly in the late March breeze. A faint fragrance of jessamine hangs in the air; the sleepy note of a cardinal echoes from a cassina thicket, while atop a tall palmetto a mockingbird salutes the coming night with a burst of melody. Silence comes and the stars appear, glinting in golden splendor through the purple gloom.

Suddenly through the air comes another sound, a sharp, clear-cut, insistent chant. Splitting the silences, it strikes clappingly upon one's ears, ringing with startling emphasis, unmistakable, thrilling and welcome. The first chuck-will's-widow has returned to the Carolina low country, and spring is definitely back again!

There is something about nocturnal birds that fascinate one strangely. Doubtless the cloak of darkness that shrouds their movements and activities has a great deal to do with it. One cannot but wonder at their comings and goings; how they pursue their hunting amid the gloom. Their voices too lend much to the fascination, for the notes seem a part of the night itself, just as the bodies of the birds themselves seem more like detached and living particles of darkness than of flesh and feathers.

All my life I have lived amid the haunts of some of these furtive kindred of the dusk, but the chuck-will's-widow above all seems to typify the mystery of the night and invests it with a sense of intangible yet satisfying tranquillity. It has always seemed to me that those beautiful lines, written of the cuckoo, might be even more applicable to the chuck-will's-widow, for, in truth, much of the time it does not seem a bird at all, but simply "a wandering voice." Without brilliant plumage or grace of form, it nevertheless possesses undoubted character, and long acquaintance with the bird only increases the interest that is bound to be aroused in any study of its life and habits.

Spring.—Generally speaking, the chuck-will's-widow arrives in the South in March. There is some indication that a few birds spend the winter in southern Florida for C. J. Pennock (MS.) has stated that "in the Charlotte Harbor district a few at least appear to winter. February 26, 1926, one heard calling; February 24, 1927, while camping, we heard one." Arthur H. Howell (1932) states that it "winters in small numbers" and gives December and February dates. R. J. Longstreet's opinion (1930) is that "the chuck-will's-widow is a summer resident in north Florida and a permanent resident in south Florida." Audubon considered the species a permanent resident in the State, and his idea was, according to Allen (1871), confirmed by "old residents," though he himself states it "is not observed till about the first of March." I am obliged to spend portions of every winter month in Florida, being constantly in the field throughout the southern Everglades and the Keys, but it so happens that I have yet to find the chuck-will's-widow during this season. The wintering population is undoubtedly small and scattered.

It reaches north-central Florida about March 18 (D. J. Nicholson, MS.) and appears in the coastal districts of southern Georgia a few days later (T. D. Perry, MS.). In South Carolina, about Charleston, it arrives anywhere from the third week in March to the first week of April. The males always arrive first, followed in a few days by the females.

Courtship.—Little time is lost by the chuck-will's-widow, after its arrival from its winter home, in seeking a mate. Almost at once it undertakes the search, and it is at such times that the observer has an opportunity to see them actively in daylight. The courtship performance is an interesting one and, all things considered, is not difficult to observe. The

outstanding characteristic is the strutting pomposity of the male. He sidles up to the watching female, his wings droop, the tail is widely spread, and he swells and swells until it really seems that the limit of inflation is reached and another fraction of distention would cause him to disintegrate like a bursting bomb. Various vocal efforts are indulged in meanwhile, accompanied by quick, jerky motions. Audubon (1840) has compared this phase of the proceedings to that of the domestic cock pigeon, while Arthur T. Wayne (1910) likened it to a turkey gobbler's antics. This latter has always seemed to me most apt, for not only does the seemingly endless inflation remind one vividly of the turkey, but the motions also suggest this bird.

After this period of intense and apparently exhausting display, a space of calm and quiet pervades the pair should the male have been successful in his suit, and he perches placidly beside her.

Nesting.—No semblance of a nest is constructed. The eggs are placed on the ground upon a carpet of dead leaves, and the sitting bird constitutes one of the finest examples of protective coloration that nature affords. Such dependence is placed in it that the closest approach is possible, and the bird flushes only when nearly trodden on. The mottling of the plumage is exactly like that of the variegated background of leaves, sun splashes, and shadow, and one may look directly at the bird without seeing it. Once the bird is flushed, however, it is perfectly easy to see the eggs, for they then stand out like huge pearls against the leaves, not having the similarity to the ground that characterizes the eggs of the nighthawk.

Mixed oak and pine woods are usually the nesting haunt of the chuck. In the large live-oak groves, which occur over so much of the plantation country of the South, the species is abundant and shows a remarkable tendency to place the eggs in nearly the same spot year after year. As a rule there is little if any undergrowth about the eggs. When under the pines, this would not be expected, and in the oak groves the ground is always covered by a veritable carpet of leaves, through which no undergrowth appears. Thus the eggs can be seen from a considerable distance should the bird be off them.

On May 8, 1926, I found two eggs on Folly Island, S. C., lying upon pine needles. Marking the spot accurately I returned the next year and on

May 12, 1927, found two eggs within 5 feet of the spot used previously.

M. G. Vaiden (MS.) writes from Rosedale, Miss., that he discovered a set of eggs on May 1, 1911. *Twelve years later*, on April 27, 1923, he returned to the same locality and found two eggs within 10 feet of the same spot! He states that the “trees were larger, the hillside washed into gullies, but otherwise about as formerly.”

Walter Colvin (MS.), of Arkansas City, Kans., reports the first nesting of the species in that State, two eggs having been found in May 1923, near Arkansas City, Cowley County. Another nest was discovered the following year (1924) also in May. Probably the chuck had been nesting there for some time previous and was also discovered in Miami County, by Mr. Colvin’s son John, in 1929.

Nesting observations from a variety of sources indicate that the chuck raises but one brood. However, if the eggs are taken, the bird will lay again and again until young are hatched. The late Arthur T. Wayne, of Mount Pleasant, S. C., once took in succession three sets from a pair near his home, and a fourth was laid, incubated, and hatched.

The chuck brooks no tampering with the eggs whatever. If they are handled, or as much as touched in some cases, the bird removes them to what is considered a safer locality. This habit was noted long ago and gave rise to much dispute and speculation as to the method employed in the transportation. It has been definitely proved that the bird takes them in the mouth. Audubon (1840) describes it so well that his account is given herewith:

When the Chuck-will’s-widow, either male or female (for each sits alternately) has discovered that the eggs have been touched, it ruffles its feathers and appears extremely dejected for a minute or two, after which it emits a low murmuring cry, scarcely audible at a distance of more than eighteen or twenty yards. At this time the other parent reaches the spot, flying so low over the ground that I thought its little feet must have touched it, as it skimmed along, and after a few low notes and some gesticulations, all indicative of great distress, takes an egg in its large mouth, the other bird doing the same, when they would fly off together, skimming closely over the ground, until they disappeared among the branches and trees. But to what distance they remove the eggs, I have never been able to ascertain; nor have I ever had an opportunity of witnessing the removal of the young. Should a person, coming upon the nest when the bird is sitting, refrain from touching the eggs, the bird returns to them and sits as before. This fact I have also ascertained by observations.

The first “apparent” recorded instance of the occurrence and nesting of the chuck in Ohio is recorded by E. S. Thomas (1932) and dated May 14,

1932. On May 21, 1932, the nest and two eggs were found and the female, with two eggs, was collected for the Ohio State Museum.

My earliest nesting record for South Carolina was made on April 13, this being nearly two weeks in advance of the next nearest date. The eggs are laid somewhat sooner than usual in forward seasons, the above record being an illustration of such an instance.

Eggs.—[AUTHOR'S NOTE: The chuck-will's-widow regularly lays two eggs, which are between oval and elliptical-oval and usually moderately glossy. Major Bendire (1895) considered these eggs as "among the handsomest found in the United States." I cannot do better than to quote his description of them, as follows:

The ground color of these eggs is of such a subtle tint that it is almost impossible to describe it accurately; it varies from a rich cream, with a faint pinkish suffusion, to a pale cream, and more rarely to pure white. They are in most cases more or less profusely blotched, marbled, and spotted with different shades of brown, tawny, fawn, and Isabel-color, underlaid and mixed with lighter shades of ecru drab, lavender, pearl gray, and pale heliotrope purple. In an occasional specimen some of the markings take the shape of irregular lines and tracings, like those of the Grackles; in others they are fine and minute, obscuring the ground color to some extent. In some specimens the darker shades predominate; in others, the lighter; in fact, there is an endless variation in the style of markings, but in the entire series there is not a single specimen which is not perceptibly marked.

The measurements of 54 eggs average 35.56 by 25.57 millimeters; the eggs showing the four extremes measure **40** by 27.5, 38.1 by **28.2**, **32.9** by 25.1, and 36.2 by **23.1** millimeters].

Young.—Some notes of unusual interest in regard to the behavior of the adult at the nest, as well as the actions of the young birds, have been sent me by Herbert L. Stoddard, of Thomasville, Ga. These are transcribed herewith, and in the general lack of such knowledge they serve to illuminate something of the home life of this interesting species:

"April 30 (1928), 7 a. m.: One egg has hatched and the chick is a queer little mite covered with a yellow-ochre down, and hops about like a frog in a very lively manner. When he is uncomfortably hot or chilled he gives a plaintive little pipe that can be heard *about 20 feet*.

"May 1: Other chick hatched out this morning or during the night and eggshells were gone.

“May 3: Chicks growing fast but still being brooded in same spot. Mother goes to sleep on a fence post after flushing, but as soon as chicks start to squeal from the heat, she becomes frantic and will nearly fly into my face. I bother her a few minutes at 3 p. m. each day, as I chase her well away, then duck into the blind and take a few feet of film as she comes back to the nest. Not much action, however, and she is a *wise fowl*. She knows perfectly when I am in the blind!

“May 6: Found the nest spot empty today at noon but finally located the old bird (she has two patches of albinistic feathers in center of upper breast, so I know it’s the same individual that performed the incubation) brooding her two chicks about 30 feet south. Their eyes have been open from the first but are now a little deeper in color and are always half closed like those of the adult in daytime.

“May 13: Have kept in rather close touch with the chuck-will’s-widow family recently. They are living under a growth of sparkleberry shrubs and have lived here within a radius of 6 feet for the last ten days.

“The place is pretty well marked by their mourning-dovelike excrement. When disturbed, the chicks hop off with elevated wings in a ‘mechanical toy’ sort of way. That is, their progress is marked by a series of rapid, toadlike hops until they are ‘run down’ (usually in 30 or 40 feet). The wings serve as balances.

“The chicks have a little complaining whine that brings the mother in frantic haste. She flies noiselessly about, every now and then lying on the ground with her wings *widely spread and reached forward*, with the primaries pressed to the ground. In this queer position, she beats them a bit and opens and shuts her huge mouth, exhaling air audibly as she does so and occasionally uttering the queer, froglike croak. Altogether an odd performance. No evidence of a mate has been seen about this location.”

Plumages.—[AUTHOR’S NOTE: The young chick is completely covered with long, soft, silky down; on the upper parts the color varies from “ochraceous-tawny” or “light ochraceous-buff” on the head to “tawny” on the back; on the lower surface the color grades from “ochraceous-tawny” on the chest to “light ochraceous-buff” on the throat and belly.

The growth of the juvenal plumage, in which the sexes are alike, is rapid. Ridgway (1914) describes it very well as follows: "Similar to the adult female in 'pattern' and coloration of tail, primaries, and primary coverts, but otherwise different; scapulars and middle wing-coverts ochraceous-buff, irregularly barred with black; pileum more grayish, with small spots, instead of streaks, of black; under parts barred with black on a light brownish buffy ground, without vermiculations, mottling, or spots, and band across lower throat indistinct or obsolete."

This plumage is worn but a short time, as a partial molt into a first winter plumage begins in July; I have seen a specimen that had nearly completed this molt on August 2; this plumage closely resembles that of the adult female, as the juvenal wings and tail are retained; I can find no evidence of a spring molt. Young birds apparently retain the first winter plumage, including the juvenal wings and tail, until the following summer; I have seen birds in this plumage during winter and as late as May in spring.

Both adults and young have a complete annual molt, mainly in July and August. At this first postnuptial molt young birds become practically indistinguishable from adults, and the sexes become differentiated. Young birds in fresh fall plumage are darker and more richly colored than adults, with more "ochraceous-tawny"; the colors have faded some by spring.

Adults have two recognizable color phases, a tawny phase, in which the ground color of the two central rectrices varies from "ochraceous-buff" to "ochraceous-tawny", with deeper ochraceous or buffy colors in the scapulars and wing coverts; and a gray phase, in which the ground color of the two central rectrices is pale buff, or pale grayish buff, and the scapulars and wing coverts are paler and grayer.]

Food.—The chuck-will's-widow, like its family relatives, is an insect eater par excellence. The semitropical nature of much of its range is highly conducive to an abundance of insects and other night-flying creatures that are the bulk and mainstay of its diet. The mouth of the chuck is enormous, a characteristic of the goatsucker tribe, and is provided with bristles that act as a sort of additional trap. The widely open mouth is as much as 2 inches at the greatest breadth.

Prey is secured at low elevations, often only a few feet from the ground. The bird works the edges of woodlands bordering open fields and often

makes sallies over the latter. The flight is silent, and the birds seem to be no more than gigantic moths. Beetles, “flying ants,” and moths make up a large bulk of the food in many localities. Small birds have frequently been found in the stomachs of this species. While seemingly incongruous, this is, after all, not difficult to understand when the conditions are considered. Many observers have concluded that this type of stomach content is taken by mistake; that the small, fluttering bird, confused by the darkness, is taken for a moth and snapped up by the cruising chuck, of course being swallowed whole.

An alternative theory exists, however, and, if true, the bird-taking habit would be removed from the realm of the accidental and fall into purposeful, predatory effort. The late Edward H. Forbush pointed out that the “goatsuckers show an anatomical affinity to the owls. They have similar, soft plumage, noiseless flight, large eyes and nocturnal vision.” It is possible that, with this structural relationship, there are other phases of likeness between the chuck and the owls. It is the largest of the goatsuckers that occur in this country, and the other representatives of the family do not seem to indulge in small bird prey. Doubtless this is because of their considerably smaller mouths, but whatever the reason the chuck remains as the outstanding example of this procedure.

That this habit is certainly not accidental sometimes is definitely proved by the observation recorded by Gerald Thayer (1899) in which he relates the instance of a chuck-will’s-widow pursuing and catching warblers near a ship off the Carolina coast. Hummingbirds, swallows, sparrows, and warblers have been among those birds found in the stomachs of the chuck, and the frequency with which this occurs lends color to the supposition that it is more intentional than accidental. More research is necessary on this subject.

Even granting the truth of it, the economic status of the chuck is on the right side of the ledger and the great percentage of its activities are beneficial, for the noxious insects which it destroys are numerous. Miss Phoebe Knappen, of the United States Biological Survey, in answer to a request of the writer has very kindly furnished a summary of specific results in the stomach analysis carried out on this species in the laboratories of that Bureau. A full stomach from Oklahoma, without date and therefore not included in the tabulation below, contained the following: *Dendroica* (sp.),

70 percent; Coleoptera (*Calasoma*, *Harpalus*, Carabidae, *Ligyris*, *Strategus*, Scarabaeidae), 22 percent; Orthoptera (*Schistocerca*, *Nesconocephalus*), 8 percent.

The remarkable percentage of bird remains shown by this stomach would seem a great argument for the support of predation by purpose. Seventy percent of the total food among the wood warblers! However, it seems also to be a most exceptional case, for nothing that even approaches it is found in the list below, which embraces a range of 45 stomachs from five States and one Canadian Province. The entire amount of the stomachs listed contained the remains of but *two* birds!

With reference to the Oklahoma stomach, I wonder whether there might be a seasonal variation in the bird-taking propensities of the chuck-will's-widow in relation to migration. It would appear reasonable to believe that when there is high activity among birds traveling through a given area, such as would take place in the spring migration, the chucks of that locality would have greater opportunity in securing them. One would not have to incline to the predatory theory to accept this, for if there are a great many small birds passing through an area for a few weeks, the chucks in their night hunting would blunder across more birds than would be the case later in the season. If the take is accidental, a higher percentage of accidents would then occur. If, on the other hand, the take is deliberate, then the chances of indulging that habit would be greater and would fall off later in the season. So, whatever impulse governs the matter, the migrations would result in more birds appearing in the diet. Unfortunately, the Oklahoma stomach was undated, so it is impossible to ascertain whether the bird secured its high percentage during a migration or not, but I incline to the belief that it was a spring specimen.

To return to the analysis, Miss Knappen states: "The other 45 stomachs taken in March (2), April (31), May (11), and November (1) were collected in Florida (37), Georgia (1), Mississippi (2), North Carolina (1), Ontario (1), and Texas (3). The annual percentages of different items in the food, which was entirely animal, equal: Carabidae, 3.64; *Phyllophaga*, 32.98; other Scarabaeidae, 25.28; Cerambycidae, 4.49; Elateridae, 1.34; other Coleoptera, 5.13; Lepidoptera (moths), 12.36; Odonata, 4.63; Aves, 7.21; and miscellaneous animals, 2.85.

“The genera most persistently eaten were *Phyllophaga* (May beetles) and *Anomala*. The birds consumed were 1 *Dendroica palmarum* and 1 *Helminthophila* sp., while the miscellaneous bracket includes various bugs, flies, a bivalve, and other animal material.”

The insect content of these 45 stomachs totals more than 70 percent in but three classes (*Phyllophaga*, Scarabaeidae, and Lepidoptera) and other kinds make up considerably more than that. Birds are represented by only 7.21 percent, and both victims were warblers. One cannot but wonder at the single “bivalve,” an item that would certainly not occur to most students as being connected with a chuck’s diet.

Behavior.—Being as close kin to the whippoorwill as the chuck is, it cannot be expected that its habits will vary extensively. Inactive by day and a persistent hunter by night, it fulfills the usual characteristics pertaining to the family. Sitting motionless on a mossy log, a branch of some forest tree, or ensconced within a natural cavity, it dozes away the daylight hours. Some observers have found it sleeping in company with bats, in an obscure hollow. I have never found the chuck among such company, all my daylight observations being connected with the bird’s occupancy of some low limb, or on the ground itself.

When flushed, it rises with easy, fitful wafts of silent wings, alternating the beats with periods of sailing. Frequently it describes a curve and swings back near the spot from which it was flushed. Fairly close approach is allowed, even after the bird has been disturbed and alighted again. Doubtless it puts a great deal of dependence upon the wonderfully protective coloration of the plumage.

In the Carolina low country the chuck is very fond of roosting in the sandy roads so characteristic of the rural districts. Passing along at dusk, one may see the reflection of the eyes plainly in the glare of a car’s headlights. At times a roosting bird is disturbed under these conditions by day, and I have had them flush and come directly at the car, swerving only slightly aside to pass. One bird on Bulls Island, S. C., was flushed two or three times in an hour, as we had occasion to pass the same spot often, and once it flew by so closely that an extended arm might have touched it. This habit of frequenting the sand roads is shared by the whippoorwill when it is present in coastal Carolina during the winter months.

Little or no distinction is made between these two birds in much of their range. The uninformed observer takes it for granted that any night bird that calls, except an owl, is a whippoorwill, and this seems the more strange in a section where the chuck is abundant and the whippoorwill comparatively uncommon, as in coastal Carolina. The latter calls but rarely during its winter sojourn in the Charleston area; indeed, I have heard it but twice in all my years of ornithological study. One of these instances was in January, the other early in March. When the whippoorwill is present in this area, the chuck is not, for the former leaves before the latter appears from the south. In spite of this fact and the overwhelming evidence of the chuck's presence and comparative absence of that of the whippoorwill, the people of the low-country are far more familiar with the name of the latter and credit the call of the chuck to the other bird.

The eyes of the chuck-will's-widow reflect light admirably. Some years ago E. B. Chamberlain, of the Charleston Museum, and I carried out a series of experiments in "jack-lighting" amid the woods of Cumberland Island, Ga. It was a revelation in many ways. The night woods were literally twinkling and sparkling with eyes! The ground and low bushes gleamed with hundreds of points of light from the eyes of spiders; the lagoons reflected the ruby-red of many alligators, while here and there along the bank a wandering raccoon stared into the light beam or a trotting gray fox paused to sniff, one foreleg upraised like a pointer. Florida screech owls were seen perched atop low stumps, hunched and motionless, and although we could work up to within 3 feet of them, the take-off when it occurred was so utterly noiseless that not a whisper of sound ensued at even that close range. Dozens of deer were seen, their eyes, of course, reflecting the light perfectly, and even grazing horses and cows along the edges of the lagoons were as plainly noted.

Now and then a very large pair of eyes close to the ground shone out. Coming closer, we could see a chuck, sitting like a stone, staring rigidly into the light. While one of us held the light, the other worked around to the side and came up on the bird from behind, and reaching out could pick up the staring bird with ease. We examined several in this way, while they uttered a hissing note of fear or anger. The birds struggled strongly while held and were very difficult to quiet.

Though sharing with the other goatsuckers the characteristic habit of perching lengthwise, the chuck occasionally departs from custom and proves the ancient adage that exceptions make the rule. N. B. Moore (MS.) writes that he has seen it perch directly across a branch when the latter is an inch or more in diameter. He once "saw one perch on a greenbrier one-quarter inch in diameter as cleverly as any bird, though it sank suddenly under its weight for 7 or 8 inches. The bird remained on it for 10 or 15 minutes." It is likely that the chuck indulges in this more than one would ordinarily suppose. I have seen it but once, when a bird was flushed in daylight and flew to a small, gnarled oak, where it alighted among the outer twigs, perching distinctly crosswise. It had two young in the near vicinity.

In the reference already made above to Thayer's (1899) account of this species capturing warblers on a ship off the South Carolina coast, he noted that, on shipboard, the bird perched crosswise on the rigging at times. Another remarkable character was that this bird was seen, on flights out from the ship, actually to alight on the surface of the ocean! This is certainly phenomenal and constitutes behavior that is utterly at variance with the bird's ordinary habits. One more instance of crosswise perching is noted by W. S. Long (1935); he saw a specimen near Lawrence, Kans., that indulged in this posture.

The chuck-will's-widow frequently roosts in the same spot day after day, and one may be fairly certain of surprising a bird regularly when once the roosting area is located. During migrations it occasionally is found in rather extraordinary situations, one of the most striking of these being noted by J. M. McBride (1933), of New Orleans. He writes that he watched one for a week, September 14 to 21, 1933, occupy an unprotected branch of a hackberry tree just even with his second-floor window. It was to be seen daily on this branch from 6 a. m. to 6 p. m. His house was in the heart of the residential district of New Orleans.

Voice.—There is no doubt that the voice of the chuck-will's-widow is its most interesting and outstanding characteristic. Indeed, it is the one thing that many ever know of the species. It is a bird easily heard but comparatively seldom seen; therefore, though the call may be a nightly sound throughout the summer, the author may be utterly unknown to many by sight. However, no one who lives within the range of the chuck can have failed to listen to the notes perforce, and only a deaf person can fail to be

aware of its presence. Though the specific name of “vociferous” has been applied to the whippoorwill, it is equally true of the chuck-will’s-widow, if not more so, but the generic name of the latter is well chosen, for the mouth is certainly “cavelike.”

The call of this species is well deserving of comment, particularly in view of the fact that there seems to be so much confusion about it in the recent literature. Why this difference of opinion should exist, and why certain positive statements have been made, are sources of wonder to me and to others who know the voice of the chuck intimately. How anyone could listen for only a few minutes to the call and then say that “the song of the chuck-will’s-widow is less vigorous than that of the whippoorwill; it consists of *three* notes ... with a slight accent on the *first* syllable” is beyond my comprehension. And yet more than one ornithologist has so stated. It seems significant that all those so describing it are northerners, that they know the chuck only by reason of short southern trips of a few days or weeks. Or, perhaps, they take the opinion of others who have as little information as themselves. If their experience with this bird covered any extended period, they could hardly fall into such error.

The call of the chuck-will’s-widow is distinctly 4-syllabled (some observers say five at times), and therein lies one of the marked differences between it and the whippoorwill, which does have a 3-syllabled call. The accent is not on the first but on the third syllable; in other words, on the “wid” of widow. Few birds “say” their names as plainly. The *chuck* is uttered on a lower tone than the rest but is distinctly audible at 300 yards or more. At some distance one might be excused for thinking the call 3-syllabled, for it may sound like *will’s-widow*, but on still nights the first syllable is plainly heard even across broad bodies of water, as occur over much of the southern coast region.

The notes are not limited entirely to the dusk of evening or night. The bird sometimes calls in full daylight, either on cloudy or bright days, and sometimes during rains. I have heard it at 1 p. m. on a bright, clear day. It is, of course, not the rule any more than is the cross perching sometimes indulged in, but it certainly occurs. However, it is during late evening and all through the night that the chuck really performs, and it sometimes calls through the entire period. In localities favored by the species, several birds may be calling at once, which results in a jumble and overlapping of notes.

Herbert L. Stoddard writes that, never having heard a chuck-will's-widow calling in the daytime, he was "greatly surprised to hear two calling back and forth at 11:30 a. m. today (May 25, 1928). It was crystal clear and the sun was *hot*, but these two called over 5 minutes, exactly as they do in the nighttime."

In rapidity and frequency, there is much variation. A bird may utter a very few calls or very many. I have counted individual calls many times, and there seems to be no established custom or sequence. The usual interval between the notes is about $2\frac{1}{2}$ seconds, when the bird is doing a string of them. I counted the calls of a bird just outside my window one night and it ran off 111 without "drawing breath" other than the short spacing between each, $2\frac{1}{2}$ seconds. The calls were uttered at the rate of 25 a minute, this series taking about $4\frac{1}{2}$ minutes. One of my longest counts is 176 calls successively uttered without a break. E. S. Dingle tells me that he has counted 300 consecutive calls. On the night of June 2, 1939, at my home in St. Andrews Parish, Charleston, I heard a chuck that beat anything I have encountered yet. I had gone to bed; the night was warm and I was lying near a window, when a chuck started up about 50 yards away in one of the live oaks in my yard. I began counting almost automatically, and kept it up, idly wondering whether it would reach my former record. It did, and then some. I continued to count, and count. Finally, I got up and sat by the window, in order not to miss any of it. The calls were perfectly continuous, and uttered at the usual rate, although twice there was a slight break in perhaps as much as a second's lateness between them. The bird shifted its perch twice, moving perhaps a few yards each time, but did not stop calling. The total was *eight hundred and thirty-four calls* (834).

The notes are clear-cut, insistent, and sharply enunciated with the exception of the first syllable. There is a ringing quality about them that is very striking, and one gets the impression of full-voiced effort. The head is moved noticeably when the call is uttered, and doubtless considerable muscular effort is put forth. Some writers have termed the notes "doleful," "monotonous," and "melancholy," but to me they have never seemed anything but soothing and dreamily satisfying. Charles Torrey Simpson (1920) says that the chucks "make night hideous" with their "terrible chatter." Thus do tastes differ!

When it arrives from its winter quarters the chuck is particularly vociferous, and keeps this up until after the eggs are hatched. There is then a cessation followed by some renewed activity before departing for the south on the approach of fall. It must be very susceptible to cool weather, for it does not remain even as far south as Charleston until early in fall. The first part of September usually sees it gone, although individuals linger longer than that.

Besides the regular, self-naming call, the chuck has another note, which is not well known and is very difficult to describe. It is not the hissing sound uttered when the bird is caught or handled but is given occasionally when about its hunting. Almost entirely, if not entirely, it is a *flight* note; at least I have never heard it when the bird is sitting. It is inadequate to describe it as a “growl,” and yet that is the only word that seems to approach it. As one flies by in the gloom, this note is heard, and it is an eerie, utterly indefinite sound, possessing a strangely unearthly quality which impresses one with wonder that it comes from a bird.

It is seldom if ever referred to in the literature but some have remarked upon it in correspondence to the writer. The late James Henry Rice, Jr., of Brick House Plantation, Wiggins, S. C., once had an army officer visiting him who remarked on this note, but Mr. Rice himself, being very deaf, was not aware of it, though he knew the chuck well. He asks, in a letter to Mr. Bent, whether anyone has noted what he termed “that clucking sound.” I should hardly describe it as a “cluck” but it may impress some as such.

One other note has been commented on by those thoroughly familiar with the chuck. It is often given just as the bird is *flushed* and, like the one above, is very difficult to describe. It can be interpreted as a “croak” perhaps, and Dr. Eugene E. Murphey (MS.), of Augusta, Ga., calls it “froglike.” His allusion to it, as well as the utterances of the regular call as given in fall, is given in a communication as follows:

“I imagine most field ornithologists are familiar with the froglike croak that the bird makes when flushed, and I am inclined to believe that this note is much more apt to be sounded when the bird has been flushed from the nest. One observation may be worthy of note, namely, that I have heard the chuck-will’s-widow singing as late as September 12 in Edgefield County, S. C. When it comes to interpreting the quality of a bird’s song, it is

impossible to get away from a personal construction, which, of course, is invalid in a scientific observation, but it seemed to me that on this occasion the song was very definitely less vehement and forceful, certainly less frequently reiterated than is the case in spring; in fact, the whole thing seemed to have a querulous and uncertain character, somewhat as if he were wondering why he should be singing at this particular time of year. I endeavored to collect the bird, but the rapidly gathering darkness made it impossible for me to secure it, although I was very close to it several times and saw it take flight. Unfortunately, it chose to fly toward the darkening east rather than the west where there was still an afterglow."

In commenting generally upon the continuity of the chuck's calls during the early part of the season, Herbert L. Stoddard, of Thomasville, Ga., has sent me the following notes:

"Spent the entire night of April 14, 1927, on the alert in the observatory at the quail pens on lookout for an owl which has been killing quail ... a brilliant moonlight night. Chuck-will's-widows called *all night*, no 5-minute period between 9 p. m. and 5 a. m. elapsing without one to eight or ten calling. They have many guttural notes of different inflection, as well as the beautiful call note, a guttural, low-toned *waugh* given in questioning tones being common. These notes are most frequently uttered when a pair of the birds are together."

Fall.—The latest record in fall for the chuck-will's-widow in lower South Carolina is September 28 (Wayne, 1910). The great majority of the birds have left some time before this date. Indeed, Dr. Murphey's record mentioned above was a late one and impressed him markedly, as his account shows.

The earliest arrival record for the whippoorwill for the same locality is September 15, 1928 (Edward S. Dingle, MS.). The latest whippoorwill record is for April 1, 1911 (Wayne, 1910). Thus, in some exceptional years there may be the slightest overlapping of the arrival and departure of the chuck and the whippoorwill, but in the main the one has gone when the other appears, and there is usually some little interval between the sojourns of the two.

Referring to the fall migration in El Salvador, Dickey and van Rossem (1938) say: "Chuck-will's-widows were seen as late as October 29 at Rio

Goascorán, where they were more common than in any other locality. * * * Most were found well up in trees, once as high as a hundred feet above the ground, and so wild that collecting them was usually impossible. At Lake Alomega one flew from tree to tree through the high forest and at no time permitted an approach closer than about a hundred yards. * * *

“The usual daytime locations were large, horizontal branches twenty feet or so from the ground and in rather heavy woods.”

DISTRIBUTION

Range.—Southeastern United States, the Caribbean region, Central America, and northern South America; casual north to Ontario and Nova Scotia.

Breeding range.—The chuck-will’s-widow breeds **north** to southeastern Kansas (Arkansas City and Independence); Missouri (Willard, Springfield, and Sulphur Spring); southern Illinois (Olney); southern Ohio (West Union); and southern Maryland (Point Lookout). **East** along the Atlantic coast to Florida (St. Augustine, Daytona Beach, Royal Palm Hammock, and Man-o-war Key). **East** to Florida (Man-o-war Key, Fort Myers, St. Marks, Lynn Haven, and Pensacola); southern Alabama (Spring Hill); southern Louisiana (St. Francisville and Urania); and southern Texas (Houston, San Antonio, and Kerrville). **West** to central Texas (Kerrville, Waco, and Commerce); eastern Oklahoma (probably rarely Norman and Copan); and southeastern Kansas (Arkansas City).

Winter range.—The winter range is not clearly defined, but at this season it has been found **north** to Cuba (Isle of Pines and San Pablo); the Bahama Islands (Andros Island and Nassau); the Dominican Republic (Catarrey and Samana); and Puerto Rico (Arecibo, San Piedras, and Vieques Island). **East** to Puerto Rico (Vieques Island); and northern Colombia (Medellin). **South** to northern Colombia (Medellin and Antioquia); Panama (Panama City and Divala); Costa Rica (Rio Sicsola and Candelaria); Nicaragua (San Juan del Sur); El Salvador (Lake Olomega and probably Barra de Santiago); and Guatemala (Guatemala City). **West** to Guatemala (Guatemala City); and western Cuba (Isle of Pines). It appears that occasionally individuals may spend the winter in Florida, as one was recorded from Orlando on December 1, 1885; one from Lake Jackson on

December 5, 1911; and another from the same general area on December 28, 1903. One also was seen at Chenier au Tigre, La., on January 2, 1934.

Spring migration.—Early dates of arrival are: Florida—Orlando, February 17; Melrose, March 3; Palma Sola, March 5; Daytona Beach, March 9; Merritts Island, March 12. Alabama—Prattville, April 2; Barachias, April 3; Greensboro, April 5; Montgomery, April 6. Georgia—Savannah, March 15; Cumberland, March 25; St. Marys, March 28, South Carolina—Charleston, March 12; Frogmore, March 31; Columbia, April 6. North Carolina—Raleigh, April 10; Louisburg, April 17. Virginia—Lawrenceville, April 12; Bowers Hill, April 26. Louisiana—Bains, April 2; Bayou Sara, April 11; Baton Rouge, April 18; New Orleans, April 28. Mississippi—Biloxi, April 9; Jackson, April 10. Arkansas—Delight, April 10; Monticello, April 10; Fayetteville, April 14. Tennessee—Chattanooga, April 10; Knoxville, April 12; Belfast, April 18. Kentucky—Covington, April 7; Bowling Green, April 25. Missouri—Valley Park, April 18; Monteer, April 23. Texas—Refugio County, March 17; Corpus Christi, March 18; Austin, March 21; Kerrville, April 8. Oklahoma—Tulsa, April 20. Kansas—Manhattan, April 26; Elmdale, April 29.

Fall migration.—Data on the autumn movement are not plentiful, but late dates of departure are: Oklahoma—Canadian River, September 5. Texas—Grapevine, September 20; Brownsville, October 1; Corpus Christi, October 22. Arkansas—London, September 1; Delight, October 7. Mississippi—Bay St. Louis, September 25. Louisiana—New Orleans, September 21. Virginia—Lawrenceville, August 24. North Carolina—Louisburg, September 19; Raleigh, September 21. South Carolina—Summerton, September 23; Charleston, September 28. Georgia—Athens, September 6; Savannah, September 23. Florida—College Point, October 19; Pensacola, October 21; Punta Rossa, October 30.

Casual records.—Among records of this species north of its known breeding range are several for Maryland—one heard at North Beach on June 28, 1930; one heard at Clements on August 14, 1932; a mounted specimen in the collection of the Cambridge High School, taken at Fishing Creek sometime prior to 1933; and one recorded at Laurel on May 12, 1935. A specimen was taken at New Haven, Conn., on May 17, 1889; another was captured at East Boston, Mass., on October 13, 1915; one was killed at Pictou, Nova Scotia, on October 22, 1890; one was taken at

Dayton, Ohio, on May 1, 1933, one was collected on Point Pelee, Ontario, on May 19, 1906; one was taken at Indianapolis, Ind., during April or May 1908; and one was obtained at Sugar Creek, in southeastern Iowa, on June 17, 1933. There are several records for Kansas north of areas where it is known to breed, among them being a specimen collected at Wichita on June 12, 1898; one taken at Hamilton on April 30, 1912; and one obtained at Lawrence on May 4, 1935.

Egg dates.—Arkansas: 11 records, May 15 to June 26.

Florida: 53 records, March 7 to June 30; 27 records, March 20 to May 13, indicating the height of the season.

Georgia: 28 records, April 25 to June 18; 14 records, May 7 to 24.

Texas: 17 records, April 4 to June 16; 9 records, May 2 to June 5.

ANTROSTOMUS VOCIFERUS VOCIFERUS (Wilson)

EASTERN WHIPPOORWILL

PLATES 21–23

HABITS

CONTRIBUTED BY WINSOR MARRETT TYLER

Almost every man, woman, and child living in the wide breeding range of the whippoorwill knows the bird by name. Those who once hear it singing, reiterating its name perhaps a hundred times or more without a pause, cannot fail to realize that they are listening to a whippoorwill, but how many of this multitude who know the whippoorwill's name ever saw the bird, or would recognize it if they did see it? Not, it may be presumed, one-tenth of 1 percent.

Yet the whippoorwill lived many long years in denser obscurity still, for, playing a part behind the scenes, so to speak, its lines were ascribed to another actor in the play; it was not recognized as a bird at all until the early part of the last century. Prior to this time the whippoorwill was supposed to be nothing more than the voice of the nighthawk, and even now in many rural districts the two birds are not clearly distinguished from each other.

William Brewster (1895) says: "They are still very generally regarded by country people throughout New England as one and the same bird."

Spring.—The whippoorwill starts northward from central Florida in the latter part of March. This northerly movement evidently represents a general migration from the southern and eastern Gulf States, and through them from points farther south. The bird arrives in the latitude of Boston, Mass., late in April or early in May, thus flying a distance of a thousand miles or more in 35 or 40 days—a migration that corresponds closely, both in time of year and in speed of travel, with that of the chimney swift. Of this journey Wilson (1831) says:

In their migrations north, and on their return, they probably stop a day or two at some of their former stages, and do not advance in one continued flight. The whip-poor-will was first heard this season [1811] on the 2^d day of May, in a corner of Mr. Bartram's woods, not far from the house, and for two or three mornings after in the same place, where I also saw it. From this time until the beginning of September, there were none of these birds to be found within at least one mile of the place; though I frequently made search for them. On the 4th of September, the whip-poor-will was again heard for two evenings successively in the same part of the woods. I also heard several of them passing, within the same week, between dusk and nine o'clock at night, it being then clear moonlight. These repeated their notes three or four times, and were heard no more. It is highly probable that they migrate during the evening and night.

F. Seymour Hersey (1923) tells of a striking instance of nocturnal migration when a multitude of whippoorwills arrived suddenly at Lakeville, Mass., in the middle of the night.

In 1901 [he says], on the evening of May 4, about eight o'clock, a single bird was heard singing. This was the first arrival noted and no others were heard that evening. At two o'clock the following morning, six hours later, I was awakened by birds singing loudly everywhere. I dressed and went out and for more than an hour the chorus continued. There were numbers of birds about the house, on the door-step and ridge-pole, others singing in the road or from the stone walls along the road side, while still others could be heard down in the pastures,—often eight or ten were singing at the same instant. I walked down the road for half a mile and the birds seemed equally as abundant on neighbors' farms. It seems probable that the migration takes place at night as these birds had just arrived.

Courtship.—Few observers have had the good fortune to watch the sexual activities of the whippoorwill. One must be very near the birds to see, in the semidarkness, the courtship in detail, and even should we catch sight of a courting pair—a rare happening—we may get but a glimpse of their actions, because, if they flit only a little way back into the gloom, they are lost to view, fading into the shadows.

Frank Bolles (1912) tells of the following experience. He was hidden under a “narrow fringe of spirea bushes, 2½ ft. high only 3 ft. from the stone”—a stone on which a whippoorwill sang every evening. He says:

It uttered its note about twenty or thirty times when to my astonishment another whip, alighted near it, on the left (W.) end of the boulder. One or two sounds like the soft popping of corn came from the new arrival, and the first bird, which had ceased its call, faced west and began a strange, slow dance, advancing a step at a time towards its mate, raising its body to the full length of its legs at each step, thus making a sort of undulating approach. The other bird remained where it alit, but seemed to be moving its body up and down or else slowly pulsating its wings. The first bird, which I think was the male, seemed to continue its dance entirely around the female. As he passed her, indescribable purring and popping sounds were made and one of the birds flew lightly away—the ♀ I think. The male resumed his first position, and remained silent. Then he rose and circled in the air, catching an insect I thought, for he came back at once to the spot on the rock which he always covers. A moment later his mate seemed to call from below the house, near the lake, and he flew, his white feathers flashing as he spread his tail, and the strokes of his wings making a distinct and quite loud sound as he passed close above my head.

Henry K. Coale (1920) reporting the observation of his neighbor, Moritz Boehm, says:

On different occasions, while the male was calling, he saw the female going through some peculiar antics, but in the dusk could not make out just what she was doing. One evening, when he was sitting on the lower step, the birds came up and performed within ten feet of him. He kept perfectly quiet. The male called from a low branch overhead, while the female strutted on the gravel path below, with wings and tail outspread and head lowered, and sidestepped back and forth, half way around to the right, then to the left, all the time uttering a curious guttural chuckle. This performance was kept up for ten or fifteen minutes.

Bendire's (1895) account of the whippoorwill's courtship is the best in the literature; it has become almost a classic, and ornithologists still deplore the regrettable incident that interrupted the observation.

While on a collecting trip in Herkimer County, New York, with Dr. William L. Ralph, in June, 1893, I witnessed a most amusing performance, which one may see perhaps once in a lifetime. I happened to be in a little outbuilding, some 20 feet in the rear of the house at which we were stopping, early on the evening of the 24th, about half an hour after sundown, when I heard a peculiar, low, clucking noise outside, which was directly followed by the familiar call of “whip-poor-will.” * * * Directly alongside of the small outbuilding previously referred to, a barrel of sand and lime had been spilled, and from the numerous tracks of these birds, made by them nightly afterwards, it was evident that this spot was visited regularly, and was the trysting place of at least one pair. Looking through a small aperture, I saw one of the birds waddling about in a very excited manner over the sand-covered space, which was perhaps 2 by 3 feet square, and it was so much interested in its own performance that it did not notice me, although I made some noise trying to fight off a swarm of mosquitoes which assailed me from all sides. Its head appeared to be all mouth, and its notes were uttered so rapidly that, close as I was to the bird, they sounded like one long, continuous roll.

A few seconds after his first effort (it was the male) he was joined by his mate, and she at once commenced to respond with a peculiar, low, buzzing or grunting note, like “gaw-gaw-gaw,” undoubtedly a note of approval or endearment. This evidently cost her considerable effort; her head almost touched the ground while uttering it, her plumage was relaxed, and her whole body seemed to be in a violent tremble. The male in the meantime had sidled up to her and touched her bill with his, which made her move slightly to one side, but so slowly that he easily kept close alongside of her. These sidling movements were kept up for a minute or more each time; first one would move away, followed by the other, and then it was reversed; both were about equally bold and coy at the same time. Their entire love making looked exceedingly human, and the female acted as timid and bashful as many young maidens would when receiving the first declarations of their would-be lovers, while the lowering of her head might easily be interpreted as being done to hide her blushes. Just about the time I thought this courtship would reach its climax, a dog ran out of the house and caused both to take flight.

Nesting.—The whippoorwill lays its two eggs on dry, well-drained ground, generally near the edge of a wood of small mixed growth—oak, beech, pine—where the floor of the wood is clear of dense underbrush and where the trees are not crowded together, but spaced far enough apart to cast an uneven shade. The eggs may lie on the open floor or under a small bush—not tucked away near the stems, but out in the shadow of its branches. The bird builds no nest, although a slight depression about the eggs may result from the presence of the parent there during incubation; for concealment it relies solely on the soft colors of the fallen leaves and the flickering light of the woodland.

It is rare to find the eggs laid in a more open situation.

Lewis McI. Terrill, in a letter to Mr. Bent, gives in detail the results of remarkably close observations on the home life of a pair of whippoorwills and their brood. His observations were made near St. Lambert, Quebec, in 1933.

On May 14 Mr. Terrill came upon a pair of whippoorwills in a patch of deciduous trees, mainly young maple and birch. A week later he flushed the female “from a single egg lying on a bed of old leaves in a small glade” near the spot where he first saw the birds. “There was no depression whatever, and the egg appeared as if it had been casually dropped there.” The second egg was not laid until the 23d, indicating “that egg deposition takes place on alternate days.”

Invariably at his subsequent visits Mr. Terrill found the female incubating or brooding, but while the young birds remained in the vicinity of the nest he saw the male near it only once (June 20).

He says: "The male spent the day in a thicket over 400 yards away. I usually heard him singing from this direction in the early part of the evening; later he sang from a point nearer the nest; and finally from its immediate vicinity. I gather from this that he visited his family regularly at night.

"On May 27 the eggs were resting in a noticeable depression made by the pressure of the bird's body. One might almost call it a nest although no extraneous nesting material whatever had been added. The female was very consistent in her behavior, usually leaving the eggs when I was 10–15 feet away and flying to a dead branch 2 feet from the ground where she uttered a few protesting chucks, which resembled a call of the catbird and to a lesser extent the *chuck* of the hermit thrush.

"On the occasion of the male's visit (June 20) both birds were very worried, and their calls, especially that of the male, resembled the *whip* note of his song, although much subdued. He sometimes called *whip-will* when excited by the distress calls of the young.

"When returning to the nest the female frequently hovered before alighting, often dropping to the ground a few feet from the nest. Even at that short distance she would not attempt to walk onto the eggs, but would fly up again, hover, and then alight directly on the nest.

"The nighthawk, we may note, progresses differently. To be sure, it occasionally flies short distances when approaching the nest, but the final approach is by walking, or perhaps I should say creeping in a 'Charlie Chaplin-like' shuffle. The different methods of approach to the nest are, I think, indicative of the different habitats of two very similar birds. The woodland whippoorwill hops or flies to avoid obstructions, whereas the nighthawk can gain its objective without leaving the ground.

"The female whippoorwill was still incubating on June 10, but on the 11th there were two young in the nest. The incubation period for the last egg laid was, therefore, at least 19 days, and possibly nearer 20.

"The first definite movement away from the nest was noted on June 18, when the female was brooding the young 50 feet away. On the 19th and 20th she was respectively 70 and 85 feet from the nest. The female often

alighted crosswise on a limb when excited, or for the purpose of facing me, but quickly assumed the lengthwise position.

“June 21–22—Female brooding young 100 feet from nest.

“June 25—One young bird flew 15 feet.

“June 26—Older chick flew 26 feet when female was flushed. Younger bird still jumped, then squatted, but when I placed it on a branch, it flew 15 feet. Both young always alighted on the ground, but perched readily. This was the last I saw of the family.”

Mr. Terrill’s report of this family of whippoorwills makes it clear that the male parent very rarely came near the nest at the times he was watching it. This accords with the experience of many observers at other nests. For example, H. E. Tuttle (1911) says: “The male Whip-poor-will I saw only once, and that was after the young were fully grown. He was very conspicuous in the dusk as he sat on a log, uttering rasping sounds in his throat and opening and shutting his tail, brilliantly marked with white at the edges. It was only a day or so after seeing the male bird that I lost sight of the young birds altogether.”

But why should we expect the male whippoorwill to come to the nest in the middle of the night—the whippoorwill’s day? There is nothing to do there at night except to keep the eggs warm, or, after they hatch, to brood the young, and his mate can do that while she sleeps on the nest. So he sleeps a little way off. But when the dark comes—when his morning breaks—when the night insects begin to fly, and food abounds, and his hungry children cry, where is the male parent then? We do not know, but we may assume, as Mr. Terrill suggests, that he joins his family and aids in feeding the young.

When a female bird is approached while she is incubating (Bendire says: “I believe the female attends to this duty almost exclusively”) the behavior varies a good deal in different individuals. In many accounts of her actions, she is reported to flop about on the ground, seemingly trying to lead the intruder away. Wilson (1831) reports that “in traversing the woods one day in the early part of June, along the brow of a rocky declivity, a whippoorwill rose from my feet, and fluttered along, sometimes prostrating herself, and beating the ground with her wings, as if just expiring.” On the

other hand, H. E. Tuttle (1911) speaks of a bird, brooding young, which was “very fearless, allowing me to touch her back and making it necessary for me to shove her gently off the young when I wanted a glimpse of them.”

Arthur C. Bent, in his notes, says that late in May he “flushed a whippoorwill from near a woodland path, where it apparently had been roosting regularly as evidenced by its droppings.” A few days later, not 25 yards from the path, he “flushed the whippoorwill from the ground and saw its single egg lying on the flat, bare oak leaves.”

C. H. D. Clarke, writing to Mr. Bent from Ontario, Canada, points out how changes in the topography of a region may affect the local whippoorwill. He says: “The common denominator explaining the local distribution of this species is, I believe, to be found in its feeding and egg-laying habits. The whippoorwill feeds in the open, like the nighthawk, but unlike it, fairly near the ground. Although both birds lay their eggs on the ground, the nighthawk nests in the open, whereas the whippoorwill always nests among trees. Hence, as a breeding bird, it is found in glades and around the edges of woodlots. Many of the woodlots, however, in this vicinity are closely grazed by cattle at the present time, a condition that prevailed less commonly in the semipioneering stage of our country. The whippoorwill does not tolerate this change; it will not breed in the grazed woodlots and, as a consequence, has been reduced in numbers here. It also seems to avoid extensive areas of conifers, possibly because of the absence of hardwood litter on which to lay its eggs. The area at Frank’s Bay, in which the bird breeds very commonly, is a sand plain that was burned over about 25 years ago and has since grown up in many places to dense stands of poplar from 15 to 20 feet high. Here the whippoorwill has plenty of shelter in the dense poplar woods, an abundance of hardwood litter, and may cruise about over the treetops not far above ground.”

Eggs.—[AUTHOR’S NOTE: The two eggs of the whippoorwill are between oval and elliptical-oval in shape and become somewhat glossy when incubated. The ground color is usually pure white, but occasionally a faint creamy tint is perceptible. The markings consist of spots or small blotches of “pale Quaker drab” or “pallid Quaker drab,” scattered over the eggs more or less irregularly; an occasional egg has large, irregular blotches of this color. Overlying these pale gray markings, or scattered among them, are often many small spots or fine dots of various browns, such as “cinnamon-

brown,” “tawny,” or “tawny-olive.” An occasional egg is almost immaculate.

The measurements of 50 eggs average 29.0 by 21.3 millimeters; the eggs showing the four extremes measure **31.5** by 21.0, 30.48 by **22.86**, **20.48** by 21.34, and 28.45 by **20.07** millimeters.]

Young.—The little whippoorwill chick, hatching out from an invisible egg, finds itself lying on the ground, with dead leaves all about. The dead leaves look like the chick, and the chick looks like the dead leaves; no one can tell them apart; practically the chick is a dead leaf, and, although hatched, it is still invisible, just as it was when hidden in the egg.

Some birds depend on speed for safety, or on agility or strength, but the whippoorwill relies chiefly on not being seen. Safety comes to the whippoorwill in dim light, half shadows, and the faint, confusing obscurity of dusk, and among these, on the borderland of invisibility, the whippoorwill lives all its days.

Nests of the whippoorwill are found almost always by accident. The old bird starts up from near the observer’s feet, and a search—sometimes a long one—reveals the eggs or the young birds. For example, A. Dawes DuBois (1911) says:

The first nest was found on May 16, 1908, in a strip of woods of medium size trees, thickly undergrown, on a high bank of the Sangamon River [Illinois]. The ground was well carpeted with dried oak leaves. Our first intimation of Whippoorwills in this place was the sudden appearance of an adult bird fluttering along the ground in front of us, apparently with a broken wing. We stopped at once and while my companion stood to mark the place, I followed the bird a short distance. She fluttered along noiselessly, feigning serious injury and leading me away from the nest as rapidly as I could be induced to follow.

A search revealed the nest within a pace of the spot we had marked. It contained one egg and the broken shell of another which gave evidence of having hatched. Although I stooped to examine the broken shell I did not see the bird that had hatched from it until my companion called my attention to it. The little fellow was crouched, motionless, upon the brown leaves not six inches from the broken egg-shell.

H. E. Tuttle (1911) speaks thus of the young birds: “The newly hatched birds were very attractive-looking little chicks so long as they kept their mouths shut. They were a uniform buff color, which matched well with the leaves, and the instant their mother left them they each ran in opposite directions and squatted. In this maneuver the old bird seemed to aid them

materially by the vigorous flip which she gave them as she rose, often tumbling them over on their backs.”

J. G. Suthard writes to Mr. Bent from Muskegon, Mich., as follows: “On June 14, 1936, I flushed a whippoorwill from an oak-leafed spot on a steep hillside overlooking a large timbered swamp. I shortly discovered two downy young with their eyes only partly opened. They made no effort to escape and were silent when handled. The parent flew around several times, uttering a *whup-whup-whur* note, and then perched on a dead limb of a nearby tree. One of the eggshells was about 6 feet below the nest on the hillside, and the feces of the young had not been moved by the parents. As this nest was only about 30 yards from the main highway, I returned several times between this date and June 24 to see if, owing to my disturbance of the young, the parents would move them. Each time I visited the nest the parent was brooding the young in practically the same spot.”

Lewis McI. Terrill, in his study of nest life, quoted under “Nesting,” says that on June 12, before the young birds were two days old, “whenever the female was flushed, the nestlings hopped or jumped several inches with the suddenness and unexpected agility of ‘jumping beans,’ then squatted in hiding posture in the manner of woodcock chicks. The entire movement was so rapid that it almost escaped notice.” He continues:

“From the 13th to the 16th the female was brooding the young either in the nest or in the shade 2 or 3 feet away. On the latter date I heard one of the nestlings give a weak, complaining *whip*, which was answered by the mother 20 feet away. It attempted to follow her, progressing by little hops, but was in difficulty when it encountered heavy undergrowth where it was unable to hop. The smaller of the nestlings remained in the nest.

“June 26—The older bird when placed lengthwise on a limb quickly turned about and perched crosswise, demonstrating youth’s objection to slavish custom! The older bird now frequently used the *whip* note, which appears to be the chief motif in the whippoorwill vocabulary. The younger bird still called in wheezy tones that I readily imitated by sucking my finger—so well that the mother bird frequently responded by flying to me and fluttering at my feet. The young at this date, nearly 16 days old, closely resembled their parents.”

Terrill definitely established the incubation period of one of the eggs in the nest he observed as not less than 19 days, "and possibly nearer 20." Burns (1915) gives the incubation period as 17 days, and Audubon (1840) gives it as 14 days.

Plumages.—[AUTHOR'S NOTE: The downy young whippoorwill is thickly covered with long, soft, silky down, shading in color from "cinnamon" on the back to "pinkish cinnamon" on the chest, and to "light pinkish cinnamon" on the crown and abdomen; it matches the dead leaves on which it is hatched.

The juvenal plumage begins to grow at an early age. Ridgway (1914) says that the young male is "similar to the adult male in 'pattern' and coloration of lateral rectrices, as well as of primaries and primary coverts, but rest of plumage quite different, the wing-coverts and scapulars deep brownish buff or clay color, the former with coarse and irregular small spots of black, the latter with very large irregular spots of black, the under parts barred with dusky on a brownish buffy ground and, like most of the upper parts, without fine vermiculations, the pileum spotted instead of streaked with black, and the band across lower throat indistinct, more or less broken by dusky barring, and buffy instead of white." The young female, he says, is "similar to the young male, but three lateral rectrices broadly tipped with ochraceous-buffy instead of having a large white distal area."

A young bird in juvenal plumage, nearly grown, collected in Massachusetts in July, is like the young male described above, except that the feathers of the interscapular region and the median wing coverts are from "ochraceous-buff" to "light ochraceous-bluff," with a narrow shaft streak and a conspicuous subterminal small spot of black.

During July and August the juvenal contour plumage is shed, the juvenal wings and tail being retained, and a first winter plumage is acquired, in which the contour plumage closely resembles that of the adult. This is worn until the following summer, when a complete molt produces the fully adult plumage. Both young and old birds have a complete annual molt between July and September.]

Food.—The earliest report on the food of the whippoorwill is that of Wilson (1831), who was the first writer to show that the whippoorwill and the nighthawk are different birds. He says: "Their food appears to be large

moths, grasshoppers, pismires, and such insects as frequent the bark of old rotten and decaying timber. They are also expert in darting after winged insects.”

Knight (1908) puts the following items on the whippoorwill’s bill of fare: “Their diet,” he says, “would seem to be entirely insectivorous and among the various things I have known them to eat are Sphinx moths of various species, *Actias luna*, *Samia cecropia*, *Samia columbia*, *Telea polyphemus*, and a great variety of species of Noctuidae, also grasshoppers, crickets, mosquitoes, caddis flies, and in fact almost any sort of insect available.”

Bendire (1895) reports that “in the Western States, which are sometimes overrun by swarms of Rocky Mountain Locusts, it also feeds largely on these when abundant.”

Forbush (1927) tells of the whippoorwill the following story which will endear the bird to all mosquito-haters. He says: “While I slept unsheltered nightly for a week in the Concord woods, rolled in my blanket, with only a head-net hung to a branch overhead to protect me from mosquitoes, I noticed each morning upon awaking just before daylight that something fluttered softly about my head. The sound was like that produced by a large night-moth, but soon I heard something strike the ground a few feet away, and then a well-known cluck convinced me that my visitor was a Whippoorwill. The bird came nightly while I remained in the woods, and each morning before daylight it flew around my head-net until it had caught all the mosquitoes there.”

Eaton (1914) says: “I have taken 36 full-grown moths from the stomach of a single Whippoorwill which was killed early in the evening, indicating that within an hour and a half he had killed and devoured these full-grown moths, each one of which contained hundreds of eggs.”

Whippoorwills secure a large part of their food by capturing night-flying insects on the wing, but Ernest Ingersoll (1920) states that they also “have a way of balancing themselves near a tree-trunk or barn-wall, picking ants and other small provender off the bark; and even hunt for worms and beetles on the ground, turning over the leaves to root them out.”

Francis H. Allen (MS.) says: "One evening I saw one take off from the branch of an oak for what was probably its first feeding flight of the night. It opened its mouth wide *before* launching into the air."

Behavior.—In order to study the whippoorwill at short range it is well to visit its haunts for a few evenings and learn how the bird we are to watch behaves when it wakes from its day's sleep. Whippoorwills move about over a considerable territory when they come into the open for their daily session of singing and feeding, they follow a route, evening after evening, that varies little, and on the circuit there are stations—a stone wall, a low branch, or a certain spot on the ground—where they are almost sure to stop and sing for a while.

If we seat ourselves near one of these stations where the light, which will be almost gone when the bird arrives, will favor our view, and where a dark background will obscure us from the bird, we shall be able to see the whippoorwill at short range, for if we sit motionless (no easy task, for mosquitoes will torture us) the bird will pay little attention to us. We must sit quiet and wait, following the song as it swings around the circuit, and we must watch the spot where the bird is about to alight, for, although in flight it looms big even in the dusk, when it comes to rest, with a flip of wings it becomes a bit of dead wood, a clod of earth, or vanishes altogether.

On several evenings late in May 1914, at Wilton, N. H., I visited what appeared to be the whippoorwill headquarters—a dry wood of small deciduous growth bordering a sloping field, on one of which was a moist alder run that ran down to the edge of the wood. When I arrived, between sunset and dark, wood thrushes and veeries were singing, but before they quieted down for the night, the whippoorwills (from one bird to two or three) began to sing, always from the dry wood. They sang intermittently, and generally after each series of *whip-poor-wills* their voices came from a different part of the wood. By the time the light was becoming uncertain (when one would have difficulty in reading print) one bird, leaving the wood, worked up the slope, passing the field either by way of the alder run or by a wood of larger growth and an apple orchard that bordered the higher sides of the field.

On each of the first evenings when I visited the ground, one bird paused in the corner of the field where it joined the alder run, and sang a few times,

and on two of these evenings I was able to approach the bird but not near enough to see it. The next evening, therefore, as soon as the bird that was singing in the wood began to change his position, I retired to this corner of the field to await him and sat down on a bank where my figure would not show against the sky. That evening was unusually dark and cloudy. The bird left the wood by the lower side, and at 7:50 I heard the song coming nearer and nearer through the alders behind me. Then, two minutes later, it came with startling suddenness from almost at my side. The bird sat on the bare ground at the foot of the bank not 6 yards from where I sat. In bringing my glass to bear upon him, I disturbed him, I think, for he flew silently away. He alighted, however, on a rock and began to sing. He was now 12 yards from me and on a level with my eyes. His side was toward me, and he faced nearly in the direction from which he had just flown. He sat flat on the stone with his head thrown slightly backward and upward and, on alighting, immediately began to sing.

The song at close range sounded like *cuck-rhip-oor-ree*, the final note accented and held longer than the other three, although the *rhip* was louder and longer than the *oor*. The song was remarkably regular; twice, however, the bird increased the tempo, and once he doubled one note—either the *rhip* or the *oor*. After a pause the *cuck* was invariably the first note given when he continued his song.

Even in the dim light the band of white across the throat was clearly visible, and twice during each repetition of the *cuck-rhip-oor-ree* this band was drawn backward—slightly at the *cuck*, markedly during the final *ree*, when, I think, the beak was open wide. Later, when the bird more nearly faced me, these movements of the white band were less noticeable. The bird sat on the rock for three or four minutes, singing almost continuously. He sat absolutely still for the most part, but twice he moved backward about an inch, as if each time he took a single backward step. His departure, with no apparent cause, was noiseless and abrupt, breaking the song at *oor*.

F. Seymour Hersey (1923), who watched with great care a whippoorwill making its nightly round, says: “The time taken to make this circuit varied from 25 to 30 minutes. I watched this bird from several places of concealment and ascertained to my satisfaction that it was the same individual that visited each of these places and that the order given above was not varied. The spot from which he sang was, in all cases, nearly the

same, i. e., within a very few feet of the place where he was seen on a previous evening.”

Frank Bolles (1912) gives a remarkable picture, seen from almost within arm’s reach, of a singing whippoorwill. Mr. Bolles, who was hidden near a stone to which the whippoorwill came nightly, says:

Suddenly I hear a rather feeble whip, 12 times *S.* of me, then silence and then a bird flies to the stone in front of my face, coming low over the bushes and alighting with its tail towards me. It squeaks or clicks three times, and I fear it suspects me and is giving a slight alarm note, but the next moment it begins the piercing *quip o’rip* slightly raising its head and dipping its tail each time it makes the sound. The head rises on the *quip* and falls on the *rip*. The wings do not move, nor the body save by a slight tipping. I could see the bird’s outline perfectly against the white background of the shingled barn on which the moonlight fell fully.

When the whippoorwill comes out in the dusk for its evening round, alighting on a stone wall, on the ground, or on a big horizontal branch high in a tall tree, we may sometimes catch sight of it against the sky, as it flies from one station to another. In the air the whippoorwill does not resemble the nighthawk at all. Its wings are broad and, compared to those of the nighthawk, short, and it moves them with an easy sweep, with none of the nighthawk’s jerkiness. When we see it flying steadily across an open field, it suggests an owl moving through the gloom on its broad, silent wings.

Taverner and Swales (1907) give a remarkable description of the flight of a whippoorwill seen under such circumstances at Point Pelee, Ontario. They say:

One evening, just as the dusk was darkening into night, a Whip-poor-will was heard near the camp. We stole out, and the bird was located in a large bare walnut tree in the open bush where, looking up against the still faintly illuminated sky, it could be plainly made out, sitting lengthwise, as is their fashion, on a rather large and almost horizontal branch. It remained perfectly motionless except for an occasional jerk of its white blotched tail, when it gave vent intermittently to a guttural “gluck.” These notes were repeated at irregular intervals of perhaps half a minute, several times and then, without start or warning, it launched away into the air, starting off immediately at full speed, with a drop that carried it in a large, even circle half way to the ground, and then up on the same curve, to vanish in the gloom of the trees. Then it appeared on the other side, swinging down on fixed wings in great elliptical curves as though whirled from the end of a cord, perfectly silent in flight and threading the dusky mazes of the tree tops with the utmost confidence and precision. Here and there it rapidly wheeled, without an apparent stroke of the wing, now coming into view in the lower arc of its great circling, and then vanishing silently again on the upward sweep on the other side. As suddenly as it started, it ceased in the middle of a swing and, while the eyes vainly searched for the dark object along the continuation of its course, it was seated again on the branch from which it first sprang, silent and still. This was repeated several times, and then it was joined by another, and the two circled about like

great soft, gliding bats until the sky above grew so dark that their movements could no longer be watched.

Several writers mention the fearlessness of the whippoorwill, or perhaps its failure to recognize man as a danger. For example, Bendire (1895) quotes E. A. McIlhenny, who says: "These birds are very tame, for on two occasions, while sitting still in the twilight to observe the movements of some Owls, I have had them come so close that I could have caught them. On one occasion one lit on my knee, and another on my foot as it was extended before me." And H. E. Tuttle (1911) says: "Once I watched two males fighting and singing at intervals on a fallen birch sapling. I was quite close to them,—within a yard—but they did not seem to regard me as dangerous, and when I tried to imitate the guttural noises they were making, they circled round my head so closely that one touched me with his wings. In the darkness I was probably no more than a charred stump."

C. W. G. Eifrig (1919) mentions "a unique experience" with a whippoorwill, which, displaying unexpected aggressiveness, darted repeatedly at his head.

It has been surmised that the whippoorwill uses its capacious mouth to carry its eggs, and even its young, out of danger when its nest has been discovered. There is no satisfactory evidence that the bird employs its mouth in this way, but it has been seen, on two occasions at least, carrying a young bird through the air held between its legs. J. H. Bowles (1895) says: "I flushed a whippoorwill that rose with a baby bird clutched firmly *between her thighs*," and Bendire (1895) quotes H. W. Flint as follows: "I once, and once only, saw a female (the male is never present at the nest) carry a young bird about a rod, but can not say she used her bill, and don't think she did, but I am almost sure the claws and legs only were used, as the young was hugged close to the body."

The whippoorwill is fond of taking dust baths. When driving after dark we sometimes catch sight of one as it starts up from its bath on a country road, and, as it flies off and our headlights pick it up, the white tail feathers, if the bird is a male, shine out for an instant. Forbush (1927) says: "Mr. Stanley H. Bromley of Southbridge, Massachusetts, tells me that a farmer there placed a large tray of dry wood ashes on the ground, and whippoorwills came there at night to dust in it."

Wilson (1831) states: "The inner edge of the middle claw is pectinated, and, from the circumstances of its being frequently found with small portions of down adhering to the teeth, is probably employed as a comb to rid the plumage of its head of vermin."

Voice.—If the whippoorwill "should sing by day, when every goose is cackling," the song might lose some of its witchery; we do not know; the bird sings in the dark, or when darkness is coming on fast, and the singer is invisible or almost invisible among the shadows. The song at a little distance comes to the ear as a penetrating whisper of the bird's name, repeated perfectly regularly, time after time with scarcely a pause between, at a rather rapid rate—about once a second. The fourth note, a *cluck* before the *whip-poor-will*, is heard usually only when the bird is fairly near us, although we may hear it at a distance of 200 yards under favorable circumstances. The syllable *will* carries farthest of all the syllables.

It is rare to hear any material variation in the song, but there are individual birds that regularly sing an unusual form, and sometimes a bird will introduce occasionally one abnormal phrase into his singing.

Simeon Pease Cheney (1891), speaking from the point of view of music, says: "In the courageous repetition of his name he accents the first and last syllables, the last most; always measuring his song with the same rhythm, while very considerably varying the melody—which latter fact is discovered only by most careful attention. Plain, simple, and stereotyped as his song appears, marked variations are introduced in the course of it. The whippoorwill uses nearly all the intervals in the natural scale, even the octave. I have never detected a chromatic tone." Describing altercations between two or more birds, he says:

These altercations are sometimes very amusing. Three whippoorwills, two males and a female, indulged in them for several evenings one season, in my garden. They came just at dark, and very soon a spirited contest began. Frequently they flew directly upward, one at a time. Occasionally one flew down into the patch near me, put out his wings, opened his big mouth, and hissed like a goose disturbed in the dark. But, the most peculiar, the astonishing feature of the contention was the *finale*. Toward the close of the trial of speed and power, the unwieldy name was dropped, and they rattled on freely with the same rhythm that the name would have required, alternating in their rushing triplets, going faster and faster, louder and louder, to the end.

The bird is remarkable on account of the regularity of its song and the great number of times it repeats the *whip-poor-will* without a pause. From

50 to 100 repetitions are not uncommon. Forbush (1927) says: "John Burroughs, however, made a count which so far as I know exceeds all others. He records that he heard a bird 'lay upon the back of poor will' 1088 blows with only a rarely perceptible pause here and there, as if to take breath."

F. Seymour Hersey (1923) writes: "The Whip-poor-will sings most continuously from dusk till about 9.30 p. m. and from 2.00 a. m. till dawn. During the intervening hours only an occasional song is heard. The song season lasts from their arrival in spring until late July or early August. Then there is a marked falling off in the number of singing birds heard until toward the end of August or early in September an increase in the number of singers is again noted. The songs of these late birds often lack the energy that characterizes the spring performance but a good many continue to sing until they leave for the south. My latest singing bird was noted September 24, 1901." These dates refer to eastern Massachusetts.

Of the possibility of the female singing, he says: "June 15 a Whip-poor-will alighted on the fence and uttered its 'chuck' note, which usually precedes the regular song, repeating it a number of times but not giving a note of the usual 'Whip-poor-will' call. It also did the same while on the wing. This bird was supposed to be a female as no conspicuous light area was visible on the tail. If so, she was capable of singing the same as the male for I later heard and saw her sing, both from the fence and while on the ground in the middle of the road. She finally flew and was followed by another bird which may have been her mate."

Of the whippoorwill's minor notes, we have seen above that the growling *gr-gr-gr* or *gaw-gaw-gaw* is presumably associated with courtship. I have never heard the note except when two (or more) birds were together, on or near the ground. This note suggests a little a note of the female woodcock, which is used under similar circumstances. The whirring *whup-whup-whirr* is evidently an alarm note.

A. Dawes DuBois (1911) mentions two other notes, evidently of alarm. He says: "She fluttered from the spot as she had done the previous day, but this time uttering a very low hissing or 'soughing' sound," and again, "She kept vigilant watch, however, at a short distance, moving about near the

ground with a remarkably noiseless flight but uttering a ‘chip’ or ‘whit’ similar to that of a domestic chick.”

If we are outdoors at the end of the day, when the sun has gone down and all the ways are darkened, if we are walking along a quiet country road fringed by woods and open fields, or, in a canoe, are drifting down a stream flowing slowly past farmland—pastures, stone walls, orchards—and if we listen, what do we hear? If it is summer, the bird songs are gradually fading away as the birds fall asleep, the robin chorus lessens when the light grows dim, and when it is almost dark a field sparrow may sing for the last time before night comes. If it is autumn we hear little bird song, only a short period of chipping and clucking before the birds settle for the night, and after that only the insects that will sing the night through. But let us listen. Was that a whippoorwill? Do we hear a whippoorwill, or do we imagine we hear one because the scene has changed to a world of shadows—the whippoorwill’s world—and association has brought the bird to our mind, and its song has come to our ears? The song is faint and comes from far away. Perhaps we did not hear it; perhaps there was no song to hear.

This is a peculiarity of the whippoorwill’s song; it is so bound up with association that we are sometimes misled. It is the same with the bluebird when we listen for its song over the brown fields of March.

Field marks.—The whippoorwill and the nighthawk appear very much alike when sitting either on the ground or along a horizontal branch of a tree, for in such situations it is difficult to see the points where the two birds differ. The whippoorwill is bristly about the mouth; the nighthawk is not. The tips of the whippoorwill’s folded wings do not come to the end of the rounded tail, whereas the nighthawk’s wings project beyond the forked tail. The whippoorwill has a narrow line of white on the upper breast. The corresponding mark on the nighthawk is broader and includes the throat. Perhaps the best mark for diagnosis is the pale, barred sides of the nighthawk. For purposes of field identification this part of the whippoorwill may be said to be unbarred.

In the air the distinguishing mark of the nighthawk is a conspicuous spot of white in the wing. The whippoorwill lacks this mark. The flight of the two birds (see above) is very different and identifies them at a glance.

The chuck-will's-widow, although similar to the whippoorwill in plumage, is a much larger bird.

Enemies.—The clearing away of a large part of the North American wilderness during the past two centuries or so has not materially affected the whippoorwill; it drove the bird back from the settlements a little way, farther and farther as the towns grew in extent and became the great cities of today, but at the present time, not far beyond the city limits, whippoorwills find miles and miles of country wild and secluded enough for their breeding purposes. Fifteen miles from the city of Boston, Mass., for example, as well as within a mile or two of many small towns in the State, the bird is still abundant, nesting on the dry wooded ridges and eskers.

This ability to flourish as man advanced into the country, when so many birds failed to hold their own, may be accounted for by the habits and equipment of the whippoorwill, which, when it moves about, is “bescreened in night” and is so obscurely colored that we may say “the mask of night is on its face” even in the daytime, as it lies motionless on a carpet of dead leaves.

If the bird should be discovered and attacked, we may imagine how often the whippoorwill, with its marvelous powers of flight, may escape hawk, owl, or fox.

Fall.—We rarely see whippoorwills in autumn, but as we hear them sing not infrequently at this season we know that they sometimes linger in New England almost to the end of September, a time when hard frosts are at hand, which will either kill the insects or hasten them into retirement.

Taverner and Swales (1907) report an unusual gathering of whippoorwills on Point Pelee, Ontario. They say: “In our various September visits we have usually found them more or less common, but at that season they are much quieter, and seldom do more than call a few times in the early evening and then cease. Sometimes one will be heard again through the night, but more often not. September, 1905, beginning the 4th, we saw from one to six until the 13th, when a great flight of them appeared on the Point. That day, in the red cedar thickets near the extremity of the Point, we flushed thirty between twelve and half-past one in the afternoon.”

Winter.—George Nelson, who has known the whippoorwill for years in its winter quarters on the east coast of Florida, tells me that the bird is pretty evenly distributed in the country about Sebastian, frequenting chiefly the ridges and hammocks where, during the day, it rests on the ground or on the trunk of a fallen tree. Not infrequently, as Mr. Nelson has been driving after dark along U. S. Route 1, a bird has started up from almost beneath the wheels of his car and has flown off in the glare of the headlights. He says that the bird is not in song during winter, but just before it starts northward, late in March, it sings for a few evenings, and that its departure invariably coincides with the arrival of the chuck-will's-widow.

Each evening during my stay at Sebastian with Mr. Nelson in mid-February 1931, just as it was beginning to grow dark, a whippoorwill appeared in the dooryard, a clearing in a dense hammock on the shore of the Indian River. The bird perched lengthwise here and there on the thick limbs of a live oak, well up in the big tree, but clearly visible from the ground, and made frequent sallies out into the air, sometimes sweeping clear away from the tree, sometimes only flitting among its branches, returning either to the perch from which it had flown, or to another one. Presumably these sallies were made in pursuit of flying insects—there was a businesslike air in the bird's behavior—but there was no sound of any snapping of the beak audible to me as I stood near the foot of the tree. Our first intimation that the bird had arrived from its day's seclusion was the sound of a low *chuck* repeated at short intervals. The bird gave this note from its perch and from the air; it was very similar to the introductory note in the whippoorwill song, but a little sharper. As the bird flew about, it sailed a good deal, wheeling around with some tilting from one side to the other, the wings held out straight and flat from the body with no, or very little, bend at the wrist joint. I was strangely reminded of the flight of a shearwater—the whippoorwill seeming to avoid the branches as the shearwater avoids the tops of the waves, tilting over them as it sails.

This was when the bird was moving slowly, but at times it increased its speed and executed the most intricate maneuvers, appearing and disappearing among the branches, ever changing its direction, either sailing or flapping its wings, swerving sharply from side to side, heeling over till one wing pointed nearly to the zenith and the other to the earth, then snapping back to an even keel. It shot straight upward, dived head

downward, and doubled back, twisting and gyrating with such rapidity that it seemed to be tumbling about in the air. The turns were so quick and the pace so reckless that the bird appeared in a frenzy and in danger of dashing itself against a limb of the tree, yet from the midst of these complicated evolutions it instantly righted itself and, with a flash of wings, settled flat and motionless on its perch.

Although there was the appearance of a lack of caution in these mad dashes among the network of branches, we were convinced, as we watched, that the bird governed its movements with perfect precision, with the acme of coordination.

The flight seemed silent; even when the bird passed within a few feet of my head, I heard no sound. It appeared to be alone, and after remaining for ten minutes or so, it flew off, and we heard or saw no more of it until the next evening.

When we flashed a light on it, the eye gleamed back a bright orange.

Mr. Nelson said that earlier in the winter the behavior of the bird had been different. It came about the house every evening for a while, visiting a small tree (*Assonia*) to which insects were attracted by big clusters of open flowers. This tree was about 10 feet tall, with large leaves but plenty of open space between the branches. The bird went to the ground after each flight into the tree, and it appeared to Mr. Nelson that the insects, as they flew among the flowers, could best be seen against the sky from this point. The bird did not return to this tree after the flowers had faded.

During a second visit to Florida, more than a month later, I saw, presumably, the same bird again. It acted exactly as it had before, perching, indeed, on the identical spot on the limb of the live oak that had been a favorite perch in February. On this occasion also only one bird visited the tree, and while feeding was silent except for the low chuck. On March 24, after the bird had been to the tree and had gone away, I heard him singing off in the hammock. This singing on his winter quarters indicated that he felt spring was here, and it was time to leave for his summer home in the north.

DISTRIBUTION

Range.—United States and southern Canada east of the Rocky Mountains, south in winter to El Salvador.

Breeding range.—The breeding range of the eastern whippoorwill extends **north** rarely to central Saskatchewan (Prince Albert); southern Manitoba (Gypsumville, Lake St. Martin, Shoal Lake, and Winnipeg); northern Michigan (McMillan and Sault Ste. Marie); southern Ontario (Sudbury, Algonquin Park, and Ottawa); southern Quebec (Montreal and Sherbrooke); rarely northern Maine (Presque Isle); and rarely southern New Brunswick (Scotch Lake). The **eastern** limits extend southward from this point along the Atlantic coast to eastern Virginia (Ashland and Lawrenceville); hence southwest through the interior to North Carolina (Raleigh and Highrock); and Georgia (Young Harris and Atlanta). **South** to northern Georgia (Atlanta); northern Alabama (Lookout Mountain and Sand Mountain); central Arkansas (Clinto and Big Piney Creek); and Texas (Troup). **West** to eastern Texas (Troup); northwestern Arkansas (Rogers and Pearidge); eastern Kansas (Ottawa and Topeka); eastern Nebraska (Peru, Omaha, and Neligh); southeastern South Dakota (Vermillion); Minnesota (St. Cloud, Fosston, and Williams); southwestern Manitoba (Treesbank); and rarely central Saskatchewan (Prince Albert).

Closely related subspecies, Stephens's whippoorwill (*A. v. arizonae*), is found chiefly in Mexico, but in summer it has been recorded **north** to Arizona (Bradshaw Mountains, Sierra Ancha, Mount Graham, and the Chiricahua Mountains); New Mexico (Turkey Creek, Blank Range, and Fort Bayard); and southwestern Texas (Chisos Mountains).

Winter range.—The winter range of the *entire* species is **north** to southern Sonora (Alamos); southern Texas (San Patricio and Port Arthur); rarely Louisiana (Baton Rouge); southern Alabama (Fairhope); northwestern Florida (Pensacola and St. Marks); and east-central South Carolina (Mount Pleasant). **East** to east-central South Carolina (Mount Pleasant); Georgia (Savannah); eastern Florida (Amelia Island, Orlando, Lake Worth, and Royal Palm Hammock); British Honduras (Toledo District); and Costa Rica (probably San Jose). **South** to Costa Rica (probably San Jose and Puntarenas); El Salvador (Rio San Miguel, Puerto del Triunfo, and La Libertad); Oaxaca (Tehuantepec); and southern Jalisco (Colima Volcano). **West** to Jalisco (Colima Volcano and Bolanos Volcano); probably western Durango (Salto); and southern Sonora (Alamos).

Spring migration.—Early dates of spring arrival are: North Carolina—Louisburg, March 19; Weaverville, March 23; Raleigh, March 28. Virginia—Variety Mills, March 29; Lawrenceville, March 30. District of Columbia—Washington, April 1. Maryland—Mardela Springs, March 29; Cambridge, April 12. Pennsylvania—Williamsport, April 21; Berwyn, April 22; Philadelphia, April 22. New Jersey—Vineland, April 11; Morristown, April 23. New York—Jay, April 14; Geneva, April 25; Rochester, April 29. Connecticut—Jewett City, April 23; Hartford, April 23. Massachusetts—Boston, April 13; Taunton, April 20; Wilmington, April 26. Vermont—Wells River, April 25; St. Johnsbury, May 5; Rutland, May 7. New Hampshire—Charlestown, April 26; Tilton, May 2. Maine—Portland, April 26; Phillips, May 2. Quebec—Montreal, April 25. New Brunswick—Scotch Lake, May 7. Mississippi—Suffolk, March 22. Arkansas—Delight, March 29; Clinton, March 30. Tennessee—Athens, March 24. Kentucky—Eubank, April 2; Bowling Green, April 6. Missouri—Palmyra, April 9; St. Louis, April 13. Illinois—Chicago, April 13; Port Byron, April 23. Indiana—Fort Wayne, April 11; Bicknell, April 12; Lafayette, April 16. Ohio—Columbus, April 13; Wauseon, April 16; Youngstown, April 18. Michigan—Detroit, April 15; Sault Ste. Marie, May 3. Ontario—Ottawa, April 25; Toronto, April 28. Iowa—Keokuk, April 11; Sabula, April 16; Iowa City, April 18. Wisconsin—Milwaukee, April 21; Madison, April 26. Minnesota—Lanesboro, April 17; Elk River, April 24; St. Vincent, May 11. Texas—Grapevine, April 7. Kansas—Onaga, April 19; Topeka, April 20. North Dakota—Larimore, May 20. Manitoba—Aweme, May 1; Raeburn, May 1.

Fall migration.—Late dates of fall departure are: Manitoba—Aweme, September 27. Minnesota—Elk River, September 23; Lanesboro, October 2. Iowa—Keokuk, October 2; Sabula, October 4. Missouri—Concordia, October 3; Palmyra, October 12. Ontario—Ottawa, October 16; Point Pelee, October 17. Michigan—Sault Ste. Marie, September 23; Detroit, October 6. Ohio—Wauseon, September 20. Indiana—Bicknell, September 21; Fort Wayne, September 22. Illinois—Port Byron, September 19. Kentucky—Eubank, October 4. Tennessee—Athens, October 3. Arkansas—Delight, October 4. New Brunswick—Scotch Lake, September 8. Maine—Phillips, October 6. New Hampshire—Ossipee, October 2. Vermont—Wells River, September 14; Rutland, September 21. Massachusetts—Harvard, October 10. Connecticut—Hartford, October 18. New Jersey—Morristown, October 10. Pennsylvania—Berwyn, October 5; Philadelphia, October 11. District of

Columbia—Washington, October 13. West Virginia—French Creek, October 2. North Carolina—Raleigh, November 6.

Casual records.—The whippoorwill has been recorded, either by its call or by the collection of specimens, at several points outside its normal range. One was taken at Port Collins, Colo., on September 14, 1903, and one was heard at Eastend, in southwestern Saskatchewan, on August 29, 1919. In North Dakota, two specimens have been obtained at Grafton, one on October 9, 1923, and the other on May 24, 1924, while it was recorded at Stump Lake in the spring of 1910 and also at Cando. It has been noted occasionally near Wichita, Kans., and one was noted at Harper, Kans., on May 1, 1936. During the period July 5 to September 22, 1905, one was heard calling on Isle Royale, Mich., and in the summer of 1922 one was heard on the Cascapedia River, Gaspé Peninsula, Quebec. In Puerto Rico a specimen was collected for Cory, probably in 1888, and Wetmore reported seeing a bird that he felt certain was this species on December 23, 1911, near the experimental station at Rio Piedras.

Egg dates.—Arizona: 12 records, May 3 to August 8; 6 records, May 22 to June 12, indicating the height of the season.

Connecticut: 9 records, May 20 to June 18.

Illinois: 39 records, May 9 to July 26; 20 records, May 18 to June 17.

West Virginia: 40 records, May 6 to July 7; 20 records, May 15 to June 3.

ANTROSTOMUS VOCIFERUS ARIZONAE Brewster

STEPHENS'S WHIPPOORWILL

PLATE 24

HABITS

This southwestern race of the whippoorwill is found in southern Arizona and New Mexico, in southwestern Texas, and southward through the mountains of northern Mexico, being replaced by another subspecies farther south. Although it is common enough in some of the Arizona mountain ranges, it is oftener heard than seen. It sits so closely, is so inactive during

the day, and is so protectively colored that it is easily overlooked. The use of a flashlight at night will often enable one to locate one of these birds by the “eye-shine,” even at considerable distance.

William Brewster (1881), in naming this subspecies, described it as “generally similar to *A. vociferus* but much larger; with the rictal bristles considerably longer; the gular crescent and a pretty well defined superciliary stripe, ochraceous; the lores and auriculars tawny ochraceous. The white of the tail barely tipping the outer web of the lateral tail feathers and on the others confined to a narrow apical space; the under tail-coverts nearly without barring.”

Mr. Brewster’s type came from the Chiricahua Mountains, Ariz., where they were evidently common, for Frank Stephens, the collector, wrote to him: “I have heard several of these Whip-poor-wills singing at one time and am told that they were heard here last year. I hear *A. nuttalli* every evening. They keep high up the mountain sides, while *A. vociferus* affects the lower part of the cañons.”

Harry S. Swarth (1904) says that this whippoorwill is a “fairly abundant summer resident” in the Huachuca Mountains, Ariz., “occurring principally between 5000 and 8000 feet; they may occasionally occur at a little higher elevation, but I have never seen any below the lowest altitude given.”

A. J. van Rossem (1936) writes:

The vertical range of Stephens’s Whip-poor-will is not limited to the higher mountains. Above 6000 feet in the Santa Ritas many birds were heard on every occasion that we stayed out after dark, but I also heard a whip-poor-will several times at 5000 feet in the Atascos and Dr. Miller collected a specimen at Peña Blanca Spring, Pajarito Mountains, in June, 1931. These two localities are well down in the Upper Sonora Zone.

I took, all told, six specimens of this common, though seldom collected, whip-poor-will and saw or heard several times that number. In the Santa Ritas they showed a decided preference for groves of oaks and sycamores in the cañon bottoms, and nearly all of those which were found at night were feeding in the immediate vicinity of running water. A pair to the mile seemed to be normal in most cañons which contained water and it was obvious that each pair had its own territory.

Spring.—Mr. Swarth (1904) saw the first arrival in 1903 “on April 28, and soon after their notes could be heard every evening, usually from some thickly wooded hillside, near the bottom of the canyon.” But Mr. Stephens wrote to Mr. Brewster (1881): “I heard the first Whip-poor-will about the

middle of May. By June 1, they were as common as I ever knew them to be in the East. Sometimes I could hear three or four whistling at once."

Nesting.—We did not succeed in finding a nest of Stephens's whippoorwill in the Huachuca Mountains, but my companion, Frank C. Willard, had previously found one there on May 24, 1899; the nest was on the ground at the base of a bush, and the eggs were lying on a few dead leaves; it was at an elevation of about 6,000 feet.

Mr. Stephens wrote to Major Bendire (1895): "The locality where I found the egg was a gulch near the summit of the Chiricahua Mountains, in a thick forest of yellow pine. The nest, if it can be called so, was a slight depression scratched in the ground, under the edge of a boulder."

Dr. A. K. Fisher sent Bendire the following notes:

The Whip-poor-will's note was not heard at Fort Bowie, Arizona, during the last three weeks of May, 1894. When we made camp at the mouth of Rucker Canyon, some forty miles south of the Post, in the Chiricahua Mountains, on the last day of the month, we heard a few, and a couple of days later found the species abundant higher up in the same canyon, among the pines (*P. ponderosa*). Here at early dusk and at dawn their notes were heard almost continuously, and numbers of birds were seen. On June 5 Mr. Fred. Hall Fowler found a nest, if the slight depression in the ground can be so designated, on a steep side hill about 50 feet above the stream. It was situated under an overhanging bush at the edge of a flat rock, and contained two young, recently hatched, and the fragments of egg shells from which they had emerged.

Mr. Fowler wrote to Bendire concerning the same nest: "The eggs were deposited on a bed of oak leaves by the side of a large rock; there was no nest excepting the bare leaves, which had been hollowed out slightly."

Mr. van Rossem (1936) found this whippoorwill nesting in the Santa Rita Mountains, Ariz., of which he writes:

Though males, and sometimes, before eggs were laid, mated pairs, were invariably found in the cañon beds, the two nests discovered were on hillsides at least a quarter of a mile from water. On the night of June 6, 1931, I caught the red eye-shine of a whippoorwill some distance away (estimated by daylight at 150 yards) and across a steep-banked cañon. With no expectation of collecting anything I followed the trail to a point where I estimated the shine to have been, but could locate nothing and supposed that the bird had gone. On my return to the original spot the eye was seen in the same location as before, and this time, after a little search, I found a female sitting under the protection of a fallen spray of leafless twigs which lay on a steep bank beside the trail. Three of us had passed within five or six feet of this sitting bird on two occasions the day before. The sitting bird made no effort to escape and was picked up by hand. There was one nearly fresh egg in the shallow depression in the gravelly soil which served as a nest, and stuck to the ventral plumage of

the incubating female were several small pieces of shell, showing that another egg had been laid and somehow broken.

Eggs.—The two eggs laid by Stephens's whippoorwill have been said to be pure, immaculate white, but this is not always the case. Mr. Brewster (1882) says of the egg sent to him by Mr. Stephens: "The egg is white with a dull gloss. At first sight it appears to be immaculate, but a closer inspection reveals a few faint blotches of the palest possible purple, so faint indeed that they might pass for superficial stains were it not for the fact that they underlie the external polish." This egg measures 29.72 by 22.10 millimeters.

Mr. van Rossem (1936) says of the egg referred to above: "This single egg was by no means immaculate white, but was clouded and mottled with brown and lilac, mostly in the nature of semi-concealed shell markings. It was similar to but very much less highly colored than eggs of the eastern *vociferus*, however." Another nest, found by him later, "contained one pure white egg and a newly hatched chick."

Philo W. Smith, Jr. (1900), received two sets of eggs, taken by O. C. Poling in the Huachuca Mountains, which "very much resemble in shape and color sets of the common Poor Will in his collection, being possibly a trifle larger, and one egg of each set has a few almost imperceptible pinkish spots on one end, the other egg in each set being unspotted."

There is a set of two eggs in the Thayer collection, taken in the Chiricahua Mountains, Ariz., on June 6, 1904, by Virgil W. Owen, that are decidedly spotted. These eggs are oval and only moderately glossy. The ground color is pure white, and both eggs are finely and irregularly marked with small spots and minute dots of "pale Quaker drab," "pallid Quaker drab," and very pale "clay color."

The measurements of 29 eggs average 28.8 by 20.8 millimeters; the eggs showing the four extremes measure **30.9** by 21.8, 29.0 by **22.6**, **25.8** by 20.1, and 28.9 by **19.9** millimeters.

Plumages.—I have not seen the downy young of this subspecies, but the two newly hatched chicks, taken by Mr. Fowler, are described as "covered with light brown down, and were not more than 1½ inches long." Mr. van Rossem (1936) describes a newly hatched chick as "clothed with a very

respectable covering of down—in color between ‘cinnamon’ and ‘orange-cinnamon’ of Ridgway.”

I have not seen enough material to work out the molts of this race, but I suppose that they are not very different from those of the eastern whippoorwill. Mr. Brewster’s type was a male, which is described above. He later received an adult female from Mr. Stephens, of which he says (1882): “This specimen differs even more widely from the female, than does my type from the male of *A. vociferus*. The ochraceous of the lores, superciliary-stripe, and neck-collar, spreads over the entire plumage both above and beneath, giving it a tawny tinge which overlies and obscures the usual dark markings. On the shoulders, breast, lores and throat this color deepens to a fine reddish-chestnut, and elsewhere it replaces the ashy, dirty white and other light tints of the eastern birds. In its general coloring the plumage strikingly resembles that of the brown phase of *Scops asio kennicotti*.”

Mrs. Florence M. Bailey (1928) says that the young male is “similar to adult male but top of head spotted instead of streaked with black, throat band indistinct, wing coverts and scapulars broadly barred with dusky, and irregularly marked with black; underparts barred with dusky on a brownish buffy ground. *Young female*: Similar to young male but outer tail feathers tipped with brownish buff instead of white.”

Behavior.—The food, feeding habits, and general behavior of Stephens’s whippoorwill are apparently similar to those of its eastern relative. Mr. van Rossem noticed a peculiarity that I have not seen in the eastern bird, of which he writes: “A, to me, surprising circumstance, was the marked erectability of the feathers above the eyes. Both of the sitting females carried these tufts constantly erect the entire time they were under observation. A male seen from directly in front alternately raised and flattened them. On one previous occasion, when night hunting in El Salvador, I had observed the eastern subspecies (*vociferus*) to have markedly erectile tufts; in fact until I picked the bird up I was certain that I had shot some small ‘eared’ owl. About 45° from the horizontal was the maximum elevation, though when viewed from directly in front the tufts appear nearly vertical.”

Fall.—Mr. Swarth (1904) says that, in the Huachucas, “they seem to remain rather late in the fall, as at the end of August their notes were heard as frequently as ever, and I have a female taken by H. Kimball on September 29, 1895. An adult male secured on August 29, 1902, had not yet quite completed its moult.”

PHALAENOPTILUS NUTTALLI NUTTALLI (Audubon)

NUTTALL'S POORWILL

HABITS

Nuttall's poorwill is the best known and the most widely distributed race of this species, occupying a wide range in western North America, from southern Canada to Mexico and from the Great Plains to eastern California. Major Bendire (1895) says of its haunts:

In some of its habits it differs considerably from the preceding species of this family which are almost entirely confined to the denser woodlands; the Poor-will, however, although frequently found in similar localities, is apparently equally as much at home on the open prairie and the almost barren and arid regions of the interior, which are covered only here and there with stunted patches of sage (*Artemisia*) and other desert plants. The climate does not seem to affect it much, as it inhabits some of the hottest regions of the continent, like Death Valley, in southeastern California, as well as the slopes of the Rocky and Blue mountains, in Oregon, where it reaches altitudes of from 6,000 to 8,000 feet. I have heard the Poor-will in Bear Valley, Oregon, in a locality where frost could be found every month in the year.

Dr. Alexander Wetmore (1932) writes: “At times poorwills are found in growths of low forests, but they are more often encountered in regions where dense clumps of brush are scattered over otherwise open ground, as is common in desert and semiarid localities, or in brush-grown, rocky canyons, where the ground is rough and strewn with boulders.”

Referring to the Huachuca Mountains, in Arizona, Harry S. Swarth (1904) says: “I found the Poor-will quite abundant during the summer months in the foothill region and in the lower parts of the canyons; but though most numerous below 5000 feet they were by no means restricted to these parts, for I saw or heard some in all parts of the mountains occasionally up to an altitude of nearly 10,000 feet.”

George F. Simmons (1925) designates its haunts in Texas as “high, gravelly flats or bits of plateau grown with post oak timber and with

occasional moist spots; gravel patches dotted with catsclaw bushes, located in post oak growth on slopes and flat tops of hills; high prairies on rough ground along the terraces of valleys; bare ground on rocky hillsides; among shrubbery or on semi-arid flats.”

Nesting.—The nesting of the poorwill is a very simple affair. The two eggs are laid on the bare ground, without any semblance of nest building; a slight hollow may be scraped in the bare earth, or the eggs may be laid on hard gravelly ground, or even on a flat rock. The exact spot chosen for a nest site may be in full sunlight, but oftener it is at least partially shaded by some bush, often a greasewood bush or a bunch of sagebrush, or some other bunch of vegetation. The only eggs taken by Major Bendire (1895) were “laid on the bare ground under a small grease-wood bush (*Obione*) and were fully exposed to the sun,” near Tucson, Ariz. Prof. D. E. Lantz wrote to him from Kansas, regarding the nesting habits of the poorwill in that region, that “with one exception the eggs taken were laid upon bare patches of gravel or on low, flat rocks, and placed usually near a bunch of weeds or a tuft of grass. The exception was a set found on the bare ground in an alley in Manhattan City. This alley was in constant use and it was strange that the eggs remained for so long a time undisturbed, for when taken incubation had begun in both eggs. The Poor-wills usually keep to the vicinity of steep hills and old dead grass. They seem to return to the same locality from year to year to breed.”

E. S. Cameron (1907), reporting from Montana, says: “On June 26, 1907, Mr. M. M. Archdale flushed a Poor-will from her two white eggs on a steep hillside in some rough pine brakes at his ranch near Knowlton. In this unfrequented place the eggs were fully exposed on the bare earth amidst the pines. On June 28, we went together to the place intending to photograph the eggs, but they had been already removed by the bird.”

Eggs.—The two eggs of the poorwill are generally said to be pure white, but Bendire (1895) says that the color is not pure white and that “on close inspection it can readily be seen that it is a delicate cream, with a faint pinkish tint which does not perceptibly fade. Eggs in the collection taken more than twenty years ago still plainly show this peculiar tint. The eggs are unspotted as a rule, but an occasional specimen shows a few faint, darker shell markings around one end, which are barely perceptible to the naked eye, and which fade considerably in time.”

In shape the eggs vary from oval to elliptical-oval; and they are only moderately glossy. The measurements of 50 eggs average 26.3 by 19.9 millimeters; the eggs showing the four extremes measure **28.9** by 19.8, 27.5 by **21.6**, **22.1** by 19.3, and 24.6 by **18.1** millimeters.

Young.—The period of incubation does not seem to be definitely known. Dr. Wetmore (1932) says: “Both birds are said to assist in incubation. When disturbed about the nest, they tumble about and with widely opened mouths make a loud hissing sound terrifyingly like the hissing of a snake.”

Robert B. Rockwell writes to me that he found a female with two young “as large as an ordinary week or ten days old chick. They were squatting perfectly motionless on the ground, about a foot apart, each in the shadow of something. Their feathers were quite well developed. They made no move when I picked them up, but opened their eyes when I put them down again.”

Plumages.—Elmer C. Aldrich (1935) describes the downy young of the dusky poorwill, less than a day old and about 2½ inches long, as covered with a rich buff down. Ridgway (1914) describes the downy covering of a young *nuttalli* as “vinaceous-buff, paler on underparts.” Two small partially downy young in my collection are showing the growth of the first plumage on the forehead, crown, nape, back, and scapulars; these feathers are dull buffy white, minutely sprinkled with grayish, and have small spots and narrow bars of black; the new feathers on the underparts and flanks are dull white, indistinctly barred with dusky; the wings are less than one-third grown, and the tail is just sprouting.

Ridgway (1914) says of the young in juvenal plumage: “Not essentially different from adults, but markings in general less sharply defined, especially on underparts, and throat patch buff instead of white.”

I have been unable to learn anything about the molts of either young or adults, but Dr. Joseph Grinnell (1908) says that a specimen in juvenal plumage, “taken August 22, shows many feathers of the full adult plumage in the throat and breast. The juvenal plumage is characterized by having the throat patch buff and the back conspicuously mixed with cinnamon-rufus.”

A young male, collected by Van Tyne and Sutton (1937) in Brewster County, Tex., on May 25, was in the midst of the postjuvenal molt; “the

crown is uniform gray, without any suggestion of a dark central patch, the feathers being only lightly speckled with black. * * * The chest lacks entirely the black patch and band, the feathers being lightly and indistinctly tipped with tawny and whitish.”

The so-called frosted poorwill (*nitidus*) is now regarded as merely a color phase of *nuttalli*.

Food.—The food of the poorwill consists, so far as it appears by the data available, entirely of insects, mostly the smaller, night-flying species, such as moths, beetles, chinch bugs, and locusts. Mrs. Bailey (1928) says that “in one stomach, 80 per cent of the contents was grasshoppers and locusts.” Many of these insects are caught on the wing in the capacious mouths of these birds, but many are also picked up on the ground.

Dr. R. W. Shufeldt (1885) noticed a poorwill “apparently amusing himself by making short jumps of two feet or more up in the air, then resting on the road to repeat the performance in a moment or so. Another was going through similar capers on the broad walk. They seemed to be perfectly oblivious to my presence, and, indeed, some children further along were trying to catch them with their hands.” He shot one of the birds, and “was much surprised to find in its mouth some four or five quite sizable moths, and the upper portion of the oesophagus was filled with a wad of a dozen or fifteen more. Fully half of these were yet alive, and two or three managed to fly away when freed from the bodies of their more disabled companions. This, then, is what the bird was up to; instead of flying about as a Nighthawk does, taking his insect prey in a conspicuous manner upon the wing, he captures it in the way I have described above.”

A. Brazier Howell (1927) writes:

August 28, 1926, I was sitting near midnight, on the observation platform of the California Limited as it stopped at Needles, California. It was with much interest that I then noted at least three poorwills (*Phalaenoptilus nuttalli nitidus*?) hawking about a powerful arc light in the railway yards close by. The observation point of one of these was upon the top of a board fence well within the circle of illumination; of the others, some point out of my direct vision and just beyond the fence. One after the other, until my train left ten minutes later, they would flutter up in their quest for insects, not just somewhere near the light but apparently right against the glass globe which inclosed the arc, returning each time to their respective stations for observation.

Mrs. Bailey (1928) says: “When hunting for food the Poor-will skims swiftly and noiselessly close over the ground with irregular turnings and

windings and rests between, and when its catch contains hard indigestible parts like the wing coverts of beetles, ejects them in the form of pellets, as do the hawks and owls, kingfishers, and others of similar food habits. A road through a forest with its abundant flies and insects is said to be one of its favorite hunting grounds.”

The stomach of one taken by Van Tyne and Sutton (1937) on May 25 in Brewster County, Tex., was crammed with four large June beetles (*Phyllophaga* sp.) and a large army ant (male *Eciton* sp.).

Behavior.—Mrs. Bailey (1928) says that a poorwill, wounded by Mr. Bailey, exhibited a surprising method of defense; he “opened his mouth wide and hissed and blew and flopped about on the ground, always facing the enemy. Blowing like a blow snake and opening and shutting his mouth, he was enough to terrify all minor enemies.”

Referring to the eye-shine, common among the Caprimulgidae, Mrs. Bailey quotes Dr. Bergtold as saying: “While motoring at night through a particularly dark canyon, I noticed far ahead in the illuminated road, two small glowing pink spots which were extinguished when a bird flew from the road on the near approach of the car. The bird alighted again, some distance ahead in the road, when the pink spots reappeared and were identified as the bird’s eyes; it was shot and proved to be a Poor-will.”

Dr. Wetmore (1932) says that poorwills “rest during the day on the ground, though after night, when feeding or calling, may seek higher perches on stones or posts or on low branches. On one occasion I saw one by bright moonlight calling from a bush, where it perched crosswise on a small limb, like any ordinary bird, though ordinarily they rest lengthwise of branches, like others of their family.”

Dr. Elliott Coues (1874) says: “Like others of its family, Nuttall’s Whippoorwill is oftener heard than seen. When flushed from its retreat in the daytime, among the shrubbery or tall weeds, it rises hurriedly with wayward flight, dashes a few yards, and re-alights. There is something about it at such times that strongly recalls the Woodcock, and the bird is quite as difficult to shoot on the wing.”

Voice.—The call of the poorwill is generally recorded as a clear pronunciation of its name, but many observers have noted a third syllable,

audible at only a short distance, making the complete song sound like *poor-will-low*, or *poor-will-ee*, when the bird is near, or *poor-will* when farther away, or even *p'-will* when still farther away. Mr. Simmons (1925) writes it *puih-whee-ee*.

Dr. Coues (1874) says: "This cry is very lugubrious, and in places where the birds are numerous the wailing chorus is enough to excite vague apprehensions on the part of the lonely traveler, as he lies down to rest by his camp-fire, or to break his sleep with fitful dreams, in which lost spirits appear to bemoan their fate and implore his intercession."

Some other writers give a less unpleasant impression of the song; for instance, Dr. Wetmore (1932) says: "Near at hand these calls are harsh, but with distance the first two assume a pleasant, somewhat melancholy cadence." And Mrs. Bailey (1928) writes: "When we were camped on the edge of a canyon in the Guadalupe Mountains, at dusk while the bats were flying down in the canyon, up along the edge came the Poor-wills so near that we could hear their syllables distinctly—*poor-will'-uck*, *poor-will'-uck*. Sometimes two would call antiphonally, faster and faster till they fairly tripped over each other. The call as it is often given is a delightfully soft, *poor-will'*, *poor-will'*, *poor-will'-uck*, which like the delicious aromatic smell of the sagebrush clings long to the memory of the lover of the west."

Henry W. Henshaw (1875) says that "their notes are most often noticed in early evening, and again just before dawn, but not infrequently their song is heard through the entire night. * * * When flying, they emit a constantly repeated clucking note, which is, I think, common to both sexes. * * * The males continue their notes till very late in the season; for I frequently heard them during the first part of October, and even as late as the 17th."

Field marks.—In superficial appearance the poorwill looks very much like a small whippoorwill, and its behavior is similar. It also somewhat resembles a partly grown young nighthawk, but its behavior is different; whereas the nighthawk flies about high in the air in pursuit of its prey, the poorwill hunts on or near the ground, flitting about like a large moth on silent wings; the poorwill is more strictly nocturnal in its activities than the nighthawk; furthermore, it exhibits no white patch in the wing, while flying, but shows white tips on the lateral tail feathers. Its note is, of course, characteristic.

Winter.—The poorwill retires from the northern portions of its range late in fall and spends the winter near, or beyond, our southern borders. Dr. Coues (1874) reports, in some notes from Ogden, Utah, that “it lingers at its summer home till the autumn is far advanced, as we found it at Ogden as late as October 6, quite far up the slope of the mountains, in the midst of a driving snow-storm—the first of the season—the snow having then already accumulated to the depth of several inches.”

In Arizona, New Mexico, and central Texas it is usually absent from late in October to early in April. It apparently winters more or less regularly in southern Texas, though its main winter range is in Mexico.

DISTRIBUTION

Range.—Central and western United States, and southern British Columbia, south to central Mexico.

Breeding range.—The breeding range of the poorwill extends **north** to southern British Columbia (Kamloops and Okanagan Landing); Montana (Billings and Terry); northwestern South Dakota (Slim Buttes); north-central Nebraska (Long Pine Canyon); and southwestern Iowa (Pottawattamie County). **East** to southwestern Iowa (Pottawattamie County); eastern Kansas (Blue Rapids, Onaga, Lawrence, and Clearwater); central Texas (Kerrville, San Antonio, and Somerset); eastern Coahuila (Sabinas and Saltillo); and probably Morelos (Cuernavaca). **South** to probably Morelos (Cuernavaca); southern Sonora (Alamos); and southern Baja California (San Jose del Cabo). **West** to Baja California (San Jose del Cabo, Miraflores, Triunfo, Pozo Grande, San Fernando, Santo Domingo, La Joya, San Telmo, and probably Todos Santos Islands); California (San Diego, Escondido, Ojai Valley, Santa Cruz Mountain, San Geronimo, Covel, and probably Yreka); Oregon (Brownsboro, Bridge Creek, and probably Netarts); eastern Washington (Crab Creek and Cheney); and British Columbia (Summerland and Kamloops).

Winter range.—The species is probably resident in the southern part of its range and during the winter season is found **north** to southern California (Berryessa Station, Paicines, Death Valley, and Laguna); rarely southern Arizona (Tucson); and southern Texas (El Paso, rarely Kerrville, Laredo, and Falfurrias).

The range as outlined is for the entire species, which has been separated into four currently recognized geographic races. The typical subspecies, Nuttall's poorwill (*P. n. nuttalli*) occupies all the United States portion of the range except that part of California west of the Sierra Nevada. The dusky poorwill (*P. n. californicus*) is found in western California from the northern part of the Sacramento Valley south to northwestern Baja California; the desert poorwill (*P. n. hueyi*) is restricted to the lower Colorado River Valley, southwestern Arizona, and northeastern Baja California; and the San Ignacio poorwill (*P. n. dickeyi*) is found in Baja California south of latitude 30° N.

Spring migration.—Early dates of spring arrival are: Texas—Kerrville, February 4; San Antonio, February 27. Kansas—Manhattan, April 7; Onaga, April 7. New Mexico—Chloride, March 31; State College, April 7; Rinconada, April 10. Colorado—Beulah, April 29; Denver, May 9. Wyoming—Laramie Peak, May 2; Lingle, May 3. Montana—Terry, May 16. Arizona—Tombstone, March 20. Utah—Kobe Valley, May 23. California—Piedra, March 5; Daggett, March 12; San Clemente Island, March 30; Lassen Peak, April 16. British Columbia—Okanagan Landing, April 22.

Fall migration.—Late dates of fall departure are: British Columbia—Okanagan Landing, September 20. California—Daggett, October 21; Los Angeles, October 25; La Verne, October 28; Garnsey, November 2. Arizona—San Francisco Mountain, September 29. Montana—Big Sandy, September 4; Bighorn River, September 19. Wyoming—Powder River, September 9; Clear Fork, September 19. Colorado—Beulah, October 8. New Mexico—State College, October 30; Sierra Hachita, November 24. South Dakota—White River, September 27. Kansas—Onaga, September 27. Oklahoma—Kenton, September 26.

Egg dates.—Arizona: 5 records, May 2 to August 2.

California: 42 records, March 22 to August 8; 21 records, May 6 to June 25, indicating the height of the season.

Colorado: 5 records, May 26 to July 27.

Texas: 11 records, April 29 to June 20.

PHALAENOPTILUS NUTTALLI CALIFORNICUS Ridgway

DUSKY POORWILL

PLATES 25–27

HABITS

The goatsuckers are now in their proper place in the A. O. U. Check-list, showing their relationship to the owls, which they strikingly resemble in several characters. The poorwills are conspicuous in this respect; they are more strictly nocturnal in their activities than some of the owls, for which they are well adapted; the eyes are very large, suggesting those of owls; the mouth is very broad, but some of the owls have broad mouths also; the plumage is fully as soft as that of the owls; and their flight is noiseless, like that of most night-flying birds. The Caprimulgidae are not predators on vertebrate animals, although the chuck-will's-widow has been known to eat birds; but they all live on animal food; and many owls live largely on insects.

The dusky poorwill does not enjoy so extensive a distribution as its inland relative, *nuttalli*, but perhaps it is equally as well known throughout its range in California.

Nesting.—The steep slopes and ridges of the foothills and the sides of canyons seem to form the favorite nesting haunts of this, as well as other poorwills. Elmer C. Aldrich (1935) gives a very good description of a locality in Tuolumne County, Calif., where a nest was found on July 5, 1934:

The altitude was about 5600 feet, and the general vegetation of the area consisted of yellow pines, incense cedars, white firs, black oak islands, a few species of ceanothus, and a little manzanita, with mountain misery covering most of the open hillside. The immediate location of the nest was in a little circular clearing about fifteen yards in diameter, surrounded by young yellow pines closely knit together by small, interwoven branches. The clearing contained three manzanita bushes and one ceanothus bush. The greater part of it was strewn with long dead pole-like logs, which appeared to be one of the basic requirements for the Poor-will's protection. The entire north side of the opening was bordered by a large decayed log of a diameter of three feet, which, because of the common use by the Poor-will, came to be called "the log."

The nest was found when we were coming from the north and upon advancing four yards after stepping over this log. When the adult flushed from the nest the observer's foot was but eighteen inches from the site. The bird flew across the clearing into the edge of the dense forest, where it lit on a small log and watched without movement for fifteen minutes while pictures of the two light buff eggs were obtained.

Dr. William L. Holt writes to me that he found a nest near Banning, Calif., on June 13, 1909, on the bare ground on the north side of a sandy hill, nearly bare of vegetation. J. E. Patterson has sent me two excellent photographs (pls. 25, 26) of poorwills' nests. The one taken in Stanislaus County, Calif., on June 23, 1934, was on the ground in open timber; the other, taken in Mariposa County, Calif., on June 21, 1933, was on the ground in a fire deadening on a hillside; both were in the Transition Zone.

A. J. van Rossem (1920) found a nest on April 18, 1919, a very early date, on the side of a canyon, where there was a heavy growth of wild lilac and white sage; "no attempt whatever seemed to have been made at constructing a nest, the eggs lying on the bare ground among pebbles, etc., in the shade of some dense brush that bordered upon a small open space." The nest has also been found on bare rock, but is usually, at least partially, shaded by some bush, loose brush, brakes, or weeds. The birds are very apt to return to the exact spot to nest each year.

Eggs.—As a rule the eggs of the dusky poorwill, two in number, are similar to the eggs of other poorwills, but van Rossem and Bowles (1920) write:

In a majority of the descriptions that are given for eggs of the various forms of the Poorwill, the color is stated as white, without markings, sometimes with a pinkish tinge. However, such was by no means the case with the set of eggs under discussion. Before blowing, the ground color was a strong salmon pink; but this, after blowing, turned to a clear, glossy, pinkish white, strongly suggesting eggs of the Merrill Pauraque (*Nyctidromus albicollis merrilli*), although the pink of the Poorwill eggs showed a closer approach to salmon. Around the larger ends was a rather dense wreath of lavender and dusky spots and dots, making the eggs look exceedingly like the marked eggs of some small petrel. * * * In the course of time many of the smaller dots have faded out, leaving only a comparatively few spots and dots to show where the heavy wreath was once located. The strong pinkish tinge has also very largely gone, in spite of the fact that the eggs have been carefully kept from exposure to the light.

The measurements of 40 eggs average 26.3 by 19.4 millimeters; the eggs showing the four extremes measure **28.4** by 20.0, 27.7 by **21.1**, **24.2** by 18.2, and 24.9 by **17.0** millimeters.

Behavior.—Poorwills are notoriously close sitters, but the pair studied by Mr. Aldrich (1935) were unusually tame. He was repeatedly able to approach cautiously to within 3 or 4 feet of the incubating bird; once, at night, eight observers managed to approach within 2 feet, while the bird sat tight all the time. The next morning, "when we were but three feet from the

nest, the bird's large clear white spots on the tail identified it to be the male that was doing the incubating. Many pictures were taken, some as close as ten inches, without the slightest sign of fear on the part of the sitting bird. * * * Experiments were made to find the extent of his 'bravery' by touching him. While touching the head the first time he flattened out his wings and spread the fan-like tail over the tips of them showing all the tail spots. The large head was then brought far back on the shoulders, the cavernous mouth opened extremely wide showing the pink interior, and a low guttural hiss was emitted at short intervals."

Later in the same day, at 2 p. m., he flushed the female from the nest and sat down 5 yards away to watch her return; "exactly ten and one-half minutes marked the reappearance of the same bird. She had flown from behind the log nearly to the top where she could barely look over and observe the surroundings. Immediately she started swaying from side to side very slowly and rhythmically for about five seconds before walking to the top of the log, each step in synchronization with the swaying. Here she paused for about twenty seconds, and then flew a few yards, within two and one-half feet of the nest, where she began swaying again. The rest of the distance to the nest was accomplished by this slow walking-swaying process, and she did not seem to get anxious and speed up as she came closer. Two short stops were made on the way."

The dusky poorwill is tame and unsuspicious at other times also. Rollo H. Beck (1897) watched one at short range while it was catching insects on a road and finally succeeded in putting his hat over it, though it escaped; he says: "After watching it a while I crawled up within four feet and had a chance to watch it in the bright moonlight. It would fly perhaps twenty or thirty feet into the air after insects and return again within four or five feet of me. One time it flew up and evidently picked an insect off the leaf of a wild cherry tree, fluttering for several seconds in its endeavor to do so. It several times flew by me after food and returning would fly within a foot or so of my head and alight just in front of me."

Voice.—H. Gordon Heggeness writes to me of the song of the dusky poorwill as he heard it in the Sequoia National Park: "Sometimes early in the morning the poorwill would be heard—the calm, liquid notes carrying far on the cool air. On August 12, 1935, a poorwill began singing back of

my cabin at 2:30 a. m. His most pleasing notes were heard continuously for the next half hour.”

Grinnell and Storer (1924) write: “It is heard most persistently at dusk of evening or in the early morning; but near Pleasant Valley on the morning of May 23, 1915, one of these birds suddenly broke out at 10 o’clock and uttered its *poor-will-o* 85 times (by count, within 2 or 3) at intervals of two or three seconds.”

R. H. Lawrence wrote to Major Bendire (1895) that, according to his hearing, “the words ‘Pearl-rab-it’ give a fair idea of its call”; and that “when startled it gave quickly, two or three times in succession, a low, soft note, like ‘pweek, pweek, pweek,’ which could only be heard a few yards away.”

PHALAENOPTILUS NUTTALLI HUEYI Dickey

DESERT POORWILL

HABITS

Donald R. Dickey (1928) described and named this pale race of the poorwill from a fine series of specimens collected in the valley of the Colorado River by Mrs. May Canfield and Laurence M. Huey. As to its subspecific characters, he says that it is “nearest in color to the light type of *Phalaenoptilus nuttalli nuttalli* (Audubon), which Brewster named *nitidus* and to which he gave the eminently fitting vernacular of the Frosted Poorwill, but averaging very much lighter. The backs of *hueyi* are pinkish tan, almost devoid of the silver frosting characteristic of more eastern birds, and with the size of the dark dorsal ‘owl’s eye’ marking greatly reduced, in many cases practically obsolete; under parts lighter throughout, with the dark band below the white collar narrower and of lighter tone, and with narrower barring of sides and flanks. Tail lighter and less contrastingly barred above and below.”

The 1931 Check-list gives the range of this race as the valley of the lower Colorado River, in southeastern California, southwestern Arizona, and extreme northeastern Baja California.

A. J. van Rossem (1936) took four poorwills near Bates Well, in south-central Arizona, that were somewhat intermediate between *hueyi* and

nuttalli, though nearer the latter; this locality probably indicates the approximate area of intergradation between the two races. "All were collected in an arrowweed-mesquite association along the borders of the dry stream bed. Conditions, both as to habitat and temperatures, closely approximated those found along the Lower Colorado River Valley, beyond the confines of which *hueyi* has not been detected."

This poorwill probably does not differ materially in its habits from neighboring races of the species.

PHALAELOPTILUS NUTTALLI DICKEYI Grinnell

SAN IGNATIO POORWILL

HABITS

Three races of the poorwill are found on the peninsula of Baja California. The present form ranges from about latitude 30° southward to the Cape region; the dusky poorwill (*californicus*) extends its range southward in the northwestern portion to about latitude 30°, chiefly on the Pacific slope; and the desert poorwill (*hueyi*) is found in the extreme northeastern portion.

The San Ignatio poorwill is a small, dark race. Dr. Joseph Grinnell, who described and named it, says (1928) that it is "similar to *Ph. n. californicus* in degree of general darkness but decidedly smaller, and with black areas on the individual feathers of scapulars, top of head and chest greatly reduced, in this respect resembling *hueyi*; terminal white of lateral rectrices greater in amount than in *californicus*; light portions of general color scheme much darker than in *hueyi* or *nuttalli*, tinged with clay color rather than 'frosted'—in this respect darker even than in average *californicus*; dark barring on posterior lower surface much more extensive and heavier than in *hueyi* or *nuttalli*."

William Brewster (1902) referred the poorwills of the Cape region to the race he named the frosted poorwill (*Ph. n. nitidus*), based on a pair collected by M. A. Frazar in the Sierra de la Laguna. Mr. Frazar said that on the mountains the poorwills did not begin singing until about the middle of May. "Their note is a *pow-wè-hoo*, the first syllable given long, the accent on the second, and the last little more than a retraction of the breath. They

were almost invariably in large oaks and very seldom on the ground. A female shot June 6 was undoubtedly mated and would have laid soon.”

Nothing seems to be known about the nesting or other habits of this poorwill.

NYCTIDROMUS ALBICOLLIS MERRILLI Sennett

MERRILL'S PAURAUQUE

PLATE 28

HABITS

Merrill's pauraque is the largest subspecies of a widely distributed species of goatsucker that ranges throughout Mexico, Central America, and northern South America, and that has been divided into six subspecies. Our pauraque extends its range from northern Mexico into the valley of the Rio Grande and northward through the Gulf coast of Texas to Nueces, Aransas, and Refugio Counties. The vernacular name, pauraque, pronounced “pou-rä'-kâ,” is derived from a fancied resemblance to one of its notes. The Mexican name, cuiejo, pronounced “coo-ya-ho,” has a similar origin.

This bird was first introduced to our fauna by Dr. J. C. Merrill, who took the first specimen at Fort Brown, Tex., on April 1, 1876. The subspecies was first described and named by George B. Sennett (1888), who gives a full account of its plumages at different ages and says of its relation to other races of the species: “This form, when compared with others of the species from Southern Mexico, Costa Rica, Panama, Guiana, and Brazil, can be distinguished, first by the prevailing gray color on upper parts, where the others have brown, rufous or cinnamon; second, by its large size, exceeding the large southern Brazilian form in length of wing and equalling it in length of tail; third, by the males having the outer tailfeather generally without white, and the white when it does occur being much restricted, while in more tropical and South American forms the rule is that males have much white on inner web of outer tailfeather.” This third point seems open to question, as this character seems to be very variable, possibly owing to age.

D. B. Burrows, in some notes sent to Major Bendire (1895) says: "During the winter the birds may be flushed from the dense thickets in the bottom lands, but as the nesting season approaches they leave these close retreats and seek more open ground."

Mr. Sennett (1878) says of their haunts: "I saw them occasionally, singly and in pairs, about the thickets and open chaparral, and once in the canebrakes close to the woods." Again (1879) he says: "They breed in the more open places among the cactus and scattered bushes along with *O. texensis*—Texas Nighthawk. On dark days I flushed them from thickets in the chaparral, or from copses near the bottom lands on the edges of the woods."

Nesting.—Mr. Sennett's nests "were found in open brush, on the bare ground. One of them was partly concealed by the branches of a low bush 6 or 8 inches from the ground."

Mr. Burrows wrote to Major Bendire (1895): "I have never found the Pauraque nesting in the dense thickets, where they hide during the winter. They seek the more open ground, the high, level spots near the river, or up some arroya, among scattering bushes and pear cactus, but never on the rocky hills, where the Texan Nighthawk is frequently found. In one instance a nest was found at the edge of a cultivated field. The eggs are placed on the bare ground, with no attempt at nest building, and usually at the foot of a clump of bushes. The bird, when flushed from the nest, quietly darts off and drops to the ground but a short distance away."

Dr. James C. Merrill (1878) writes: "On the 15th of May, 1876, I found a set of eggs near camp at Hidalgo, and on returning in about fifteen minutes to secure the parent, who had disappeared among the thickets, I found that she had removed the eggs, although they had not been touched."

Eggs.—The pauraque lays two very handsome eggs, quite unlike the eggs of any other species in the family. They vary in shape between oval and elliptical-oval, with a decided tendency toward ovate or elliptical-ovate, one end often being slightly more pointed than the other. The shell is smooth, with little or no gloss. The ground color varies from "light ochraceous-salmon" to "pinkish buff". Some eggs appear to be nearly immaculate, but in most cases they are more or less evenly covered, some sparingly and some profusely, with small blotches, spots, or dots of pale

“clay color,” pale “cinnamon,” or, more rarely, with deeper shades of “cinnamon-rufous”; often there are underlying blotches or spots of “ecru-drab” or “light cinnamon-drab.” Occasionally the markings are concentrated around the larger end, but usually they are quite evenly distributed.

The measurements of 50 eggs average 29.80 by 22.36 millimeters; the eggs showing the four extremes measure **35.10** by 23.10, 31.50 by **24.64**, **27.18** by 20.57, and 28.96 by **20.32** millimeters.

Young.—In his unpublished manuscript on the birds of the Caribbean lowlands of Central America, kindly lent me, Alexander F. Skutch devotes considerable space to his studies of the care and feeding of the young of the local race of the pauraque (*Nyctidromus albicollis albicollis*). As the habits of the different races of this species probably do not differ materially, it seems pertinent to include here some of his remarks on this subject. He says that “during the day male and female take turns on the eggs, relieving each other every two or three hours. The male is usually found incubating in the early morning. At night, I have found only the female covering the eggs.” Other species of goatsuckers have been reported as moving their eggs to a place of safety after being disturbed, but he found no evidence that this species does so, saying: “I have given pauraques all kinds of incentives to move their eggs to a safer place, but they have never shifted their positions even a few inches as a result of my interference. The case was quite different when I moved the eggs from the spot which the birds had selected as their nest.” In this case, he had moved the eggs twice, once only a few inches and once a foot away from an area infested by fire ants; but each time the parents moved them back again to their chosen spot; the result was that, as soon as the eggs began to hatch, the chicks were destroyed by the ants. He did not see how the eggs were moved; probably they were pushed along with the feet and body, as the nighthawk has been seen to do.

Both parents took turns in brooding the young throughout the day, relieving each other at intervals of two or three hours, as they did on the eggs. As the young were fed in the dusk or deeper darkness of evening, it was only after several trials that he was able to see the performance clearly. He writes: “It was dark now, but at the critical moment I pushed the button of a powerful flashlight. The mother was resting on the ground in her customary position, looking toward me with eyes that shone like rubies in

the beam of light. The little ones stood on tiptoe in front of her, their necks stretched up, bringing their heads on a level with her mouth. One was being fed while the other waited impatiently for its turn. Though her *mouth* was so big she could easily have swallowed her nestling whole, her little *bill* was inserted into its widely open mouth, just as with a hummingbird, and with distended throat and convulsive movements of the body she regurgitated into it the insects she had captured. When she had given sufficient to the first one, she paused with head erect, clearly much alarmed by the light, but the second continued its silent importunities and the parent yielded despite her fears, feeding it in the same manner. Then she went off to hunt more insects.

“Presently one of the parents entered the thicket again, but instead of going to the youngsters it settled on the ground behind the blind, fully 20 feet from them, and began the low croaking-clucking call, which draws them as the magnet draws the floating needle. The two- and three-day-old bantlings drew themselves up and began hopping bravely toward the voice, peeping as they went through the darkness. Directly in their path was a young banana plant, which I had felled in clearing a place for the blind. The large, slippery leaves lay in a tangled mass that loomed above them; the Alps formed no more insurmountable obstacle to the advance of Hannibal’s army than this confusing barrier to the young pauraques. Dauntless as the renowned Carthaginian, they pushed resolutely onward, lured by the continued calls. I completely lost sight of them in the obscurity, but for many minutes their weak cries emerged from the dark mass. At length they descended victorious into the plains of Italy, and henceforth their path, although not free from obstacles, was comparatively easy. Finally they found a haven beneath their parent’s sheltering wings.”

On this and previous occasions Mr. Skutch discovered that the young pauraques were perfectly capable of locomotion, even at an early age, in response to warning calls from their parents. He saw no evidence that the adults ever carried them in their capacious mouths, as other species of the Caprimulgidae have been reported to do, or assisted them to move in any other way. He says that “during the day male and female take turns on the eggs, relieving each other every two or three hours. The male is usually found incubating in the early morning. At night, I have found only the female covering the eggs.”

Plumages.—The downy young pauraque is as unique among the Caprimulgidae as are the eggs. It is completely covered with fine, soft down, leaving nothing exposed but the tip of the bill and the feet. The colors are rich and contrasting; the forehead, crown, and nape are “pinkish buff”; the lores, cheeks, and auriculars are “mikado brown”; the back and rump are “sayal brown,” with a central band of “pinkish buff”; the chin and throat are pale “pinkish cinnamon,” and the remainder of the under parts are “pinkish buff.”

Ridgway (1914) describes a female nestling, not yet fully grown, as follows: “General color of upper parts pale brownish gray, very minutely vermiculated or stippled with darker; pileum and scapulars with scattered roundish and subtriangular small spots of black; interscapular region clouded or blotched with black; under parts light grayish buff, narrowly barred with dusky on chest, more broadly barred with the same on throat, breast, and sides, the abdomen and under tail-coverts immaculate; thighs uniform light fawn color; remiges and rectrices (not fully grown) apparently as in adult female.” Of a young female, fully grown, he says: “Primaries and rectrices as in adult female; pileum spotted, instead of streaked, with black, the spots mostly of broadly triangular form; back also heavily spotted, or blotched, instead of streaked, with black; scapulars without buff margins; barring of under parts much less sharply defined, less dark in color; throat band light dull buff, barred with blackish.”

Mr. Sennett (1888) says of the young male: “White wing-patch of male mixed with buff; white on tail not so sharply defined, nor does it extend so near to base as in mature birds. The outer tail-feathers are barred and streaked with brown and buff. White throat-patch smaller and barred sparingly with black and buff. In this immature stage the sexes are not easily determined aside from the difference in the white tail-patches; in the males the white on second feather from outside is never less than 2.5 inches long, while in the females the corresponding patch is about one inch in length.”

The above descriptions evidently refer to the juvenal, or first plumage, which is largely replaced in summer, July and August, by a first winter plumage. In this the contour plumage is much like that of the adult, the spots on the crown and back being replaced by streaks; but the juvenal

wings and tail are retained; the sexes are much alike in this plumage but can always be distinguished by the amount of white in the tail.

I have seen birds in this plumage in October, December, and March, from which I infer that it is worn until the first postnuptial molt the following summer. Mr. Sennett (1888) thought that "at least two years must elapse before the perfect plumage is attained"; but it seems more likely that the first postnuptial molt produces a plumage that is practically adult, except for some dusky mottling on the tips and edges of the white tail feathers, and some buffy intrusion of the white wing patch, which may indicate that the full purity of these white areas is not attained at the first postnuptial molt.

Dickey and van Rossem (1938) say: "Young birds which have assumed the postjuvinal plumage may be distinguished from adults by the narrower, more pointed, and usually impurely colored tail feathers and by the buffy edgings and mottlings on the tips of the primaries. The juvenal primaries which are worn until the bird enters the second fall (first annual) molt, are from 5 to 10 mm. shorter than those of adults. Males, at least, breed the first spring as shown by dissection of several specimens. The annual molt of adults takes place in August and September."

There are two distinct color phases in the adult plumage, a gray-brown phase and a tawny phase, mainly recognizable in the coloration of the upper surface. Ridgway (1914) says of the former, "pileum, hindneck, and interscapular region grayish brown," and of the tawny phase, "general coloration of upper parts much more tawny or ochraceous, the general color of the pileum varying from fawn color to mars brown or russet."

Food.—Bendire (1895) says that "the food of Merrill's Pauraque, like that of the rest of the *Caprimulgidae*, consists mainly of night-flying insects, such as moths, beetles, etc. The crop of a specimen shot by Mr. H. P. Attwater, near Rockport, Texas, was filled with fireflies, *Photinus pyralis*?"

On several occasions, Mr. Burrows (Bendire, 1895) observed a pauraque "perched upon the extremity of a low, dead limb, or on the top of a bush. At these times I found that the bird was watching for food, and at intervals would leave the perch, dart off a short distance as if in pursuit of an insect, and as quickly return, in very much the same manner that the Whip-poor-

will is accustomed to do. I believe that these birds are strictly insectivorous; but I have never made an examination of the stomach except in one instance. The stomach of this bird, killed in January, showed parts of the feet and wings of beetles and other insects.”

Three stomachs examined by Cottam and Knappen (1939) contained parts of ant lions, stink bugs, soldier bugs, locusts, click beetles, tiger beetles, ostomid beetles, twig borer, carrot muck beetle, long-horned beetles, and traces of butterflies, moths, bees, or wasps. Beetles made up 89 percent of the whole food.

Behavior.—Mr. Burrows (Bendire, 1895) writes:

During the daytime the birds were commonly found on the ground at the foot of a growth of bushes or among fallen branches, and I have occasionally found them perched, nighthawk-fashion, upon a low branch not more than a foot from the ground. When closely approached, they dart quickly forward in a zigzag course, dropping as suddenly to the ground. This flight is short, usually not more than 50 or 80 feet, and when settled they commonly remain perfectly quiet until again flushed. They have a peculiar way of turning or facing about as they strike the ground, so that they can better watch the approaching danger. They lie so close that it is with great difficulty that they can be detected, unless the spot is carefully marked. In a number of instances, where the bird seemed to feel that it was observed, I have had them go through a peculiar bowing movement, resembling that of the Burrowing Owl, except that the body is raised from its completely prostrate condition.

The pauraque has a relatively longer tarsus than our other members of the goatsucker family; hence it is more active on its feet. Dr. Frank M. Chapman (1896) observes: “I was surprised to learn how rapidly these birds can run. On one occasion two lit within a few feet of me when it was light enough to distinguish their movements. They crouched close to the earth, sometimes running quickly and with unexpected ease for a few steps, then turned their heads sharply from side to side as though looking for insects. They would also spring suddenly fifteen feet into the air to catch a passing insect.”

Voice.—Dr. Merrill (1878) writes: “Their notes are among the most characteristic night sounds of the Lower Rio Grande, and are constantly heard at evening during the summer months. They consist of a repeated whistle resembling the syllables *whew-whew-whew-whew-whe-e-e-e-w*, much stress being laid upon the last, which is prolonged. The whole is soft and mellow, yet can be heard at a great distance. The preliminary *whews* vary somewhat in number, and late in the season are often omitted altogether.”

Mr. Burrows wrote to Major Bendire (1895): "At the approach of the breeding season and as early as the middle of March the peculiar whistle of the Pauraque becomes general, and along the lower Rio Grande, where they are common, it may be heard on all sides. * * * The birds begin to call as the dusk of evening comes on. The commonest call is a long-drawn 'ko, whe-e-e-e-w', much prolonged, and raised to a high pitch toward the last. This is repeated often and with great energy, and on a still night may be heard at a long distance. At other times the first syllable is omitted. Again it is varied by a repetition of the first syllable, as 'ko, ko, ko, ko-whe-e-e,' the first syllables repeated deliberately, and the last not so long-drawn and suddenly cut short." He also refers to a short *put, put* note, resembling one of the notes of a wild turkey.

The name of the pauraque is said to have been derived from a fancied resemblance to one of its notes; the same is true of its Mexican name, *cuiejo*.

Winter.—The pauraque is a permanent resident in southern Texas and farther south, and even as far north as the Nueces River. Mr. Burrows found it "to be fairly common" during the winter and spring of 1894. He met with it during the winter in Starr County, southern Texas, and says: "When the nesting season is over the birds remain quiet, and their peculiar whistling note is not heard with regularity, and yet, on very warm nights during each of the winter months, I have occasionally heard them. During the winter the birds may be flushed from the dense thickets in the bottom lands."

DISTRIBUTION

Range.—Southern Texas and northeastern Mexico; not regularly migratory.

The range of Merrill's pauraque extends **north** to southern Texas (Refugio County and Aransas County). **East** to Texas (Aransas County, Corpus Christi, Rio Hondo, and Brownsville); eastern Tamaulipas (Aldama, Altamira, and Tampico); and Veracruz (Mirador and Jalapa). **South** to central Veracruz (Jalapa); and Puebla (Metlaltoyuca and Tehuacan). **West** to southeastern Puebla (Tehuacan); western Tamaulipas (Santa Leonor and Rio Cruz); and Texas (Lomita, Neuces County, San Patricio County, and Refugio County).

Closely allied races of this bird are found in Central and South America.

Egg dates.—Mexico: 9 records, March 22 to June 22.

Texas: 66 records, March 5 to June 26; 33 records, April 15 to May 16, indicating the height of the season.

CHORDEILES MINOR MINOR (Forster)

EASTERN NIGHTHAWK

PLATES 29–33

CONTRIBUTED BY ALFRED OTTO GROSS

HABITS

The nighthawk, because of its piercing calls and the extraordinary evolutions and gyrations of its flight, attracts many persons, even the casual observers who ordinarily pay no special attention to birds. A bird so unique and striking, one that during its breeding season plays such an important role in our experiences out of doors, is destined to be the recipient of many common names.

Long before the white man came to America the nighthawk was well known to the Indians, and we find it taking a prominent place in their myths and traditions. Apparently the notes of this bird appealed most, since the names chosen by the various tribes were usually graphic allusions to the calls or to the characteristic booming noise heard during the courtship season. To the tribes along the Connecticut River this booming was the sound of the Shad Spirit announcing to the shoals of shad, about to ascend the river, of their impending fate. The nighthawk was known to the Seminoles of Florida as “Ho-pil-car.” In the Milicite Indian Natural History there is the name “Pik-teis-k wes,” and according to W. W. Cooke (1884) the Chippewas not only had the name “Besh-que” for the nighthawk but recognized it as a species distinct from the whippoorwill, to which they gave the name “Gwen-go-wi-a.” That the Chippewa Indians differentiated these two species is all the more remarkable when we recall that this distinction was confused by Catesby and the American ornithologists of the next 50 years who followed him. It was Alexander Wilson who first noted that they were distinct species.

When the first European settlers came to our coast they compared the nighthawk and the whippoorwill with the nightjar of their old homes, and hence we find this name in the earlier ornithological writings used as a synonym for the American bird. In certain districts of England and Scotland the nightjar is called the goatsucker, a name that originated from the queer superstition that this bird with its enormous mouth sucked the teats of goats. Like the name nightjar the name goatsucker also crossed the Atlantic, as is manifested by such names as long-winged goatsucker and Virginia goatsucker to be found in the older books and papers dealing with American birds. The name goatsucker is still applied to the order and family but is seldom used today in designating the species.

In parts of the United States, especially in the South, nighthawks are known as bats, since the birds are usually seen at dusk when their erratic flight resembles somewhat that of the common mammal. This resemblance linked with the bellowing or booming sound produced by the wing feathers during the courtship plunge has given source to the commonly used name bullbat. Audubon (1840) used the synonym Virginia bat and stated that the French Creoles of Louisiana knew the nighthawk by the metaphorical French name "crapau volans," or flying toad. In the Bahamas, as well as in certain localities of America, a common local name is "pick-a-me-dick," a crude imitation of one of its notes. The name mosquito hawk was well earned by one individual that, according to the Biological Survey, had eaten 300 mosquitoes. Other names sometimes applied to the nighthawk but less frequently than some of those previously mentioned are pisk, pork and beans, will-o'-wisp, burnt-land bird, and bird hawk.

The commonly accepted name nighthawk probably originated because of the bird's resemblance to the smaller hawks when observed in flight. However, it has sometimes proved to be an unfortunate choice, since to the layman it suggests a bird of the true hawk type. This name on occasions has been a source of trouble to an innocent bird as illustrated in the following cases. According to B. H. Warren (1890) the Pennsylvania Game Commissioners, in their interpretation of the Scalp Act of 1885, took the stand that they were obliged to allow bounty on all nighthawks because they were known as hawks. Mr. Warren also states that there is a somewhat prevalent idea that nighthawks are destroyers of young poultry, the name doubtless having given origin to the absurdity. The following clipping taken

from the Portland (Maine) Press is another example of the way the name has misled well-meaning persons. "The Press building acquired a new claim to distinction yesterday as the haunt of wild fowl when janitor Phillip Ward, upon making an ascent to the roof, discovered two hen hawk's eggs there. Apparently some bird of the genus so despised by farmers had found the top of Portland's skyscrapers the right kind of a nesting place, and had laid her plans to hatch a few juvenile hen hawks up there. Janitor Ward's unexpected arrival put an end to such plans, however, and farmers may rest assured that the breed of distasteful birds who pillage their chicken yards is two the less." Arthur H. Norton, director of the Portland Society of Natural History, investigated the story and found the victim to be an innocent nighthawk. Adverse criticisms of the name nighthawk have been numerous but the name although inappropriate is destined to persist.

Spring.—The vanguard of the nighthawks in the spring migration reaches Florida and the Gulf States about the middle of April. There have been March arrivals reported, but these are exceptional. Large flocks, some of them numbering thousands of individuals, have been seen during May. Many of these southern records may be representatives of the southern form, *Chordeiles minor chapmani*.

It is not necessary to search the isolated retreats away from the habitations of man for the first nighthawk arrivals. In fact, they are more likely to appear in the midst of our populous cities and towns, where they may be seen flying high above the graveled roofs that later are to be the scene of their nesting activities.

Courtship.—The courtship of the nighthawk is an ardent and amorous performance on the part of the male. He may be seen at twilight or early dawn uttering his sharp *peent* calls as he flies in wide circles sometimes hovering or soaring in the air high above the proposed nesting site. At more or less regular intervals he swoops down often within a few yards of his mate. Just as he seems about to dash into the ground he makes an abrupt upward turn, the vibrating primaries producing the well-known boom. After these preliminary aerial performances the male alights on the ground or graveled roof near the resting passive female. He now stands on his feet instead of resting his body on the ground. His tail is widely spread like a fan and wagged from side to side while the body is given a peculiar rocking motion. The throat is frequently puffed out, displaying a large white patch,

which is ordinarily concealed when he is at rest. Synchronized with the throat distension is the uttering of guttural croaking notes. In producing these notes the bird holds the beak tightly closed, so that the throat membrane is distended by the pressure of the air emitted from the lungs. Under these conditions the sound waves originating at the syrinx beat against the tense membrane, producing notes of a striking and peculiar resonant quality. These notes are not uttered under ordinary conditions and constitute a very important part of the courtship.

The female, as far as outward manifestations are concerned, does not seem to be at all impressed by these extraordinary antics. The male at times in seeming desperation flies directly over the female uttering a sharp *peent*. He may then circle the female several times but even to this she seems unmoved. At times, when he approaches too near, the female will take a short flight, alighting a few yards away. The male follows and the performance described above is repeated. Eventually the courtship terminates in copulation.

The aerial evolutions of the male, including the downward plunges and “booming,” are continued throughout the nesting period, but after the young are hatched his effervescent energies are directed in part to securing food for his offspring. With regard to the courtship of the nighthawk, Charles W. Townsend (1920b), who gave the subject of the courtship of birds careful study, stated: “The rapid headlong plunges of the nighthawk may be classed as a display of motion, a form of the dance. Incidentally, and perhaps accidentally at first, a loud booming sound is produced by the rush of air through the wing feathers. This instrumental music is now the important feature, although the dance is by no means a negligible one.”

Nesting.—The nesting site, according to the procedure of a pair of nighthawks studied by me at Brunswick, Maine, is chosen by the female. A banded female returned to the same nesting site on the graveled roof of the high school building for four successive years, although the males, during at least two seasons, were different individuals.

The nighthawk is solitary in its nesting habits, though there have been instances where the nests of this species have been placed very near together, approaching a gregarious tendency. For example, E. A. Samuels (1872) states that on “a ledge of rocks back of the settlement known as

Wilson's Mills, which seemed a favorite breeding-place for these birds, and, in the space of every four or five rods, a female was sitting on her eggs." B. H. Warren (1890) writes that he has found several nighthawks breeding within a few yards of each other. A similar case of this kind was noted at Gardner, Maine, where five pairs of birds nested in a very restricted area of an old deserted dock. I am inclined to believe that the existence of a good nesting site rather than a social tendency is the more important factor in causing these birds to nest in proximity. In general, each pair of birds has a relatively large nesting territory, which is vigilantly guarded. I have never noted any marked tendency of the birds to flock together until after the nesting season prior to and during the migrations.

The nests of the nighthawk located on the ground are found in a diversity of situations as far as the surroundings are concerned. It prefers gravel beaches or open barren areas of rock or soil unobstructed by tall shrubbery or trees. It never builds a nest in the seclusion of a forest. In regions where forests abound it nests in places where vegetation is sparse or preferably where forest fires have left a barren waste. C. F. Batchelder (1882) found the nighthawk frequenting burnt lands in the region of the upper St. John River near Fort Fairfield, Maine, and it has also been found to be common in the burnt lands of the Restigouche Valley, New Brunswick. R. W. Chaney (1910) found a nest in a burnt-over area near a partly burned log in Mason County, Mich. I found three nests of the nighthawk in a burnt-over area near the Biological Station, Douglas Lake, northern Michigan, and C. E. Johnson (1920) saw two young on a scantily moss-covered and stick-strewn rock outcrop in Lake County, Minn., in a district previously burnt over. It has been noted that in regions that have been burnt over the nighthawk population increases; hence the burning of timber land, which is so destructive to many species of birds, is not a hindrance but possibly an aid to the general welfare of the nighthawk.

The nighthawk has also been found to breed in cultivated areas. I. E. Hess (1910) noted its nesting in plowed fields in central Illinois, and T. G. Gentry (1877) states that old stubble fields are frequently selected as nesting sites. H. J. Rust (1911) has reported the western nighthawk in cornfields in Idaho and California. The Texas nighthawk, according to Sharp (1907), is a common inhabitant of the vineyards of San Diego, Calif., where the eggs are placed on the ground under or near the vines. Nests have

also been found in potato fields and even in gardens near the houses of man. C. B. Ressel (1889) reported the eggs of the nighthawk placed upon the loose soil thrown up by the woodchuck (*Marmota monax*), indicating that barren places even though they are composed of loose soil may be preferred to sites covered with vegetation, twigs, or other debris.

In Virginia, according to H. H. Bailey (1913), the nighthawk often departs from its usual ground nest to a place on a stump, fence rail, or tops of drifts formed on the islands off the Virginia coast. Nests on stumps have been reported from various localities. I have found nests on the flat surface of top rails of fences in central Illinois; one of these nests was 8 feet from the ground. V. Max Kemery (1925) reports an unusual nesting site between the rails on a railroad track that was in daily use. The bird would fly when the cars or engine approached but returned as soon as the train had passed. Another unique nesting place that has come to my attention is an old robin's nest located in a tree near Farmington, Maine. This nighthawk, according to Mr. Jewell (1908), nested in this unusual site for five successive years, the tree being destroyed by fire at the end of that time.

The nighthawk in increasing numbers is availing itself of nesting sites provided by the graveled roofs of our cities and towns. In certain sections of the country this tendency is so marked that nighthawks are now seldom seen remote from graveled roofs during the nesting season, according to Lynds Jones (1909). The chimney swift, a not distant relative of the nighthawk, long ago forsook its primitive nesting sites in hollow trees, and today we no longer associate them with such places. Occurring simultaneously with the destruction of the giant forest trees was the erection of chimneys in connection with the homes of civilized races. The chimney swift, deprived of its original nesting site, was quick to make the radical but necessary change. Doubtless this adaptability to a changing environment has been an important factor in the preservation of the species. Chimneys have been used in America for centuries, but the graveled roof on which the nighthawk builds its nest is a comparatively recent development. It was not until the middle of the nineteenth century that the mansard and the flat type of graveled roofs were introduced. It was not long after these first roofs were built that the nighthawks discovered the possibilities of a new and admirable type of nesting site. As early as 1869 W. P. Turnbull wrote that the nighthawk was often seen high in the air above the streets of

Philadelphia and that their nests were frequently found on the roofs of the warehouses of the city. Louis A. Zarega (1882) reported finding them breeding on a roof on the north side of 71st street, Philadelphia, in June 1882. In 1870 and 1871 Dr. T. M. Brewer (1874) found a number of instances of this bird nesting on the flat mansard roofs of Boston, and a few years later it was discovered nesting on the flat roofs of Montreal, Canada, by William Couper (1876). In 1879 a pair of the birds built a nest on a roof in the heart of the city of Cleveland, Ohio, and a few years later E. Sterling (1885) observed three pairs nesting on a slate roof of a large building near his study. Since 1880 there are innumerable records of nighthawks that have deserted the rural districts to take up their residence in the city. Today roof-nesting sites of the nighthawk seem no more unusual than the nests of the chimney swift built in our sooty chimneys. It is difficult to determine the factors that have been instrumental in this radical change, for unlike the case of the swift's hollow trees, there are just as many rocky knolls, pebbly beaches, and barren fields as there were prior to the appearance of the flat-roofed buildings. Although certain insects such as flies and mosquitoes are abundant about our cities I do not believe that the food supply has an important bearing on this question, as has been maintained by certain observers. Even in the small country villages, wherever there are graveled roofs there is the usual quota of nighthawks. In these villages the environment, as far as food for the nighthawk is concerned, is not different from regions isolated from civilization.

W. E. Saunders (1917) states that the young after their first flight often land on the ground and he asks the question: "What chance of survival is there for a young nighthawk on a city street or vacant lot?" Mr. Saunders believes that the nighthawk has steadily decreased in numbers since the bird has taken up its abode in the city and states further that immigration is the only thing that keeps up the city population. To the contrary my observations of the past 25 years in Maine indicate that the birds are not only maintaining but are increasing their numbers. Furthermore, the mortality of the birds that nest on roofs is much less than among those that choose nesting sites on the ground. In the latter place the birds are constantly exposed to the ravages of predatory animals including the cat. In the country hundreds of young birds meet with a tragic end without any of us being the wiser, while in the city such cases are more likely to be brought to our attention. On the city roof there is freedom from natural enemies. In

the study of a large number of nighthawk families I have noted relatively few young that left the nesting roof prematurely, and it has been a common experience to see them return to the roof many times after the initial flight. One bird, the account of which is related in a subsequent division of this paper returned to the roof every day until it went south on its migration at the age of 52 days. Many of the birds that nest on the roofs are never disturbed by human beings during the entire nesting period, and under such circumstances the young do not leave the security of the roof until they are able to fly well. It is obvious that the young of parents that nest on roofs are the ones most likely to survive, and they in turn will nest in similar situations the following seasons. It seems reasonable to suppose that after the habit is established natural selection and heredity play an important part in the general departure from the old to the new nesting environment.

No attempt is made by the nighthawk to construct a nest, and no materials are added to those already present on the nesting site chosen by the birds. The eggs may be in a slight depression, but no material is excavated or removed by the bird except that incidentally shoved aside by the incubating bird.

Eggs.—Normally two eggs are laid, and there is seldom a departure from this number. In a letter dated January 20, 1936, F. W. Rapp, of Vicksburg, Mich., writes that he found a nest of the nighthawk containing three eggs on May 22, 1889.

The eggs are elliptical-ovate or elliptical-oval, one end being slightly smaller than the other. The shell is strong, closed grained and moderately glossy. The ground color varies from pale creamy white to shades of cream olive-buff and olive-gray. The eggs are marked and speckled with shades of slate, black, drab, smoke and lilac gray, and tawny-olive, and some of the eggs have shades of pearly gray, lavender, and plumbeous. In some eggs the markings are fine and uniform in size, almost obscuring the ground color; in others they are less numerous but larger and more prominent. There is an endless variation in the details of the markings and colors, but the eggs in general exhibit a coloring that blends effectively with their surroundings.

The average measurements of 81 eggs in the United States National Museum are long diameter 29.97, short diameter 21.84 millimeters. The largest egg of the series measured 33.53 by 22.86 and the smallest 27.68 by

20.57 millimeters. The average capacity of nighthawk eggs, according to Walter Hoxie (1887), is 0.448 cubic inch.

The eggs do not necessarily remain in the position in which they are first laid, especially when the nesting site is on a comparatively level surface such as that provided by graveled roofs. During the period of incubation of two nests under daily observation, the eggs were gradually moved for a distance of 5 to 6 feet. This shift in position comes about by the habit of the female pulling or pushing the eggs under her breast, thus moving them a short distance each time she settled on the eggs. As the direction of the bird's approach was more or less constant, the eggs were moved in the same general direction. This resulted in a distinct trail being formed, giving an appearance that one would expect to see if a giant snail had traveled over the graveled surface and forced the larger pebbles to one side. B. H. Warren (1890) writes that the eggs of a nighthawk in one instance were moved 200 feet by the bird, which carried them in her mouth. I have never been able to verify this extraordinary behavior of a nighthawk transporting an object as large as an egg in its mouth.

The following experiment is of interest as it suggests an interpretation of the manner in which a nighthawk locates its eggs. One evening when the female nighthawk was away feeding I moved the eggs to a place on the graveled roof about 6 feet from their original position. When the bird returned she alighted in the accustomed place and waddled up to the spot where she had left the eggs. No eggs being there, she went by a few inches, turned around, and recrossed the spot. This was repeated several times, and finally much bewildered she flew away. The eggs were in plain view yet were not discovered. Fearing she might desert her nest, I returned the eggs. In about 10 minutes the female returned, alighted in the usual position and without hesitation went directly to the eggs. This experiment was repeated on this and other birds with essentially the same results. It is evident that some factor such as a hypothetical sense of location or orientation is important, whereas sight plays a minor role for the nighthawk in locating its eggs. The same factor is probably important to the nighthawk in finding its way over thousands of miles during migration and its ability to arrive punctually not only in the same State and the same town but to the identical nesting site.

Incubation.—Both the male and female have been reported as sharing the duties of incubation. George H. Selleck (1916) states that the male nighthawk incubated the eggs during the daytime, whereas the female took charge of the nest at night in the case of a pair of birds he observed at Exeter, N. H. Dr. A. A. Allen (1933) states that the male takes his place on the eggs in the evening while the female is away feeding. Forbush (1927) states that both male and female share in incubation. On the other hand, J. H. Bowles (1921), who made an intensive study of the nighthawk, records only the female incubating the eggs. T. G. Gentry (1887) states that incubation “is the exclusive labor of the female” but that the young are cared for by both parents. It is evident, if the above statements are all true observations, that there is considerable individual variation in the behavior of the male in regard to incubation. In my own intensive studies of several pairs of nighthawks that nested on graveled roofs at Brunswick, Maine, I have never seen the male incubating the eggs. In the case of one nest under daily observation the birds were subject to study day and night by a relay of observers for a considerable part of the incubation period. The same held true for a pair of nighthawks studied in northern Michigan. Furthermore, visits made to numerous nests revealed in those cases that only the female was incubating the eggs, although the male was often very near to the female or to the eggs. My observations agree with the statement of Gentry that the female does all the incubating, but the male in most instances assists in caring for the young.

The behavior of the birds in relation to the nest and eggs and correlated with the time of day, the weather, environmental conditions, and the activities of other birds can be illustrated by field notes taken on June 20, 1921. These notes are representative of observations taken throughout the nesting season. The nest under consideration was located on the graveled roof of a two-story high-school building at Brunswick, Maine, a village of about 7,000 inhabitants located on the Androscoggin River. There are numerous giant elms and other shade trees along its streets, and the spacious yards and gardens present an environment attractive to a large number of birds. The times given throughout the following notes are eastern daylight saving:

June 20, 1921. 2:50 a. m. Cloudy, moon hidden by clouds, clear near the eastern horizon where the stars shine brightly. Entered the blind on the roof

at 3 a. m. Female incubating the eggs. Her eyes were wide open when viewed with the flash light.

3:20 a. m. The female has been quiet during the past 20 minutes but has now shifted her position and is facing northeastward directly toward the first faint light of dawn. (Daily observations revealed that the nesting bird usually faced the sunrise in the morning and the sunset in the evening. In other words, she oriented herself toward the source of light preceding the times she anticipated leaving the nest to feed. It became evident through repeated experimental tests that her leaving the nest was considerably influenced by the factor of the intensity of light.)

3:24 a. m. The male gives a loud *peent* call as he awakens on his perch in the elm tree. (The elm stands in the school yard and some of its branches extend over the roof wall. The male's favorite perch was on a large horizontal limb just above the level of the roof. When he was through with his activities in the morning or evening he usually retired to this limb. He generally perched lengthwise on this limb, but it was not rare for him to depart from the conventional nighthawk position and perch crosswise to the limb.)

3:40 a. m. Light growing brighter in the east, the female can be seen without the use of the flashlight. She sits with her eyes closed, but from time to time she opens them, then raises her head and peers around, as if testing the intensity of the light prior to her leaving the nest.

3:42 a. m. The male nighthawk alights on the roof and gives a sharp *peent* call, followed by a series of guttural *awk-awk-awk* notes.

3:44 a. m. The male shifts his position to the roof wall but continues the guttural calls, each time displaying his conspicuous white throat patch.

4:04 a. m. The male leaves the roof wall and alights on the graveled surface within 8 inches of the female. He utters a single sharp *peent* note, then continues with the striking guttural *awk* calls. The female shows no outward signs of being impressed by the ardent attentions of her mate.

4:08 a. m. The male leaves the roof but soon returns and is joined by a second male. Female leaves her nest and flies toward the males, and all leave under great excitement, uttering sharp, piercing calls. The female lent a hand in driving away the strange male.

4:10 a. m. The female returns and pulls her eggs beneath her in the usual manner.

4:14 a. m. The male alights on the roof about 30 feet from the nest and utters a series of *awk* calls in rapid succession, as if much excited as the result of his combat.

4:30 a. m. The male is chasing a strange nighthawk and, as he pursues he utters a series of *yap-yap-yap-yap* calls in rapid fire succession, a note I have not heard before.

4:33 a. m. The male is now going through his hair-raising dives and producing the so-called boom notes, which to me resemble *swo-o-o-onk*, with the accent on the last syllable.

4:46 a. m. The sun has not yet risen. The male nighthawk flies to his perch in his favorite elm tree, and thus end his activities for the morning.

9:00 a. m. The nighthawks have been quiet since 4:46 a. m. For the greater part of the time the female has been motionless, with her eyes closed, but at intervals she opens her eyes wide in response to some unusual disturbance. She did not leave to feed, as is usual for her to do each morning.

9:10 a. m. The female shifted her position and turned her eggs. I left the blind at 9:15 a. m. and my place was taken by student observers, who took notes in relays until my return in the evening. They reported no activity on the part of the male. He remained on his perch in the elm throughout the day. The female did not leave the nest but merely shifted her position slightly from time to time in adjusting her eggs. When the heat was excessive she panted vigorously in order to adjust her body temperature.

7:30 p. m. When I arrived at the roof the sun was shining on the nesting female. Her eyes were opened as I approached the nest but were again closed after I had entered the blind.

8:05 p. m. The first *peent* call of the nighthawk is heard from the male nighthawk, perched in the elm tree near the school building, where he had retired at 4:46 a. m.

8:09 p. m. Male leaves his perch in the elm. Simultaneously another nighthawk appears from the elms farther down the street.

8:46 p. m. The male alights near the nesting female.

8:47 p. m. The female, facing toward the western sky, opens her eyes wide and turns from side to side. With this preliminary action repeated several times, she leaves the nest. The male remains within a few inches of the nest but makes no attempt to incubate the eggs. He merely serves as a guard while the female is away.

8:48 p. m. The male leaves the roof for a moment, uttering the sharp *peent* calls, and then returns to take up his position near to the nest.

8:50 p. m. The female returns to the nest after being away only three minutes. The male was quiet while on the roof alone, but as soon as the female returned he started his guttural notes.

8:58 p. m. The male takes leave without any ceremony and flies to his perch in the elm tree.

9:01 p. m. The female leaves the nest quietly.

9:22 p. m. No nighthawks have been seen or heard during the past 20 minutes. The female unannounced returns, after her quest for food, to brood her eggs for the remainder of the night.

Activities began again at 3:14 a. m. the next day, and the program was in all essentials similar to that recorded for June 20. Although a large series of such observations were made the male never was seen to incubate the eggs, but he regularly visited the nesting female each morning and evening. Sometimes, especially in very cloudy weather, the male left the elm tree during the day and flew about uttering the usual *peent* calls. He was never seen on the roof during the middle of the day.

The female was always faithful to her task regardless of a parching sun or torrential downpours of rain. At one time the thermometer placed on the graveled roof reached 130° F., although it was only 98° F. in the shade on the street level. At such times her enormous mouth was wide open, and she panted incessantly. The female's presence was needed to protect the eggs from the excessive heat fully as much as it was to supply warmth during the cool evenings when the temperature sometimes dropped as low as 45° F. If the female should leave her eggs when the temperature rises as high as 130° F. the embryos inside of the eggs would be killed in a few minutes.

It was common for the heat to melt the tar beneath the gravel so that it oozed to the surface of the pebbles. Milton Goff (1932) relates an interesting experience of a nighthawk that nested on a school roof at Rockford, Ill. One of the two eggs became firmly embedded in some tar melted by the sun. The female was unable to move the egg after nine desperate attempts made during the course of two hours.

Young.—In 1922 the nighthawks arrived at Brunswick, Maine, on May 15. On June 3 the first egg was laid and was marked No. 1. It was not incubated the first day. On June 4 the second egg was laid at 11:05 a. m. and marked No. 2. The laying of the second egg was observed. It was seen when it first appeared and watched until it emerged from the cloaca two minutes later. The egg was moist at first but quickly dried in the warm air, after which the female turned around and tucked it under her breast. Incubation began immediately after the second egg was laid.

On June 23, 1922, egg No. 1 was pipped at 8:30 a. m. and at this time the peeps of the confined embryo could be heard distinctly. At 11:00 a. m. the chick had emerged. The egg shells of the first egg were removed about noon. The female took them in her beak and dropped them during flight at a point about a hundred yards from the building. Egg No. 2 was pipped at 2 p. m., and at 3 p. m. portions of the shell were broken away. The female seemed little concerned about the first chick but lavished all her attentions on the unhatched bird. At times she elevated her body, peered at the egg, and responded to the calls of the embryo with an assuring note. She frequently rocked her body over the egg as if to assist in removing the shell. At 6 p. m. the shell was cracked latitudinally and soon thereafter the cup of the larger end was slightly lifted. At 6:30 p. m. the young was completely freed from the shell, which broke away in two parts. As the shell membrane dried the two cups closed again forming a complete empty shell case. At this time the first young was being brooded while the second, in a wet bedraggled condition, awkwardly and weakly wavered its head beneath the bill of the female. The faint peeping notes uttered by the youngster were answered with guttural purring notes of the proud and triumphant mother. After the down of the second youngster was dry it joined its fellow under the breast of the mother. The egg shell was not removed until the following morning. The incubation period for this set of eggs was definitely established to be 19 days.

The next day the female attempted to brood the young continuously. One of the young, however, persisted in its attempts to break away from parental care in spite of the warning notes of its mother. In one instance the mother dragged the one young under her body until within reach of the unruly youngster, which was quickly and violently tucked under her breast with something of an attitude of rebuff.

On the second day the young frequently appeared in the open and at such times often pecked at the mother's beak as if recognizing it to be the door of a well-filled cupboard. The young had not yet been fed, as they were still dependent on the yolk provided by the egg and stored in their bodies at the time of hatching. The female remained with the young throughout the day, whereas the male did not make his appearance until 8:50 in the evening. He announced his arrival with a sharp call as he landed on the roof. His call was immediately answered by the female, whereupon he made his way to the nest and without ceremony or delay delivered the first food received by the young.

I was not prepared to see the male feed the young because the male of the preceding summer (another individual) never assisted in the care of the young. This striking difference in individual behavior emphasizes the point that it is not safe to generalize on observations of a single individual.

The next morning the female was brooding her young, and being accustomed to my daily presence she allowed me to reach under her breast to remove the young without exhibiting the least bit of fear. A female nesting on another roof when visited for the first time scooted away hurriedly, fluttered helplessly on the roof, enacting a perfect imitation of a crippled bird, a ruse to attract my attention away from her young. When I followed her she quickly flew away. On subsequent visits this bird stood her ground, elevated her wings in an upright position, and hissed at me in defiance of my approach.

The following notes are from observations made of the young that were hatched on the high-school roof the preceding year (1921). The female of that year was the same bird that nested there in 1922. The identity of the female was established by banding and also by a slight deformity of the maxilla, the tip of which was broken off. The males of the two broods were different individuals.

The young were fed during the early morning before sunrise and again in the evening after sunset. During the heat of the day, when the temperature frequently went soaring above 110° F., the young kept well concealed under the breast of the brooding bird in close contact to her abdominal air sacs. When the temperature was less torrid the young peeked their heads through the feathers and often came out entirely. At such times they sometimes amused themselves by picking at the mother's beak and rectal bristles. If the annoyance became too great the female would thrust her head beneath her breast. The young then proceeded to pick at the feathers of her crown and nape.

The female left the nest regularly about 8:30 p. m. (daylight saving time) to obtain food for herself and young. In the case of this brood the female delivered all the food required by the young. The food was delivered by the process of regurgitation. The beak of the female was thrust well into the large and widely distended mouth of the young when the transfer of food was made. After feeding, the adult brooded her young in the earlier stages of their development.

For the first three days the young remained near the spot where they had been hatched, but on the fourth day after a heavy rain they had moved to a slightly raised portion of the roof, which was free from excessive dampness. The female never left the young during a rain at this stage of their growth, even if it meant depriving them of food. The male regularly visited the roof early in the morning and again in the evening throughout the breeding season, but he was never seen to deliver food to the young. The male was usually stationed on the roof when the female was away and thus served as a watchdog in her absence. More than once he was called upon to chase away a strange nighthawk that had inadvertently alighted in his territory on the high-school roof.

On the fifth day (June 29) the young had wandered to the southern end of the building where the shadow of the roof wall protected them from the direct rays of the burning sun. The heat during the middle of the day was frequently excessive. On July 7, when the young were 13 days old, a thermometer placed on the gravel of the roof registered 140° F. The heat was too much for one of the young, which succumbed and was found dead near the middle of the roof. Its fellow nestling, hidden in the shadow of the roof wall and behind a clay ventilating pipe, escaped death but was in a

serious condition. A sponge soaked with cold water was placed near the youngster, whereupon it ignored its parent to enjoy the comfort provided by the cool sponge. This act probably aided the survival of the young nighthawk through the remainder of the record-breaking hot day.

As the nighthawk grew older he became very aggressive and pugnacious and never hesitated to pester his mother whenever he was hungry. At times he seemed mischievous. It was not a rare experience to see him crouch under his mother's breast and then by standing up quickly and rigidly topple his mother so that she was forced to extend her wings to keep from falling over. During the period of the rapid growth of the wing feathers the youngster was continually extending and stretching his wings as if to relieve the uncomfortable feeling caused by growing pains.

When the young nighthawk was three weeks old he was able to make short flights from one place to another on the gravel roof. At this stage of his growth he had become so large that it was difficult for the female to cover him adequately while brooding. This was strikingly demonstrated on the evening of July 17, when he was 23 days old. The mother had left him alone on the roof while she was away on her regular evening search for food. It was so cool and damp that the young bird uttered notes that clearly indicated discomfort. I placed him in a woolen bag, and this, combined with the warmth of my hands, was very satisfying, as indicated by his change to notes of contentment. He remained there in comfort until his mother alighted on the roof and gave the characteristic call note, announcing supper. The little nighthawk struggled out of my bag and ran directly to his mother to be fed. After he was gorged with insects, the female attempted to brood him, but to her apparent dismay he rushed back to the woolen bag, clearly recognizing that I could do a much better job at brooding.

On July 19, when the nighthawk was 25 days old, I discovered that he was no longer solely dependent on his parent for food. I found him busily engaged catching some white moths that had collected about a drain pipe of the roof. He was flying this way and that, catching the moths with great delight. As I sat there watching I chanced to pull a small handkerchief from my pocket, whereupon he dashed at the white object with the ferociousness of a tiger. Evidently to the nighthawk this was some giant moth large enough to provide for an entire meal.

The female now made frequent attempts to entice the young bird away from the roof by first offering food but flying away before it was delivered. The youngster would follow her in extended flights but invariably returned to the roof.

On July 24, when the bird was 30 days old, the female was photographed with the young for the last time. Thereafter she forsook her offspring for a roost in the nearby elm tree. She then visited the roof only for short intervals at feeding times. On each successive day the young bird took longer and longer flights, and each day I anticipated it would be my last opportunity to photograph and to observe him. Much to my delight he continued to return, seeming to enjoy my companionship.

On August 15 he left with the other nighthawks of the vicinity on their migration to the south. This unusual experience of having the bird return to the roof gave me an unprecedented opportunity to make a continuous set of daily observations and measurements of a nighthawk living under normal natural conditions up to the time it was 52 days of age. By this time the growth of the juvenal plumage was completed and exchanged in part for the first winter plumage.

Plumages.—On the first day the young are able to stand upright and are very active from the time of hatching. The eyes are open, iris bluish black; skin darkly pigmented darker above than below; bill “pale mouse gray”; tarsus and toes brownish drab. Down present on both dorsal and ventral parts of the body. Down of the ventral tracts pale gray shading to “pallid neutral gray” on the belly; chin gray, malar stripes and patch on the throat “dark mouse gray,” approaching black. Upper parts mottled and marbled, made up of patches of pale gray and “dark mouse gray.” In the region of the nape and scapulars the down has a distinct “pale olive-buff” tint. At the base of the beak the down has a tinge of buff. A circular area about the anus, the outer part of the shanks, and fore arm have patches of darker colored down. Patterns of dark and white vary considerably in different young. The average length of the down on the various parts of the body varies as follows: Crown 9, base of bill 4, region over eye 6, throat 12, belly 15, wing 11, and region of anus 8 millimeters.

On the third day the color of the bill has changed to a “deep neutral gray,” and the tarsus and toes become a dusky drab. Down at the base of the

beak, scapular region, and irregular patches of the back has faded from the colors present in the day-old chick to a "tulle buff." Iris is now a clear brown instead of the bluish black of the freshly hatched chick.

At the age of 10 days the tarsus and toes are "deep Quaker drab," bill "dark neutral gray," eyelids "light neutral gray." Exposed portions of the eyes are noticeably greater. Down of the back much worn and matted down, the feather papillae in the region of the crown, wing coverts, scapulars, and rump now more conspicuous than the down. The tips of the feathers of the back are unsheathed and exhibit a black and cinnamon color. The color pattern of the back is completely lost since the appearance of the feather papillae and their unsheathed tips. The papillae of the tail feathers are only 8 millimeters long and as yet do not show through the down. The feather papillae of the head and tail are the last to unsheath.

In chicks 13 days old the "pinkish cinnamon" of the tips of the feathers of the dorsal tracts has faded to a "pinkish buff," and some of the feathers approach a "tulle buff." Freshly unsheathed feathers of the crown are "pinkish cinnamon." Black markings have faded to a "fuscous-black" or "dusky neutral gray"; yellowish color of the breast feathers of the younger stages now faded to gray tinged with yellow. The juvenal plumage is rapidly replacing the natal down in all parts of the body. The yellowish-gray breast feathers are barred with "dusky neutral gray." As previously noted, one of the two birds died from exposure to great heat at this time and the remaining descriptions are based on one bird.

At the age of 15 days the down is ragged in appearance but is still prominent on the breast, sides of the head, wing coverts, and region of the tail. The down is worn off the feathers in the region of the crown and scapulars. Feathers of all tracts of the juvenal plumage are partially unsheathed. When the young are handled there are numerous particles of the feather sheaths that scale off. The wings exhibit a marked development. The tips of the under tail coverts recently unsheathed are ivory colored. The down persisting at the tips of the feathers comes off with the slightest pull. The pupil of the eye appears bluish black and the iris has changed to a "Van Dyke brown."

When 17 days old the sixth primary has now proceeded with unsheathing so that the white patch is 13 millimeters in extent. Tarsus and

toes are “blackish plumbeous,” bill is a “dusky purplish gray,” eyelids are “dark olive-gray”; otherwise the markings are similar to those of the 15-day-old bird.

When 20 days old a relatively small amount of down remains, but a few filaments can be seen on the tips of some of the crown feathers, sides of neck, and breast. Feathers of the wings are growing fast and are so heavy from the large amount of blood and large sheaths that the wings rest on the surface of the roof, the bird being unable to support them. The wings are frequently outstretched, apparently to relieve the uncomfortable sensation produced by rapid growth of the feathers. Bristlelike feathers now appear around the base of the beak. The fifth primary is grown so that the white patch is beginning to unsheath, and the base of the fourth primary shows white area on sheath, which is destined to form part of the white patch on the wing of the fully grown young. The white on the sixth primary made its first appearance in the 17-day-old chick.

At this age the crown is “dusky neutral gray” or black, the feathers tipped, barred, and spotted with “vinaceous-buff”; some of the tips approach “avellaneous.” The feathers of the upper parts are dark, or dusky, neutral gray variously mottled with shades of gray and “avellaneous” many of the feathers tipped with “pinkish buff.” Auriculars “light cinnamon-buff” and black, tipped with lighter shades. Primaries and secondaries blackish warm gray, some of the feathers having an olivaceous-black appearance. All the remiges are tipped with “tilleul buff” or “vinaceous-buff.” Some of the secondaries and inner primaries spotted with “vinaceous-buff.” Coverts of primaries unspotted “blackish mouse gray” or black. Coverts of secondaries spotted with “tilleul buff” and “vinaceous-buff.” Lesser wing coverts variously mottled with colors mentioned above. Feathers of the breast barred with “dark mouse gray” and shades of gray and white. Throat and chin with a crescentic band of feathers, which are barred with dark gray and white, the white predominating. Band of feathers along the edge of mandible extending below the eyes to the auriculars “neutral gray” and marked with “vinaceous-buff.” Under tail coverts “cartridge buff,” narrowly barred with black. Middle of the belly a heavy mat of down. Eyelids “light olive-gray,” tarsus deep metal gray.

At 25 days the down has been lost, except that of the middle of the belly and small patches on the legs above the heels.

At 28 days the colorings are about as described for the 20-day-old chick, but the colors are subdued and faded because of exposure to the intense sunlight. The down is now entirely replaced by the feathers of the juvenal plumage. Although the outer primaries are only partially unsheathed, the young bird is capable of long flights and frequently leaves the roof, but invariably returns.

At 30 days the secondaries and the primaries, except the outer ones, are now unsheathed. The barred feathers lining the wings are now in process of unsheathing.

At 35 days the prevailing color of the upper parts is olivaceous-black, glossed with greenish, but this base color is very much broken by irregular markings, spotting, and marbling of buffy gray and whitish; many of the larger spots approach a light vinaceous-buff, which is especially evident on the crown. Spots of the crown are larger than those of the back. The throat patch now is clear white, an indication of its sex.

At 40 days, all the primaries except the outer two are completely unsheathed. The white patch of the second to fifth unsheathed, that of the first is 11 millimeters.

The bird, although able to fly as well as an adult, returned to the roof each day and allowed me to make daily measurements and weighings without the least resistance. It never attempted to run or fly away when I approached to pick it up. When I arrived on the roof wall it gave a series of calls, which seems to be a sign of recognition. The bird placed before me in a natural sitting pose measured 187 millimeters from the tip of the bill to the tip of the wings. Distance from level of the crown to the board on which it is seated is 90 millimeters. Tip of tail to front of toe in natural position 150, and tip of folded wing to toe 165 millimeters.

The bird at this age is very active and vivacious and captures all its own food. It offers no difficulty for me to capture it for measurements and poses perfectly for photographs.

On Monday, August 15, there was a great flight and departure of nighthawks, and after that date the nighthawk was seen no more. Presumably it went southward on its migration. The bird was 52 days old,

and I had an unexcelled opportunity to observe the bird up to the time of the completion of its growth.

The completed juvenal plumages of both sexes is similar to that of the adult female except that the throat patch is not so well defined; in some it is replaced by blackish and buffy bars. The barring of the underparts is more extensive and the coloration in general is paler than that of the adults. There are whitish tips on all the primaries.

The postnatal molt has already been described in detail under the account of the young. There is a partial molt that does not include the wings and tail in September. There is a partial or complete prenuptial molt in spring when the young attain the plumage of the adult. The adults have a complete postnuptial molt before they return for their nesting activities the following year. I have not been able to ascertain the time of the postnuptial molt.

Albinistic phases of plumage, in which there is an absence of dark pigment, may appear in any species of birds. W. A. Strother (1886) reports a perfect albino taken at Lynchburg, Va.

Food.—The nighthawk is insectivorous in its eating habits. Since the major part of the insects it destroys are destructive to useful vegetation or are otherwise adverse to human welfare, the nighthawk ranks high in the list of birds beneficial to man.

The nighthawk captures the insects chiefly during its flight. The birds sweep up in their capacious mouths all types of insects from the large moths and beetles to the tiniest of flies and mosquitoes. Some of the stomachs examined have contained no less than 50 different species of insects, and some of the smaller insects are at times represented by thousands of individuals.

One of the most conspicuous elements of the food is flying ants. In the examination of 87 nighthawk stomachs, the United States Biological Survey reports that ants comprised nearly one-fourth of the total food eaten by the birds. In 24 of the stomachs the number of ants ranged from 200 to 1,800, and in all the stomachs examined there were not less than 20,000 ants (Beal, 1897). In the stomach of a nighthawk that met with accidental death at Brunswick, Maine, on August 20, 1925, there were 2,175 ants. The

mass and weight of these insects were so great that they constituted a serious handicap and probably a factor in the bird's untimely ending. Charles Drury (1887) obtained a specimen in August that contained 320 insects, chiefly winged ants. W. L. McAtee (1926) found more than one hundred carpenter ants (*Camponotus herculeanus*) in one nighthawk stomach. In most all instances the ants captured are the mating winged individuals, which fly in immense swarms during the late summer. These ants are killed at a time when they are preparing to propagate their kind, and hence the death of every female means the destruction of thousands of the next generation.

According to examinations of the United States Biological Survey (Beal, McAtee, and Kalmbach, 1916) beetles comprised one-fifth of the food eaten by 87 nighthawks examined. May beetles, dung beetles, and others of the leaf-chafer family were in greatest numbers. Beal (1897) reported finding the remains of 34 May beetles (*Phyllophaga*) in a single nighthawk stomach, in another 23, and in a third 17. In the stomach contents of one specimen no less than 17 species of beetles were identified. Chester Lamb (1912) reports that all nighthawks collected by him had eaten enormous quantities of beetles. Thomas G. Gentry (1877) reports eight species of beetles in food examined by him. McAtee (1926) reports various leaf chafers, sawyers, wood borers, bark beetles, weevils, and plant lice in the food eaten by nighthawks in the course of his study of the relation of birds to woodlots in New York State. In the examination of hundreds of droppings of nighthawks obtained from various nesting sites chiefly at Brunswick, Maine, a large percentage of the identifiable remains consisted of parts of various species of beetles.

Nighthawks, especially in the Middle West, have been known to eat a considerable number of grasshoppers and locusts. According to Ernest Harold Baynes (1915) seven Nebraska specimens were found to have eaten 348 Rocky Mountain locusts; five specimens collected in Indiana reported by A. W. Butler (1898) had eaten 9 grasshoppers, 19 beetles, 23 Heteroptera, and 4 Neuroptera. B. H. Warren (1890) reported that grasshoppers were an important element of the food eaten by nighthawks collected in Pennsylvania. F. E. L. Beal (1897) states that one nighthawk contained the remains of 60 grasshoppers. A male killed on July 7, 1882, was reported by Everett Smith (1883) to have "an ichneumon fly, a black

cricket, about twenty small grasshoppers, and many small, hard insects” in its crop.

Flies, plant lice, and mosquitoes frequently form an important element of the food of the nighthawk. A nighthawk examined by the Biological Survey had eaten more than 300 mosquitoes, and E. H. Forbush (1907) reports finding 500 mosquitoes in the stomach of one bird. McAtee (1926) reports that he found 650 plant lice in the stomach of a single nighthawk. Phoebe Knappen (1934) found stone flies in 21 nighthawks. The insects were chiefly adults but also a few larvae, nymphs, and eggs were present.

In the South various observers have noted that the nighthawk is an important factor in the control of the cotton-boll weevil. F. H. Herrick (1901) writes of a nighthawk that had been feeding on fireflies; the wide open mouth of an adult observed feeding its young was brilliantly illuminated like a spacious apartment all aglow with electricity. F. H. Carpenter (1886) relates a unique experience with nighthawks that darted at the artificial flies on his line when he was casting for trout. It is not an unusual experience to see nighthawks after dusk flying about electric lights of city streets (Knowlton, 1896) or about campfires of remote districts where they capture myriads of insects attracted by the lights.

During the migration nighthawks frequently fly near the ground and at such times may take advantage of any insects that appear in their path. During an afternoon late in August I observed a flight of several hundred nighthawks near Urbana, Ill. As I sat in a car alongside a large meadow I noticed that the birds were ravenously feasting on grasshoppers. Some of the birds lingered long enough to capture half a dozen of the insects before passing on to make place for other nighthawks in the migrating procession. A. Dawes DuBois, of Excelsior, Minn., relates a similar experience he had in the vicinity of Salt Creek, Logan County, Ill., as follows: “At dusk on the evening of May 22, 1913, as I walked along a cloverfield, we witnessed an assemblage of nighthawks in pursuit of low-flying insects. They skimmed over the clover like swallows; and their dusky forms were so numerous that they seemed to be weaving an intricate pattern in the gray twilight. They were so intent on their bountiful repast that they paid no heed to our presence, but sometimes darted past us only a few feet away.”

Nighthawks not only capture food on the wing, but they also have been observed to drink, in the manner of swallows, as they skim near the surface of the water of lakes and streams. F. Stephens (1913) observed a nighthawk drinking from a watering trough. This bird dropped its lower mandible into the water, rippling the surface of the water as it passed along. A. Dawes DuBois writes of the following experience he had at Springfield, Ill., on July 29, 1923: "Mr. R. B. Horsfall and I were walking along the margin of a small pond when a nighthawk swooped down, touched the surface of the water and rose again; but we could not tell whether it was scooping up a floating insect or a drink of water."

Voice.—The nighthawk has no claim as a singer, but nevertheless its notes are of great interest and attract fully as much attention as the voices of our more gifted songsters.

The loud piercing calls uttered by the nighthawk during flight are simple yet, like all bird notes, extremely difficult to represent in written words so as to enable a reader unfamiliar with them to gain a clear conception of their character and quality. This fact is at once emphasized if we compare the interpretations of a few of the many authors who have attempted a written version of its calls. For example to C. G. Abbott (1914) it sounds like a "grating 'beedz, beedz,'" and to Charles Bendire (1895) it is a querulous and a squeaky note resembling "aek-aek, aek-aek" or "speek-speek, speak-speak." W. E. Grover describes this note as a sharp "mueike"; and E. H. Forbush (1907) states that "the note is s-k-i-r-k or s-c-a-i-p-e, a little like the call of Wilson's snipe,—rather a startling squeak when heard close at hand." W. L. Dawson (1903) interprets the note as "mizard, mizard," and E. H. Eaton (1914) describes it as a "loud nasal 'peent, peent.'" N. S. Goss (1891) writes that its voice is a "squeak" or a "pe-up" note, and Arkansas Hoosier (1890) states that "the note is best produced by speaking the word 'beard' in a whisper." To G. R. Mayfield (1921) the call is a shrill "B-e-e-r-b" and to H. Nehrling it is "Brirrr-brirrr." H. Tullsen (1911) interprets it as a sharp penetrating "Spe-eak," and H. H. Bailey (1913) thinks it sounds like "Queek-queek."

There was probably only the slightest variation in the notes of the different nighthawks as heard by the authors mentioned above, yet how strikingly different are the interpretations as represented in the written or printed words. The note described above is the one most frequently heard,

and it is uttered independently of the seasons. It is the note that announces the arrival of the nighthawk in spring, and it is the call uttered at the time the nighthawks are congregating in fall in preparation for their departure to the south. I have been unable to ascertain whether this call is heard at their winter home in South America.

Frequently, during the courtship season, the males in their competition for a mate vigorously pursue each other. At such times both birds utter a series of sharply accented calls recorded in my field notes as resembling *Dick-a-dick-a-dick-dick-dick-dick-dick*, given in rapid succession.

There is another note of the nighthawk very different in character, an aeolian sound, produced by the rush of air through the primaries of the wings at the termination of the extraordinary downward plunge executed during the courtship season. In order to see this performance and to hear this peculiar and unique note to the best advantage it is necessary to visit the vicinity of the nesting site. Alexander Wilson (1828) described this note as a “loud booming sound very much resembling that produced by blowing strongly into the bunghole of an empty hogshead.” E. H. Eaton (1914) offers a modification of Wilson’s description in stating that the note is like that produced “by blowing across an empty bottle.” T. G. Gentry (1877) describes it as a sound resembling that “produced by a tense cord set in vibration by a sudden gust of wind.” T. Jasper (1878) states that it is a hollow whir like the rapid turning of a spinning wheel, and F. A. Hartman (1914) describes it as a “guttural ‘woof.’” The note reminds W. A. Stearns (1883) of the sound produced by a bellowing bull. In my field notes I have described this note as a muffled *wr-r,r,r,r-oonk*, but sometimes more nearly approaching *sw-r,r,r,r-ooonk*, the last syllable decidedly accented and produced with great resonance. There is nothing about this note suggesting an explosive boom or bellowing.

It has long ago been well established that this peculiar note is not a vocal sound but one produced by the vibration of the primaries. As keen an observer as Alexander Wilson (1828) stated that it is “produced by the sudden expansion of his capacious mouth.” Others shared Wilson’s view or thought that it was a sound produced by the syrinx. After one has observed the performance it can be readily understood how such an erroneous interpretation was made by the earlier observers. The plunge takes place so quickly that it is only by repeated observations made under the most

favorable conditions that the observer is convinced that the primaries are involved. Audubon (1840) was the first to arrive at a correct explanation. He writes that the source of the singular noise is “the concussion caused, at the time the bird passes the centre of its plunge by the new position of its wings, which are now brought almost instantly to the wind, like the sails of a ship suddenly thrown back.” This observation with variation of its details has been made by numerous subsequent observers. J. B. Canfield (1902) gives a description of the performance as follows: “He suddenly paused and came soaring toward me like an arrow. About fifty feet in front of me his wings were lowered below his body, throwing them forward with the flight feathers spread wide apart * * * His speed was so great that the flight feathers vibrated like a loosely-stretched rubber band when snapped with the fingers. This performance was repeated in front, back, and beside me twelve times in all, never more than fifty feet away, and as near as fifteen. In all cases the wings were in the same position, and his mouth never open.” F. A. Hartman (1914) writes, “It is very evident that the mouth plays no part, otherwise the sound would be produced at other times [than on the downward glide]. * * * The bird threw its wings far to the front at the end of his downward glide, so that the uppermost quill feathers were pointed exactly in the direction of his glide. Going at such headlong speed, these quill feathers when thrown edgewise to the air vibrated strongly, causing the ‘woof’.” Alden H. Miller (1925) while at Camp Lewis, Wash., during June and July succeeded in attracting nighthawks within 10 feet of himself by merely waving his hat in the air. He noticed that if the wings, during the downward plunge, were held in the upturned V-shaped position, a normal pose when soaring, no boom was heard, but when the wings were bent downward near the end of the dive the boom sound was produced. The intensity of the sound, according to Mr. Miller, is more or less proportional to the speed attained. The main explosive boom seems to be preceded by a brief, lesser vibrating sound, which bursts forth into the full bellow. Both parts seem to have a distinct element of pitch, but the latter part is lower, with greater resonance and depth of quality.

The sound produced by the wings described above is a part of the courtship performance and is usually produced near the nesting site. After the young no longer require the constant attention of both parents, the male loses the glamour of romance and performs less frequently and soon after

ceases almost entirely until the courtship season of another year. However, birds migrating in August sometimes “boom.”

Another note associated with the courtship season is a guttural call uttered only when the male is at rest and in the presence of the female. It may be described as an oft repeated *auk, auk, auk* or *awk, awk, awk*. This note is produced by the syrinx, but as it is uttered the bill is tightly closed and the gular membrane is tightly distended each time the note is produced. The distension is caused by the expulsion of air from the respiratory system. The distended membrane, although feathered, acts as a resonator and modifies the note, giving it a peculiar quality. Such a mechanism is present in other birds. It is especially highly developed in the prairie chicken, in which large lateral vocal sacs are present that give the “booming” notes of this grouse great carrying power.

The notes of the female nighthawk are simple calls uttered in response to the those of the male or of the young. She may utter a purring, pacifying note when brooding the young, but often these notes are so weak that they cannot be heard by the observer unless he is stationed very near in a blind.

Game.—Today we do not think of the nighthawk as a game bird, yet 60 years ago large numbers of them were killed by gunners and sportsmen, especially in the Southern States. M. G. Elzey, writing on September 18, 1876, stated: “Bull-bats (nighthawks) are the best of the minor game of this country for sport or table; have been very abundant and in superb condition here (Blacksburg, Virginia) for the past two weeks. I have killed several hundred. On one occasion took out 28 cartridges and brought in 23 birds besides 2 which fell out of bounds and were recovered by boys. Killed 17 in succession. The bats are quite as fat and better game than the reed birds.” Dr. E. Sterling (1885) wrote: “Their rapid and irregular flight makes them a difficult mark for the young sportsman to practice on, as he never fails to make a target of them when the opportunity offers. I can now understand the object for which this bird was created.” Dr. F. M. Chapman (1888), writing of conditions at Gainesville, Fla., stated, “‘Bat’ shooting is here a popular pastime, great numbers being killed for food, and in August, when the birds have gathered in flocks, favorite fields may be occupied at nightfall by as many as a dozen shooters.” Stockard (1905) lamented the fact that the birds “are foolishly slaughtered by pseudo-sportsmen who

shoot them merely to watch the bird's graceful fall or to improve their skill as marksmen."

The practice of killing nighthawks was stopped through laws and by educational methods initiated by the National Association of Audubon Societies. William Dutcher (1902), in writing of conditions in Florida, stated, "It is believed, on very satisfactory evidence that the new law has stopped to a large degree the disgraceful practice of shooting 'bull-bats' or Nighthawks (*Chordeiles virginianus*) for sport." Bird-Lore for September-October 1903 published the following note: "The Night-hawk, or Bullbat, has been so long considered a legitimate target for shotgun practice, in the south, that a report of prosecution for killing these birds at Greensboro, North Carolina, marks a new era of bird protection in our southern states."

Enemies.—The greatest enemy of the nighthawk has been man. In the past great numbers of the birds were killed for food and often for mere sport or for satisfying a lust for killing. This was especially true in the Southern States, where the birds were slaughtered during the great flights of the annual migrations.

Nighthawks nesting on the ground are subject to the same enemies experienced by other species of ground-nesting birds. Night hawks nesting on roofs are usually free from such molestation, but Albert F. Ganier informs me that he has known of sparrow hawks invading the cities and preying upon the nighthawks, especially the young. Cats and dogs become enemies of the city dwellers in the event the young leave the nest prematurely and land on the streets below.

Nighthawks have a remarkable protective coloration and have developed methods of deception such as imitating a wounded individual when an enemy approaches the nesting site. At times they assume an attitude of aggression; i. e., raising their elongated wings in a vertical position and hissing in defiance at an intruder.

DISTRIBUTION

Range.—The Western Hemisphere generally, from Yukon to Patagonia.

Breeding range.—The breeding range of the nighthawk extends **north** to Yukon (Sixty-mile River and probably Lapierres House); Mackenzie

probably Fort Goodhope, (probably Fort Franklin, Fort Resolution, and Hill Island Lake); northern Saskatchewan (Methye Portage, Stanley, and Reindeer River); northern Manitoba (probably Du Brochet Lake, Grand Rapids, and probably Churchill); northern Ontario (Martin Falls, probably Fort Albany, and probably Moose Factory); and Quebec (Lake Mistassini, Godbout, and Mingan Island). The **eastern** boundary of the range extends from this point southward through the coastal regions of the Maritime Provinces, the Eastern United States, the Bahama Islands (Nassau), to Puerto Rico (Mayaguez). **South** to Puerto Rico (Mayaguez); Haiti (Hinche and Jean Rubel); Jamaica (Port Henderson and Grand Cayman); Cuba (Isle of Pines); southern Texas (Brownsville, San Antonio, Kerrville, and Pecos); Chihuahua (Babicora); Sonora (Oposura and Los Nogales); and southern California (San Bernardino Mountains and Pine Knot). **West** to California (Pine Knot, Mona Lake, Grass Valley, and Eureka); western Oregon (Medford, Eugene, Corvallis, Portland, and St. Helen); western Washington (Bumping Lake, Gig Harbor, Seattle, and Blaine); British Columbia (Errington, Courtenay, Fort St. James, and Hazelton); southeastern Alaska (Wrangell); and western Yukon (Carcross, Whitehorse, and Sixty-mile River).

Winter range.—When the numerical abundance and extensive breeding range of this species are considered, it is surprising that knowledge concerning the winter range should be extremely limited. It seems certain, however, that it is entirely in South America **north** to Colombia (Antioquia); Venezuela (Orinoco Valley); and central Brazil (Matto Grosso). **East** to southeastern Brazil (Matto Grosso, São Paulo, and Rio de Janeiro); Uruguay (Concepcion); and Argentina (Barracas and La Plata). **South** to central Argentina (La Plata and Cordova). **West** to Argentina (Cordova, Santiago, Tucuman, and the Gran Chaco); western Paraguay (Asuncion); western Brazil (Chapada); western Ecuador (Portoviejo); and western Colombia (Bogota and Antioquia).

The outline presented is for the entire species, which has, however, been separated into seven subspecies, or geographic races. The typical eastern nighthawk (*C. m. minor*) breeds in the eastern part of the continent west to the edge of the Great Plains from Oklahoma to Minnesota and in Canada to British Columbia and southern Yukon; the Florida nighthawk (*C. m. chapmani*) nests in the Gulf and South Atlantic States north to Arkansas,

southern Illinois, central Alabama, and central North Carolina and west to eastern Texas; Howell's nighthawk (*C. m. howelli*) breeds in the southern part of the Great Plains region from northeastern Utah and western Kansas south to Oklahoma, central Texas, and New Mexico; Cherrie's nighthawk (*C. m. aserriensis*) is found in south-central Texas and the Mexican state of Tamaulipas; Sennett's nighthawk (*C. m. sennetti*) occupies a breeding range in the northern Great Plains from northeastern Montana and North Dakota south to eastern Wyoming, Nebraska, and northwestern Iowa; the western nighthawk (*C. m. henryi*) breeds in southwestern Colorado, eastern Arizona, New Mexico, Chihuahua, and Sonora; and the Pacific nighthawk (*C. m. hesperis*) nests in the western part of the continent from southeastern British Columbia and California east to southwestern Saskatchewan, northwestern Wyoming, and central Utah. In no case is it possible to indicate subspecific winter ranges, but it appears that all forms winter together in South America.

Spring migration.—Early dates of spring arrival in North America are: Florida—Orlando, March 20; Palma Sola, March 29; Pensacola, April 8. Georgia—Savannah, April 1; Kirkwood, April 19. South Carolina—Frogmore, April 12; Columbia, April 17. North Carolina—Hendersonville, April 13; Raleigh, April 15. Virginia—Variety Mills, April 10; New Market, April 24. District of Columbia—Washington, April 18. Maryland—Mardela Springs, April 19. Pennsylvania—Philadelphia, April 23; Beaver, April 27. New Jersey—Morristown, May 7. New York—New York City, April 20; Ballston Spa, April 29; Rochester, May 7. Connecticut—Hartford, April 24; Jewett City, May 6. Massachusetts—Harvard, April 19; Amherst, May 10. Vermont—St. Johnsbury, May 6; Rutland, May 12. New Hampshire—South Manchester, May 2; Concord, May 14. Maine—Dover-Foxcroft, April 29; South Portland, May 1. New Brunswick—St. Johns, May 13. Quebec—Quebec City, May 4; Montreal, May 9. Louisiana—New Orleans, April 8. Mississippi—Biloxi, April 8; Rodney, April 19. Arkansas—Monticello, April 11. Tennessee—Athens, April 20. Kentucky—Lexington, April 19; Eubank, April 23. Missouri—St. Louis, April 22; Kansas City, May 2. Illinois—Chicago, April 21; Odin, April 27. Indiana—La Fayette, May 1; Fort Wayne, May 6. Ohio—Cleveland, April 20; Oberlin, April 21. Michigan—Sault Ste. Marie, April 30; Detroit, May 1. Ontario—Toronto, April 30; Ottawa, May 9. Iowa—Keokuk, April 23; National, May 6. Wisconsin—Madison, May 2; Milwaukee, May 3. Minnesota—

Minneapolis, April 26; Lanesboro, May 7. Kansas—Onaga, April 30; Manhattan, May 1. Nebraska—Omaha, May 8; Red Cloud, May 3. South Dakota—Vermillion, May 10; Forestburg, May 16. North Dakota—Grand Forks, May 1; Argusville, May 18. Manitoba—Margaret, May 10; Aweme, May 12. Texas—San Antonio, April 2; Kerrville, April 22. Colorado—Denver, May 4; Yuma, May 9. Wyoming—Cheyenne, May 8; Torrington, May 24. Montana—Great Falls, May 15; Columbia Falls, May 29. Saskatchewan—Indian Head, May 18; Eastend, May 26. Arizona—Oracle, April 20; Santa Rita Mountains, May 1. Utah—Kobe Valley, May 24. Idaho—Ruper, May 18; Meridian, May 20. Alberta—Flagstaff, May 27; Banff, May 28. California—Red Bluff, April 1; San Diego, April 22; Eureka, May 7. Oregon—Coos Bay, May 1; Klamath Lake, May 26. Washington—Tacoma, May 22. British Columbia—Burrard Inlet, May 26; Okanagan Landing, June 1.

Fall migration.—Late dates of fall departure are: British Columbia—Okanagan Landing, September 15. Washington—Tacoma, September 11. Oregon—Weston, September 9. California—San Francisco, September 21. Los Angeles County, October 27. Alberta—Banff, September 17. Idaho—Meridian, September 15; Rupert, September 16. Arizona—Chin Lee, September 25. Saskatchewan—Eastend, September 15. Montana—Columbia Falls, September 28; Great Falls, October 6. Wyoming—Yellowstone Park, September 15. Colorado—Yuma, September 25; Denver, October 14. Texas—Taylor, October 2; Palo Pinto, October 21; Corpus Christi, November 6. Manitoba—Winnipeg, September 19; Aweme, September 24. North Dakota—Charlson, September 22. South Dakota—Forestburg, September 21; Sioux Falls, October 1. Nebraska—Blue Springs, October 5; Red Cloud, October 10. Kansas—Topeka, October 6; Onaga, October 22. Minnesota—Minneapolis, September 15; Lanesboro, September 30. Wisconsin—Madison, October 6. Iowa—National, September 30; Keokuk, October 8. Ontario—Ottawa, September 23; Toronto, October 11. Michigan—Detroit, September 17. Ohio—Wauseon, September 30; Cleveland, October 8. Illinois—Chicago, October 7; Odin, October 14. Missouri—Concordia, October 18. Kentucky—Eubank, October 9. Tennessee—Athens, October 5. Arkansas—Rogers, October 4; Dardanelle, October 25. Louisiana—New Orleans, November 3. Quebec—Montreal, September 17. New Brunswick—St. Johns, September 27. Maine—Portland, September 26; Livermore Falls, October 2; Vermont—Rutland,

September 12; St. Johnsbury, September 13. Massachusetts—Amherst, September 27; Harvard, October 3. Connecticut—Hartford, September 26. New York—Rochester, September 28; New York City, October 1. New Jersey—Morristown, October 11. Pennsylvania—Renovo, September 23; Philadelphia, October 15. District of Columbia—Washington, October 11. North Carolina—Raleigh, October 6; Hendersonville, October 13. South Carolina—Columbia, October 22; Mount Pleasant, November 9. Alabama—Autaugaville, October 11. Florida—Orlando, October 13; Pensacola, October 20.

Casual records.—With a range that covers most of the Western Hemisphere, it is not surprising that comparatively few nighthawks should be recorded beyond the known normal limits. A specimen was taken by an Eskimo at Allakaket, on the Koyukuk River in northern Alaska, late in September or early in October 1923. One was seen at Grand Falls, Labrador, on May 31, 1895, and a specimen was collected at Makkovik in June 1929. The species is a fairly regular visitor at Bermuda during migration, sometimes being very common.

Egg dates.—California: 36 records, March 20 to July 26; 18 records, June 18 to July 1, indicating the height of the season.

Colorado: 11 records, May 31 to July 18.

Florida: 40 records, April 7 to July 17; 20 records, May 15 to June 3.

Georgia: 16 records, May 17 to July 13; 8 records, May 24 to June 10.

Kansas: 28 records, May 31 to July 7; 14 records, June 13 to 28.

Maine: 30 records, May 24 to July 3.

Michigan: 13 records, May 15 to July 19; 7 records, June 2 to July 3.

New York: 9 records, May 29 to July 8.

South Dakota: 9 records, June 10 to July 1.

Texas (Brownsville region): 28 records, April 5 to June 28; 14 records, May 10 to June 5.

Texas (elsewhere): 62 records, April 11 to July 29; 31 records, May 15 to June 1.

CHORDEILES MINOR HENRYI Cassin

WESTERN NIGHTHAWK

HABITS

Dr. Gross has contributed such a full life history of the eastern nighthawk that it hardly seems necessary to say much about the other subspecies, as they are all much alike in general habits.

The above name formerly covered most of the western races that are now recognized as subspecifically distinct. This race is now supposed to be confined to the southern Rocky Mountains and adjacent high plains, from Colorado southward into Mexico, Chihuahua, and Sonora. Other races are recognized in the more lowland plains.

Cassin (1862), in his original description of this bird, named it as a species in honor of Dr. T. Charlton Henry and said of it: "This bird may be distinguished from other American species by its color, which is lighter and of a different style of variegation, the prevailing tone being a dull, pale reddish and yellowish, somewhat approaching what is called buff or drab color. It is larger than *Chordeiles virginianus*."

Dr. Harry C. Oberholser (1914), in comparing it with some more recently described races, says that it is "similar to *Chordeiles virginianus howelli*, but upper surface, both ground color and markings, much darker, the latter more tawny and also coarser; lower parts posteriorly more buffy, anteriorly more rufescent." It is lighter and more brownish than *hesperis*, darker and more rufescent than *sennetti*, and lighter and more brownish than *virginianus*.

Nesting.—The nesting habits of the western nighthawk are very similar to those of the eastern bird. Dr. Edgar A. Mearns (1890) says of his experience with it in the mountains of Arizona: "I have never known this species to infringe on the territory of the Texan Nighthawk during the breeding season; each keeps to its own ground, the latter being confined to the region below the pines, and the former residing in the pines and spruces, breeding in great numbers in these limited areas. * * * Two fresh eggs were taken at Flagstaff on June 18, 1887, in a level place, bestrewn with volcanic scoria, beneath the pines."

We found only the western nighthawk in the mountains and only the Texas nighthawk in the lower valleys. After I left, Mr. Willard found a nest on July 9, 1922, at an elevation of 5,200 feet in the Huachuca Mountains, Arizona. The eggs were lying in the open on gravelly soil; just before the eggs were taken, a heavy hailstorm occurred, during which nearly 6 inches of hail fell; evidently the female had covered the eggs during the storm, or the eggs might have been broken.

Eggs.—The two eggs of the western nighthawk are indistinguishable from those of other nighthawks, showing the same variations. In shape they are between oval and elliptical-oval, usually with little or no gloss, though some incubated eggs are quite glossy. The ground color is dull white, or grayish white, rarely “dark” or “deep olive-buff”. Usually they are quite evenly marked; some are closely sprinkled with fine dots, but oftener they are covered with small spots, streaks, or small blotches of “olive-brown,” “sepia,” or “mummy brown”; occasional eggs are marked with brighter browns, such as “russet” or “hazel”; very often there are underlying spots or blotches of “pale Quaker drab.” The measurements of 19 eggs average 30.2 by 21.6 millimeters; the eggs showing the four extremes measure **33.0** by 21.1, 30.0 by **23.1**, **27.3** by 21.5, and 27.5 by **19.7** millimeters.

Food.—Mrs. Bailey (1928) says of the food of the western nighthawk in New Mexico: “Ants in large proportion and also beetles which are the adult forms of noted pests. Specimens taken at Fort Stanton—caddice flies and gnats, together with injurious insects, including ants, plant bugs, leaf hoppers, crane flies, click beetles, wood-boring and engraver beetles, clover root weevils, and nut weevils.”

Beal, McAtee, and Kalmbach (1916) say that these birds are “so expert in flight that no insects can escape them. They sweep up in their capacious mouths everything from the largest moths and dragon flies to the tiniest ants and gnats, and in this way sometimes gather most remarkable collections of insects. Several stomachs have contained fifty or more different kinds, and the number of individuals may run into the thousands.”

Dr. Mearns (1890) writes:

In our summer camp, near the summit of the Mogollon Mountains, a small beetle was annoyingly abundant, flying into our tents in great numbers during the day, and at night swarming around our log fires. As the twilight gathered, hundreds of these Nighthawks appeared upon the scene, preying upon the troublesome insects. Careless of our presence at

the fires and of the noisy hilarity of camp, they flitted through the smoke with astonishing freedom from diffidence, capturing myriads of the hated beetles, as they passed and repassed above, between, and around us, until their flickering forms were as familiar as the stirring of the pine boughs overhead, and the fanning of their wings almost as little heeded.

Field marks.—There are two other races of this species, *hesperis* and *howelli*, that are likely to be seen within the range of the western nighthawk on migrations; these three races are not easy to recognize in life, though *henryi* is lighter than *hesperis* and darker than *howelli*.

The western nighthawk can be distinguished in flight from the Texas nighthawk by the position of the white wing band; in the western nighthawk this band is about halfway between the bend of the wing and the tip; in the Texas nighthawk the white wing band is nearer the tip than the bend of the wing.

Fall.—Mrs. Bailey (1928) says that after the breeding season and in the fall the western nighthawks range up to higher altitudes in the mountains, even as high as 12,600 feet. “They all desert the State for the winter, beginning their southward journey soon after the middle of summer. They are most numerous in migration during August and have nearly all left [New Mexico] by the end of September. * * * On the return journey in the spring they are among the very latest migrants, seldom reaching northern New Mexico before May 10.”

CHORDEILES MINOR CHAPMANI Coues

FLORIDA NIGHTHAWK

HABITS

This small race occupies the southern Atlantic and Gulf States, from North Carolina to eastern Texas. In general appearance it differs but little from the eastern nighthawk, except in size. Dr. H. C. Oberholser (1914) characterizes it as “like *Chordeiles virginianus virginianus*, but decidedly smaller; upper parts averaging a little more mottled and spotted with whitish and buffy, particularly on back, wings, and scapulars, and the ground color averaging slightly less deeply blackish (more grayish or brownish); posterior lower parts usually more purely white.”

Nighthawks are tender birds and spend their winters in South America, from Colombia to northern Argentina. Even in Florida they usually do not arrive until some time in April. Arthur H. Howell (1932) says of the haunts of the Florida race: "The Florida Nighthawk frequents open pine forests, old fields, pastures, prairies, cultivated lands, marshes, and ocean beaches. Although most active early in the morning and late in the evening, the birds frequently may be seen flying about in search of food in bright sunshine."

Nesting.—Mr. Howell (1932) says on this subject: "The two eggs are deposited on the ground, with no semblance of a nest, in an open situation, often in a pasture, broomsedge field, or wood lot, or in palmetto scrub in open forest or on the prairie, or sometimes among sparse grasses on or near the ocean beaches."

Major Bendire (1895) says that Dr. William L. Ralph took several sets of eggs in Putnam County, Fla., and that here "during the breeding season at least, the Florida Nighthawk frequents mainly low, flat pine woods, especially such as have recently been burnt over, the eggs generally lying on the bare ground. Sandy soil seems to be preferred for nesting places. One set of eggs was found by him under a small orange tree in an orange grove on the side of a sandy hill; three others were taken in flat pine woods, and in one instance the eggs laid on a few fragments of charcoal left where a fallen tree had been partly burnt, between the remaining part of the tree and the stump, about 3 feet from each."

On Cat Island, off the coast of Mississippi, I found the Florida nighthawk very common in the extensive forest of tall, long-leaf pines, with which the island is largely covered. Here, on June 16, 1910, I saw two pairs with young, the old birds fluttering along the ground to toll us away from their little ones; and I found one nest with two eggs on the bare, sandy ground among the pines.

The only other nests of this subspecies that I have seen were on some low, sandy islands in Galveston Bay, Tex., where several pairs were breeding on May 4, 1923; two nests were seen, the eggs lying in slight hollows in the bare sand, in such situations as might be chosen by least terns. In some of the southern cities this nighthawk has developed the habit of nesting on the flat, gravel roofs of buildings, after the manner of the species elsewhere.

Eggs.—The eggs of the Florida nighthawk are similar to those of the eastern nighthawk but will average somewhat more heavily marked with darker colors and are somewhat smaller. The measurements of 38 eggs average 28.86 by 21.23 millimeters; the eggs showing the four extremes measure **31.75** by 21.59, 31.59 by **22.61**, **26.42** by 20.32, and 28.20 by **20.10** millimeters.

The habits of the Florida nighthawk are not essentially different from those of its northern relative; the two are not easily recognizable in life; and the two forms are seen on migrations all through the Southern States, where heavy flights often occur. R. J. Longstreet noted an unusually heavy flight at Daytona Beach on May 11, 1926; “in about an hour, from 6 to 7 p. m., he estimated that about 3,000 birds passed north over the beach” (Howell, 1932).

Heavy flights also occur on the fall migration, mainly during August, when formerly large numbers of “bullbats,” as they were locally called, were shot for food or sport. Dr. Chapman (1888), writing of Gainesville, Fla., in 1888, said: “‘Bullbat,’ or as it is more frequently termed, ‘Bat,’ shooting is here a popular pastime, great numbers being killed for food, and in August, when the birds have gathered in flocks, favorite fields may be occupied at nightfall by as many as a dozen shooters.” This bad practice is now outlawed, and very wisely, as the small bodies of the nighthawks have very little food value, and the nighthawk is one of our most useful birds; it has no objectionable food habits and is so valuable as a destroyer of troublesome and injurious insects, such as grasshoppers, beetles, gnats, mosquitoes, and the destructive cotton-boll weevil, that it ought to be rigidly protected.

CHORDEILES MINOR SENNETTI Coues

SENNETT’S NIGHTHAWK

HABITS

The nighthawks of the northern plains were given the above name by Dr. Elliott Coues (1888), who gave as the characters of the subspecies “silvery grayish-white predominating above, the white below greatly in excess of the narrow, irregular or broken, dark bars, and little or no rufous anywhere.”

It is a bird of the treeless plains and prairies, ranging from northern North Dakota and northeastern Montana south to eastern Wyoming, northwestern Iowa, and northern Nebraska. It migrates southward through Oklahoma and Texas, and probably to South America.

F. A. Patton (1924) says of its haunts in South Dakota: "The Sennetts Night Hawk, is a bird of a barren, isolated region, moving ahead and away from settlements, to a thinly settled section, preferring a dry, almost a desert region.

"Through the eighties and up until about 1890 this bird could be found much over the entire State of South Dakota. At present its range is confined to that part of the state from the Missouri River to the Black Hills, frequenting the Bad Lands section, well up in the foot hills, preferring barren knolls and stony places. Never have I seen the bird or its eggs in any way concealed by grass or vegetation."

Nesting.—The same observer writes: "A favorite nesting place is in a cattle or horse trail over a stony knoll destitute of vegetation and what sets I find are found mostly by the bird flying from beneath my saddle horse. * *

"The eggs are laid on a rock or stones. No depression or formation towards a nest whatever, the eggs are plain gray in color and so blend with the rock that though they may be almost at one's feet are difficult to see."

Eggs.—The Sennett's nighthawk lays two eggs, the usual number for the species, which are practically indistinguishable from those of the eastern nighthawk, though they may average a trifle paler with somewhat smaller markings. The measurements of 9 eggs average 31.1 by 22.3 millimeters; the eggs showing the four extremes measure **33.8** by 21.6, 31.5 by **23.3**, **29.3** by 23.1, and 29.6 by **21.5** millimeters.

Young.—Ernest T. Seton (1890) made an interesting observation on two young nighthawks, which were apparently about three days old and still retained the shell tooth on the tip of the bill; he says:

I gently touched one of them, whereupon it crouched down more closely to the ground; but its companion, rising, hissed with open beak and snapped savagely at my fingers. On being further teased they ran off, exactly in the manner of young ducks, with outstretched wings and with neck and body at an angle of 45 degrees. After running a few feet they stopped, squatted as before, and closed their eyes. This they repeated several times, but at

best they only made little progress, and each time on being overtaken the bold one was always ready to fight. * * *

In the light of these observations it seems likely that in some of the cases in which the Night-hawks are supposed to have carried off their young, the latter had really run from danger, or were led away by the parent birds.

Plumages.—Dr. Louis B. Bishop (1896) has called attention to the fact that the females of this race do not show the characters of the race as well as the males. He collected a series of 13 birds in North Dakota, of which he writes: “Three females from the same locality taken in June and July—one of them a breeding bird taken with a typical male and two eggs—are similar but with the upper parts darker and the entire lower parts tinged with buff, which becomes ochraceous-buff on the throat. Two other female Nighthawks from the same region, one taken on June 11, and the other with two eggs on June 24, are quite different, the prevailing tint of the entire plumage, except the greater wing-coverts, wings and tail, being ochraceous-buff. These birds might readily be referred to *henryi*, but all the males taken or seen during the breeding season were unmistakably *sennetti*.”

He says further: “The pale colors of the male protect him admirably, harmonizing with the dull gray of the fences and rocks, perched on which he passes the day, while the darker colors of the female render her less conspicuous when seated over her eggs on the black soil.”

Behavior.—F. A. Patton (1924) says that “these birds will perch on a hot rock in the blazing sun the hottest summer day, the rock so hot one cannot hold a hand on it. You would think they would cook, they however seem stupid, they seem to lack the pep of other Night Hawks, to get out and soar at twilight, but are a quiet retiring bird, and I have known them on a hot day to let one approach close enough to strike them with a stick.”

CHORDEILES MINOR HESPERIS Grinnell

PACIFIC NIGHTHAWK

PLATES 34, 35

HABITS

What was formerly part of the range of the western nighthawk (*henryi*) is now assigned to this more recently described race. According to the 1931

Check-list, it “breeds from southeastern British Columbia to southwestern Saskatchewan, central Montana, central Utah, and northwestern Wyoming south along the Pacific coast to northern California and in the Sierra Nevada south to the San Bernardino Mountains, southern California.”

Dr. Joseph Grinnell (1905a) named this race and described it as “most nearly resembling *Ch. v. virginianus*, but:—outer surface of closed wing grayer toned; lower tail-coverts and feathers of belly region more narrowly and sparsely dark-barred; and, in the male, white patches on throat, wings, and tail more extensive.” He remarks further:

In tone of coloration the male is much darker than in either *Ch. v. sennetti* or *Ch. v. henryi*. Altho somewhat lighter than *virginianus*, this is in the direction of pale gray and white rather than ochraceous. The extended mottling of the otherwise blackish feathers dorsally is responsible for this lighter tone, and especially notable is the extensive silvery gray mottling on the wing coverts. Yet this dorsal tone does not nearly reach the paleness of *sennetti*. * * * The [white] patch on the outer primary invades across the shaft to include the outer web; and the same is observable of the patches on the tail-feathers. * * * The female of *hesperis* is gray rather than tawny, and is thus at once distinguishable from the female of *henryi*.

Nesting.—Major Bendire (1895) writes:

A set of eggs of this subspecies was found by me on July 3, 1875, in the foothills of the Blue Mountains, some 6 miles northeast of Camp Harney, Oregon, laid among some pebbles on the bare ground under a little sage bush. The sitting bird allowed me almost to touch it, and was very reluctant to abandon its eggs, which were but slightly incubated. On my approach, it ruffled its feathers and emitted a hissing sound, resembling somewhat the spitting of a cat when mad. Their favorite nesting places in that vicinity were the crests of gravelly ridges, always selecting a well-drained spot, where the rains could not chill the young or eggs. Bare, rocky table-lands are also frequently resorted to for similar purposes, and less often the flat tops of boulders. Extensive burnt tracts also furnish favorite abiding places for them in the more northern portions of their range; in fact, in such localities they are fully as abundant as on the more open sagebrush plains.

Henry J. Rust (1911) tells of a nest that he found near Coeur d’Alene, Idaho, in an open space in a cornfield, where the scattered cornstalks and a few weeds furnished scanty shade for the young. D. E. Brown tells me that “on the prairies of Pierce County [Washington] the bird nests on small patches of gravel, where the two eggs blend perfectly and the bird itself is hard to see. In King County, I find most of the nests in burnt-over sections. The eggs are usually placed among the remains of a rotten and charred log, where both egg and bird blend with the surroundings.”

A. Dawes DuBois has sent me some notes on three nests found by him in Flathead County, Mont.; in one of these the eggs lay on a thick bed of

coarse moss which covered the rocky ground for some distance around it; this was on the top of a high foothill. Of this nest he says: "The bird was sitting when I approached, July 16. When she saw me coming she closed her eyes and appeared to be asleep. By covering the large shiny eyeballs with the lusterless lids the nighthawk makes herself much more a part of her surroundings. The eyes are not shut tight; a narrow slit is left through which to peek. While I was in the tent blind, placed near the nest, her eyes were usually wide open; but if I slipped out at the rear and looked around the corner of the tent she slowly closed her eyes until only a narrow slit remained."

Dr. Grinnell (1908) found the Pacific nighthawk "to be a common species of the Boreal and upper Transition zones" in the San Bernardino Mountains. On June 18 he flushed a female from her two eggs at an altitude of about 9,000 feet. "These were laid on the bare ground in an open place among the pines. Nothing but a few pine needles separated them from the granite gravel."

Eggs.—The eggs of the Pacific nighthawk are similar to those of the eastern nighthawk, but will average somewhat paler. The measurements of 38 eggs average 30.23 by 21.6 millimeters; the eggs showing the four extremes measure **32.5** by 21.8, 31.5 by **23.1**, and **24.6** by **19.0** millimeters.

Food.—Dr. Grinnell (1908) says: "A specimen shot at dusk, July 4, 1906, was skinned at 11 o'clock the next day. The capacious throat and gullet were found to be crammed with large winged white ants. By actual count there were forty-three of these and many of them were still alive, although it was at least fifteen hours since they had been captured by the nighthawk."

Like other nighthawks this subspecies feeds mainly during the dusk of early morning and evening, but all observers seem to agree that it hunts largely during the day, even in bright sunlight. Its food consists of a great variety of insects, such as beetles, moths, ants, grasshoppers, and other flying insects.

Voice.—This nighthawk indulges in all the characteristic notes of the species, but Mr. DuBois mentions a note that he heard on August 20, which is somewhat different; he writes: "For two weeks or more some of the nighthawks flying over the hill by the ranger station have been uttering a

click-click-click-click—rapidly repeated and continued at considerable length. It is somewhat like a series of high-pitched quacks. Other nighthawks, flying with these birds, are calling the characteristic *peënk* at the usual intervals. I suppose the rapidly uttered quacks to be the notes of the young.”

CHORDEILES MINOR HOWELLI Oberholser

HOWELL’S NIGHTHAWK

PLATE 36

HABITS

The 1931 Check-list gives the range of this subspecies as “southern Great Plains and central Rocky Mountain regions; from Wyoming to middle Texas, Colorado, Oklahoma, middle and western Kansas, northeastern Utah, northeastern New Mexico, and southwestern Nebraska, casually to North Dakota.”

In naming this bird in honor of his friend Arthur H. Howell, Dr. Harry C. Oberholser (1914) characterized it as “somewhat like *Chordeiles virginianus sennetti*, but male with upper parts more rufescent and somewhat paler, the dark brown color more rufescent, less grayish, and the light markings much more buffy or ochraceous; posterior lower parts more buffy, and the anterior dark brown areas more rufescent.”

Robert B. Rockwell writes to me that this nighthawk is common in the region about Colorado Springs and that he has seen it in various other parts of Colorado, on the plains, in the valleys, and in the mountains up to 10,000 feet altitude.

George Finlay Simmons (1925) says that it reaches about its southern limit in the vicinity of Austin, Tex., and that its habitat includes “the plains; barren fields; cowtrails in pastures; roadsides and along railroad tracks; barren gravelly ridges, and gravelly surfaces in the open, barren hills; backland plowed fields; rocky hillsides; edges of woodlands; flat, gravelly roofs of office buildings in town.”

I cannot find anything to indicate that the habits of Howell’s nighthawk differ essentially from those of the other neighboring subspecies. Mr.

Simmons (1925) says of the fall migration in Texas: "Gathers in flocks in late summer, mid-July to third week in August; migration begins early in August, and by the end of that month the local breeding birds have departed; hundreds pass through, August 10 to August 30, but the last have not gone until late in October. During one fall migration flocks containing as many as 200 to 300 birds were observed on the open prairies northeast of Austin. During both migration seasons the various subspecies of nighthawks mix up on their way to and from South America, so that identification of migrants is hopeless without a specimen in hand."

The eggs are similar to those of other nighthawks. The measurements of 26 eggs average 30.2 by 21.7 millimeters; the eggs showing the four extremes measure **33.0** by 21.9, 31.7 by **23.4**, and **26.3** by **20.2** millimeters.

CHORDEILES MINOR ASERRIENSIS Cherrie

CHERRIE'S NIGHTHAWK

HABITS

George K. Cherrie (1896) described this race from a bird collected on November 2, 1893, in the valley of the River Aserri, Costa Rica. It is a small, pale race, about the size of *chapmani*. Nothing was known about its distribution at that time, but it has since been found to be the breeding form of southern Texas, from San Antonio southward and into Tamaulipas.

Dr. H. C. Oberholser (1914) describes it as "resembling *Chordeiles virginianus sennetti*, but decidedly smaller; general tone of upper parts lighter and more ochraceous, the light areas more buffy or ochraceous (less grayish or whitish) and somewhat more extensive; and the dark brown areas of anterior lower surface less grayish." He remarks further: "This heretofore unrecognized race seems to be in color most nearly like *Chordeiles virginianus sennetti*, though in this respect also near *Chordeiles virginianus howelli* and probably in reality most closely resembling this form, which it adjoins geographically."

Mr. Cherrie's type specimen seems to have been lost; at least, neither he nor Dr. Oberholser have been able to locate it, but the full description of the type convinced Dr. Oberholser that "there is no doubt that Mr. Cherrie had in hand a male nighthawk of the *virginianus* style in fully grown juvenal or

first autumn plumage, not adult, as he supposed. Allowing for this immaturity, the characters that Mr. Cherrie gives are just those distinguishing the small, pale race which summers in southern Texas.”

The nesting habits, eggs, food, and behavior of this nighthawk probably do not differ materially from those of other races living in similar environment. The measurements of 43 eggs average 29.3 by 21.4 millimeters; the eggs showing the four extremes measure **32.5** by **22.5**, **27.1** by 21.3, and 28.4 by **20.6** millimeters.

Plates

PLATE 1
GROOVE-BILLED ANI.



Tela, Honduras, July 5, 1930.

Female on nest.

A. F. Skutch.



Ulúa Valley, Honduras, August 5, 1930.

A. F. Skutch.

Nest of two pairs.

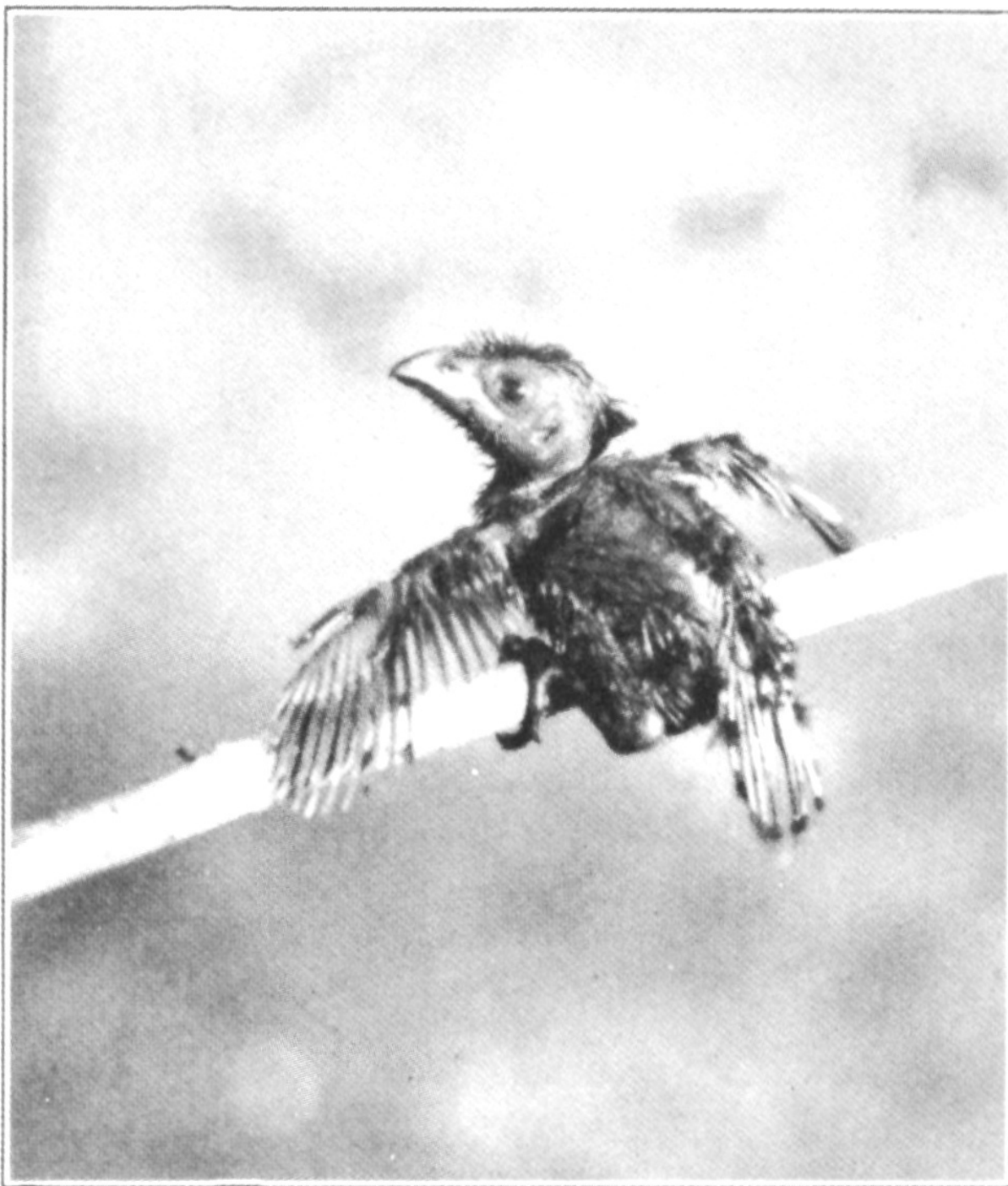
PLATE 2
GROOVE-BILLED ANI.



Tela, Honduras, July 21, 1930.

A. F. Skutch.

Young 11 days old.



Tela, Honduras, August 15, 1930.

A. F. Skutch.

Young about 6 days old.

PLATE 3
ROADRUNNER.



Fairbank, Ariz., May 18, 1922.

A. C. Bent.

Nest in a willow.



San Diego County, Calif., April 9, 1929.

A. C. Bent.

Nest in a pricklypear.

PLATE 4
ROADRUNNERS.



Adult taking lizard to feed young.

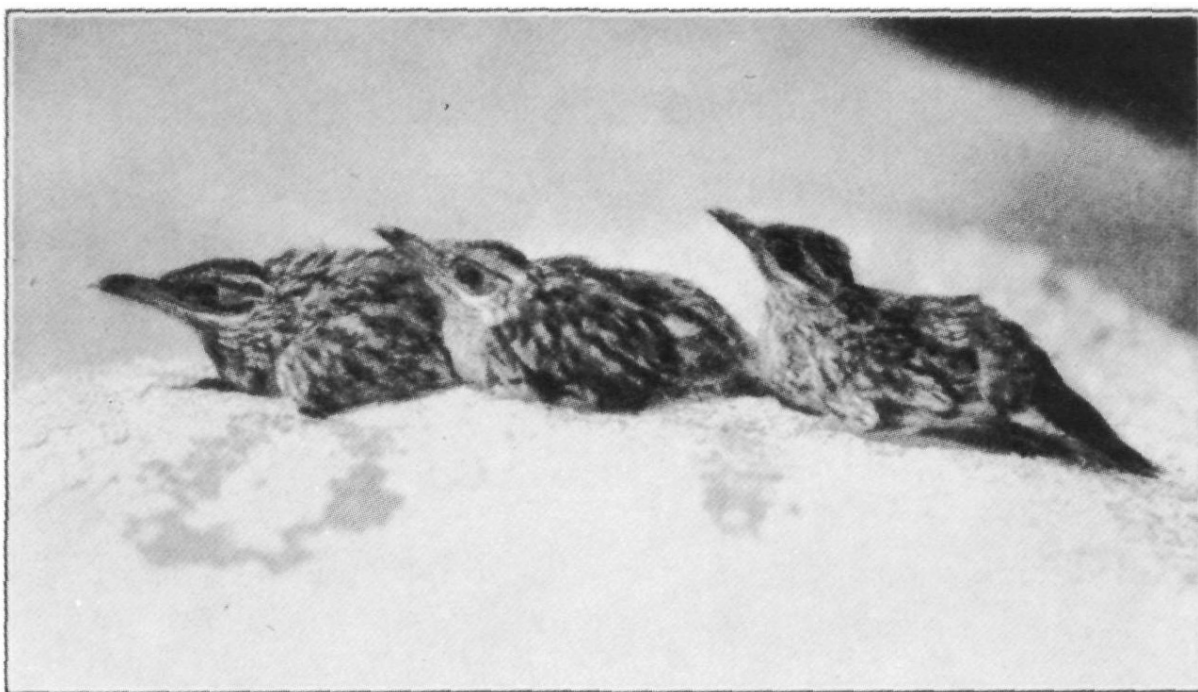


Arizona.

W. L. and Irene Finley.

Small young in nest.

PLATE 5
ROADRUNNERS.



Mojave Desert, Calif., May 28, 1916.

W. M. Pierce.

Young nearly grown.



Arizona.

Adult on nest.

W. L. and Irene Finley.

PLATE 6



Logan County, Ill., June 21, 1913.

A. D. DuBois.

NEST OF YELLOW-BILLED CUCKOO.



Arizona.

F. C. Willard.

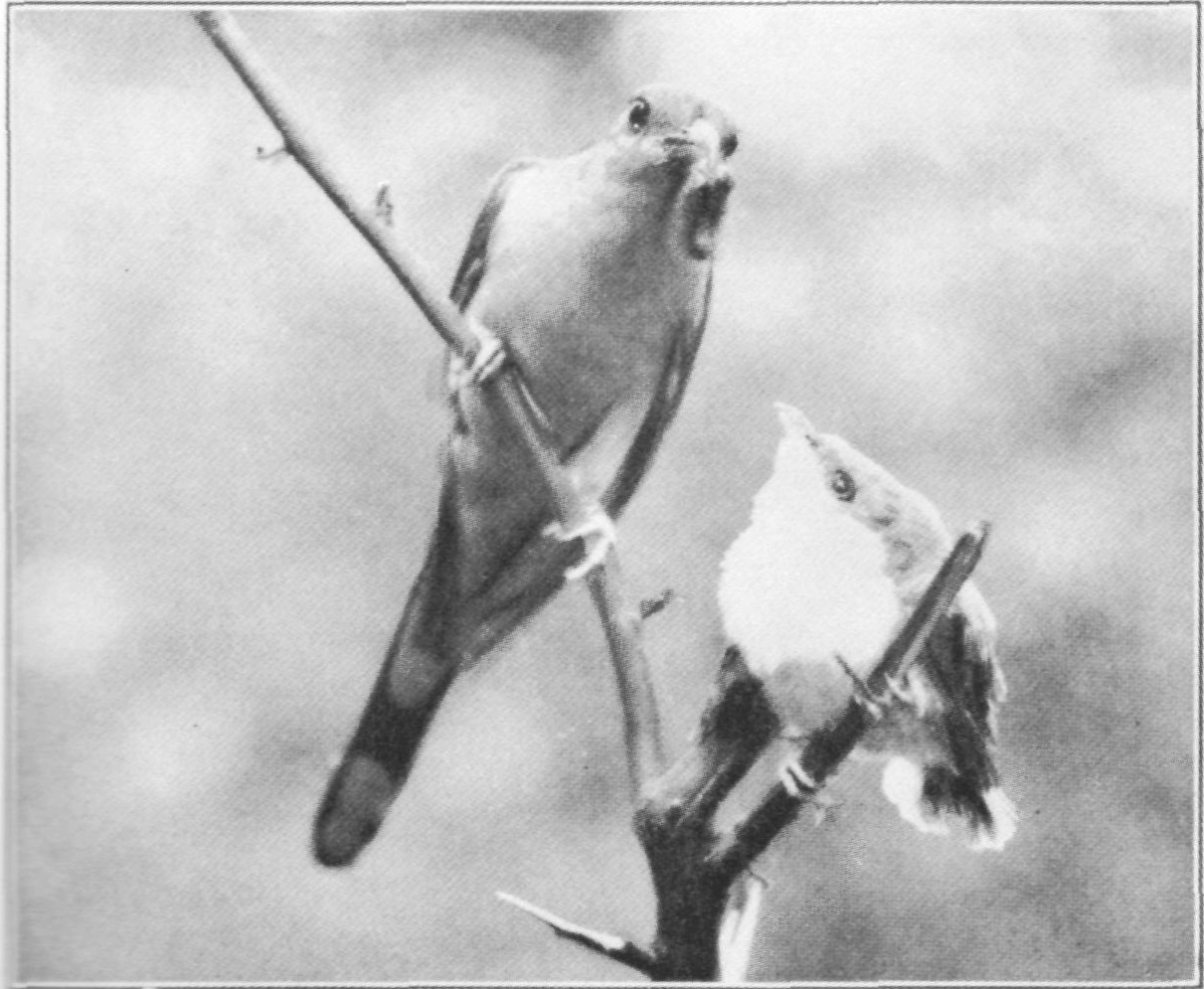
NEST OF CALIFORNIA CUCKOO.

PLATE 7
YELLOW-BILLED CUCKOOS AND YOUNG.



Battle Creek, Mich., August 30, 1936.

L. H. Walkinshaw.



Ithaca N. Y.

A. A. Allen.

PLATE 8
NEST OF BLACK-BILLED CUCKOO.

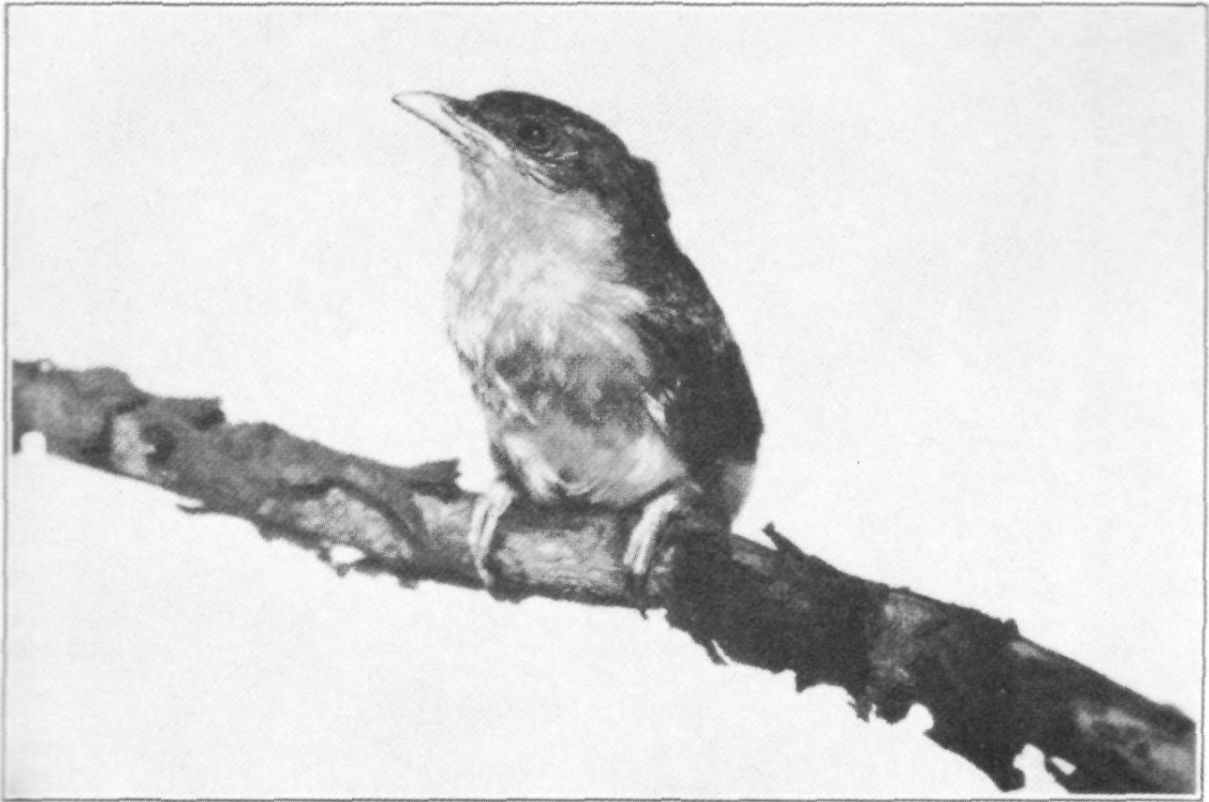




Sioux City, Iowa.

T. C. Stephens.

PLATE 9
BLACK-BILLED CUCKOO.



Omaha, Nebr., July 1, 1901.

F. H. Shoemaker.

Young in juvenal plumage.



Hennepin County, Minn., June 11, 1935.

A. D. DuBois.

A well-made nest.

PLATE 10
YOUNG BLACK-BILLED CUCKOOS.



Showing the sucking pads.



Ithaca, N. Y.

A. A. Allen.

Showing the unopened feather sheaths.

PLATE 11
ADULT BLACK-BILLED CUCKOO ON NEST.



Allen Frost.

PLATE 12
EASTERN BELTED KINGFISHERS.



Carver County, Minn., June 29, 1936.

A. D. DuBois.

Feeding young at entrance.



Ithaca, N. Y.

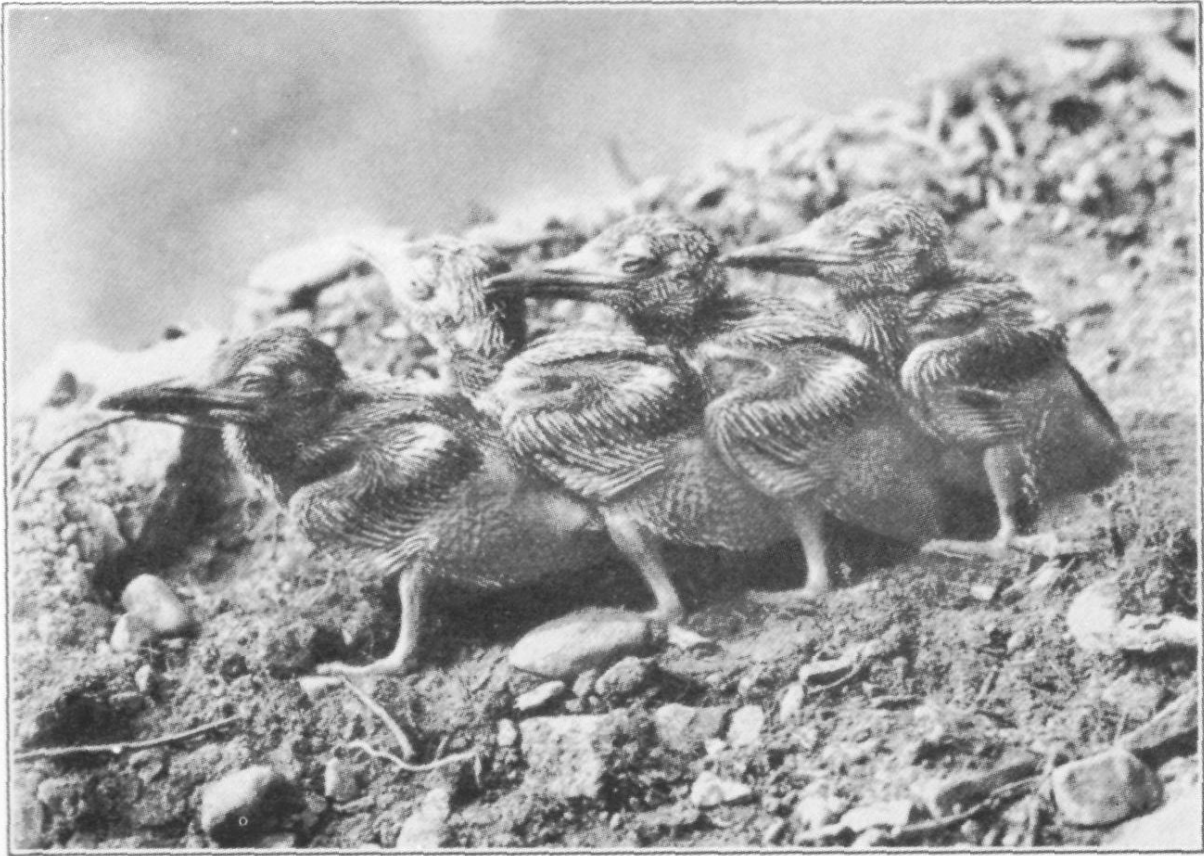
A. A. Allen.

Brooding newly hatched young.

PLATE 13
EASTERN BELTED KINGFISHERS.



Young just hatched.



Ithaca, N. Y.

A. A. Allen.

Older young.

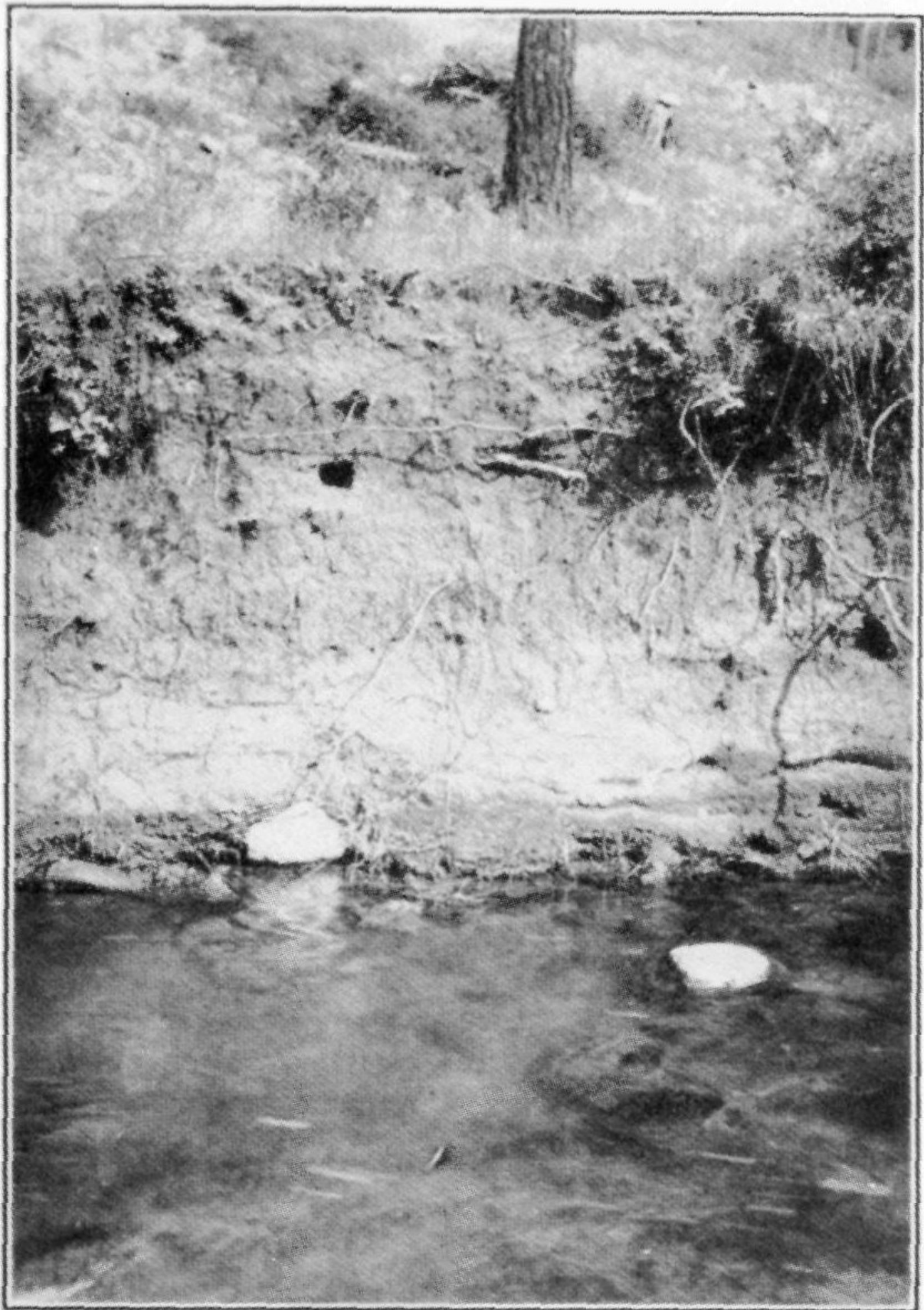
PLATE 14
EASTERN BELTED KINGFISHER.



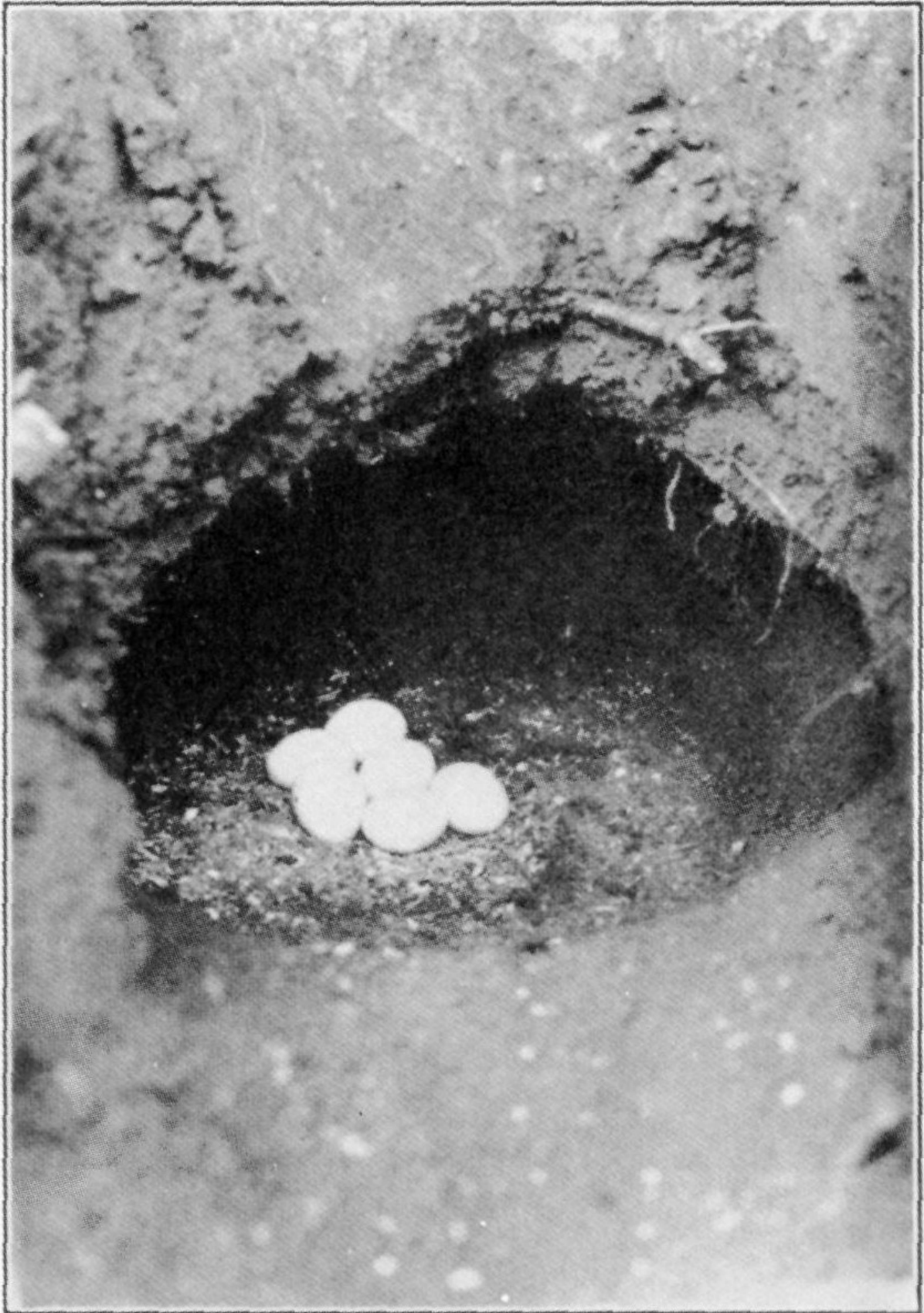
Ann Arbor, Mich.

F. N. Wilson.

PLATE 15
NESTING OF WESTERN BELTED KINGFISHER.

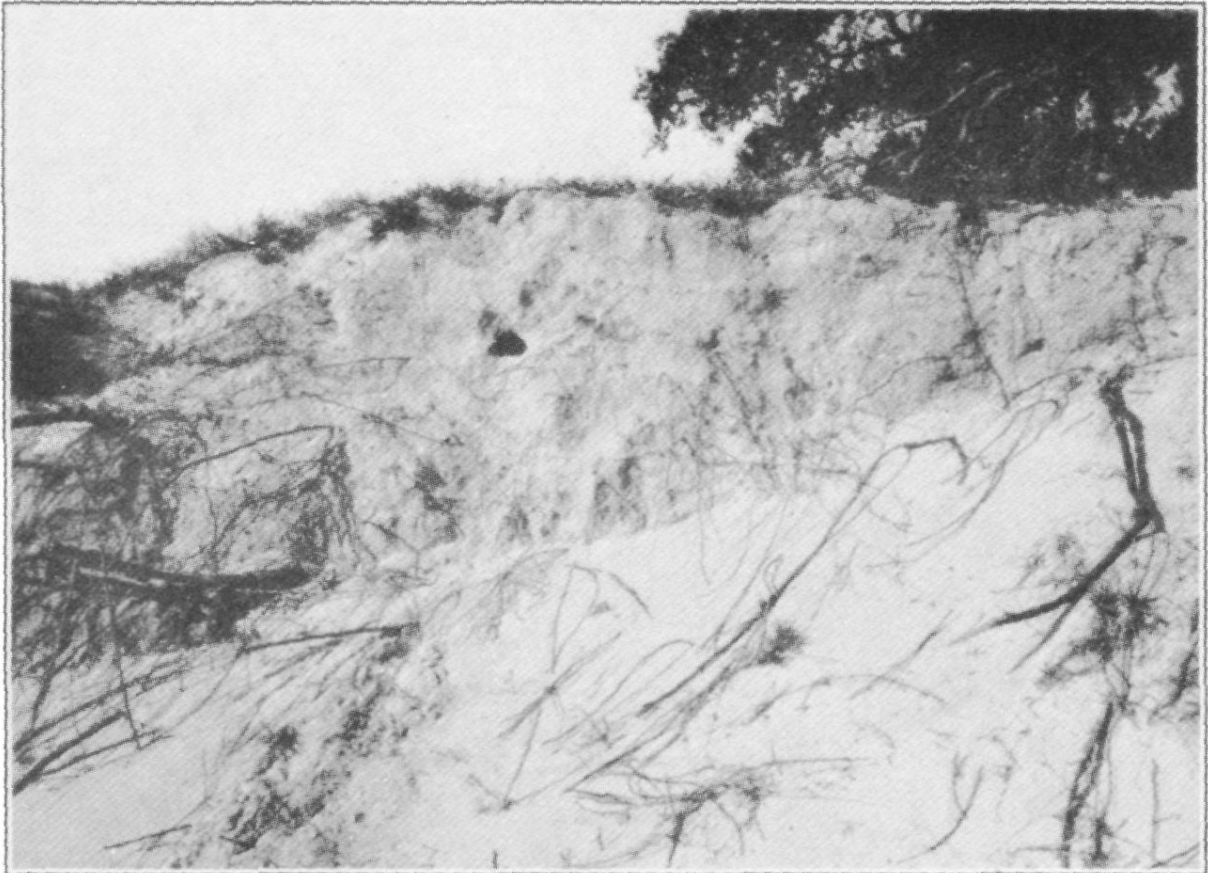


Jackson County, Oreg., May 24, 1931.



J. E. Patterson.

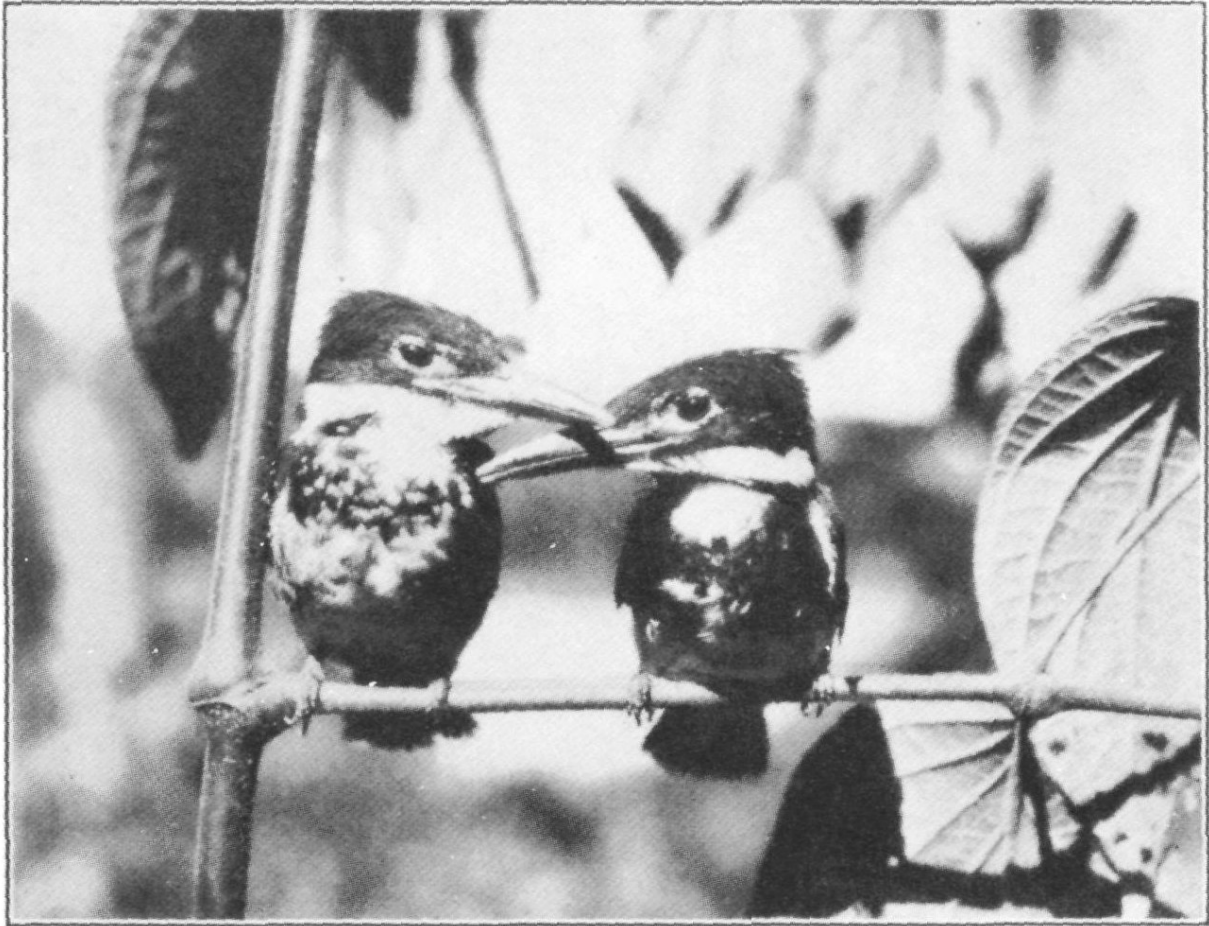
PLATE 16
TEXAS KINGFISHERS.



Copana Bay, Tex.

G. F. Simmons.

Nesting site.

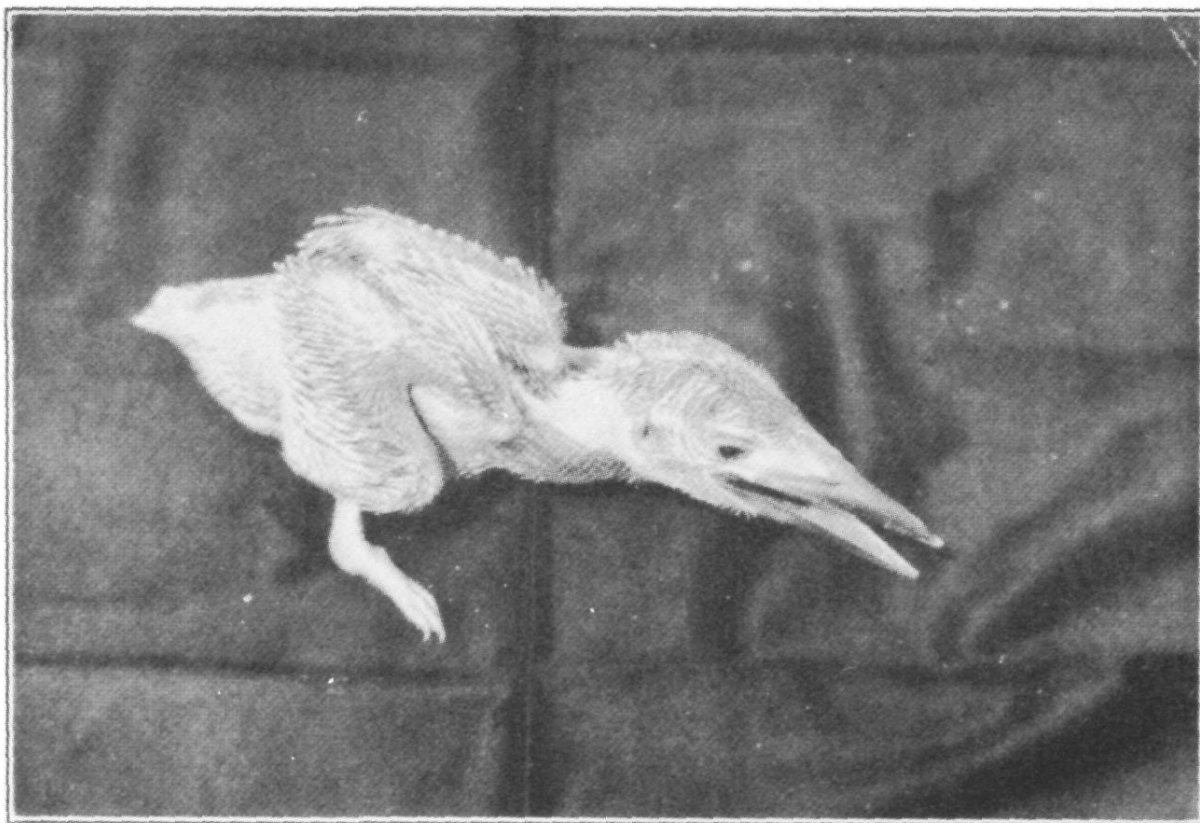


Near Los Amates, Guatemala, May 5, 1932.

A. F. Skutch.

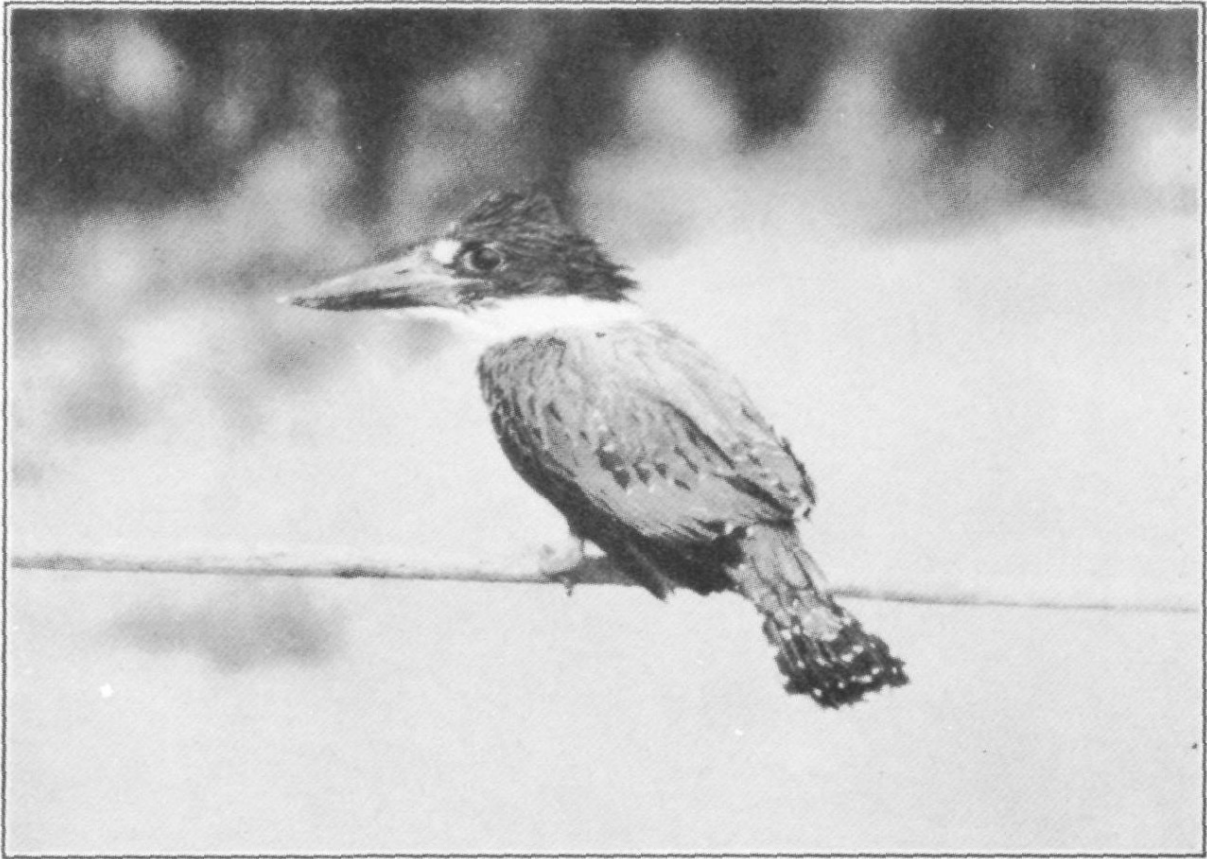
Young ready to leave nest.

PLATE 17
YOUNG RINGED KINGFISHERS.



April 10, 1932.

Nestling 2 weeks old.



Near Los Amates, Guatemala, April 26, 1932.

A. F. Skutch.

Juvenal 29 days old.

PLATE 18
NESTING OF CHUCK-WILL'S-WIDOW.





Near Dania, Fla., April 14, 1927.

D. J. Nicholson.

PLATE 19
CHUCK-WILL'S-WIDOW.



Plant City, Fla., May 1934.

A. M. Bailey.
Courtesy of Chicago Academy of Sciences.

Female and young.

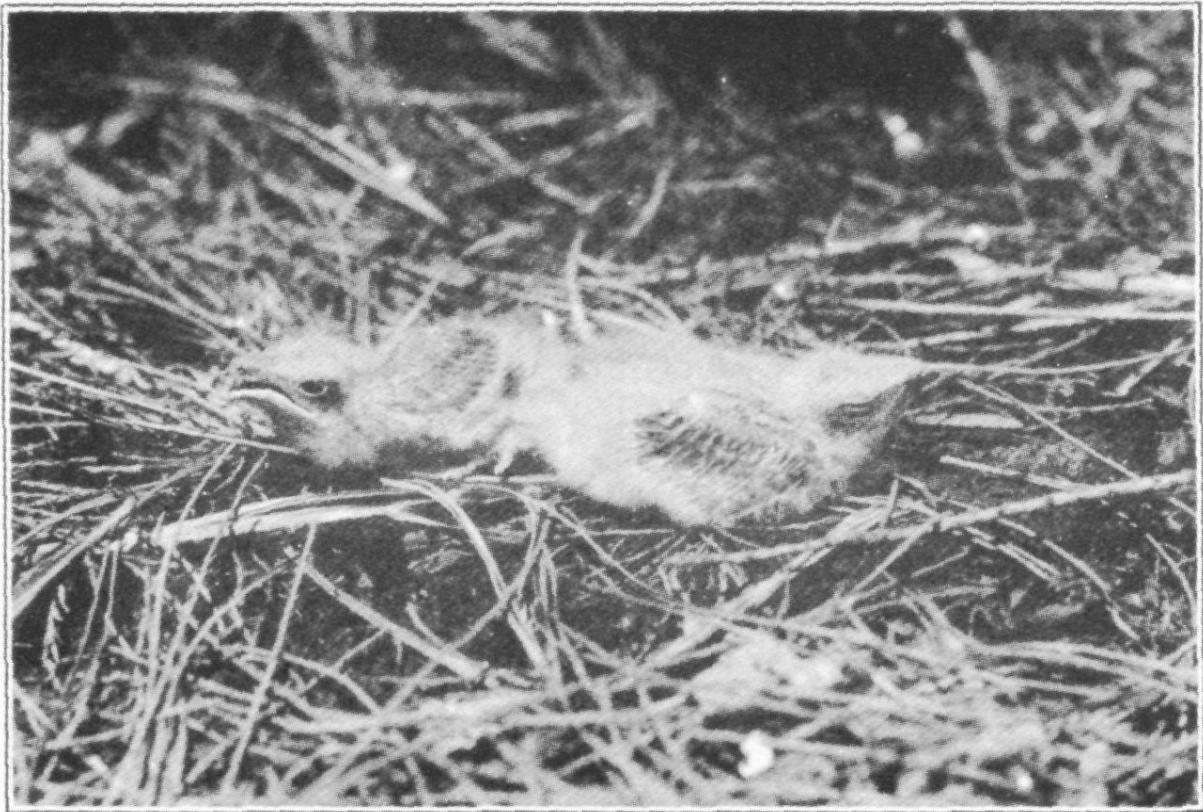


Miami County, Kans., May 24, 1929.

Walter Colvin.

Eggs in nest.

PLATE 20
YOUNG CHUCK-WILL'S-WIDOWS.]



Duval County, Fla., May 22, 1932.

S. A. Grimes.



Dade County, Fla.

F. N. Irving.

PLATE 21
NESTING OF EASTERN WHIPPOORWILL.

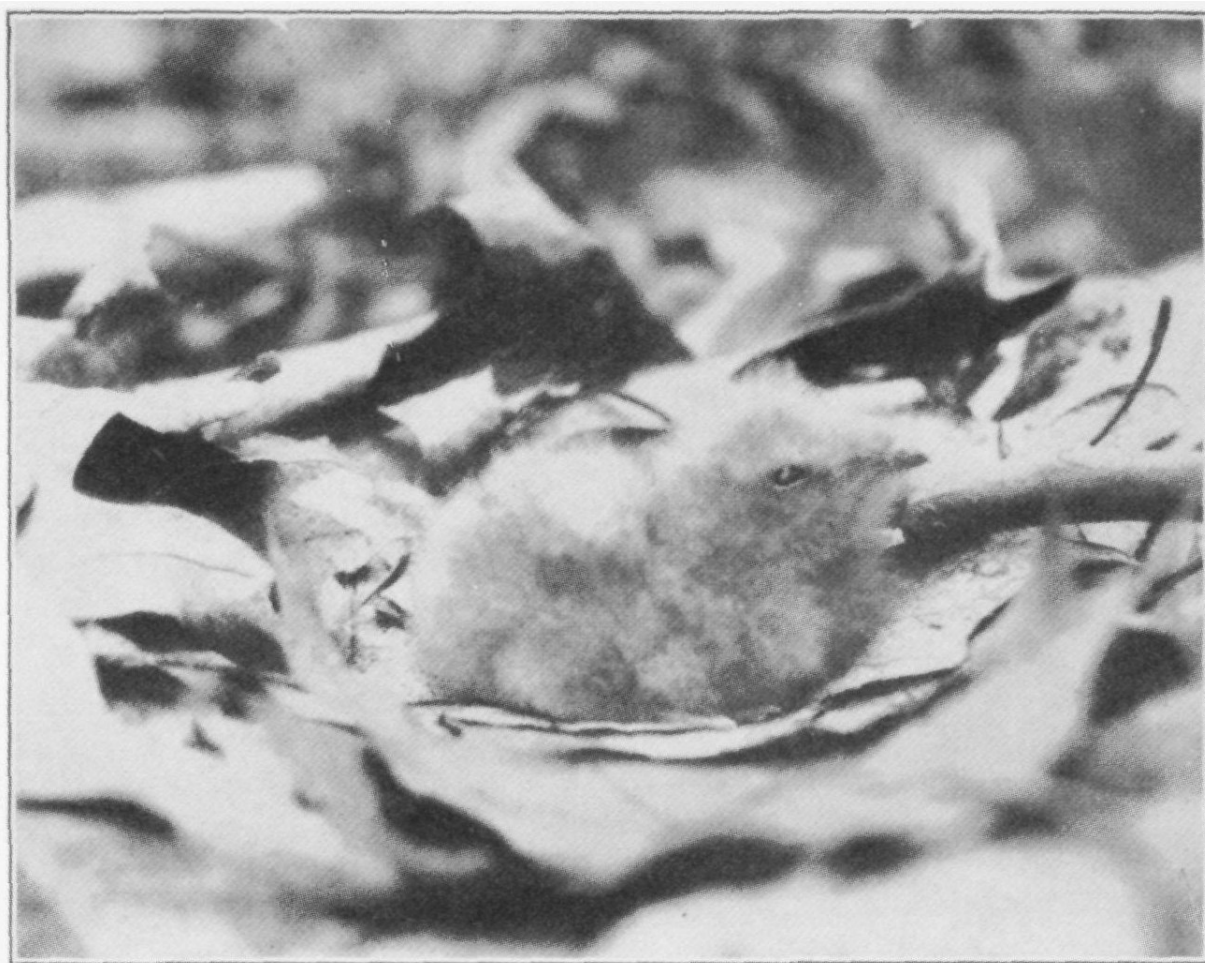




Raynham, Mass., June 5, 1903.

A. C. Bent.

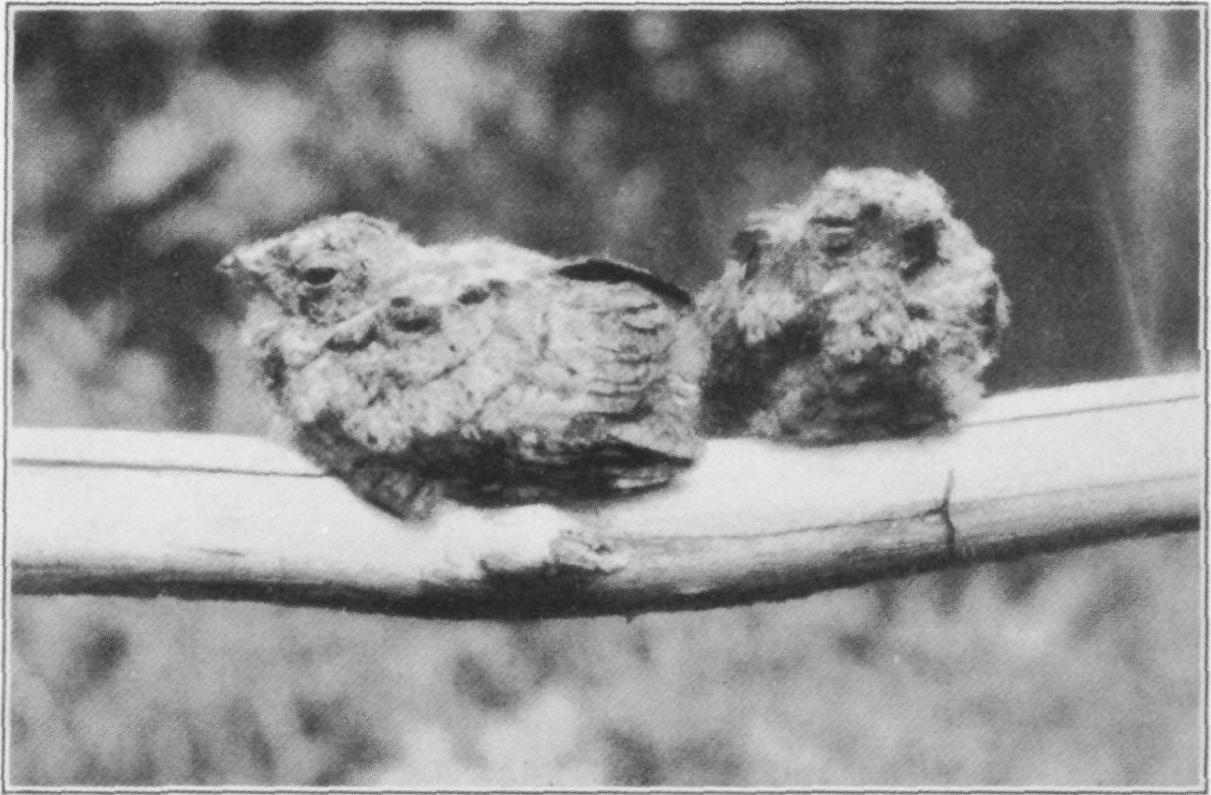
PLATE 22
EASTERN WHIPPOORWILLS.



Sangamon County, Ill., May 17, 1908.

A. D. DuBois.

Downy young.



St. Lambert, Quebec.

Young 16 days old.

L. M. Terrill.

PLATE 23
EASTERN WHIPPOORWILL.



St. Lambert, Quebec, June 11, 1933.

Female brooding young.

L. M. Terrill.

PLATE 24
NESTS OF STEPHENS'S WHIPPOORWILL.



Huachuca Mountains, Ariz.

F. C. Willard.



Santa Rita Mountains, Ariz., June 27, 1932.

A. J. van Rossem.

PLATE 25
NESTS OF DUSKY POORWILLS.



Mariposa County, Calif., June 21, 1933.

J. E. Patterson.



Stanislaus County, Calif., June 23, 1934.

J. E. Patterson.

PLATE 26
DUSKY POORWILL ON NEST.



Stanislaus County, Calif., June 23, 1934.

J. E. Patterson.

PLATE 27
EGGS AND YOUNG OF DUSKY POORWILL.



San Bernardino County, Calif.

W. M. Pierce.



Longbard, Calif.

E. C. Aldrich.

PLATE 28
PAURQUE (NYCTIDROMUS ALBICOLLIS ALBICOLLIS).



May 20, 1930.

Nest and eggs.

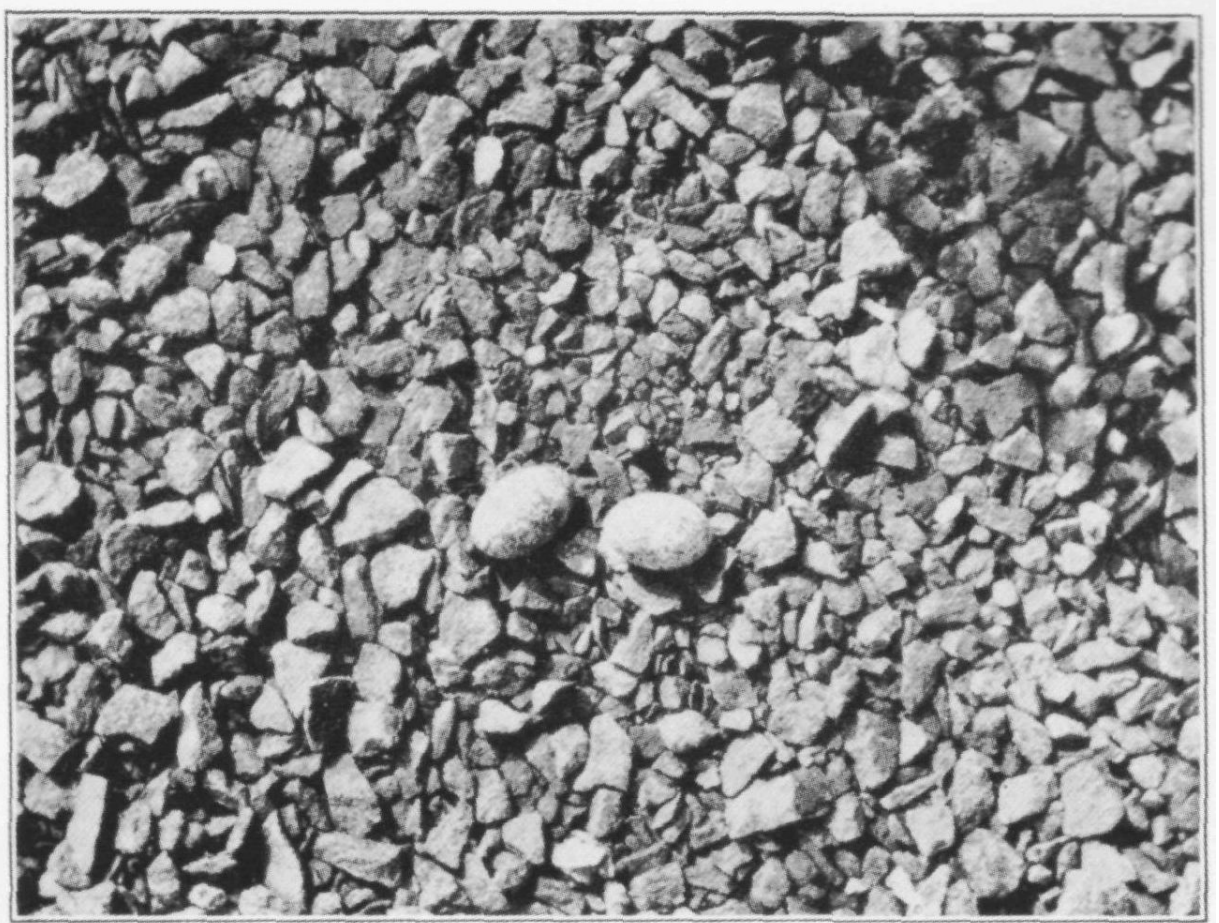


Lancetilla Valley, Honduras, May 22, 1930.

A. F. Skutch.

Young 1 and 2 days old.

PLATE 29
EASTERN NIGHTHAWK.



Michigan, June 18, 1924.

K. Christofferson.

Roof nest.

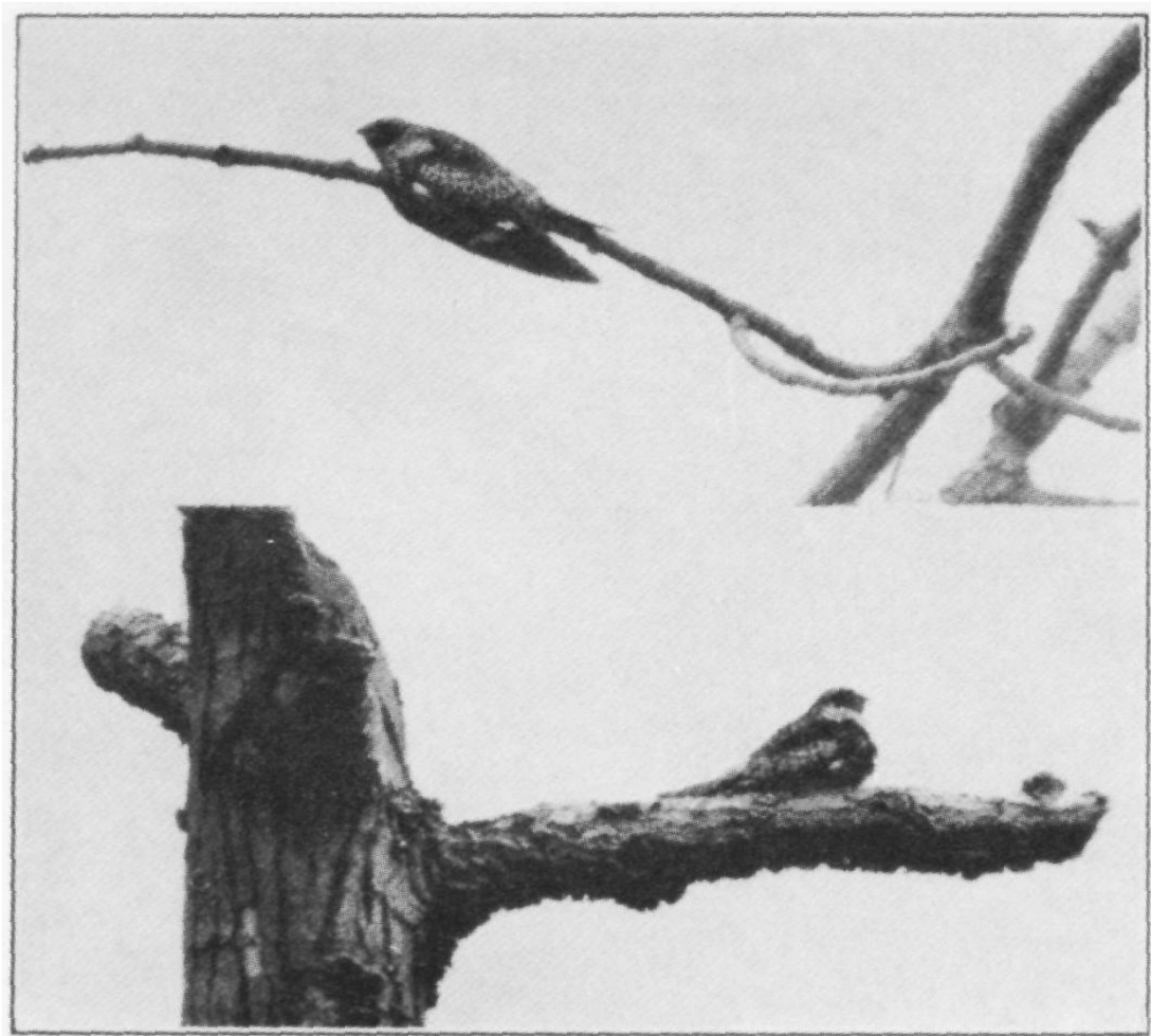


Near Church Creek, Md.

F. R. Smith.

Ground nest.

PLATE 30
EASTERN NIGHTHAWK.



Roosting poses.

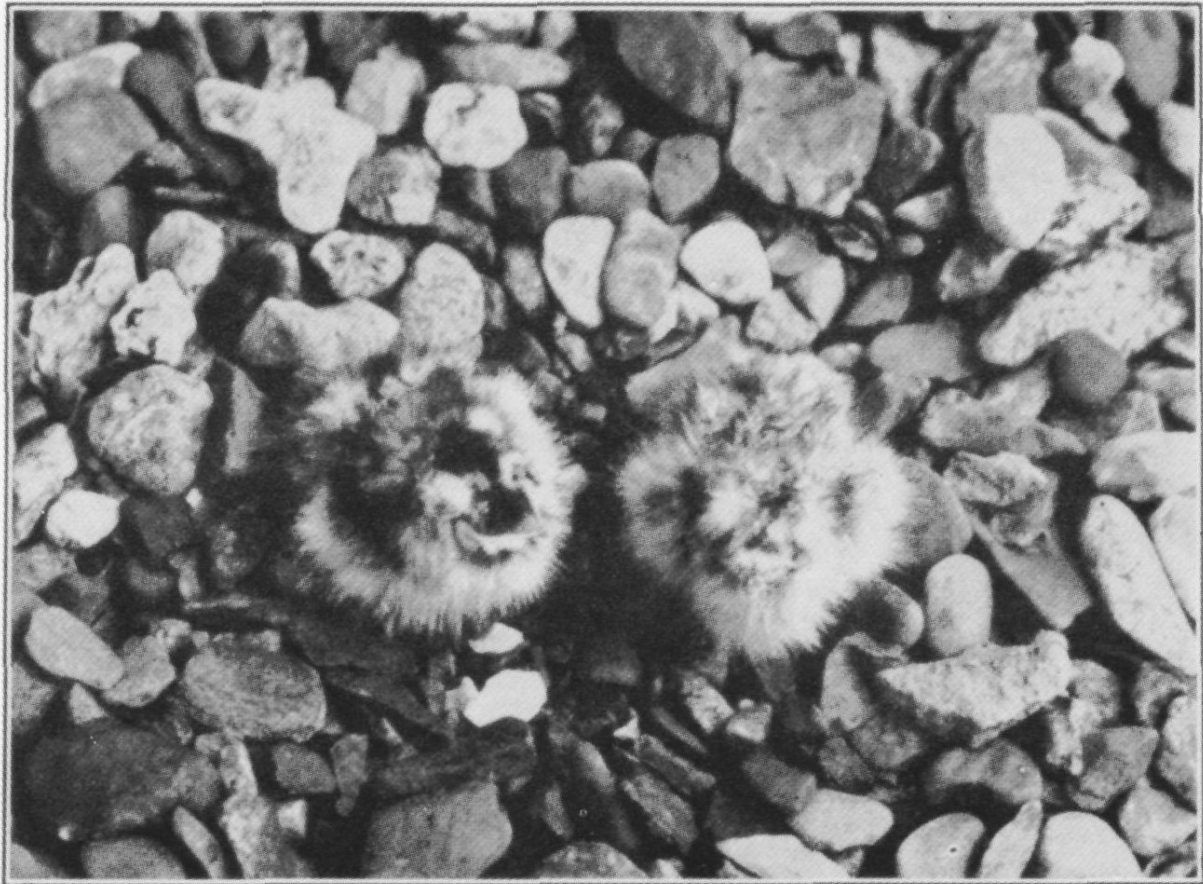


Charlevoix, Mich., June 20, 1925.

W. E. Hastings.

A well-made ground nest.

PLATE 31
EASTERN NIGHTHAWKS.



Brunswick, Maine, June 25, 1921.

Young 1 day old.

A. O. Gross.

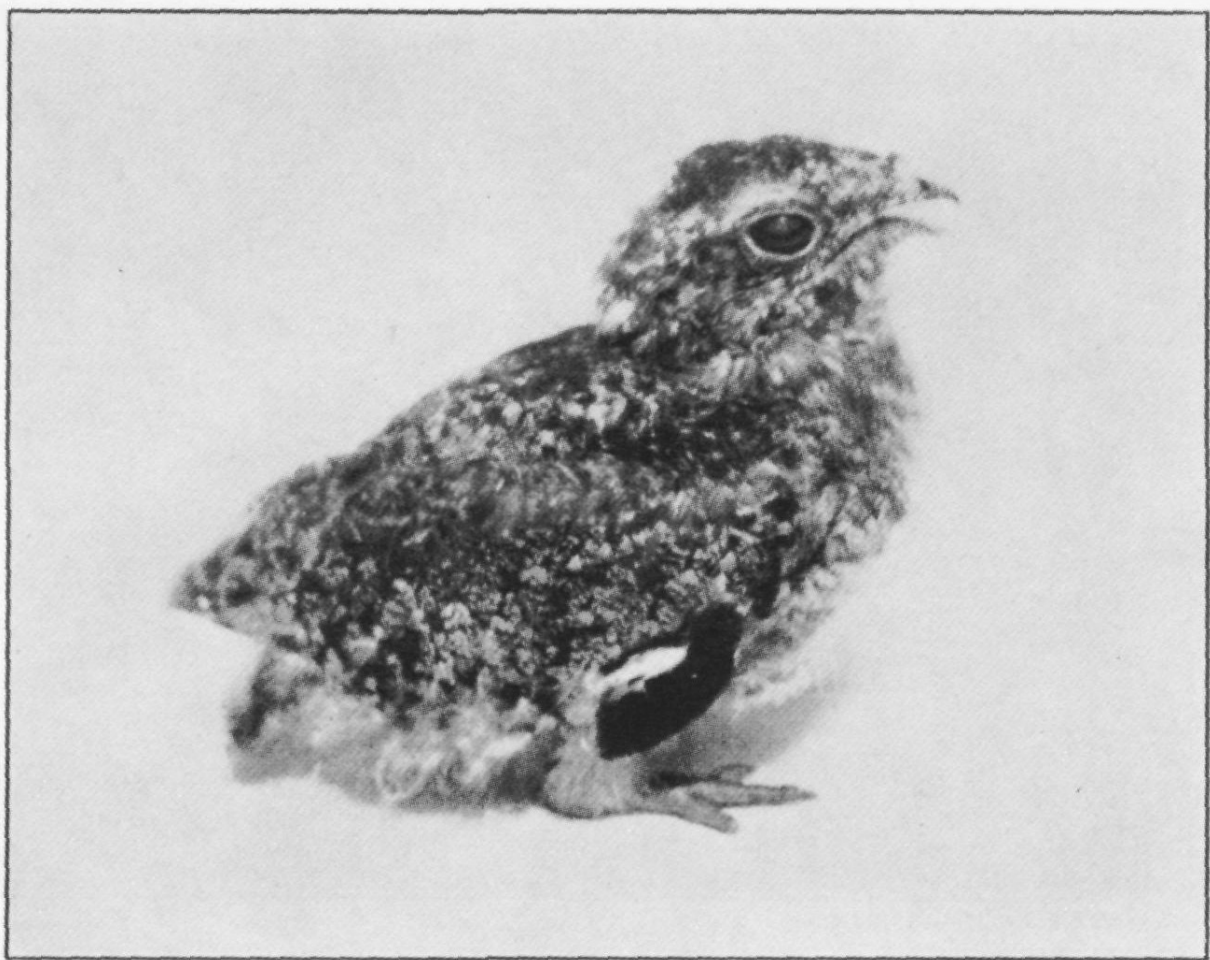


Brunswick, Maine, June 27, 1922.

A. O. Gross.

Young 12 days old.

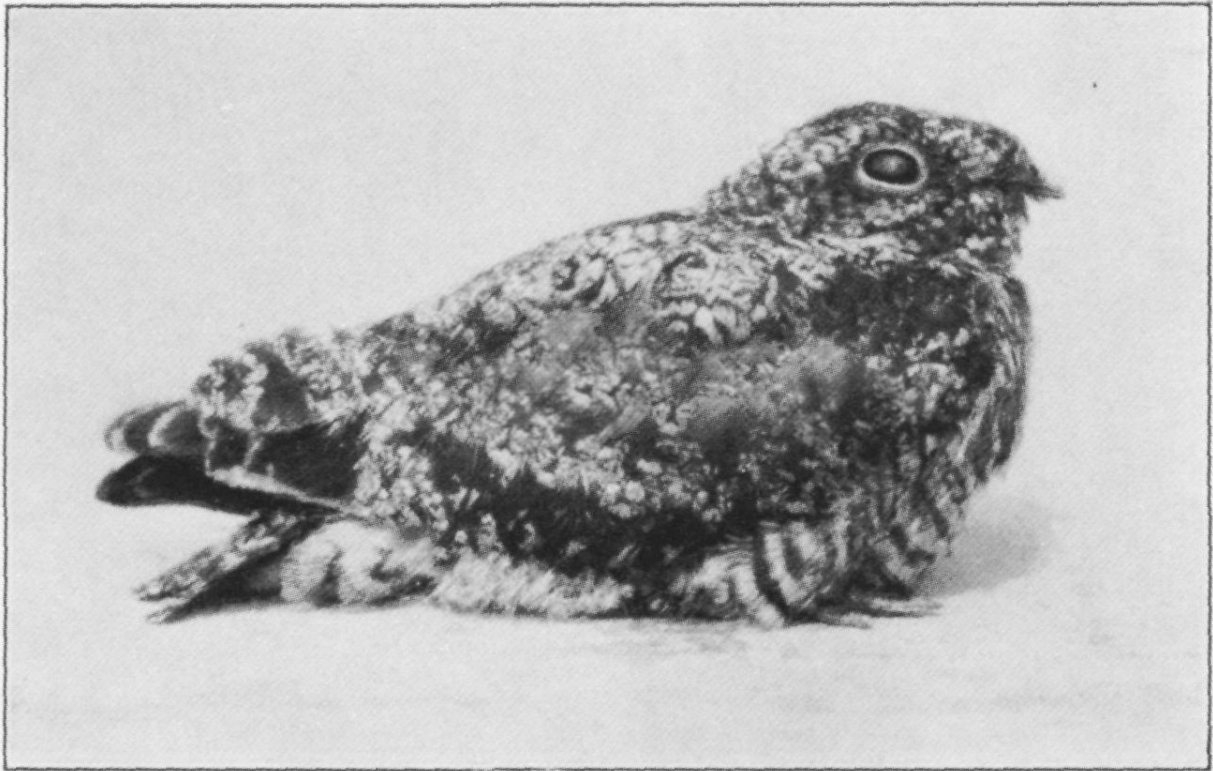
PLATE 32
EASTERN NIGHTHAWKS.



Brunswick, Maine, July 11, 1921.

Young 17 days old.

A. O. Gross.



Brunswick, Maine, July 18, 1921.

A. O. Gross.

Young 24 days old.

PLATE 33
EASTERN NIGHTHAWKS.



Brunswick, Maine, June 16, 1922.

A. O. Gross.

Adult male roosting.



Brunswick, Maine, June 16, 1921.

A. O. Gross.

Adult female incubating.

PLATE 34
NESTING SITES OF PACIFIC NIGHTHAWKS.



Jackson County, Oreg., June 22, 1924.

J. E. Patterson.



Flathead County, Mont., June 30, 1914.

A. D. DuBois.

PLATE 35
PACIFIC NIGHTHAWK.



Flathead County, Mont., July 12, 1914.

A. D. DuBois.

PLATE 36
NESTING OF HOWELL'S NIGHTHAWK.



July 4, 1911.



North Park, Colo., June 28, 1911.

E. R. Warren.

Transcriber's Note:

Words may have inconsistent hyphenation in the text. These have been left unchanged. Three asterisks separated by spaces indicates text was omitted by the author or publisher.

Most of the plates contain two photographs. The plate number and title were combined as a header. Details specific to each photograph are presented as captions below the image.

Obvious printing errors, such as backwards, upside down, reversed order, missing or partially printed letters and punctuation, were corrected. Final stops missing at the end of sentences, abbreviations and plate identifications were added. Duplicate letters at line endings or page breaks were removed. Twelve misspelled words were corrected.

Changes:

- Page numbers in the Table of Contents were adjusted to match related text.

- Added 'Plates' to the Table of Contents

- Added missing word 'by' to [protected from invasion by the wasps](#).

- Changed 'South' to '[East](#)' in description of breeding range of chuck-will's-widow.

*** END OF THE PROJECT GUTENBERG EBOOK LIFE HISTORIES
OF NORTH AMERICAN CUCKOOS, GOATSUCKERS,
HUMMINGBIRDS AND THEIR ALLIES, PART 1 [OF 2] ***

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